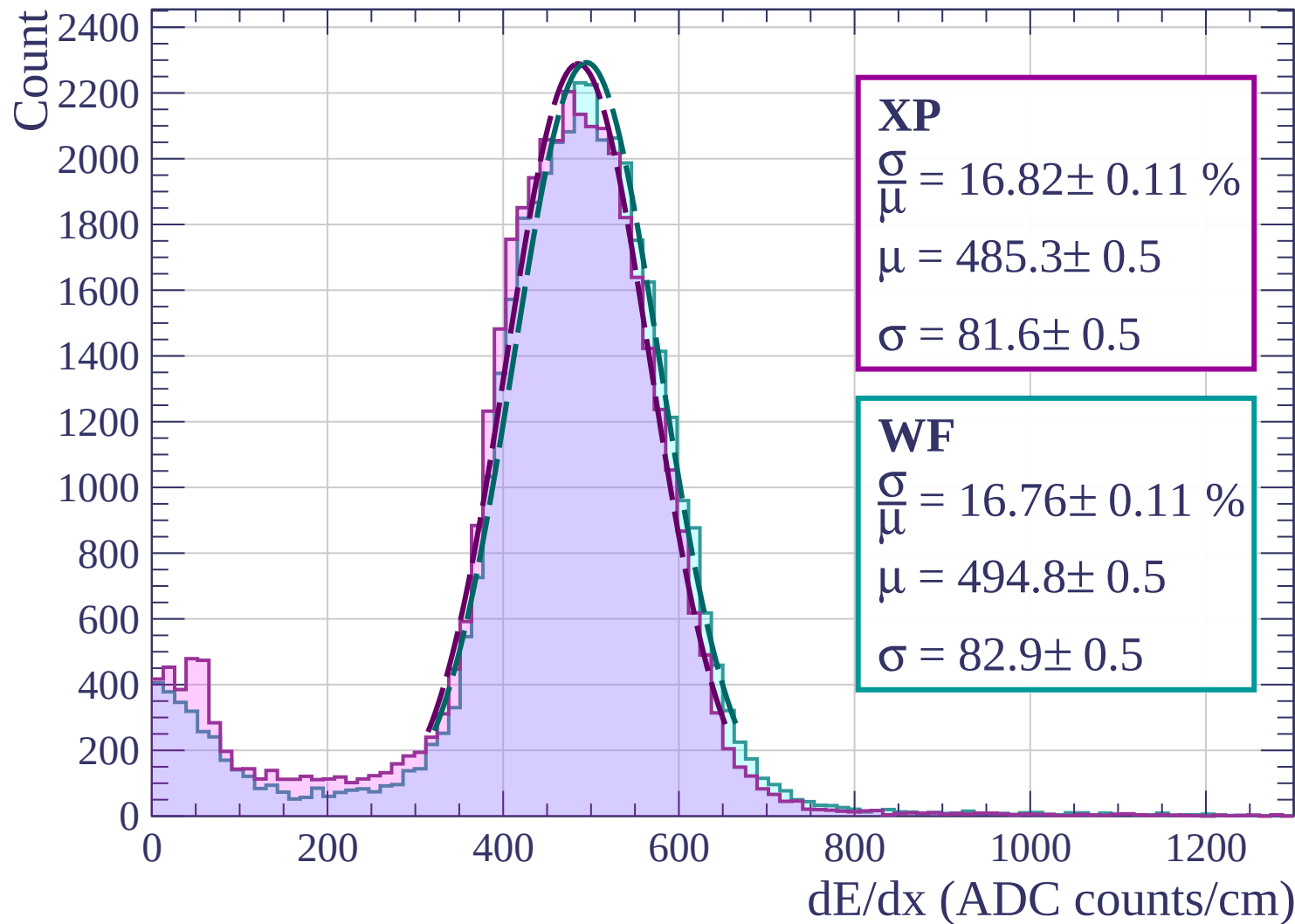
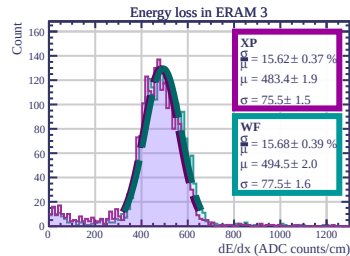
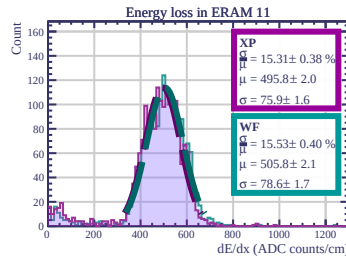
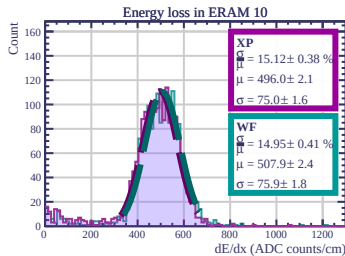
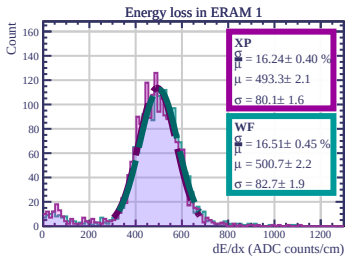
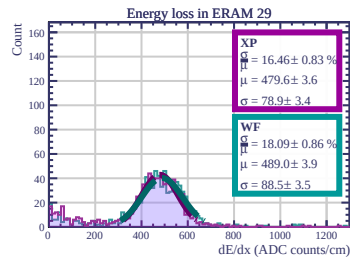
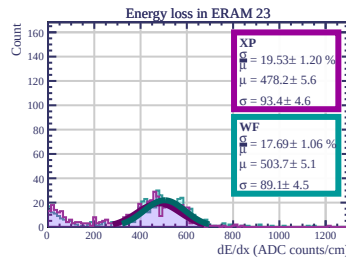
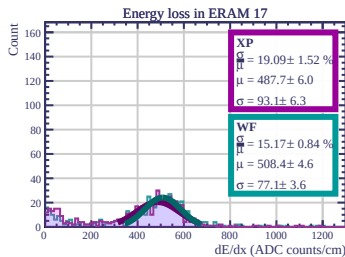
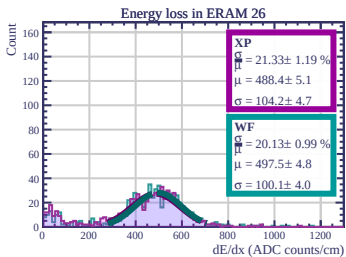
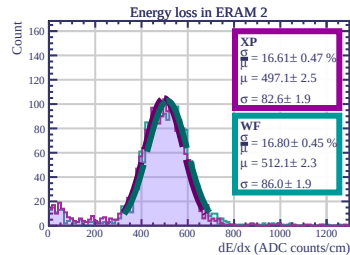
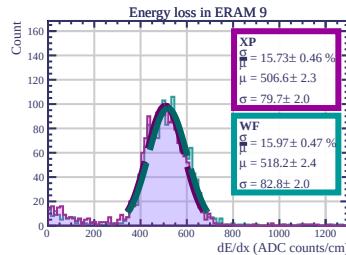
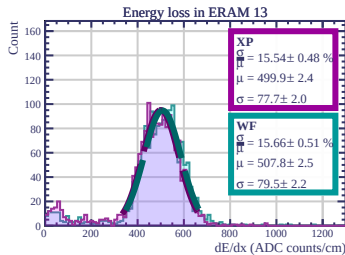
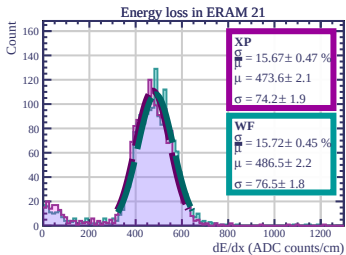
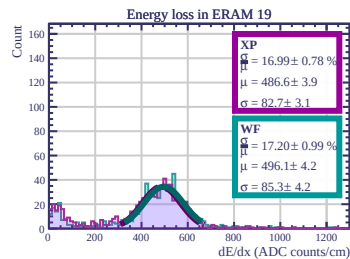
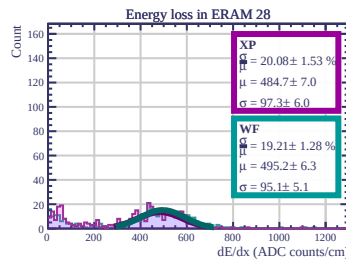
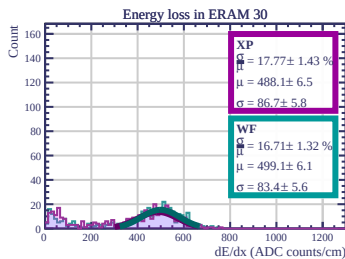
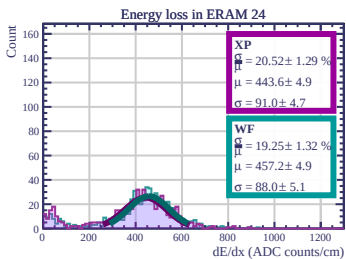
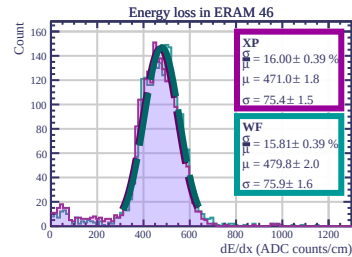
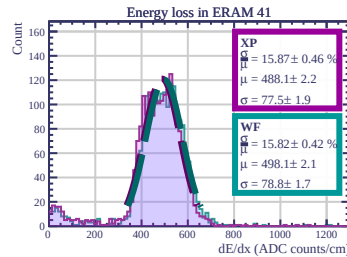
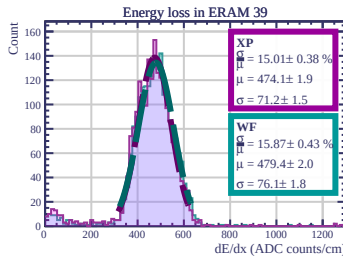
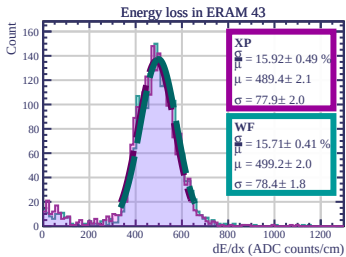
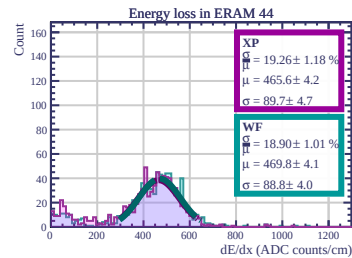
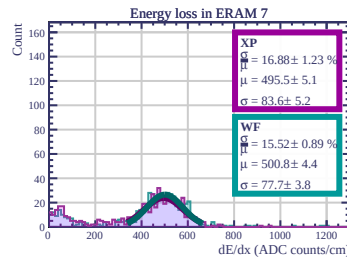
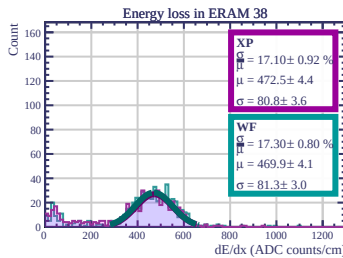
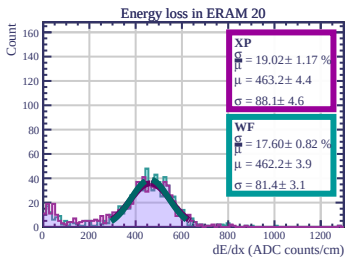
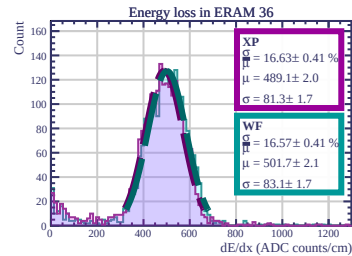
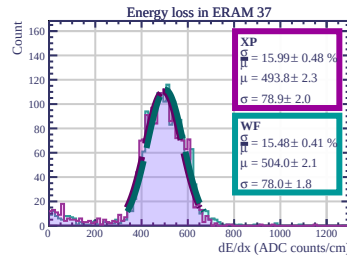
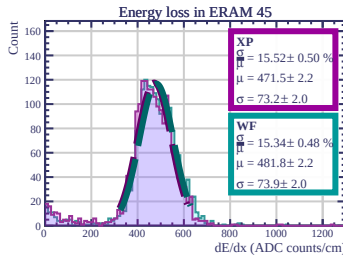
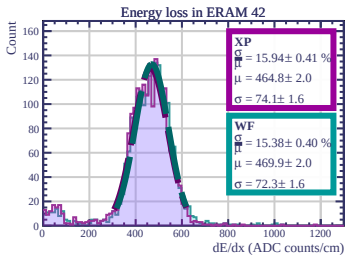
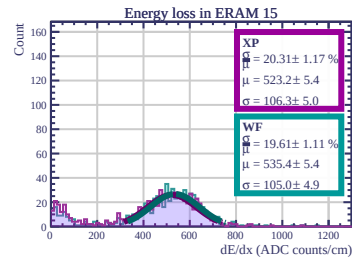
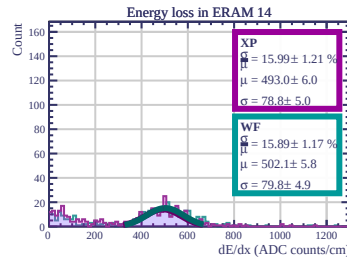
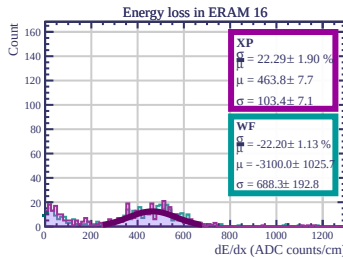
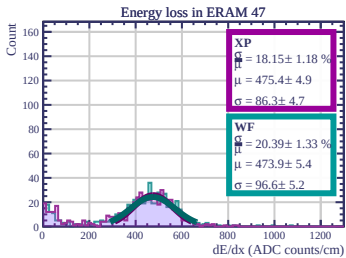
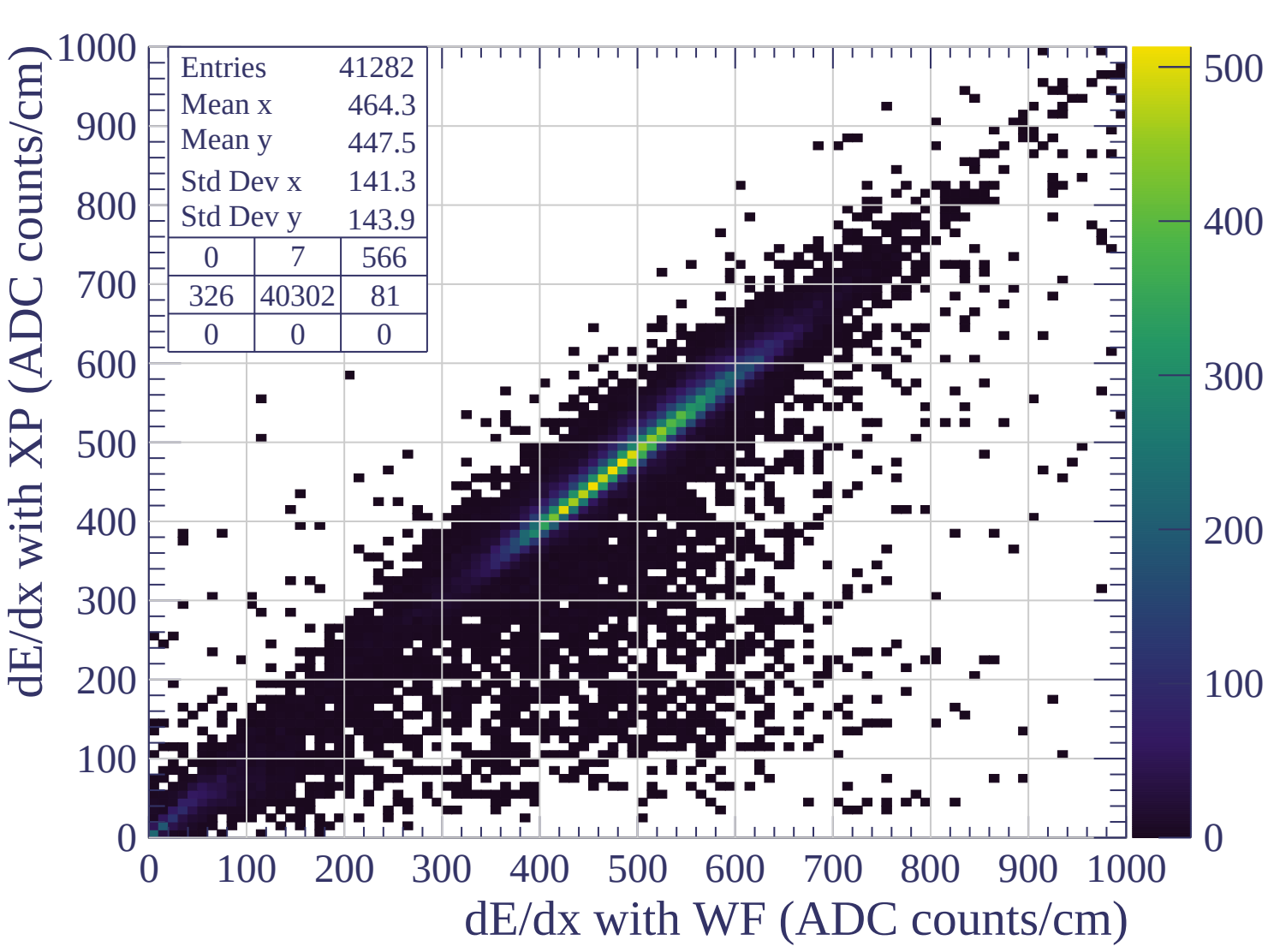
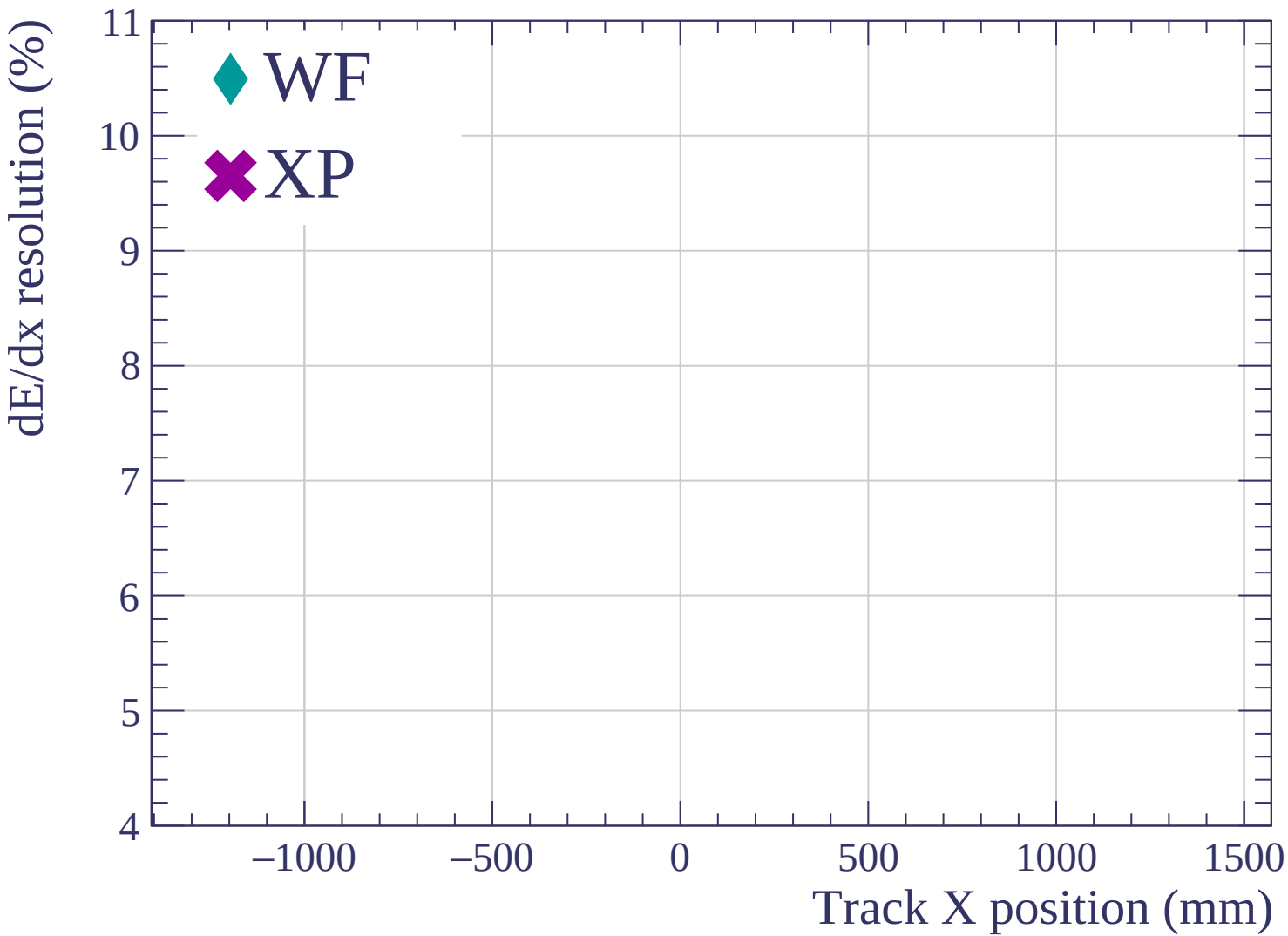


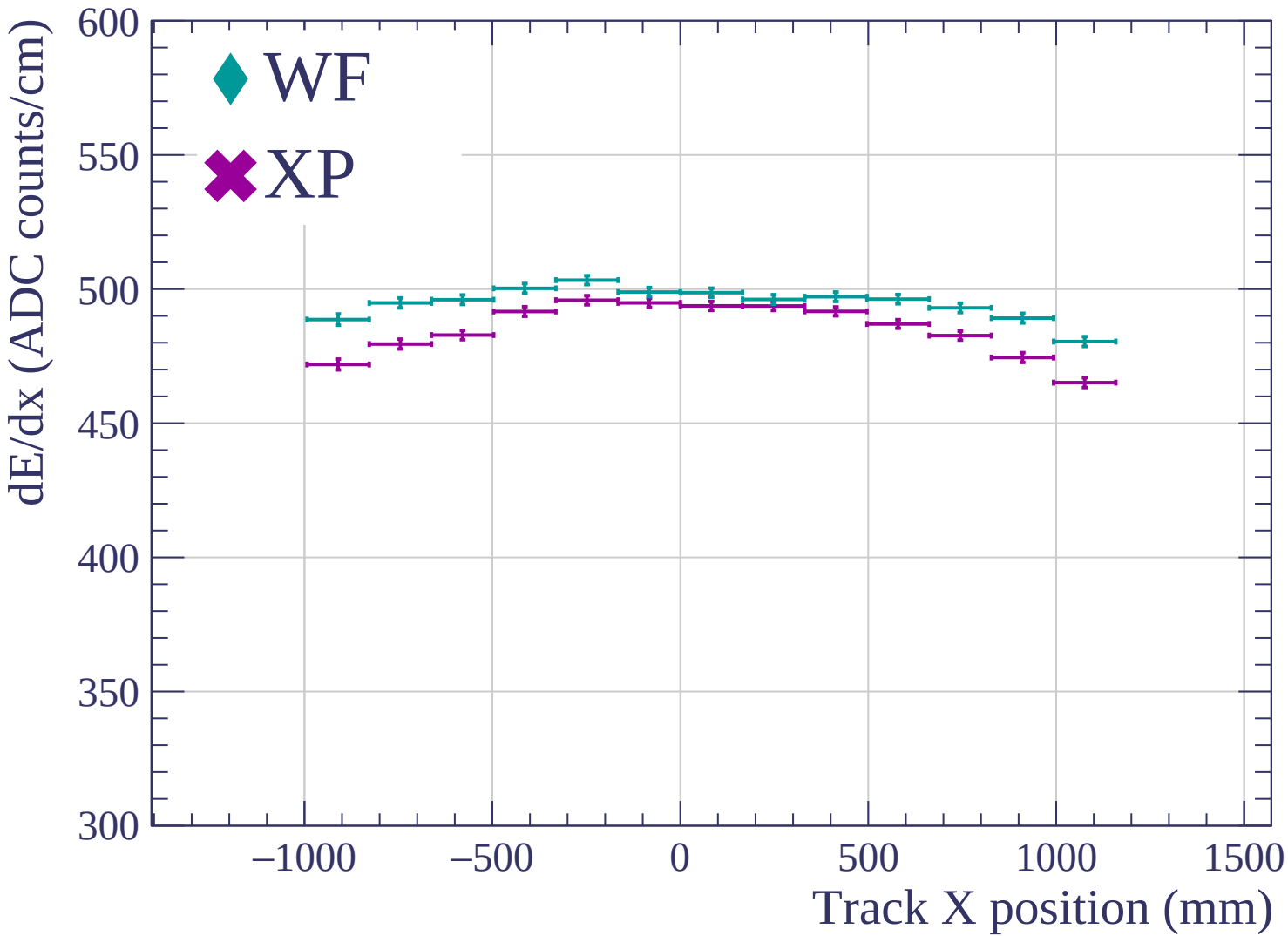


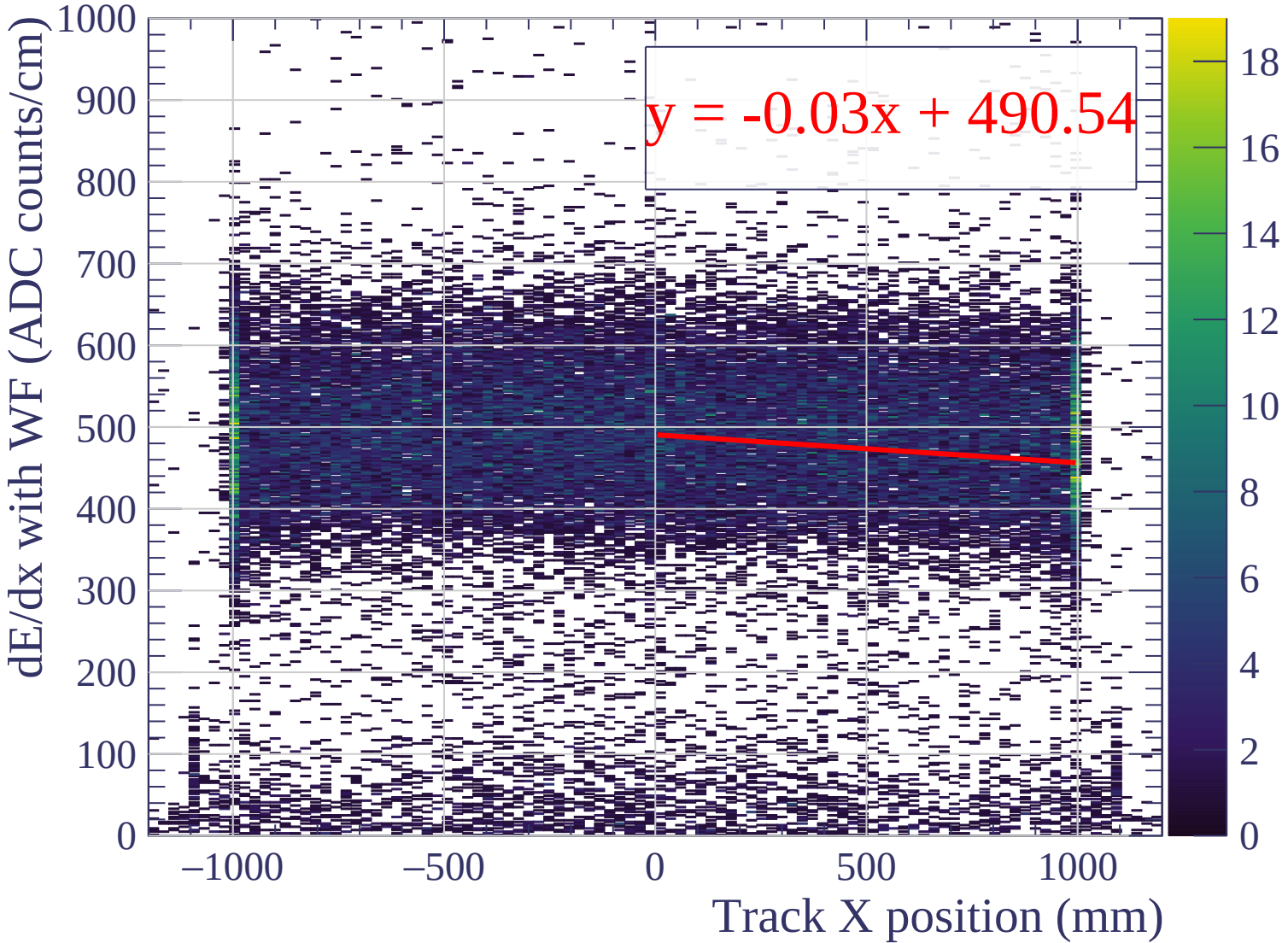


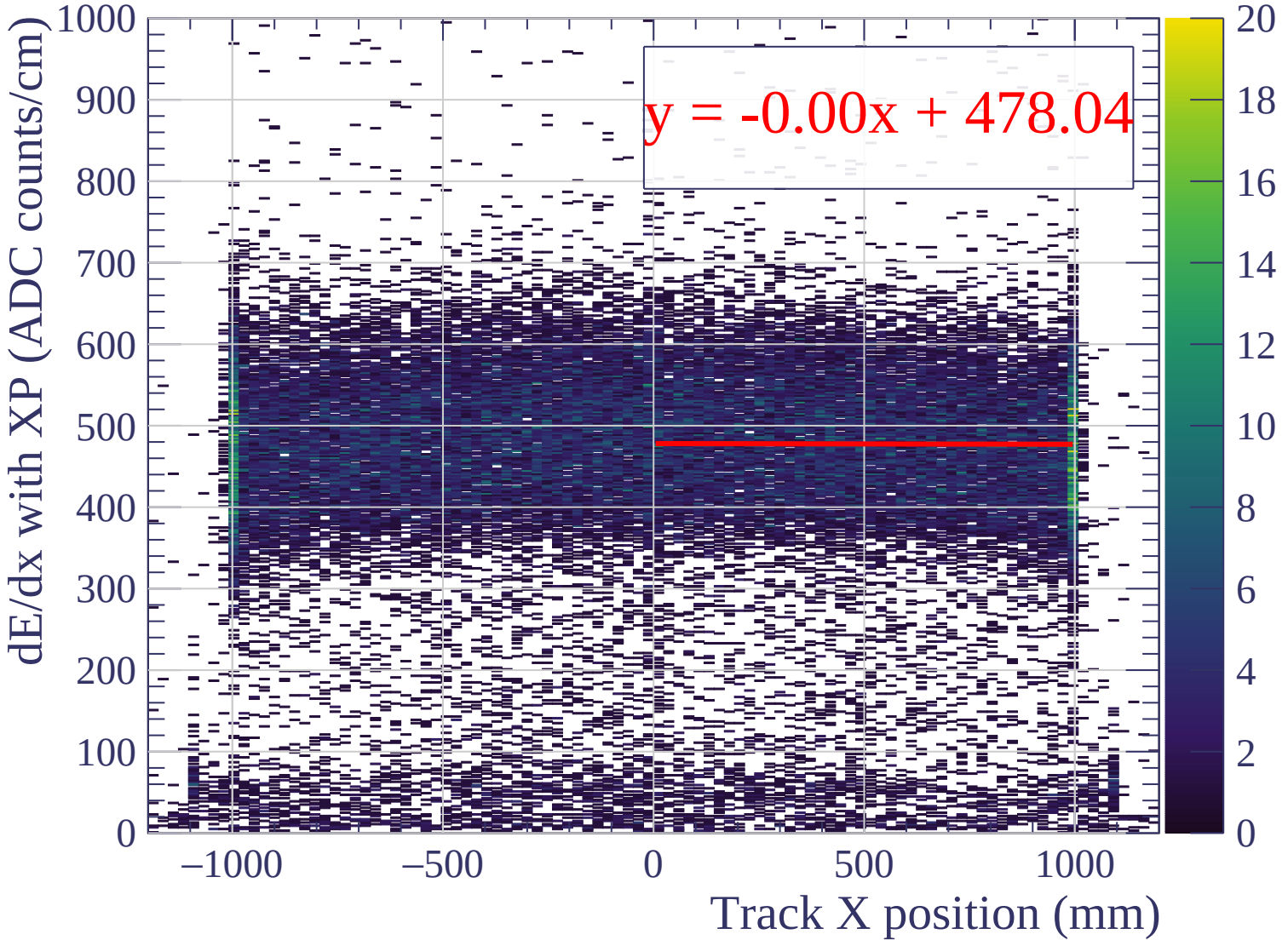


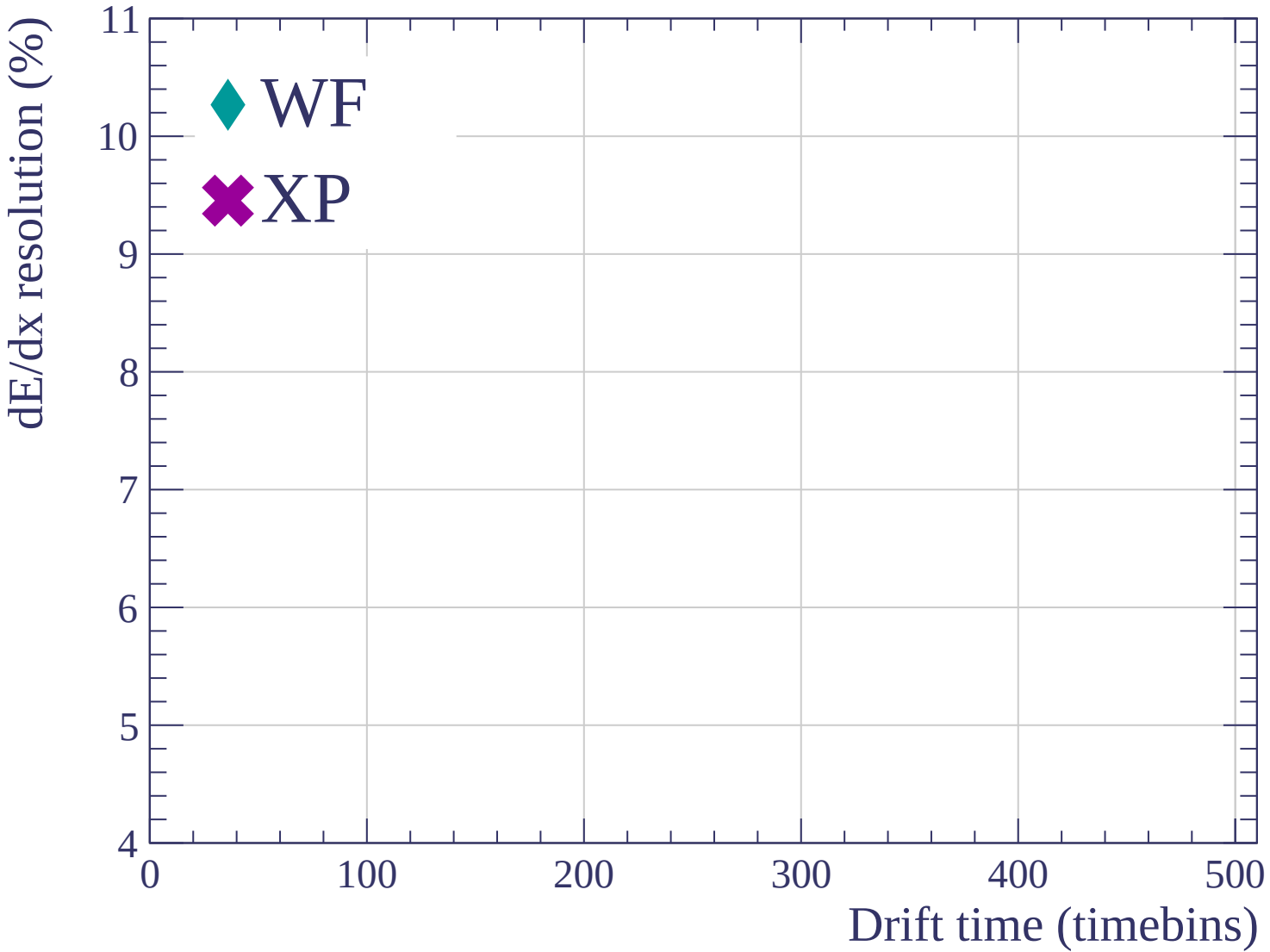


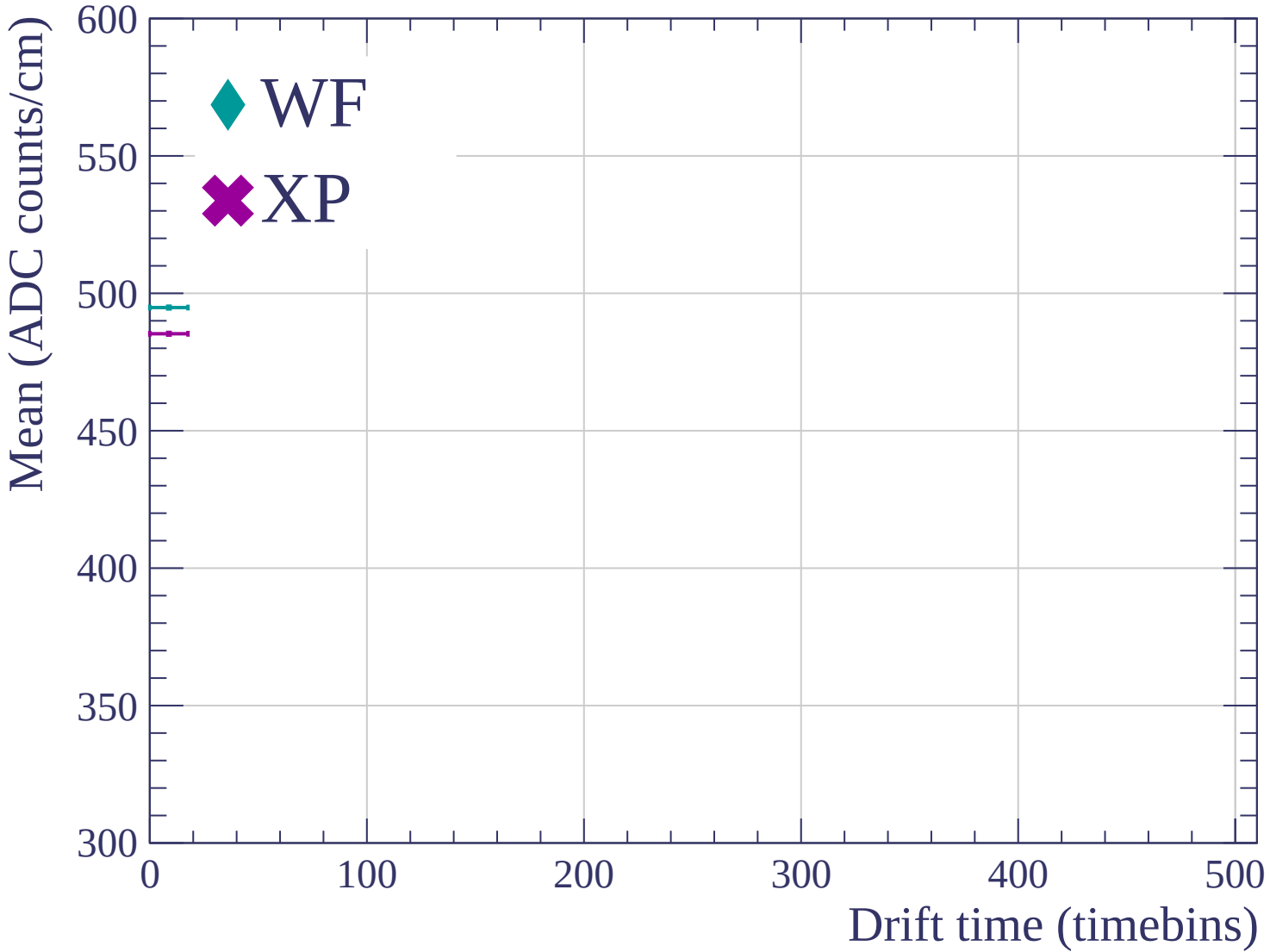


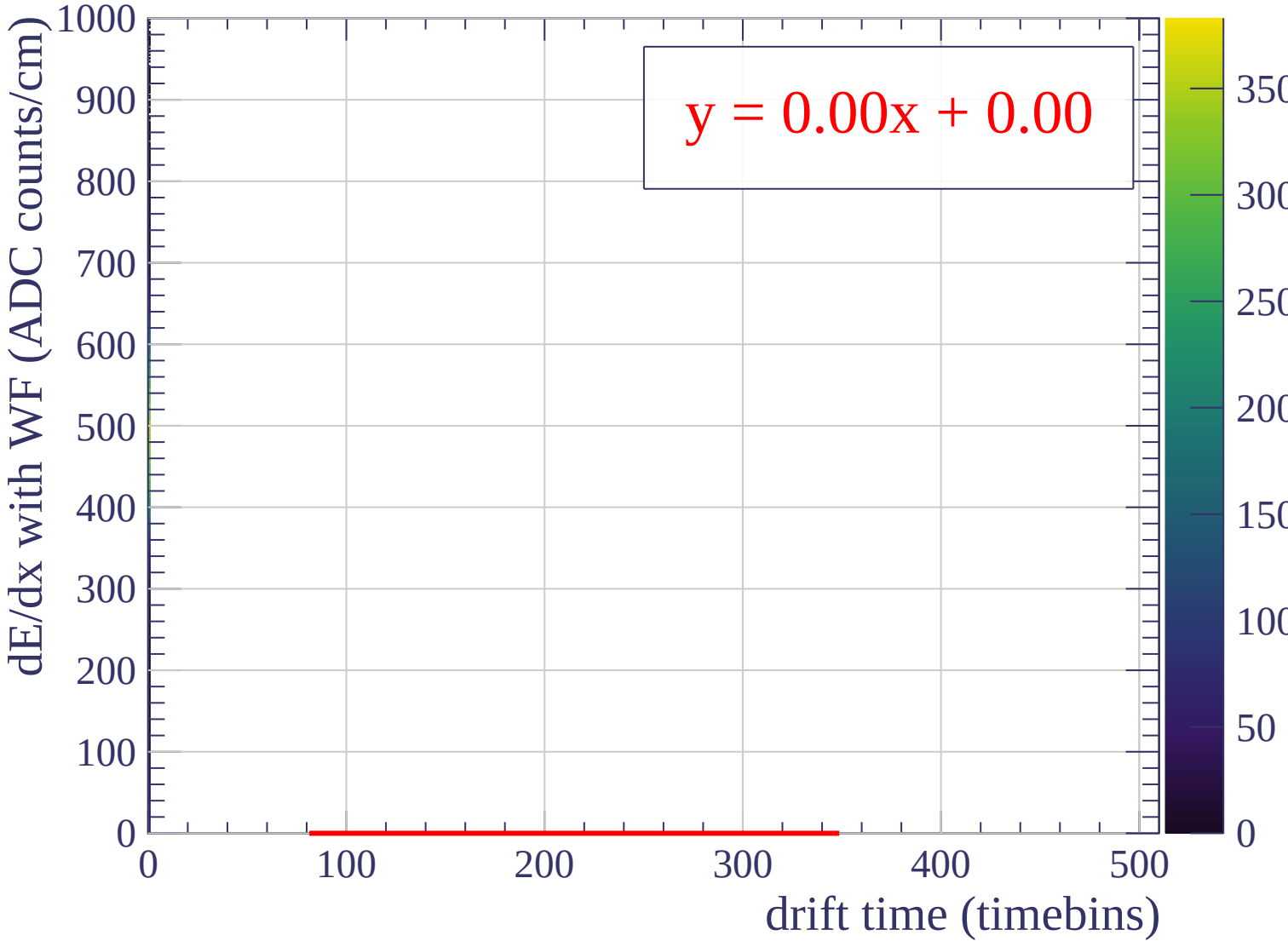


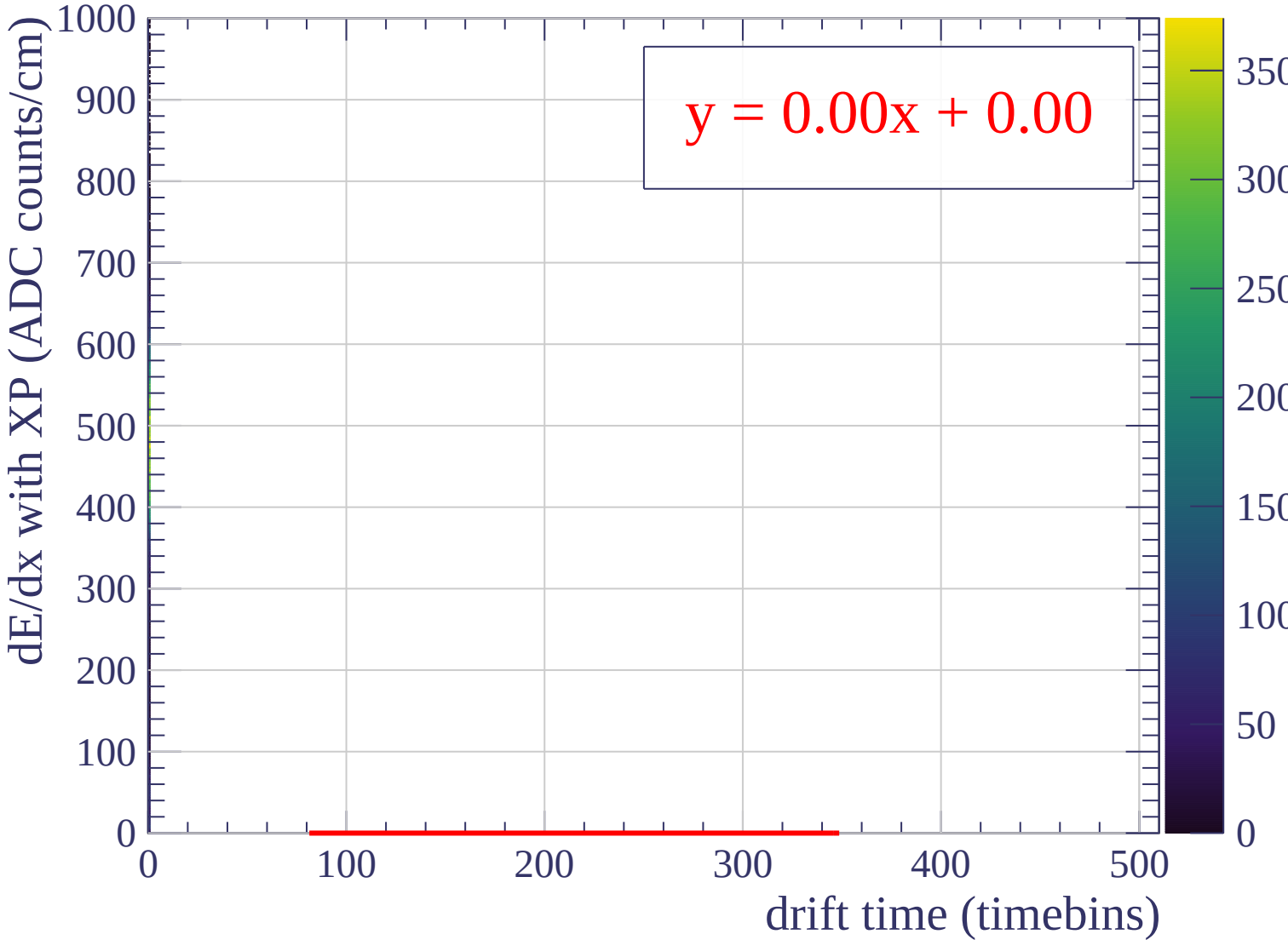


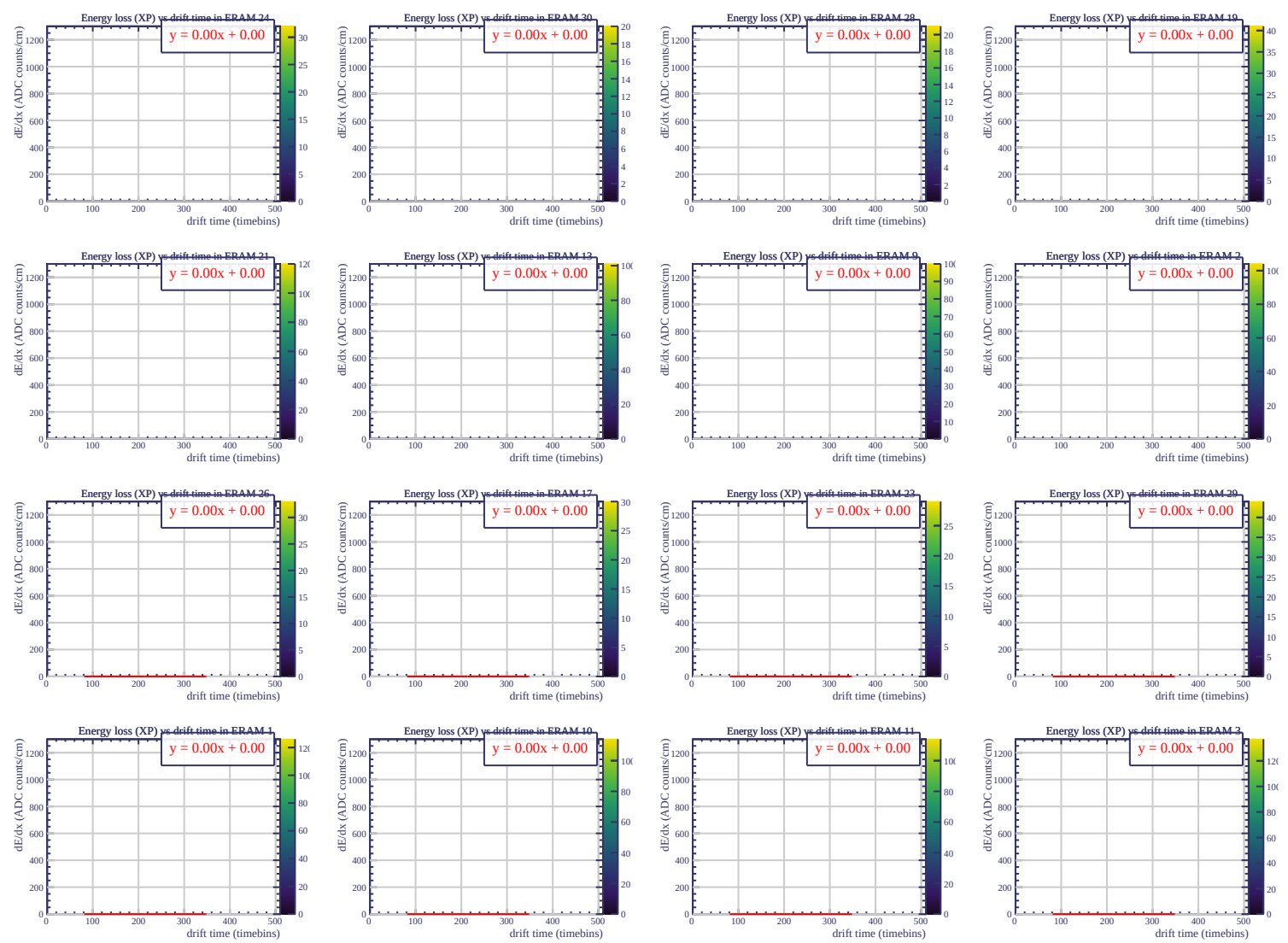


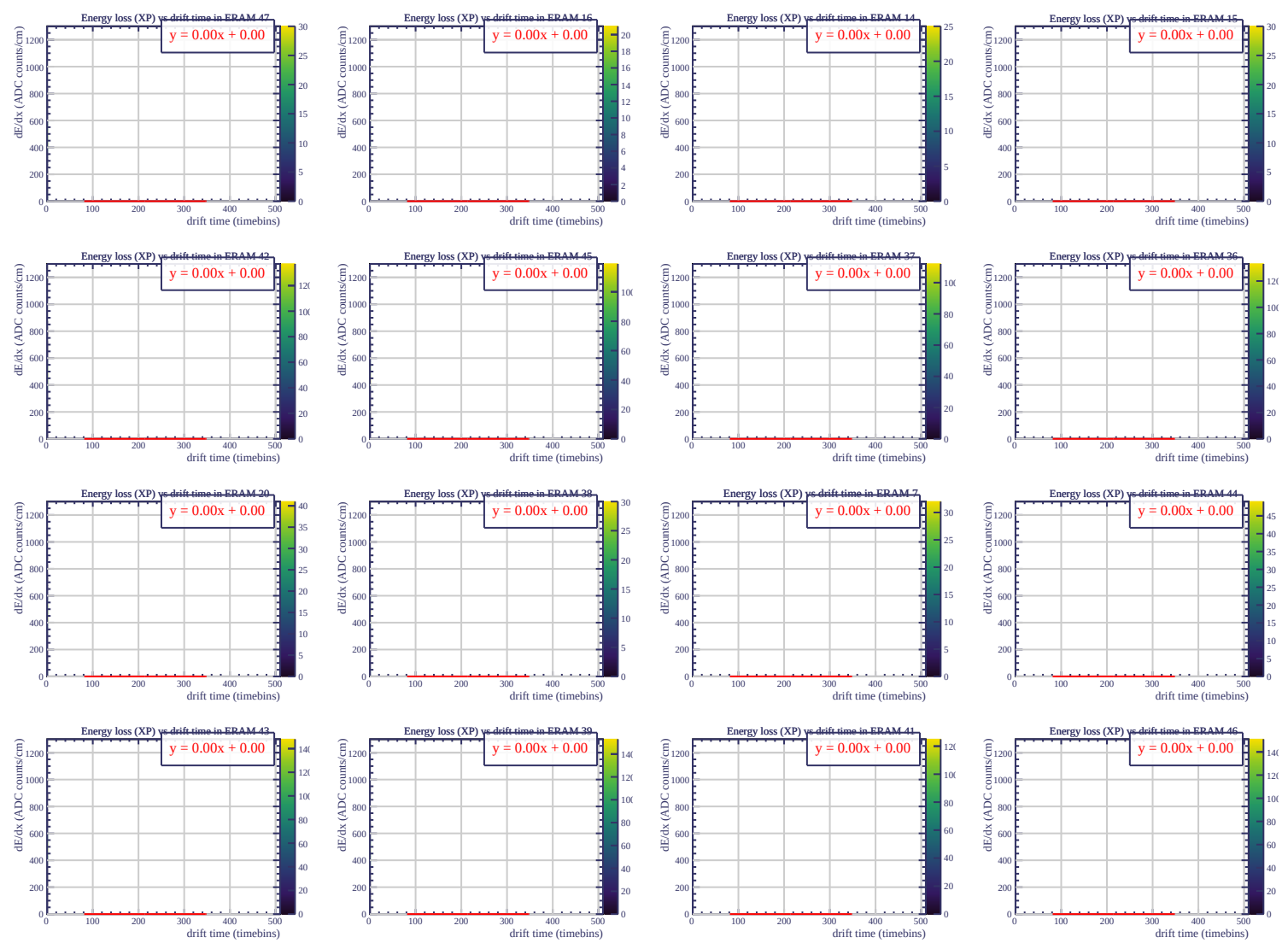


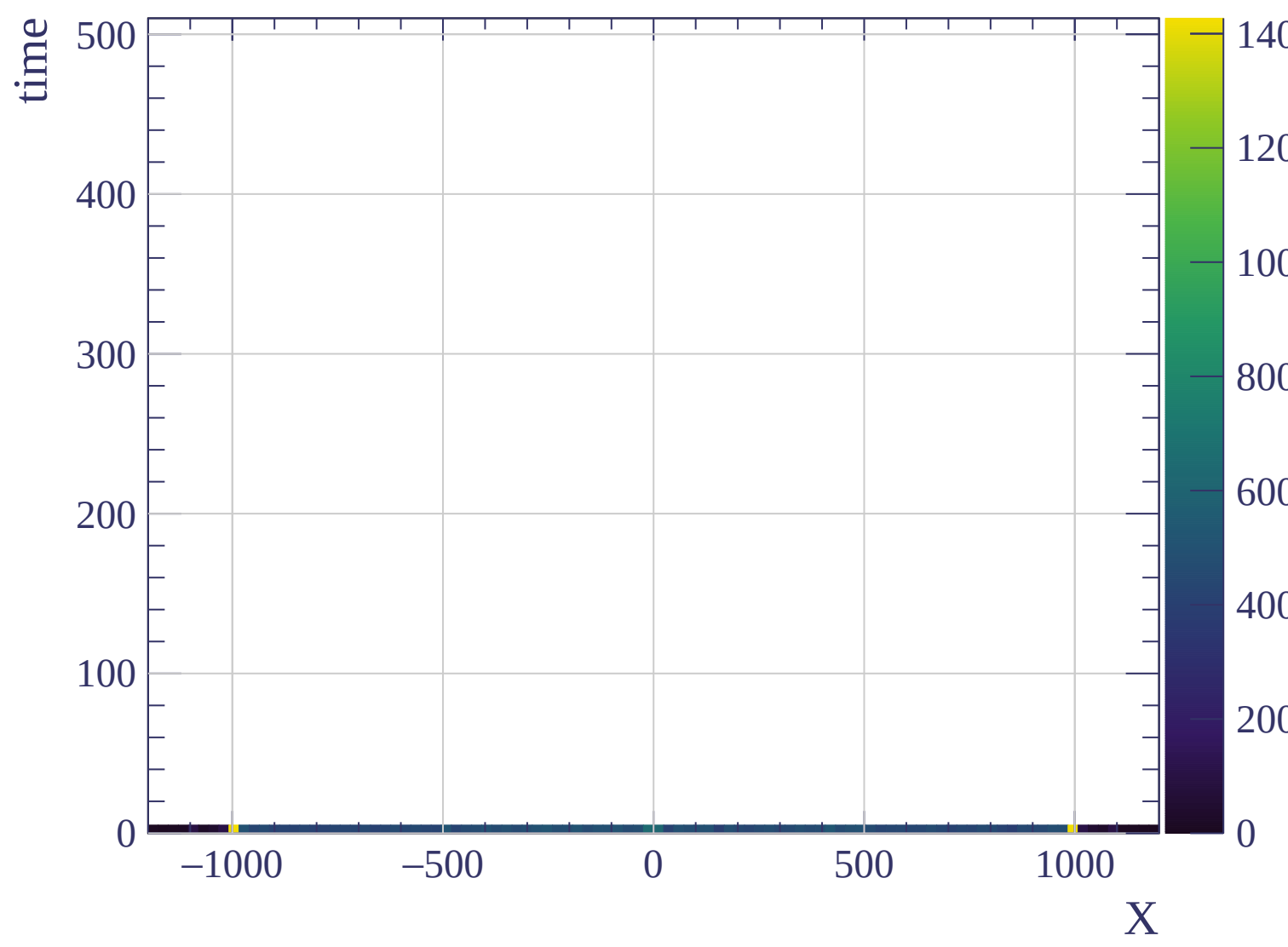


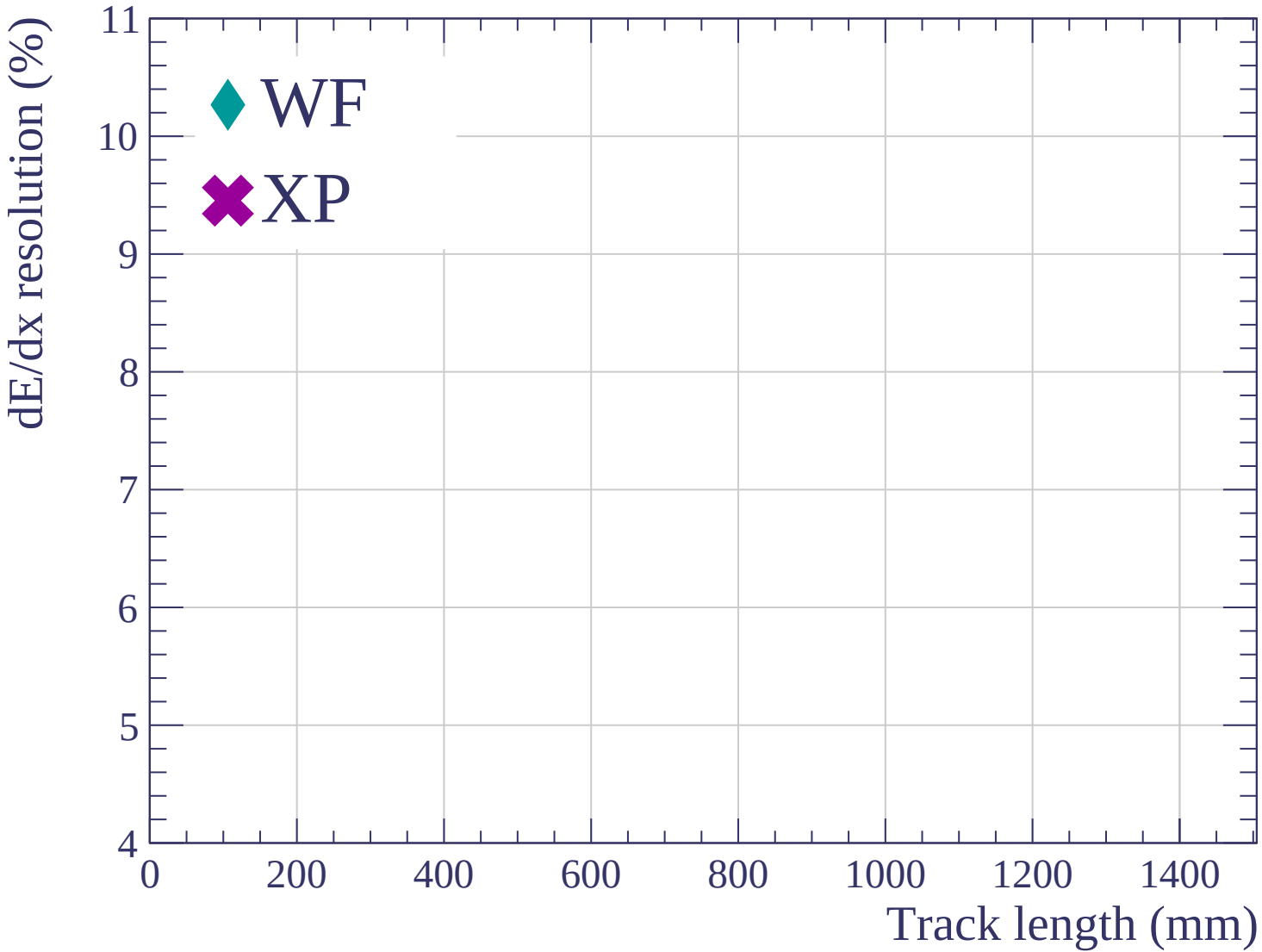


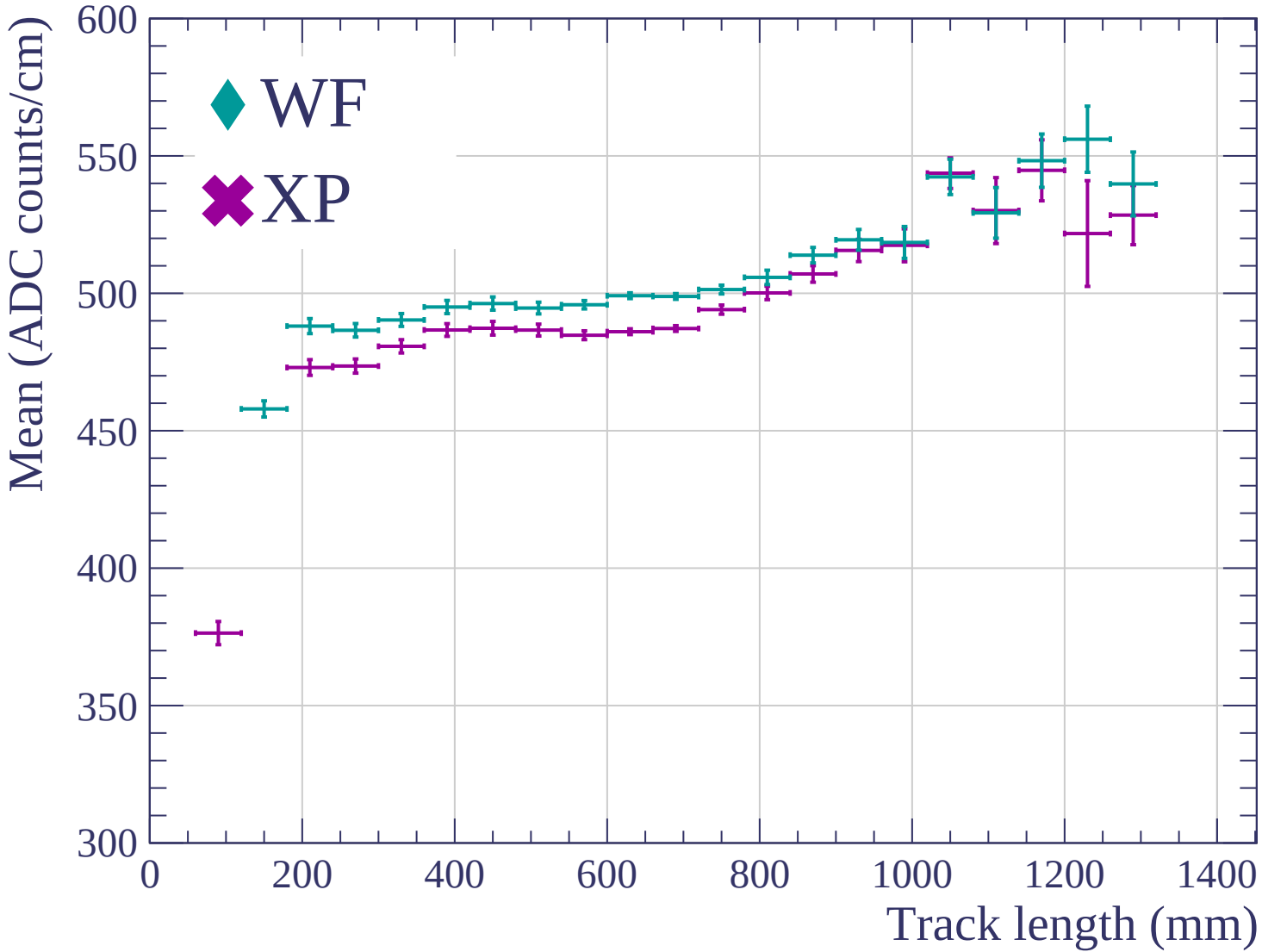


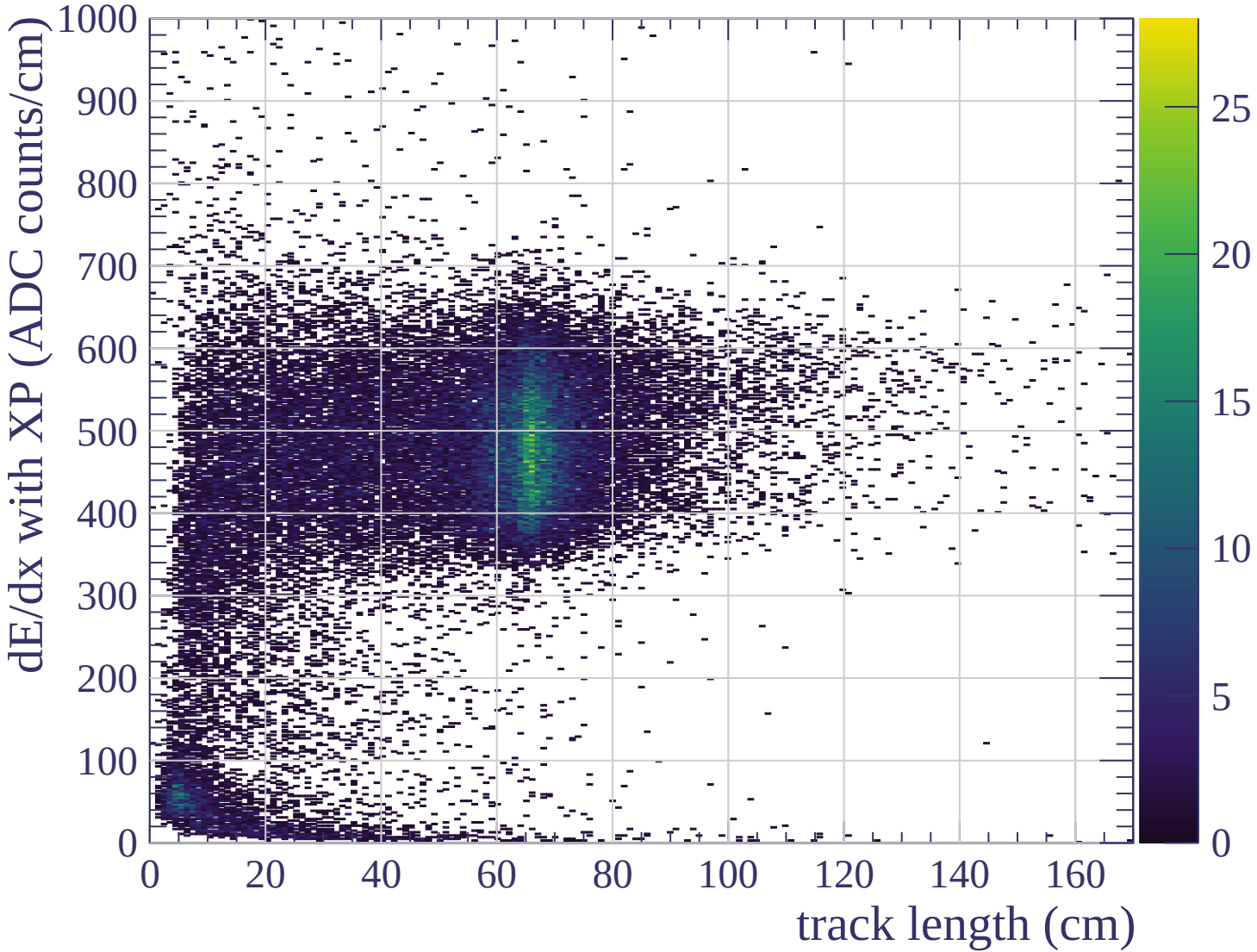


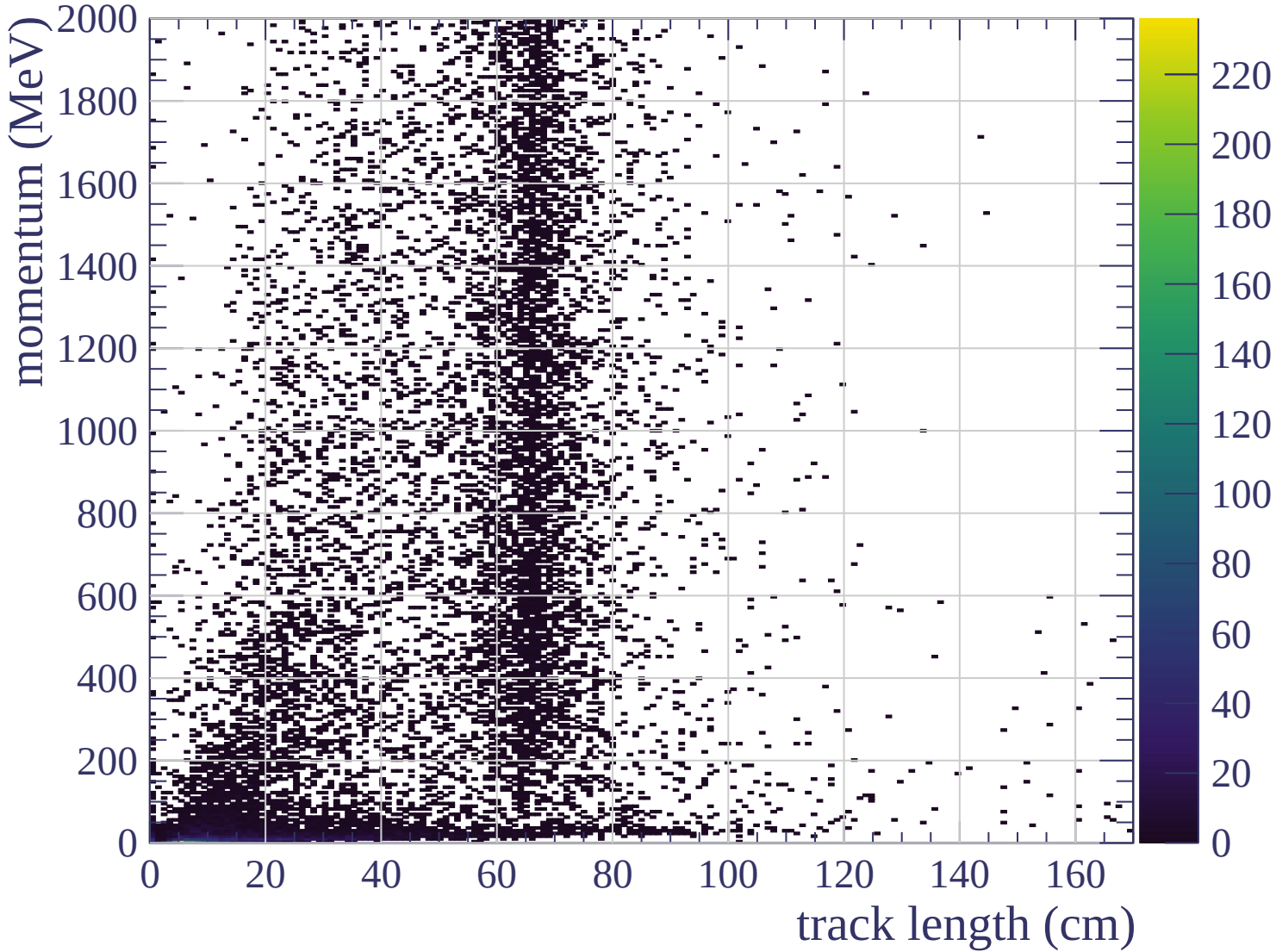


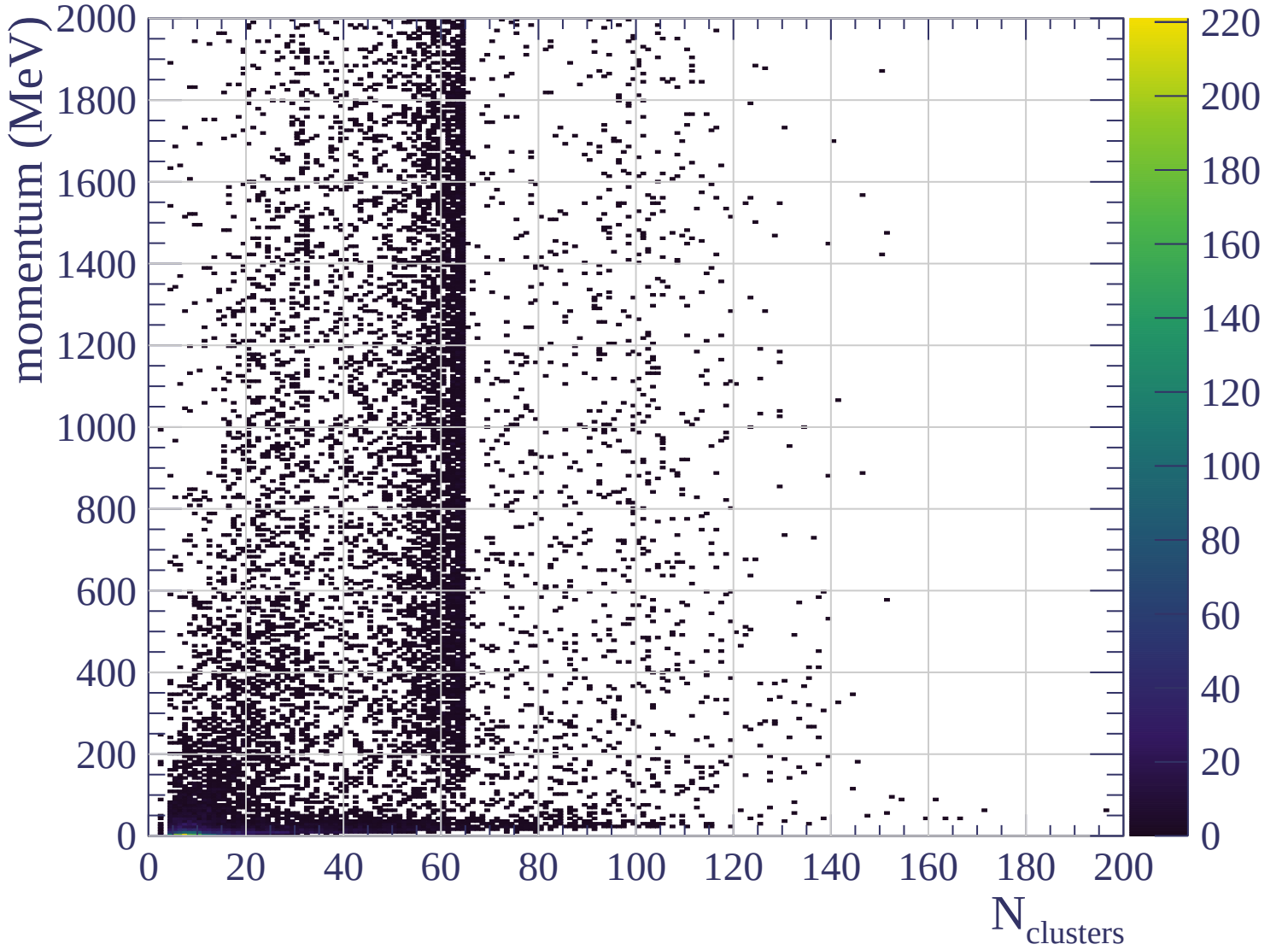


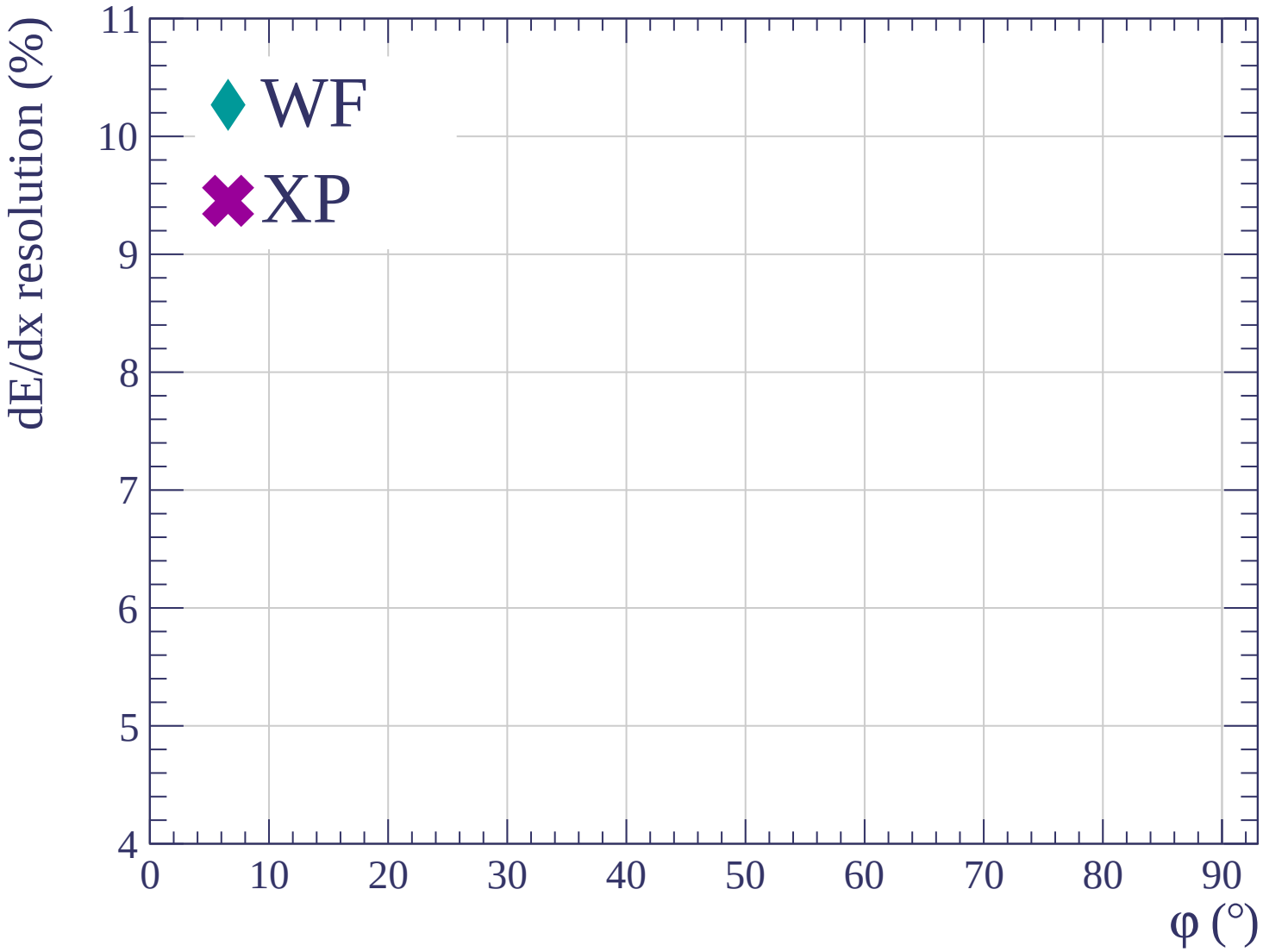


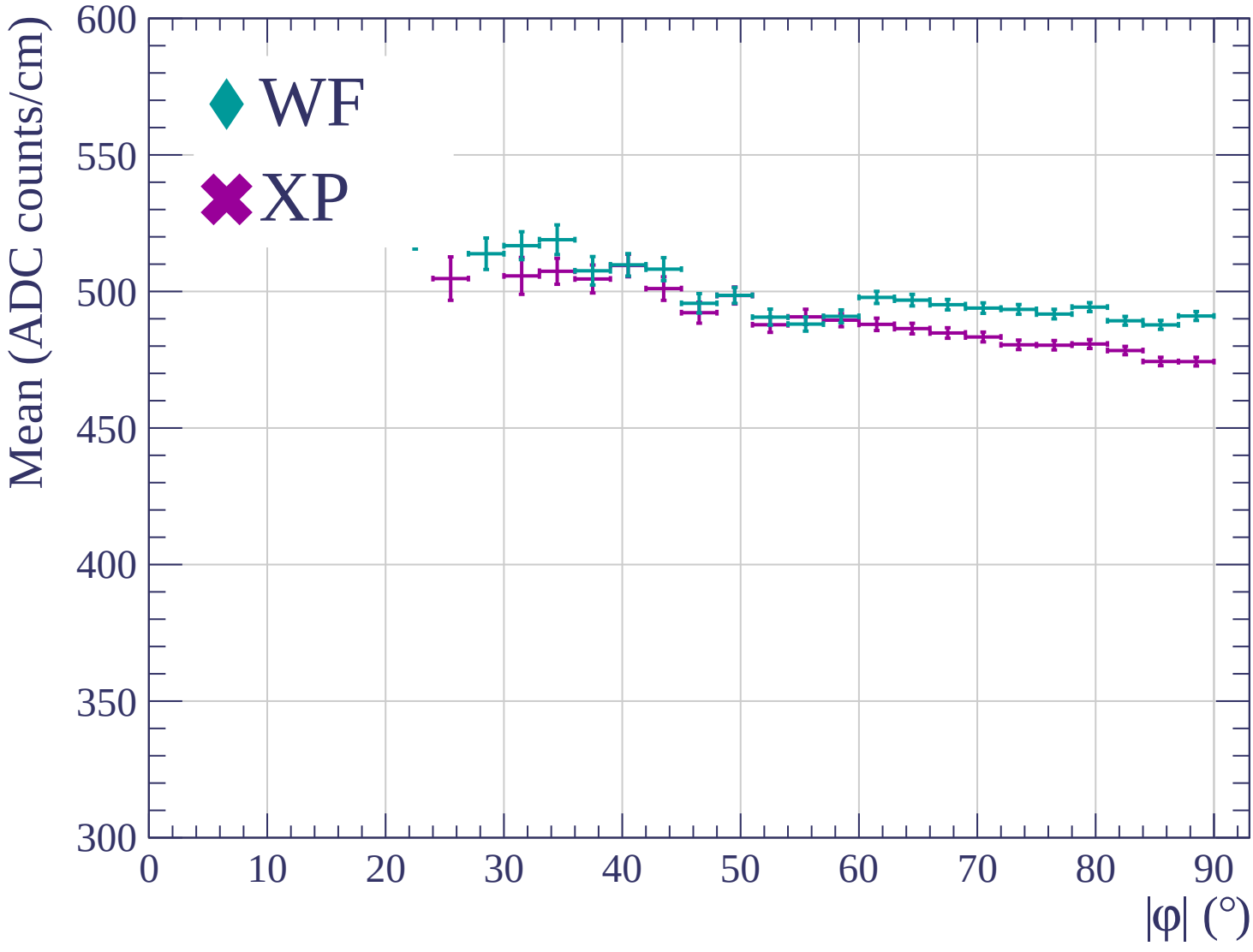


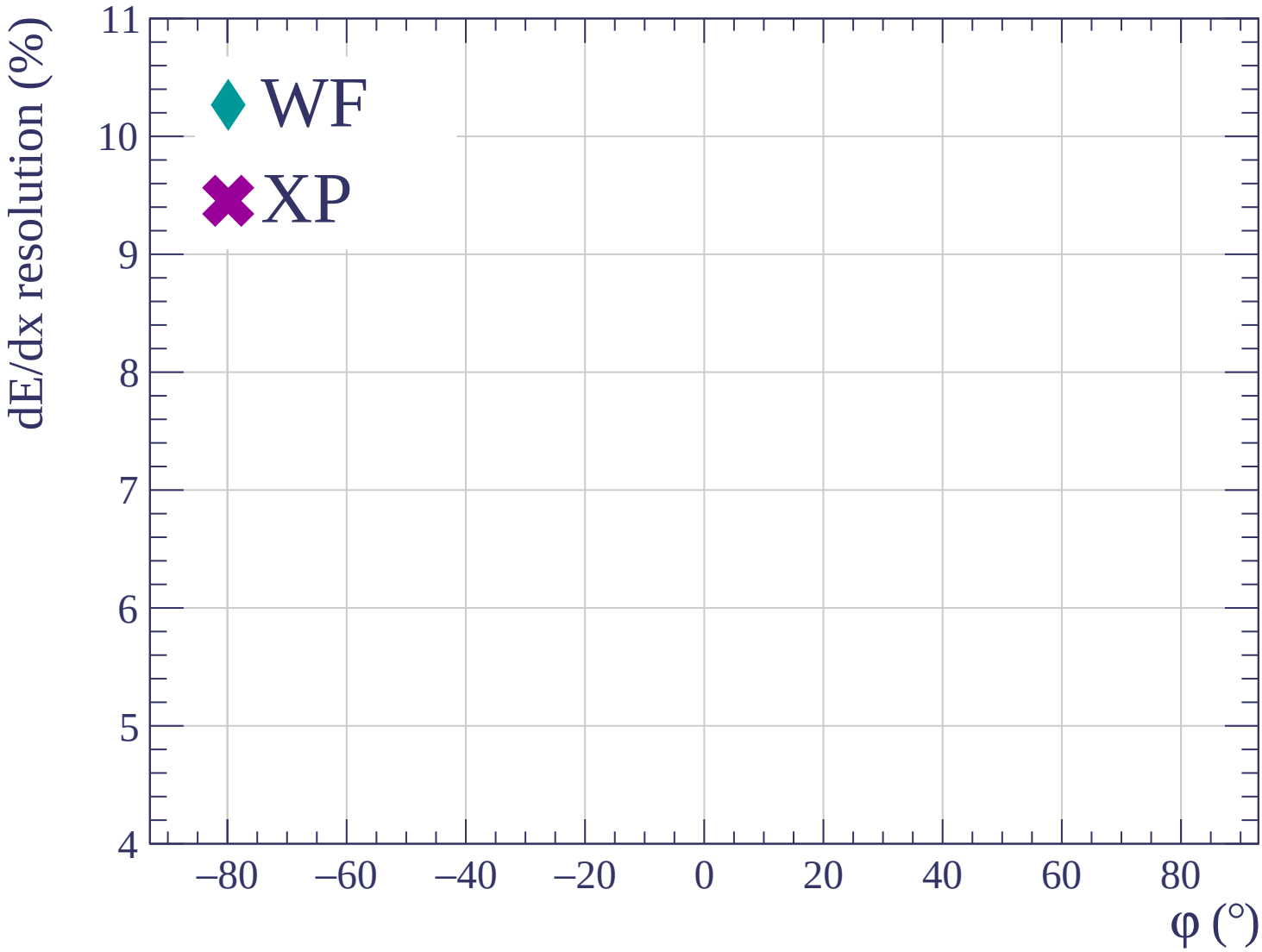


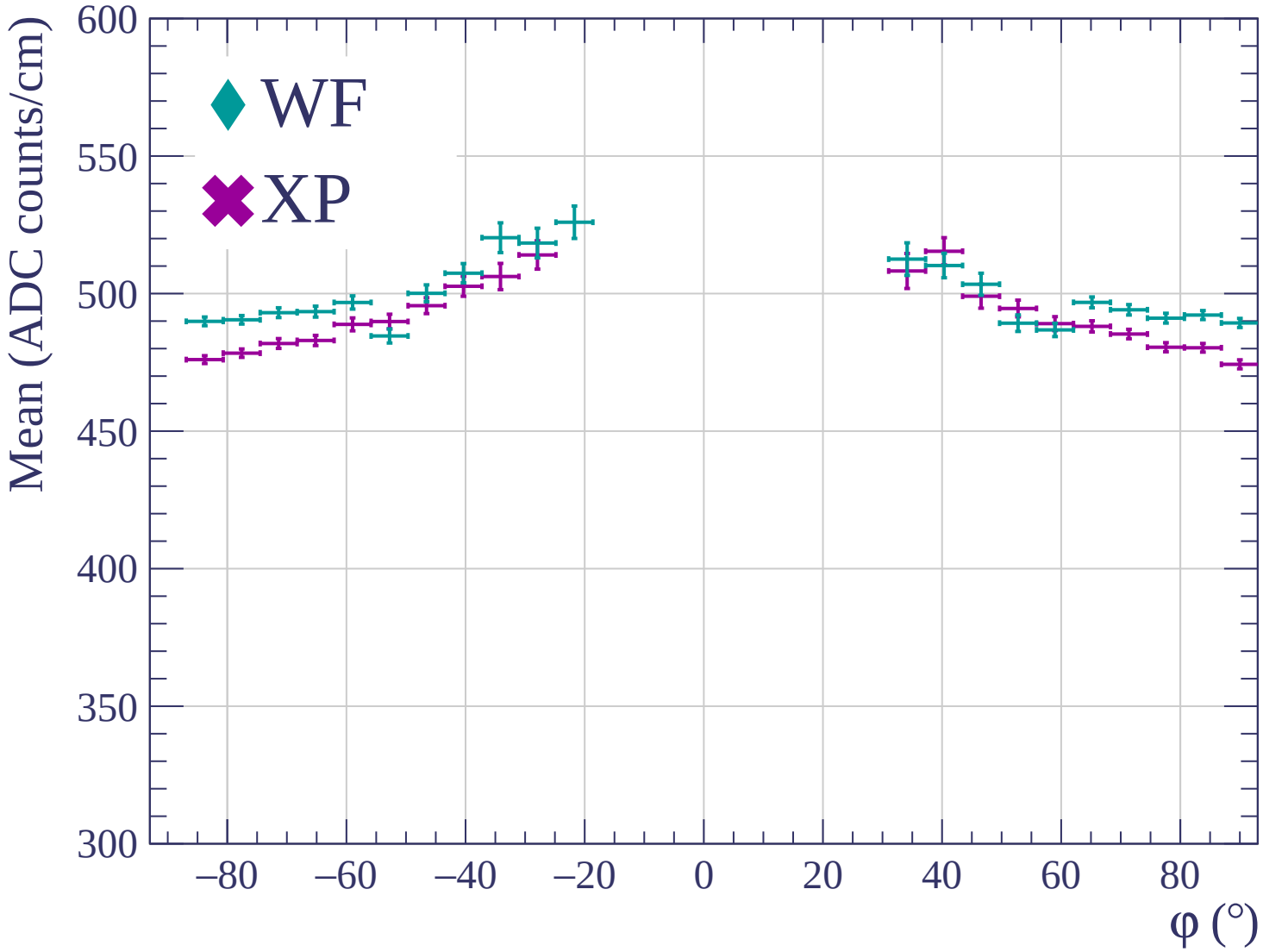


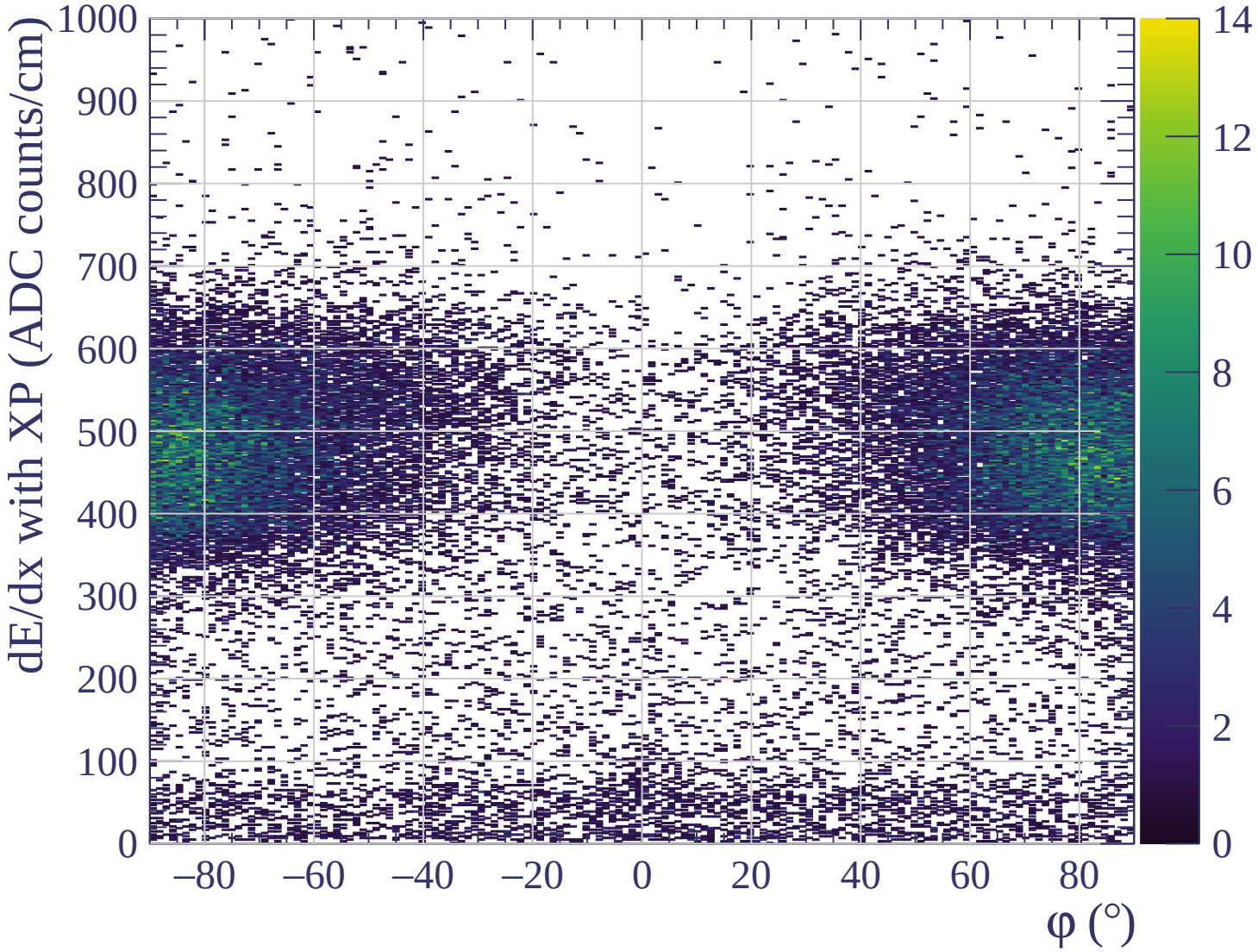


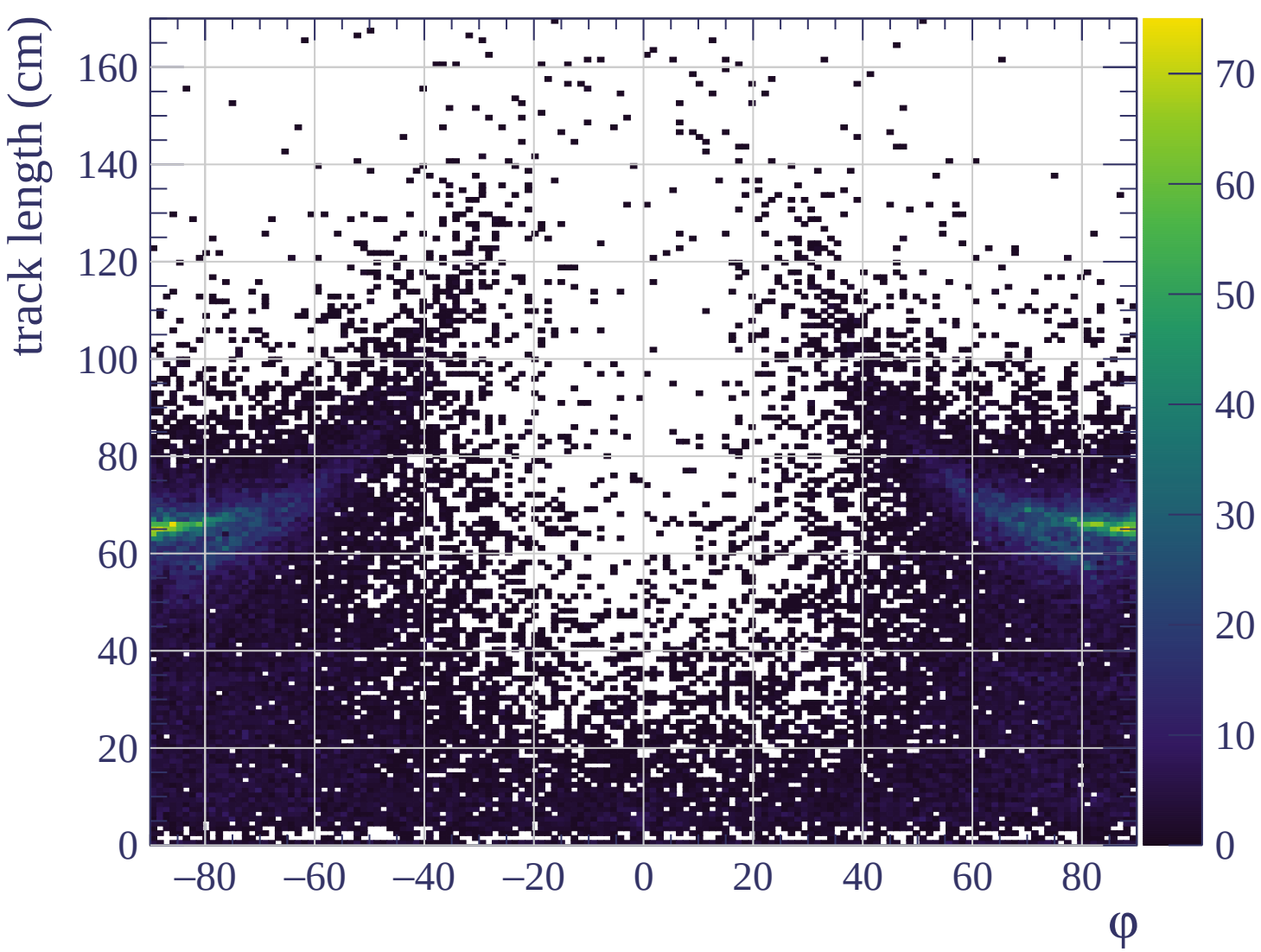


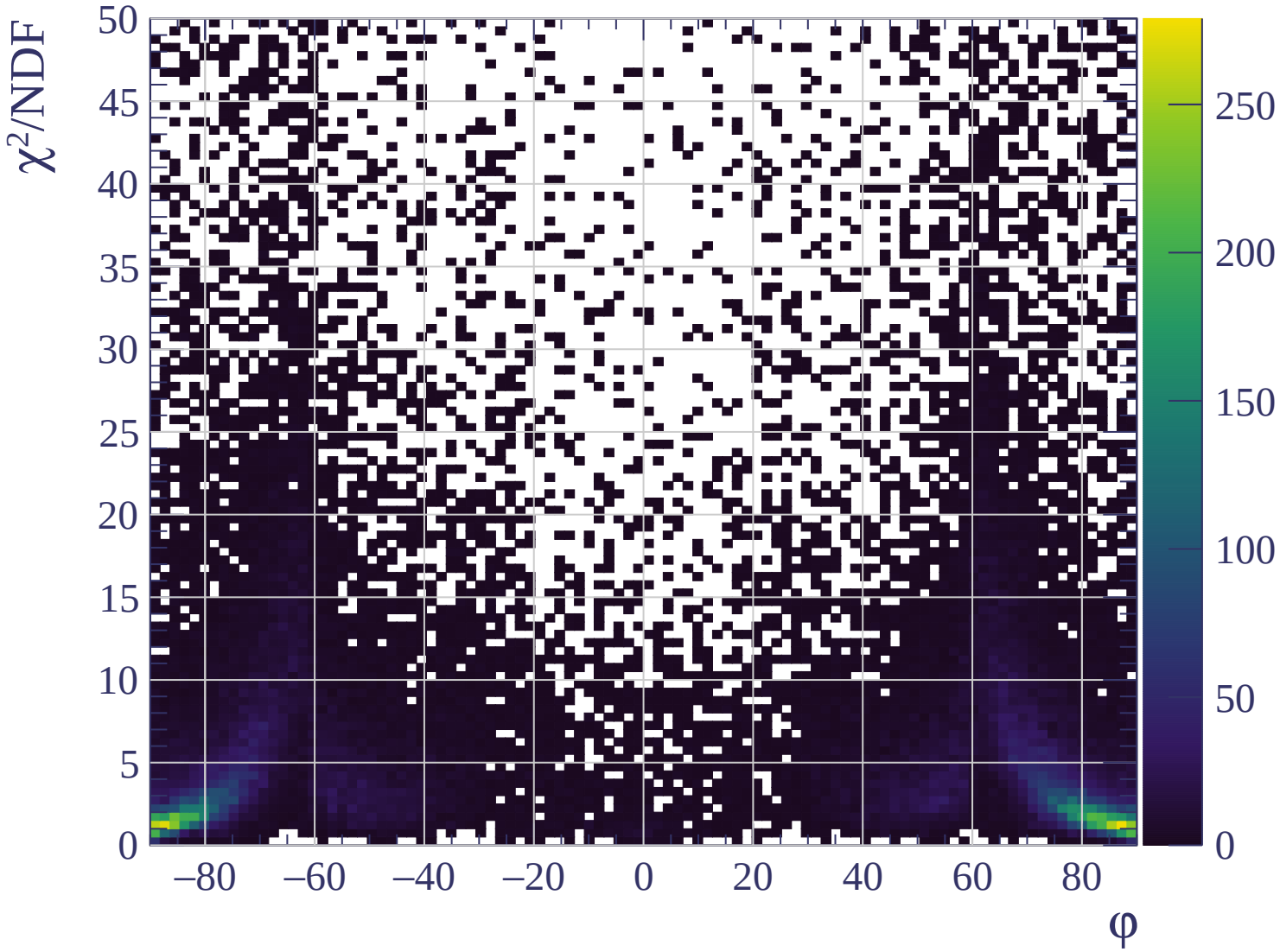


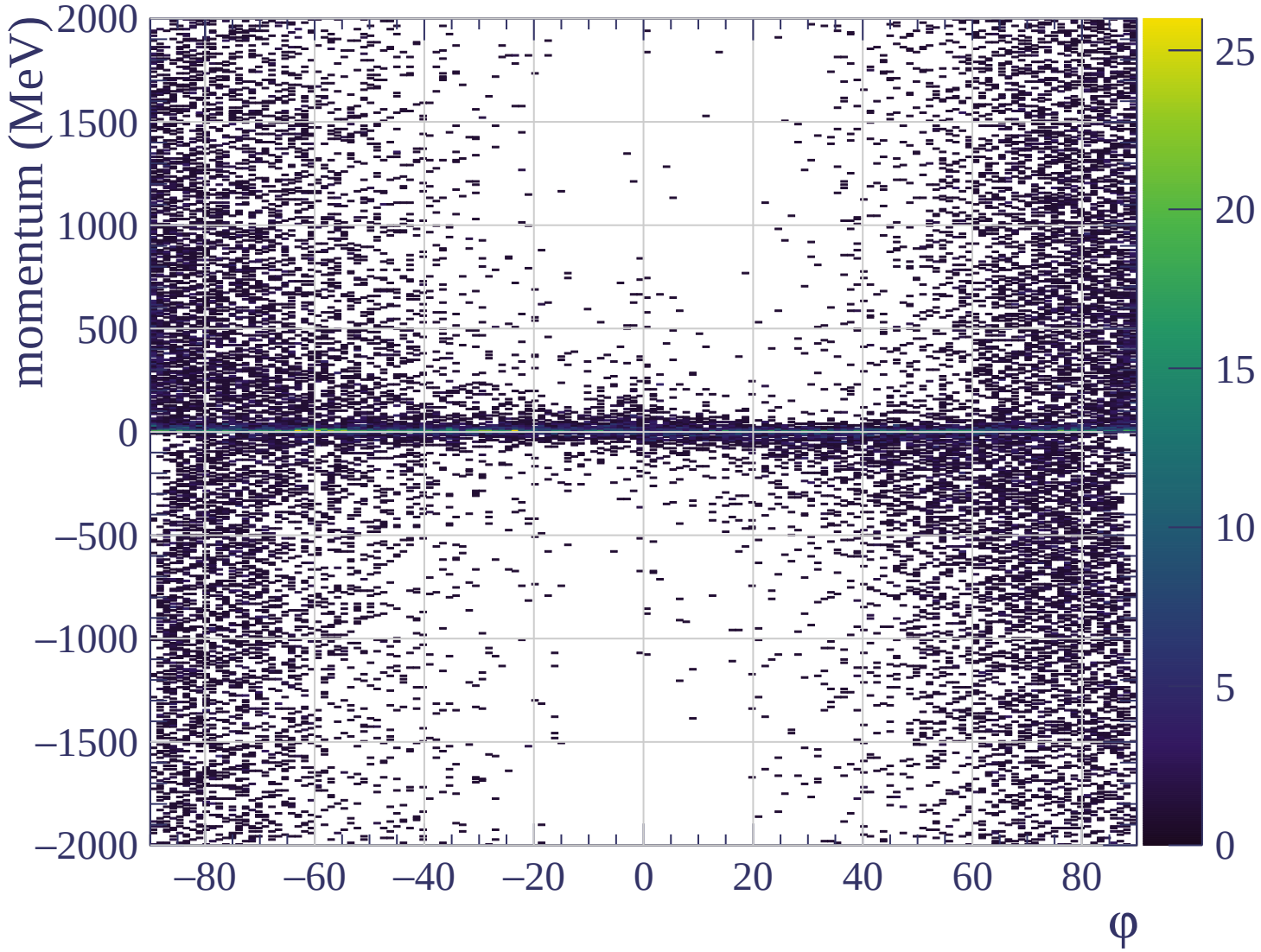


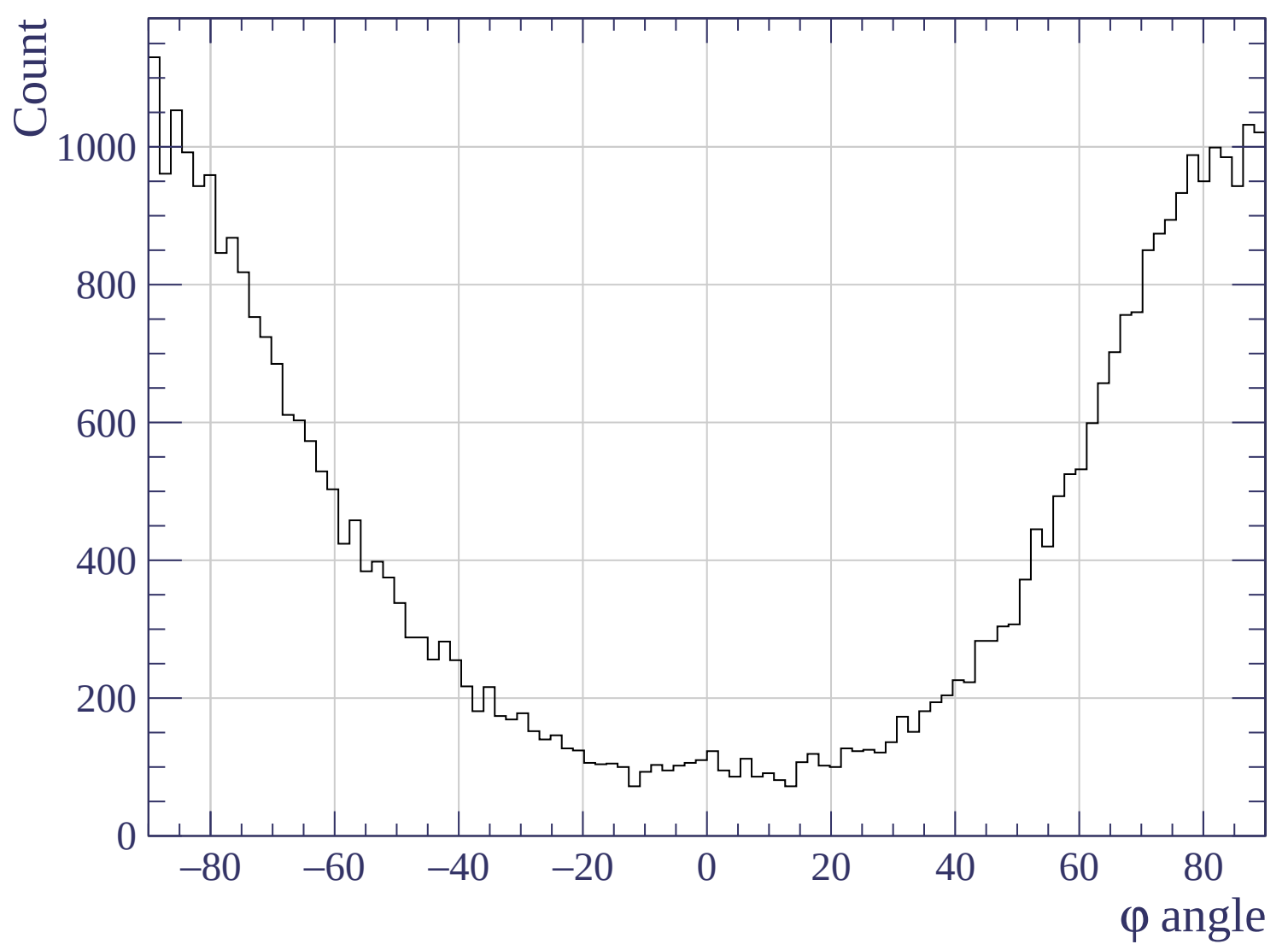


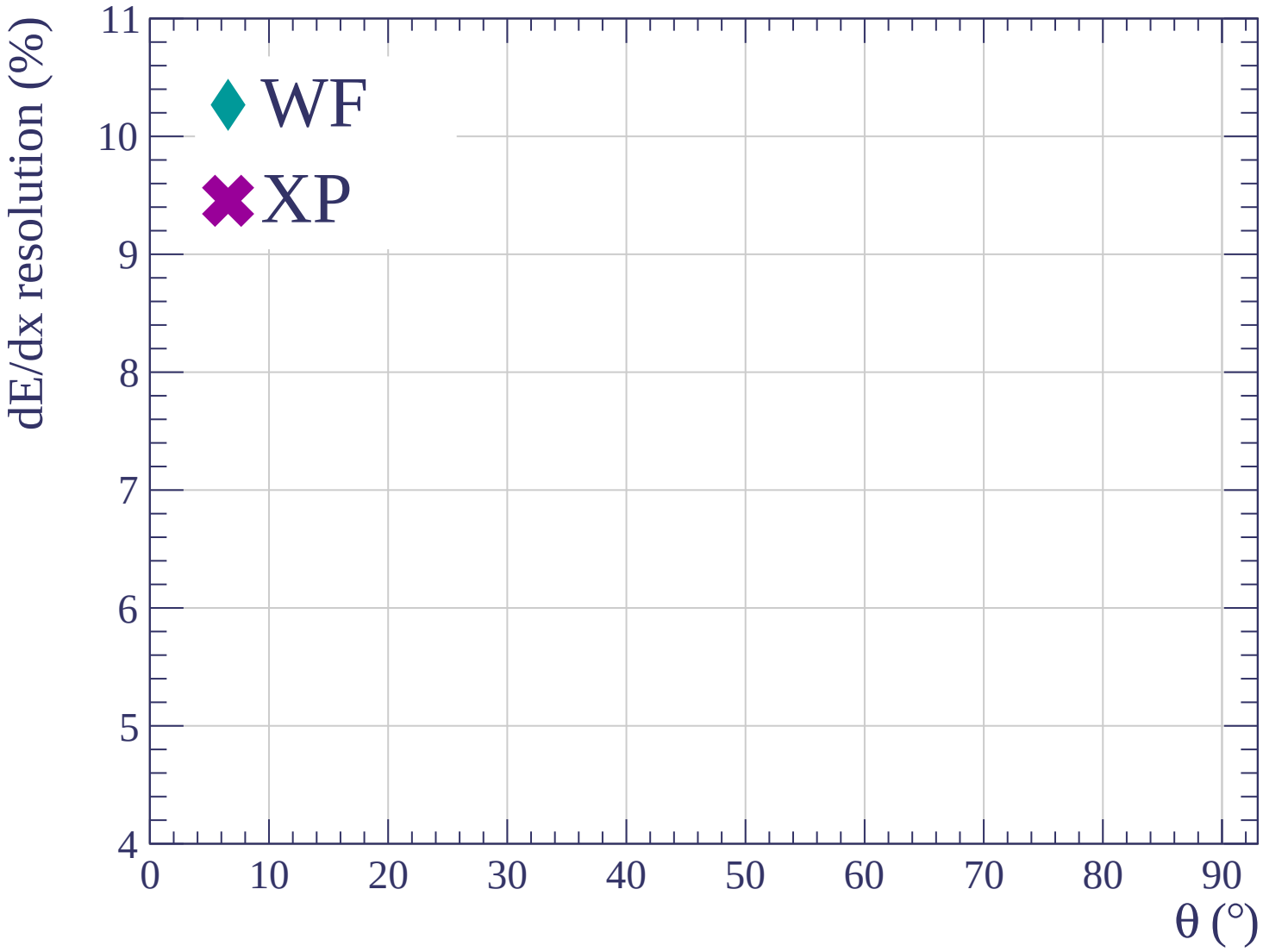


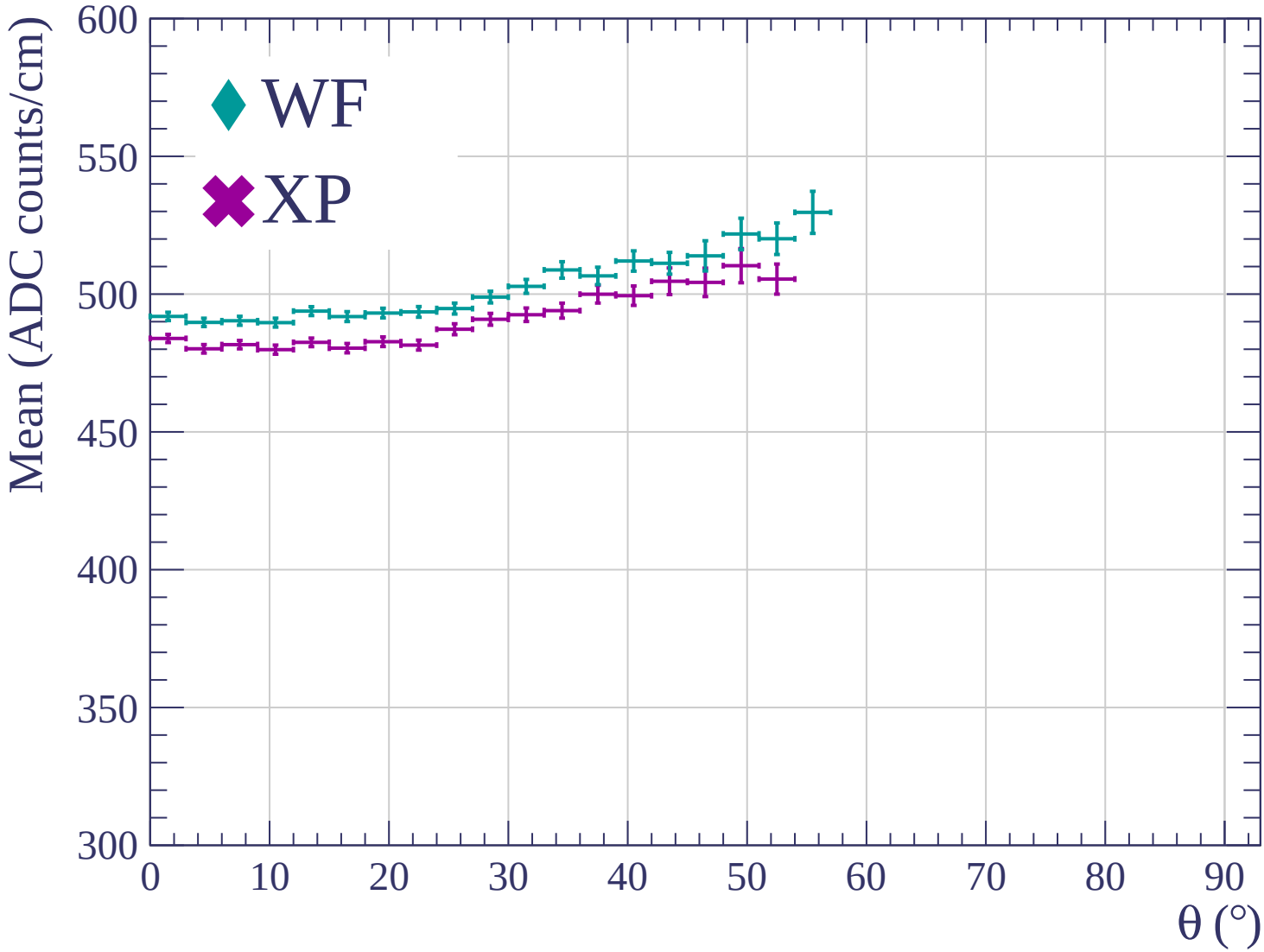


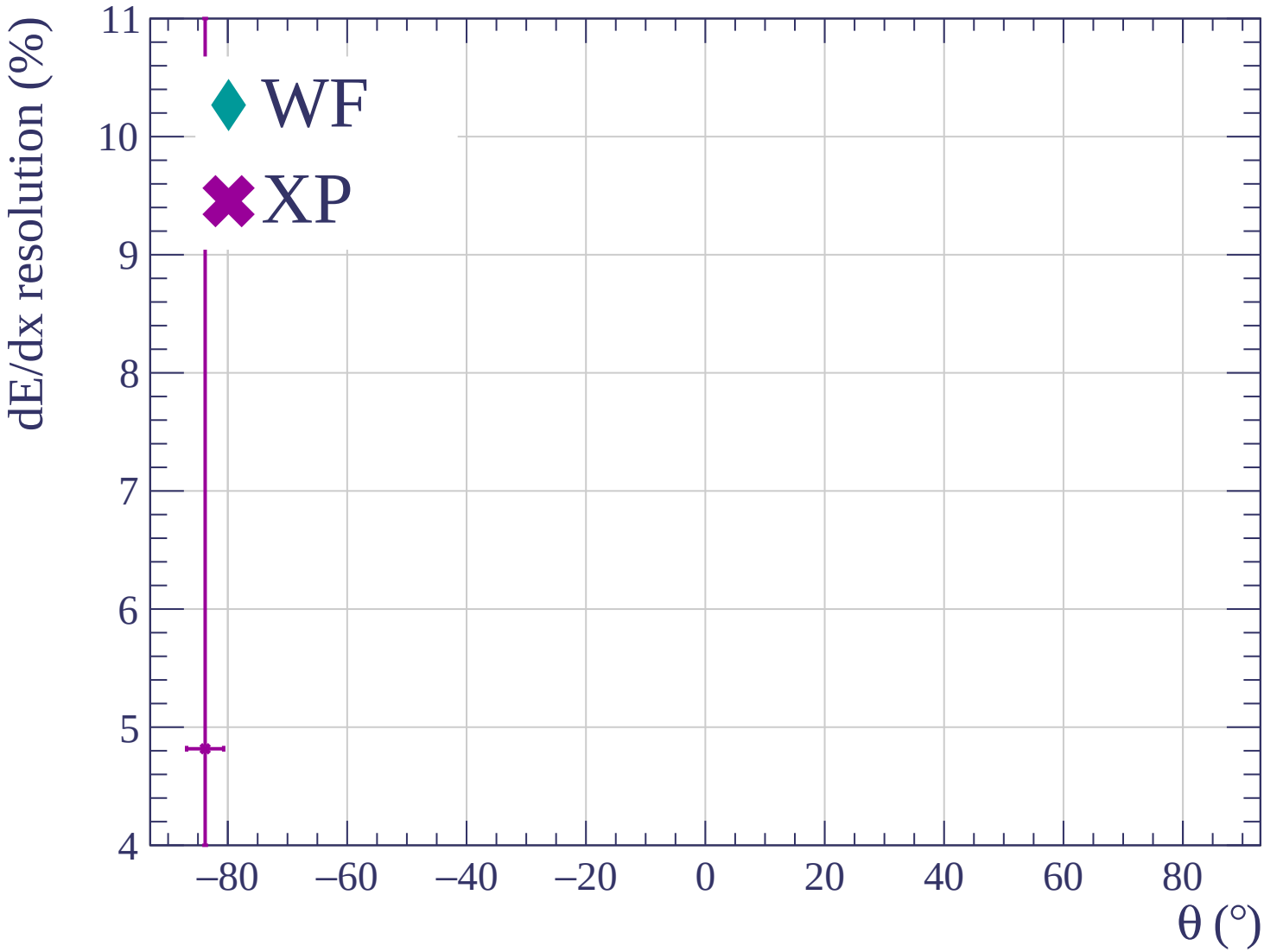


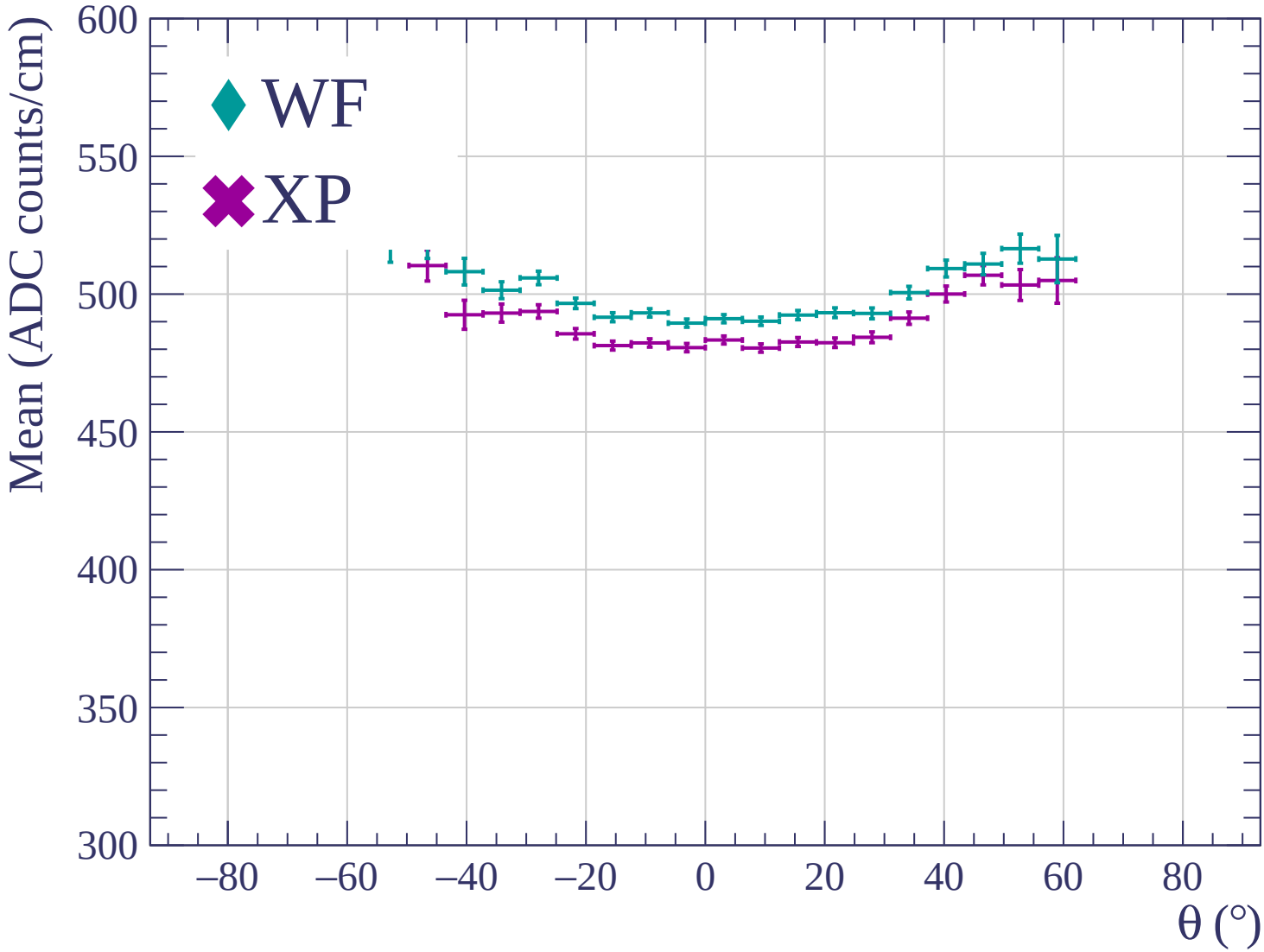


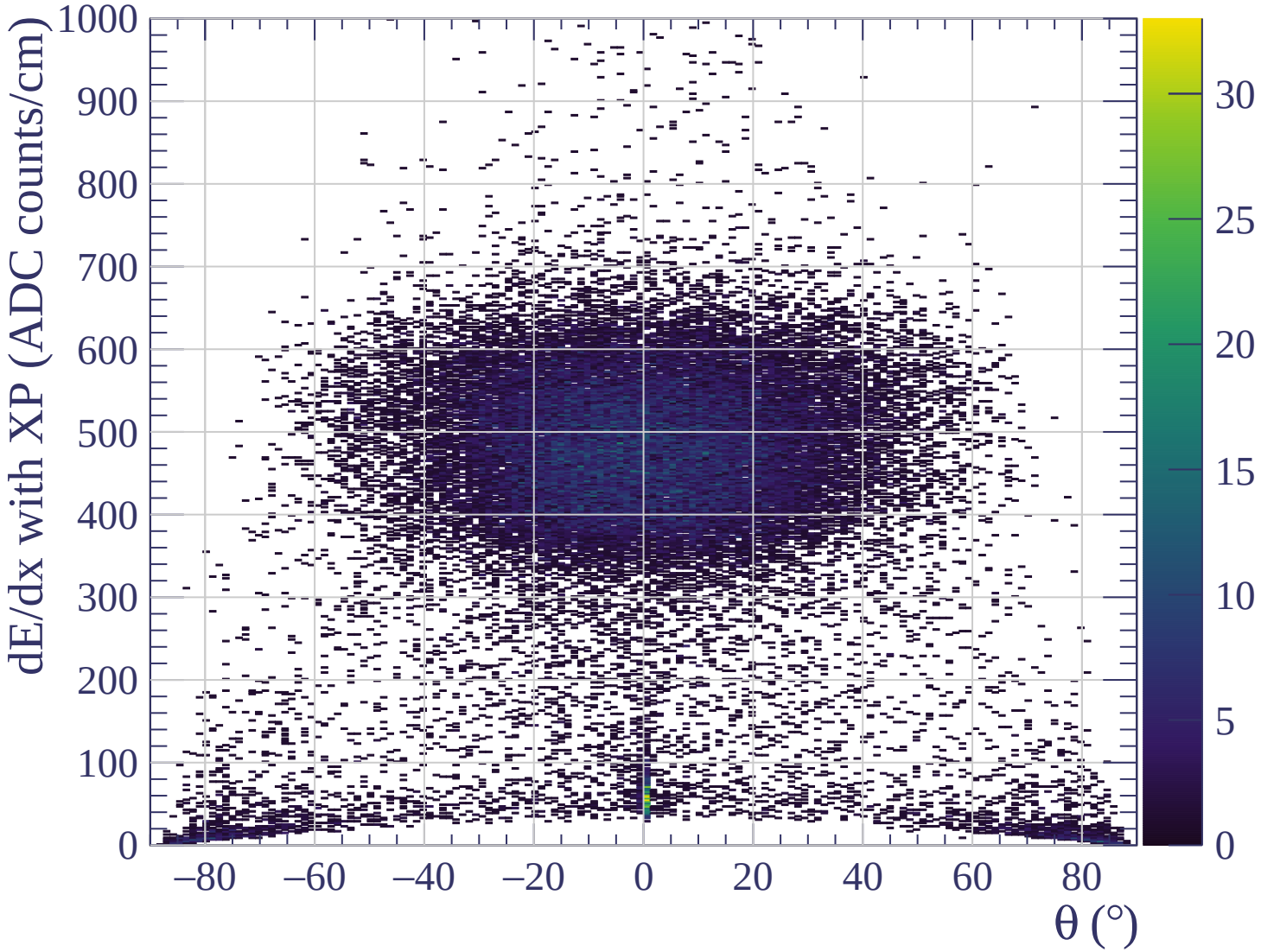


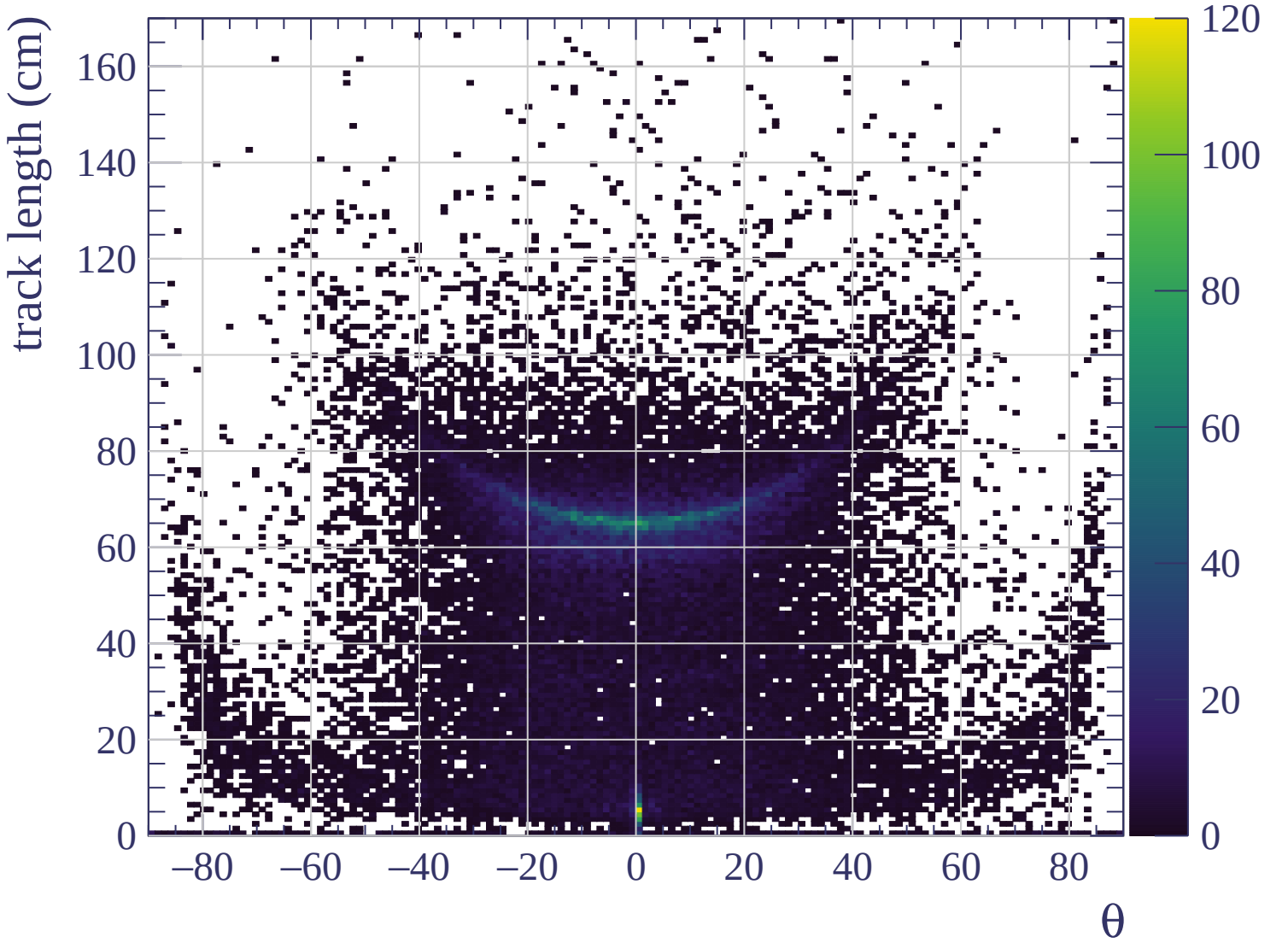


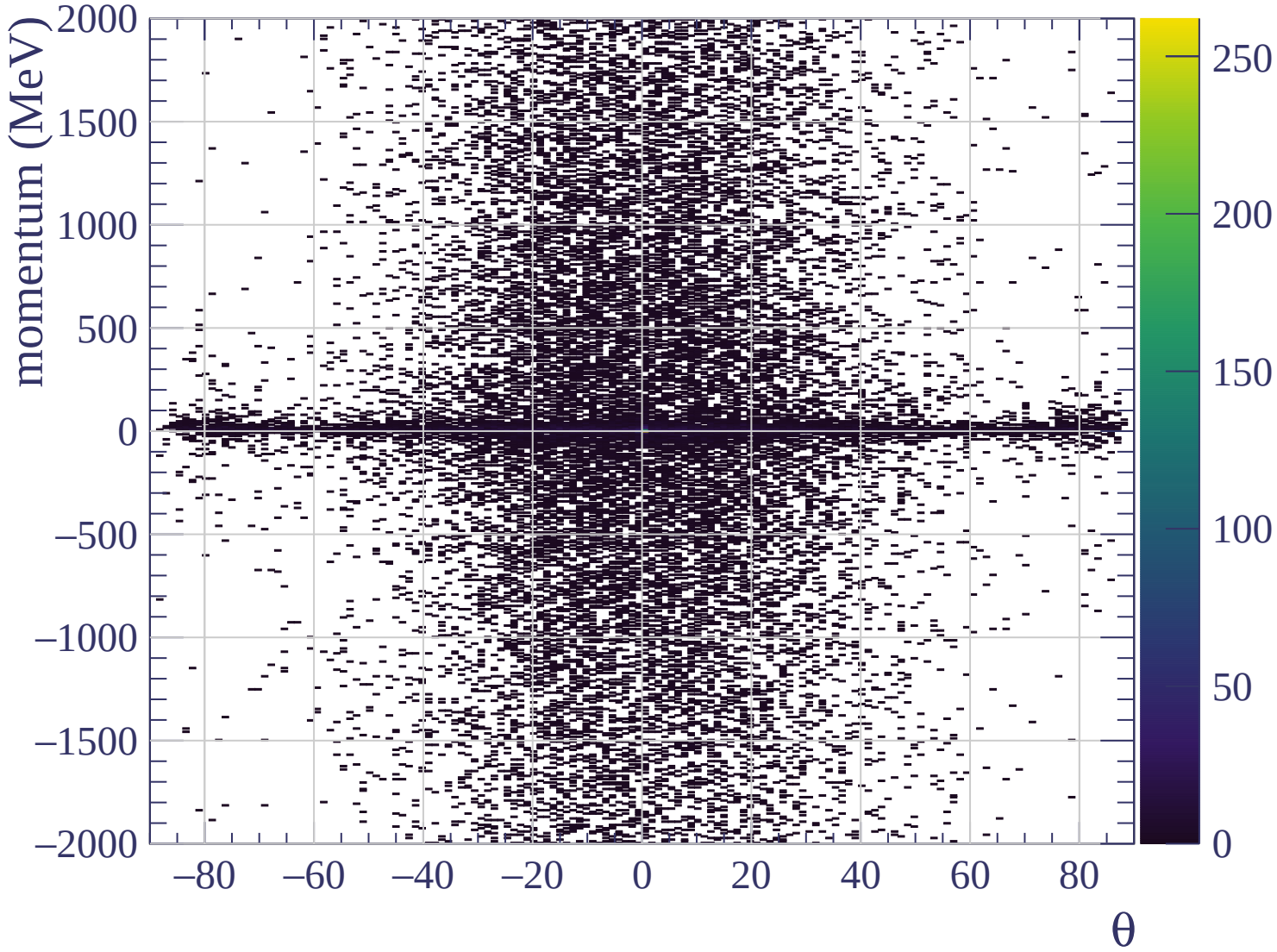


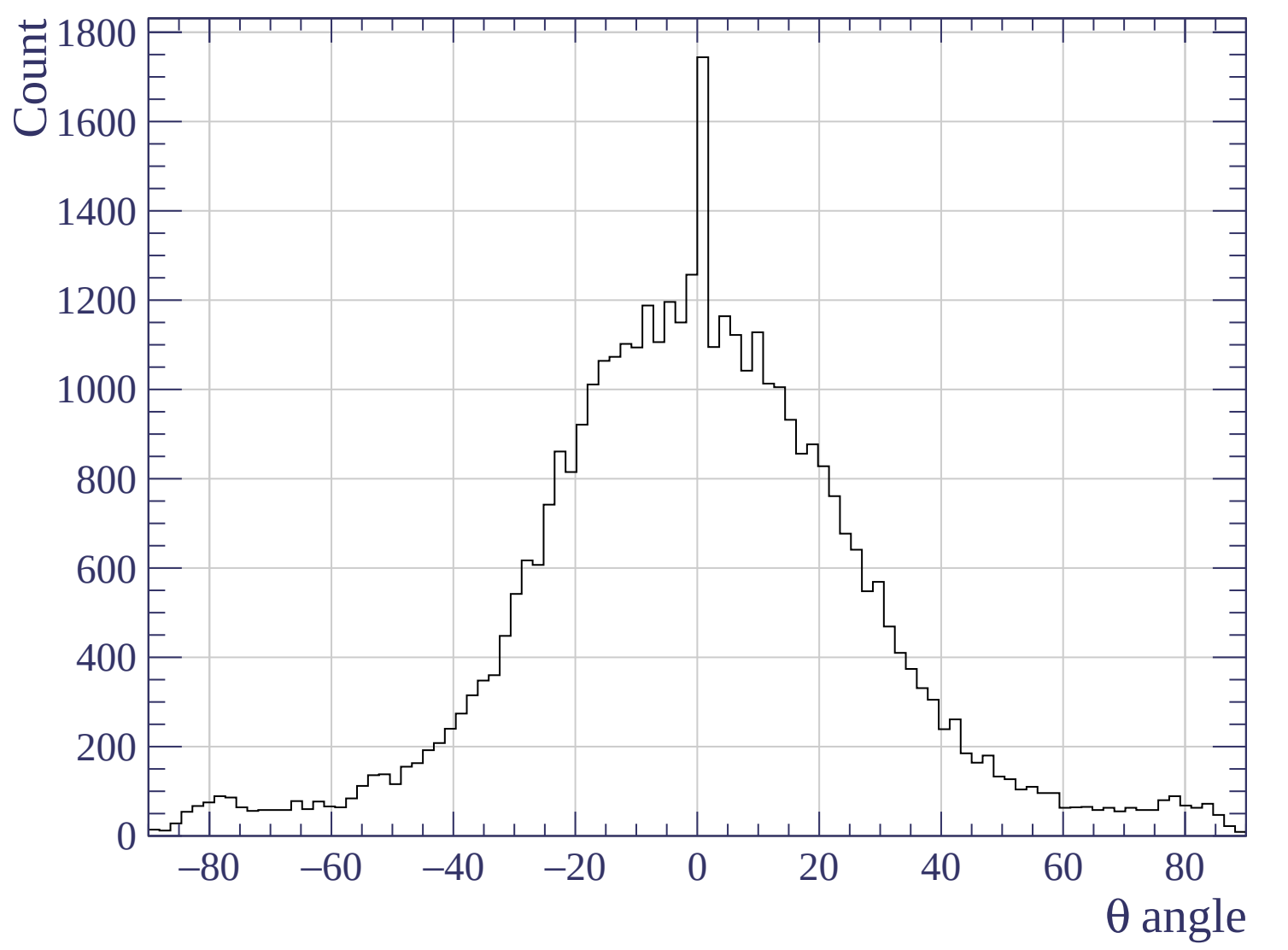


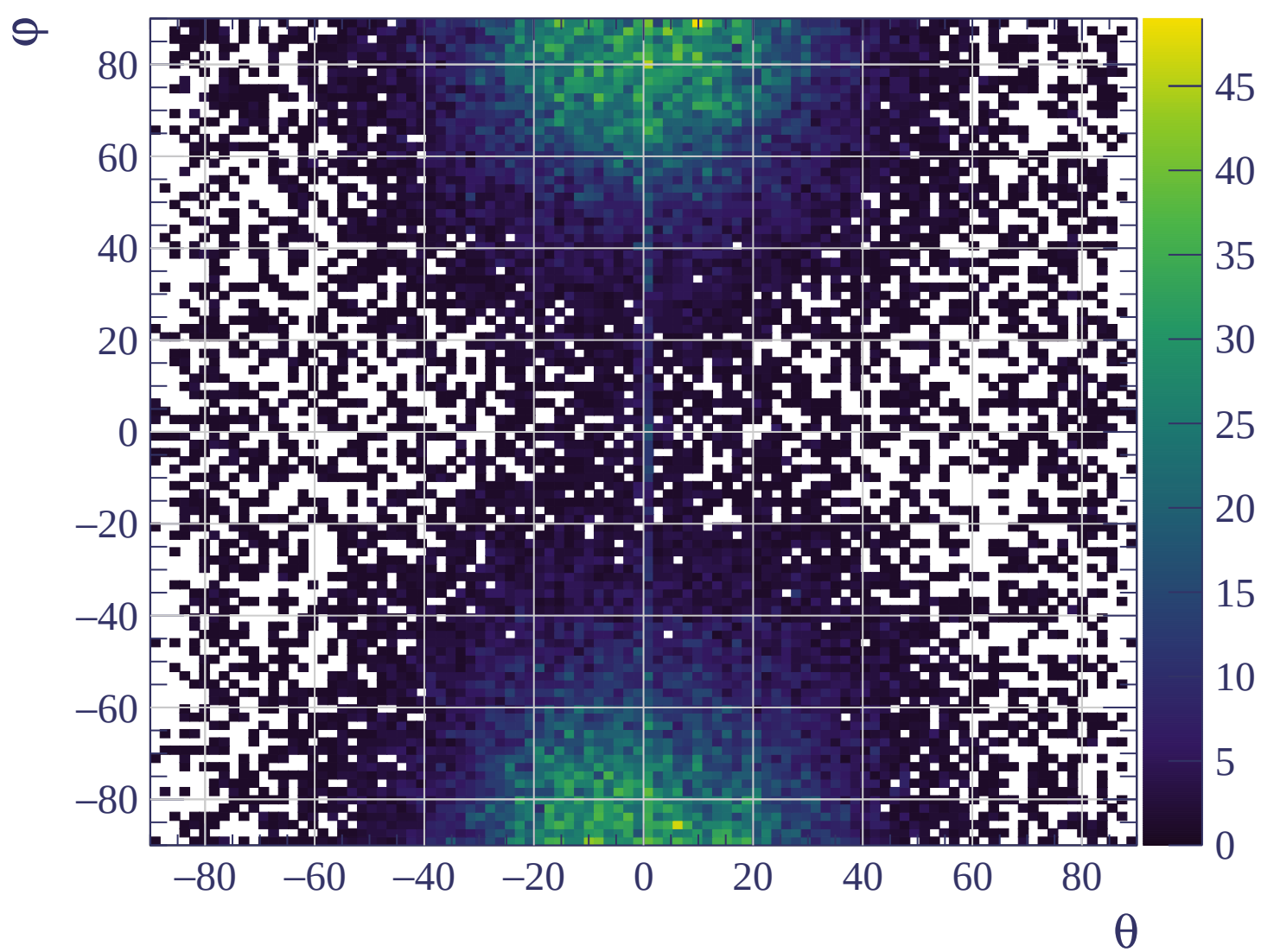


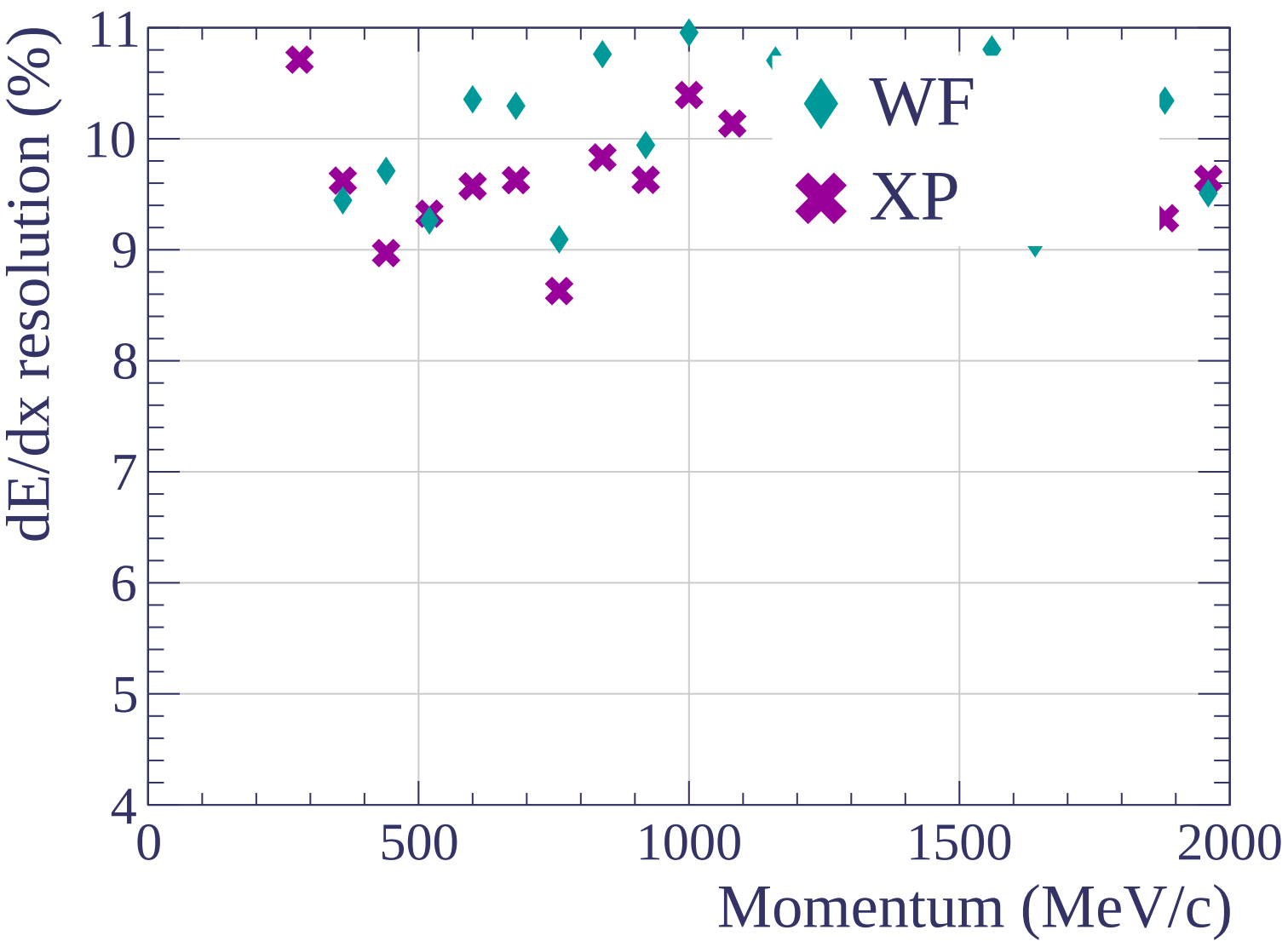


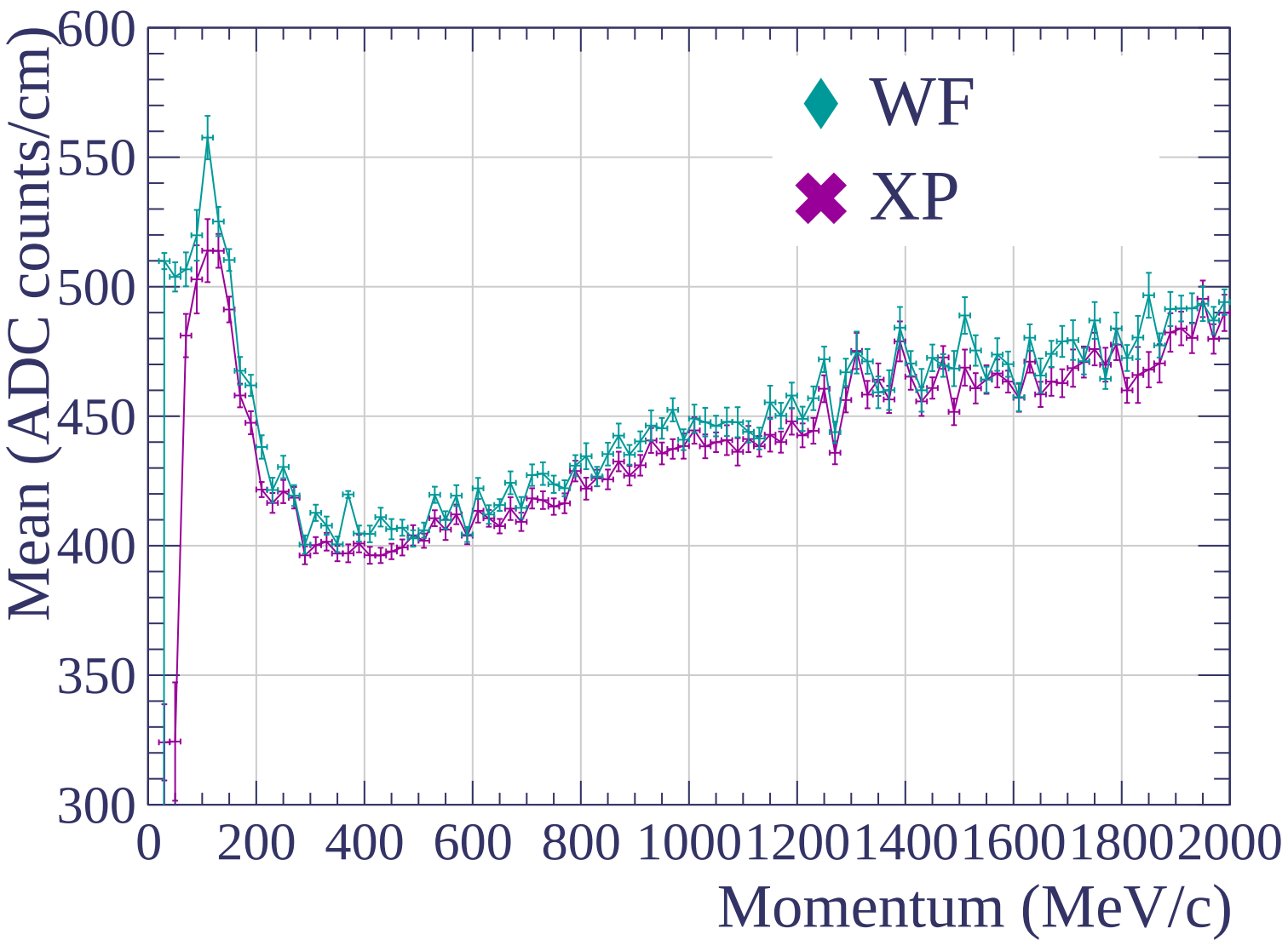


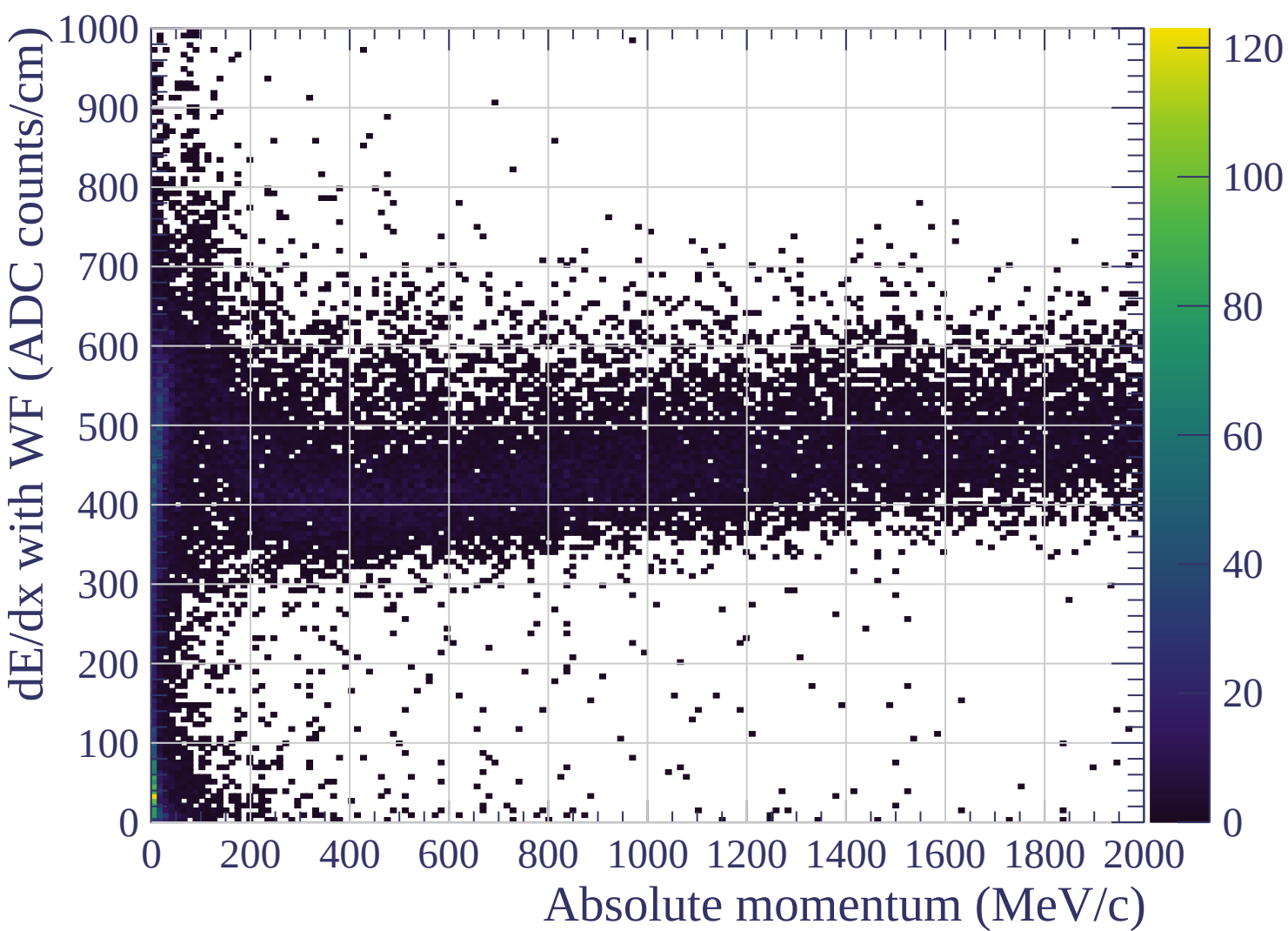


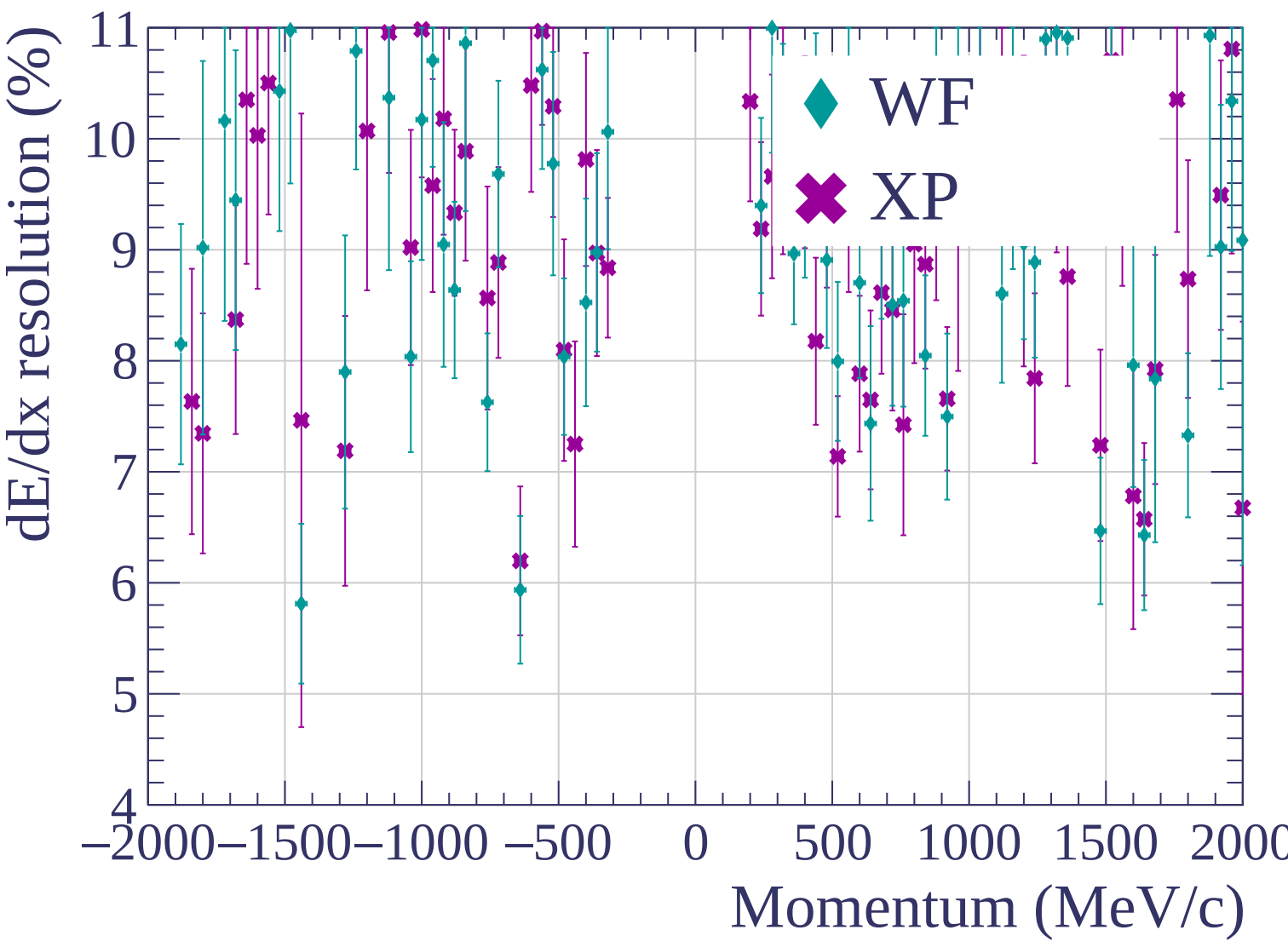


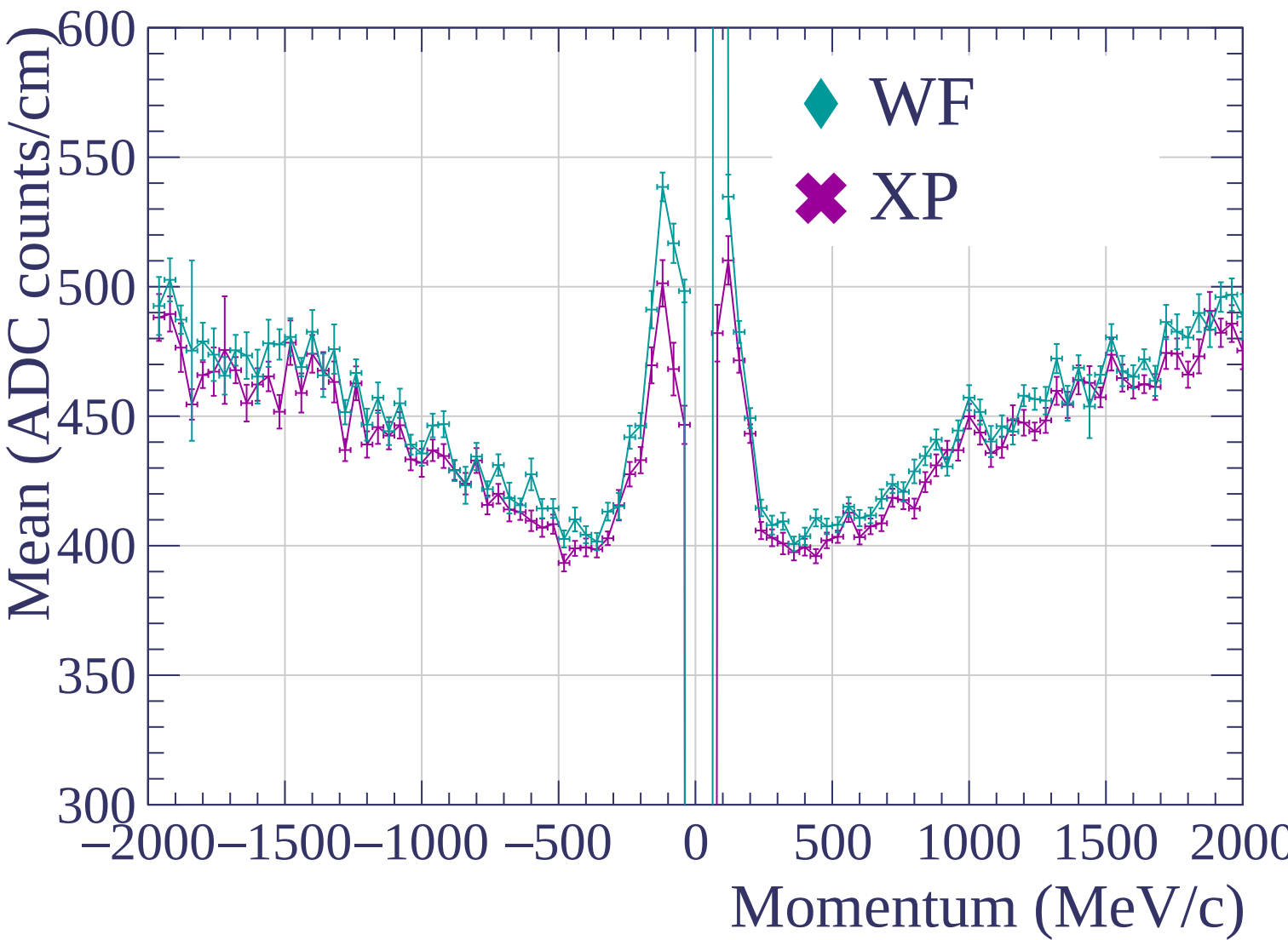


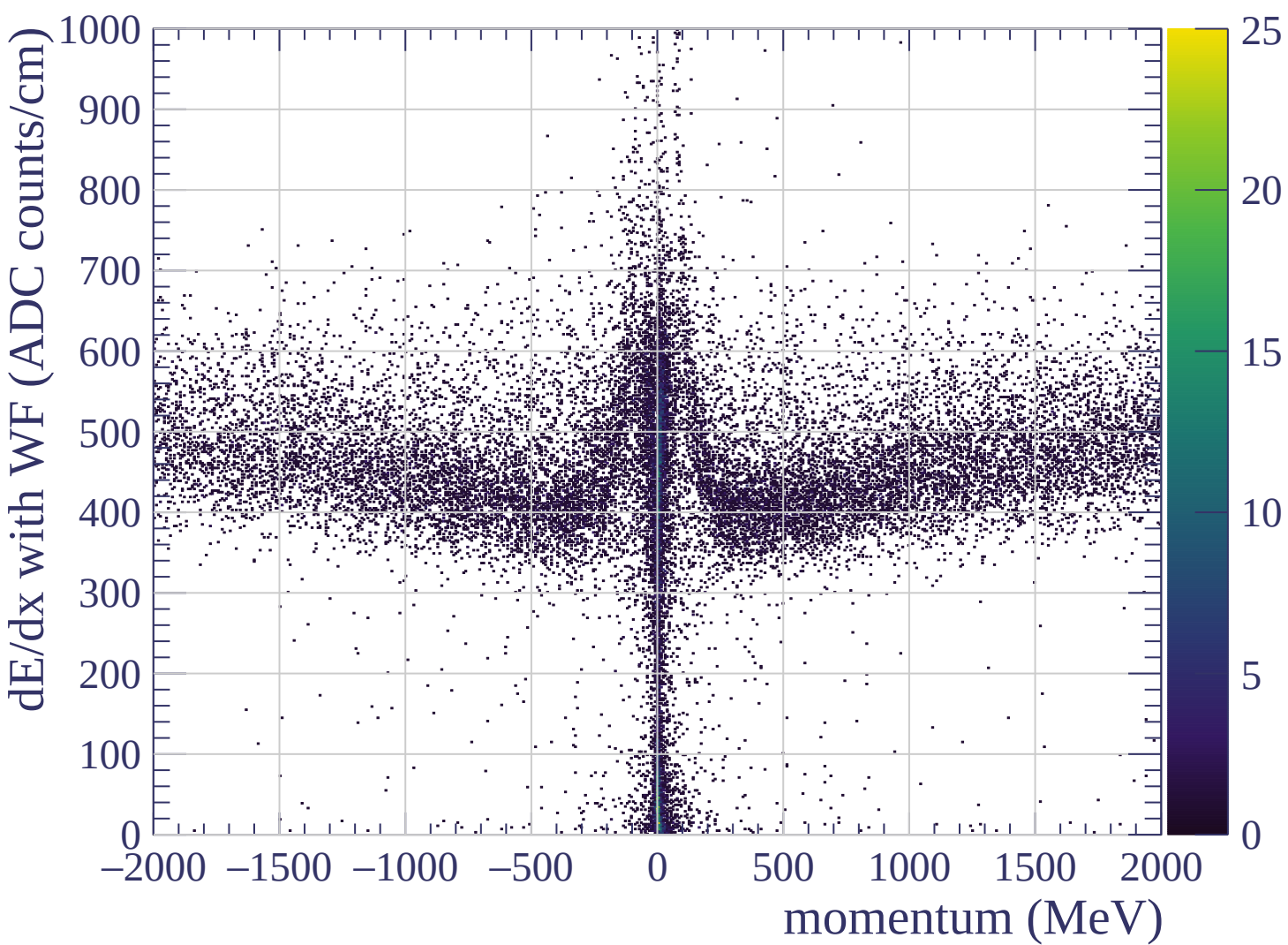


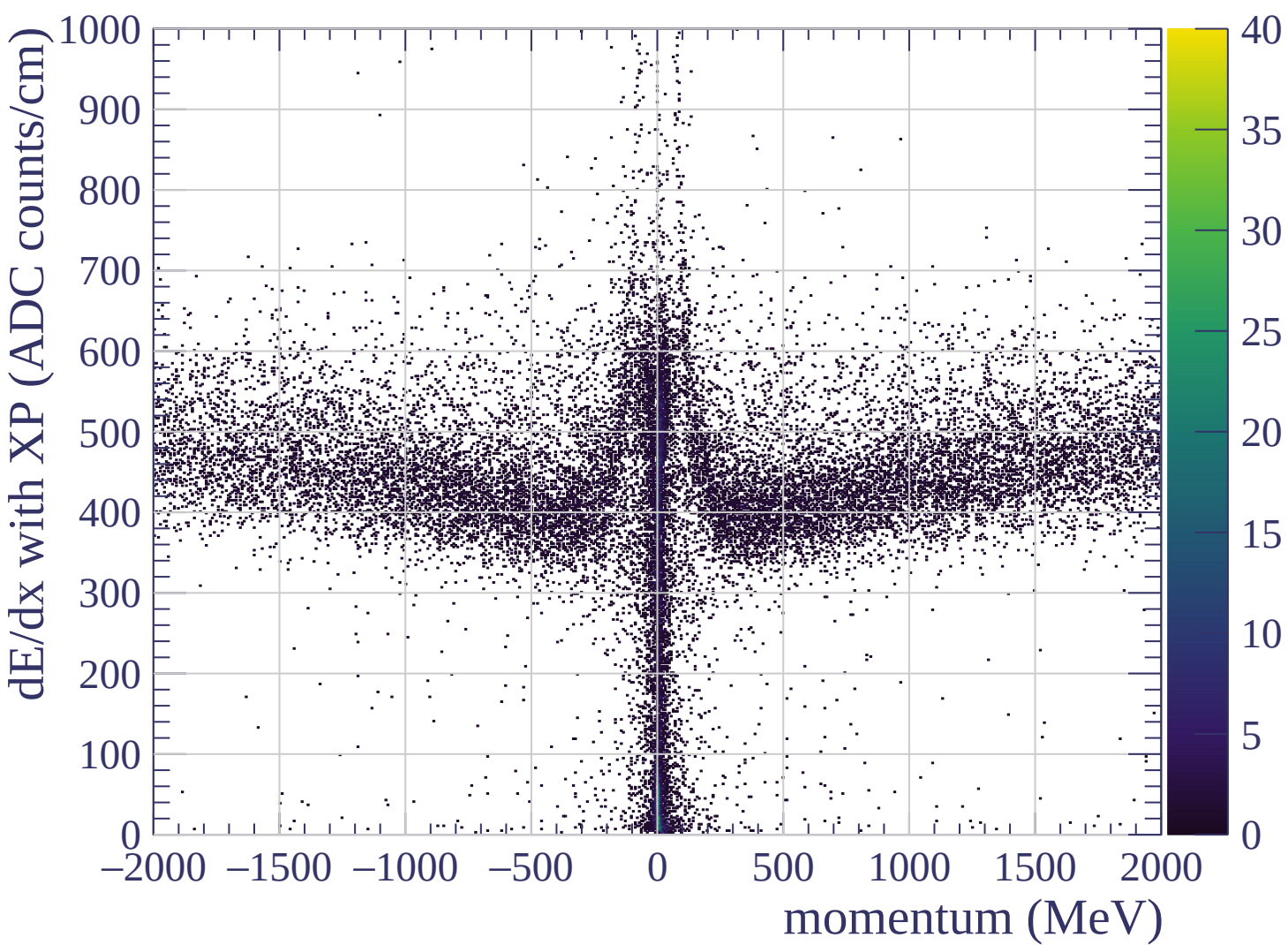


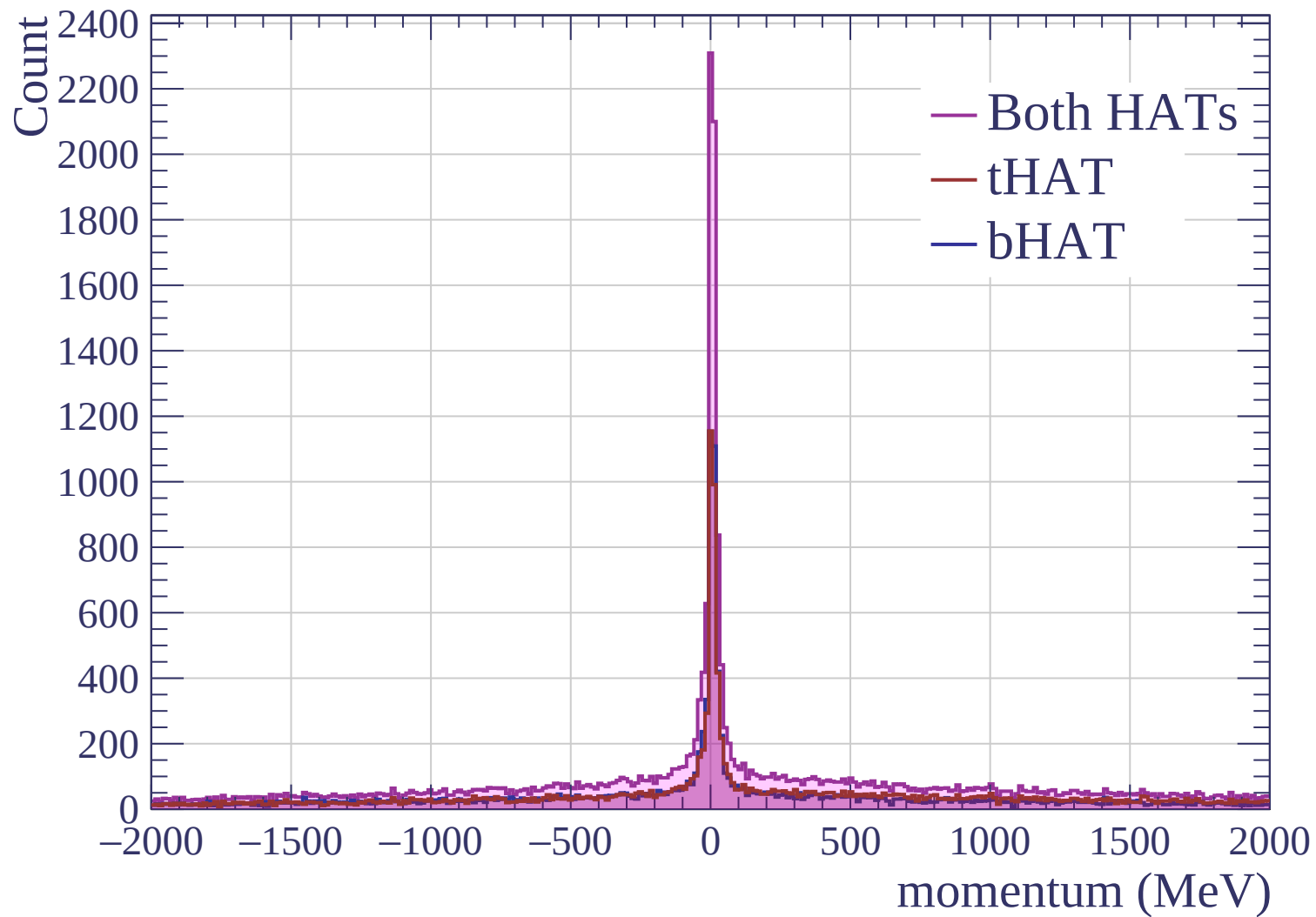


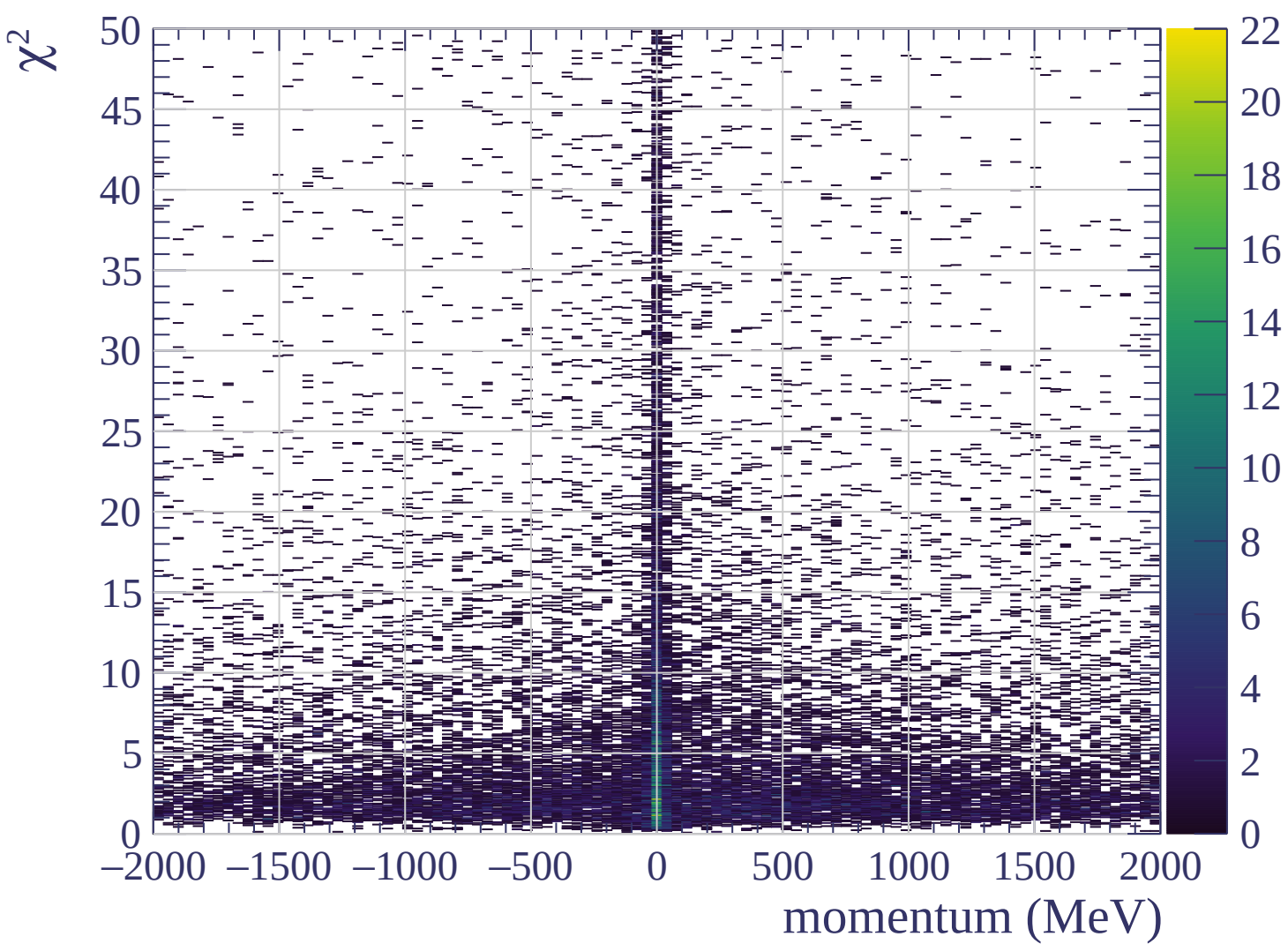


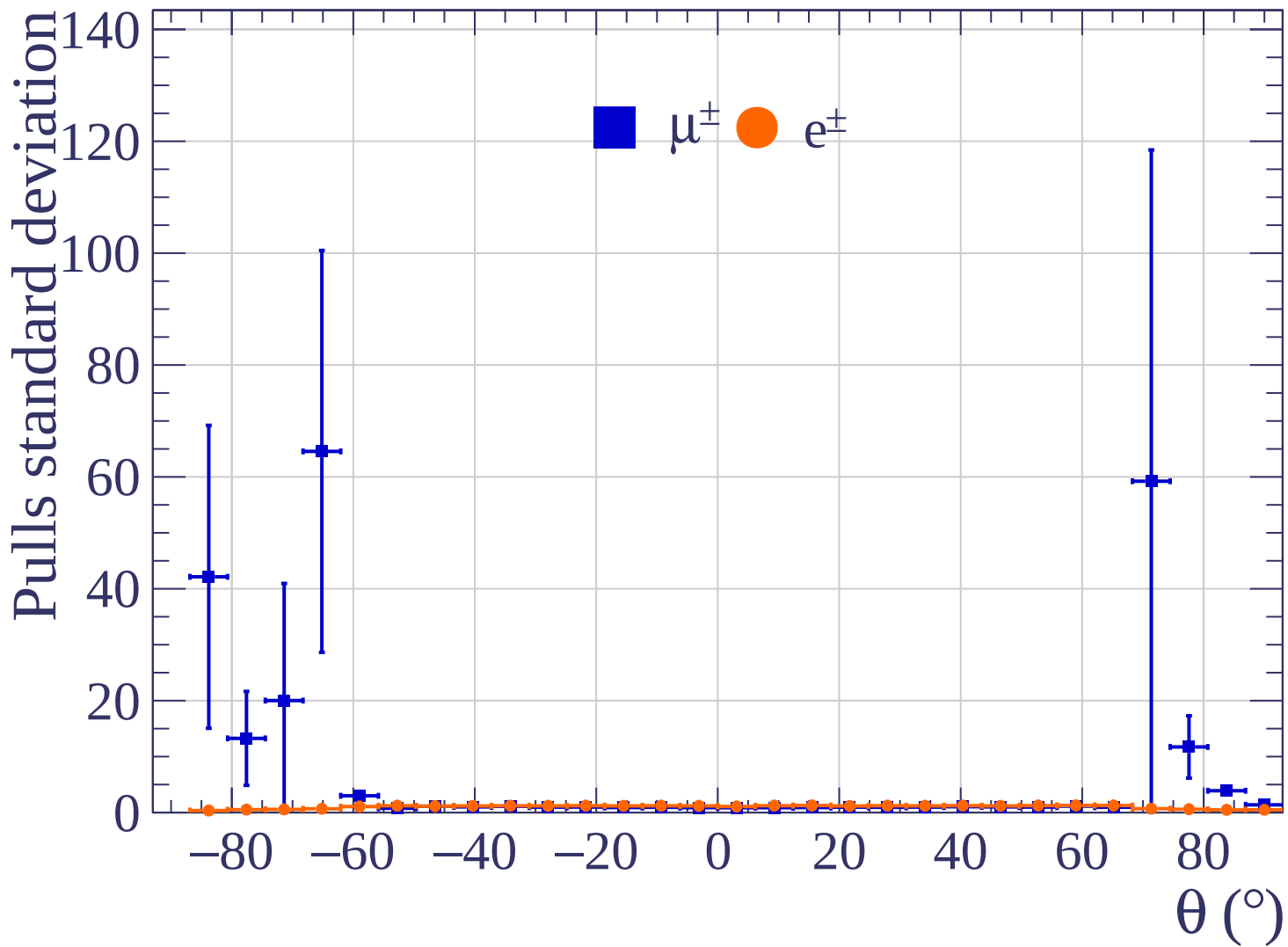


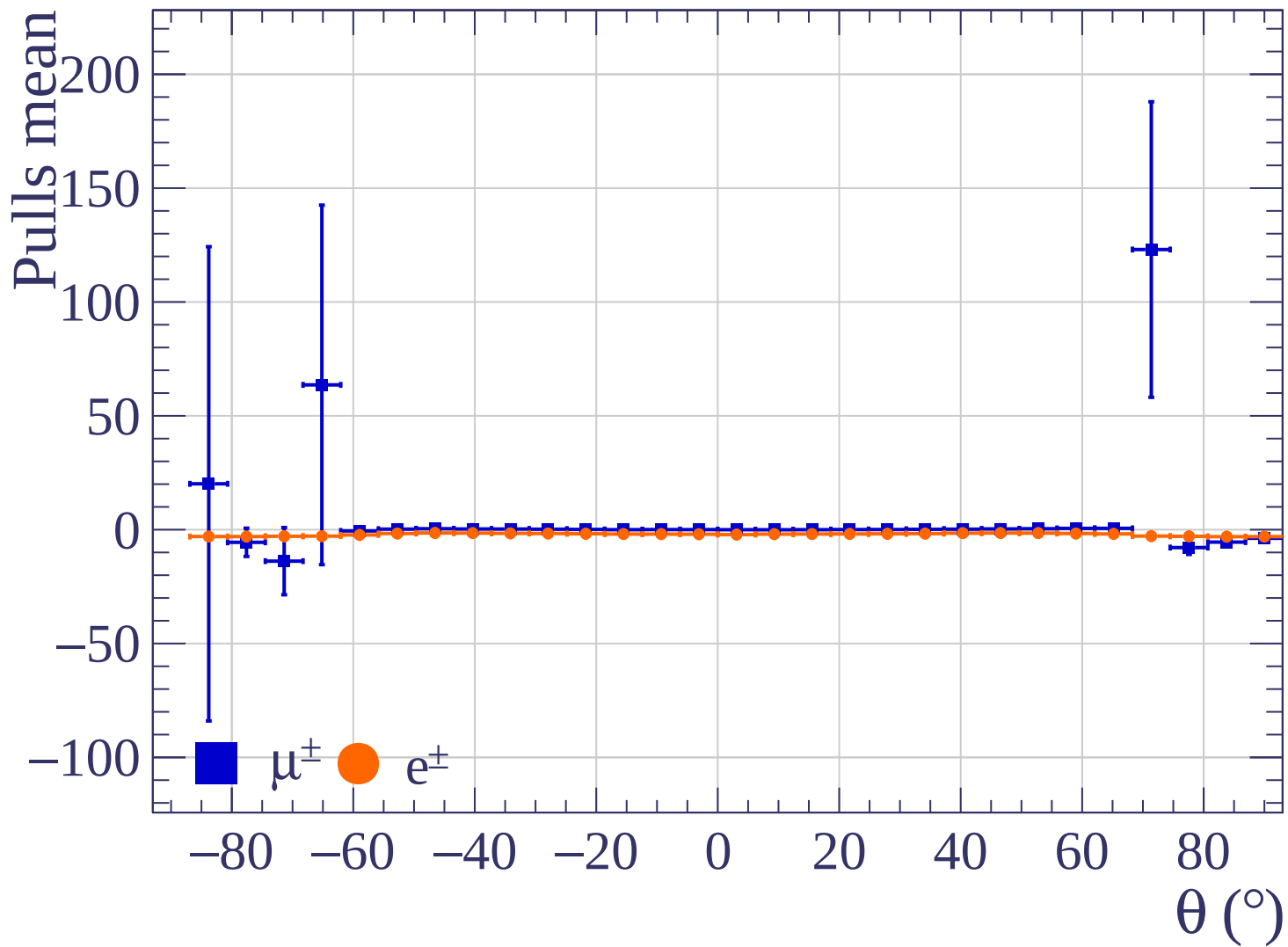


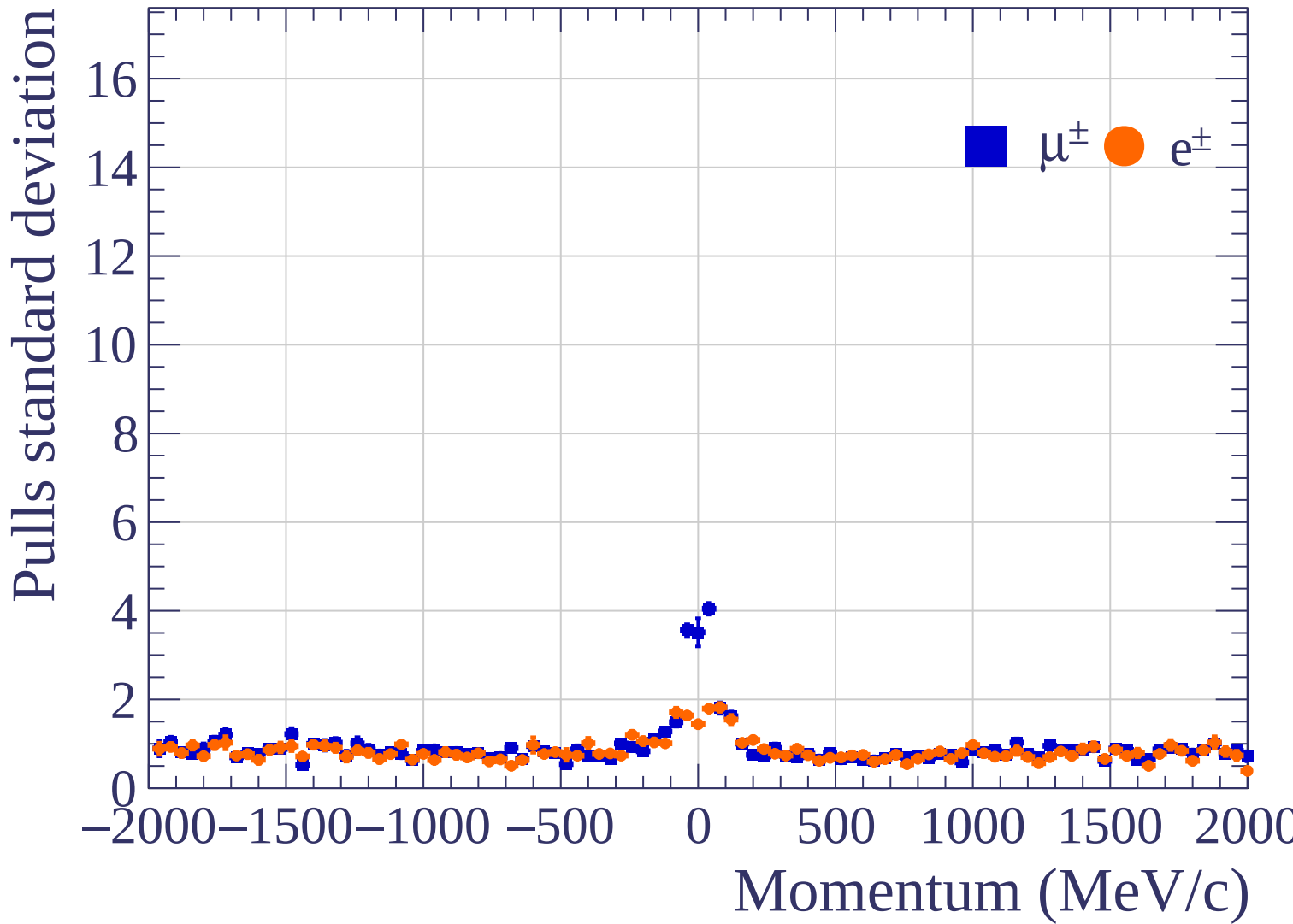


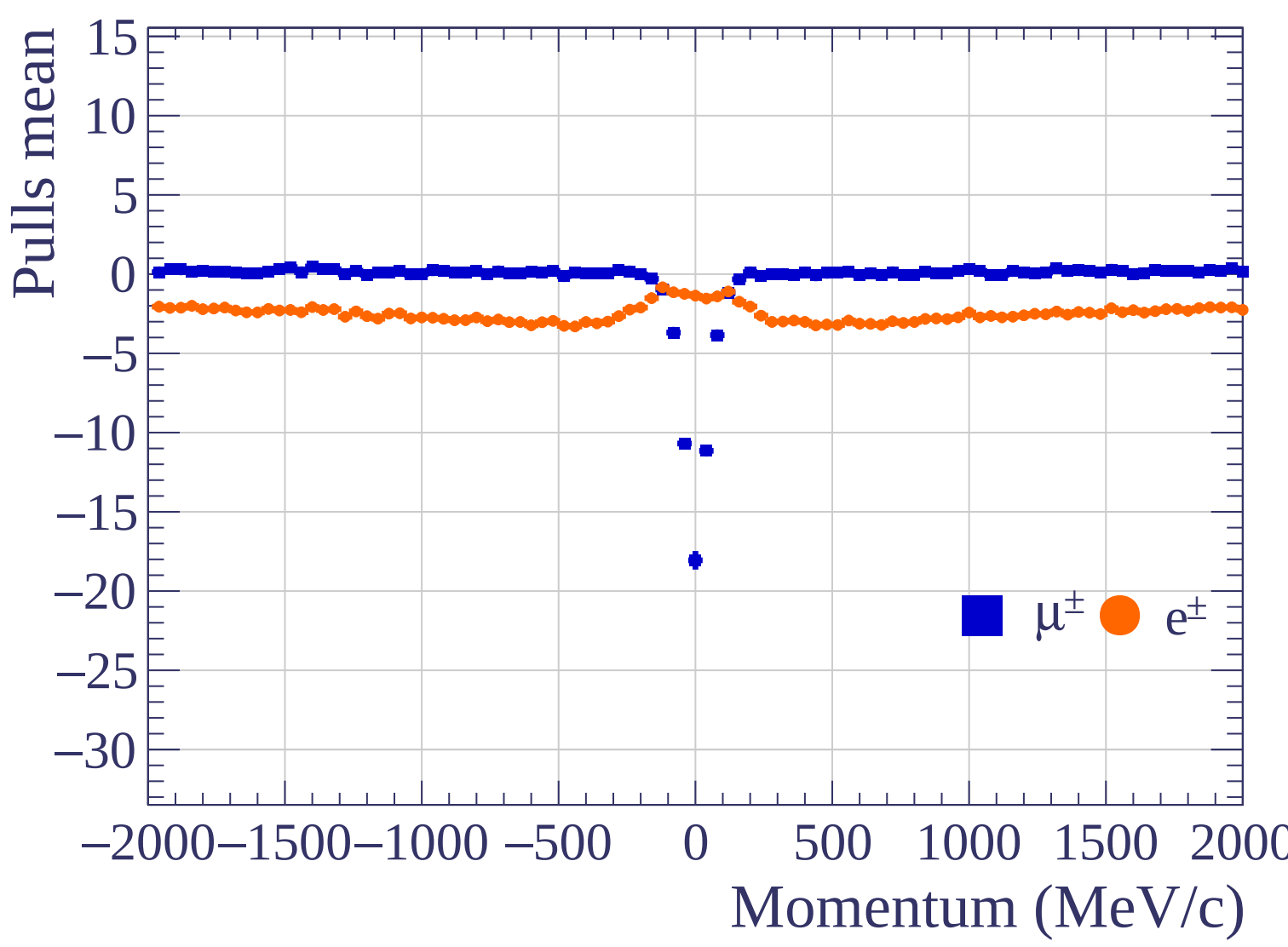


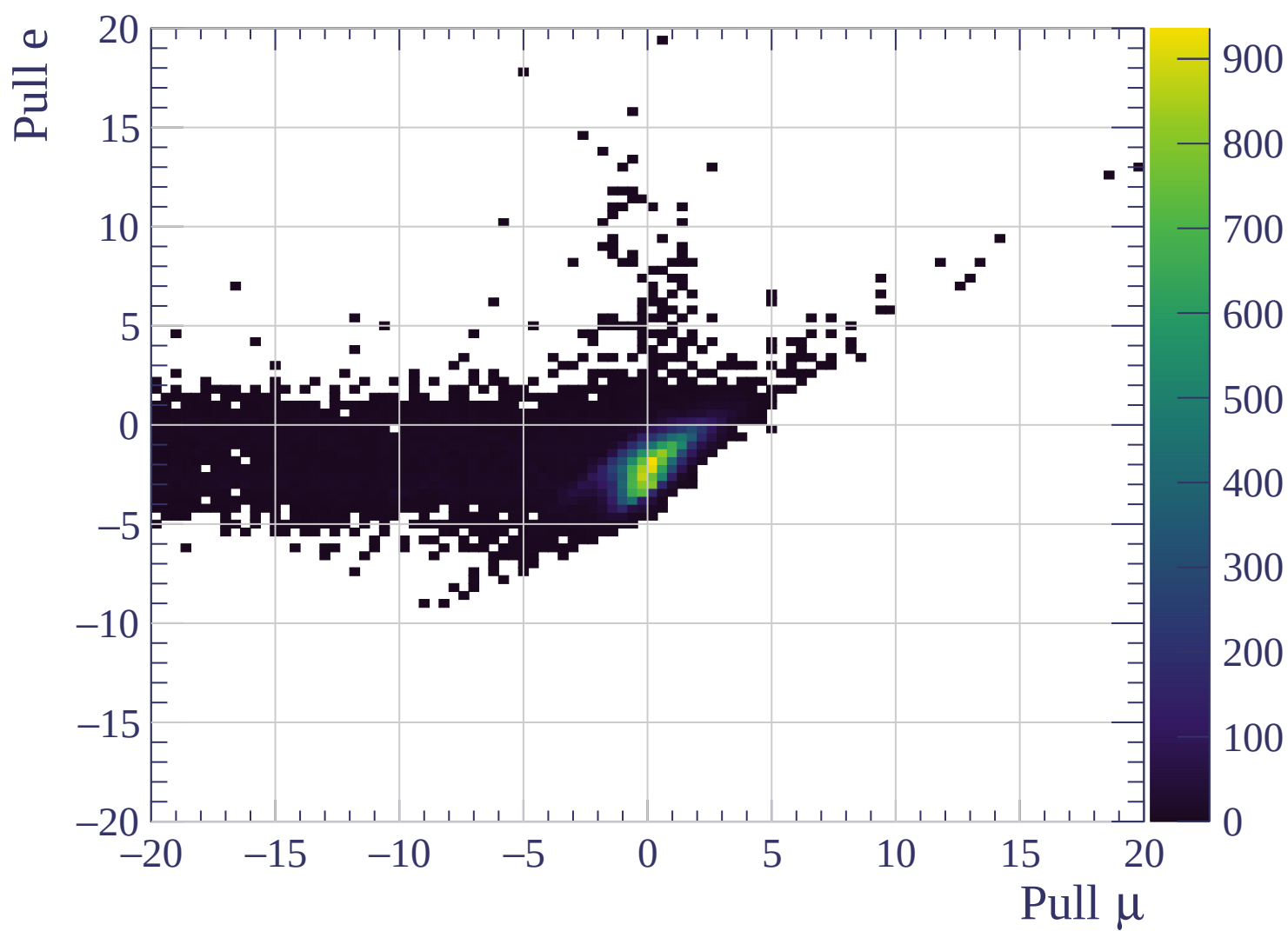


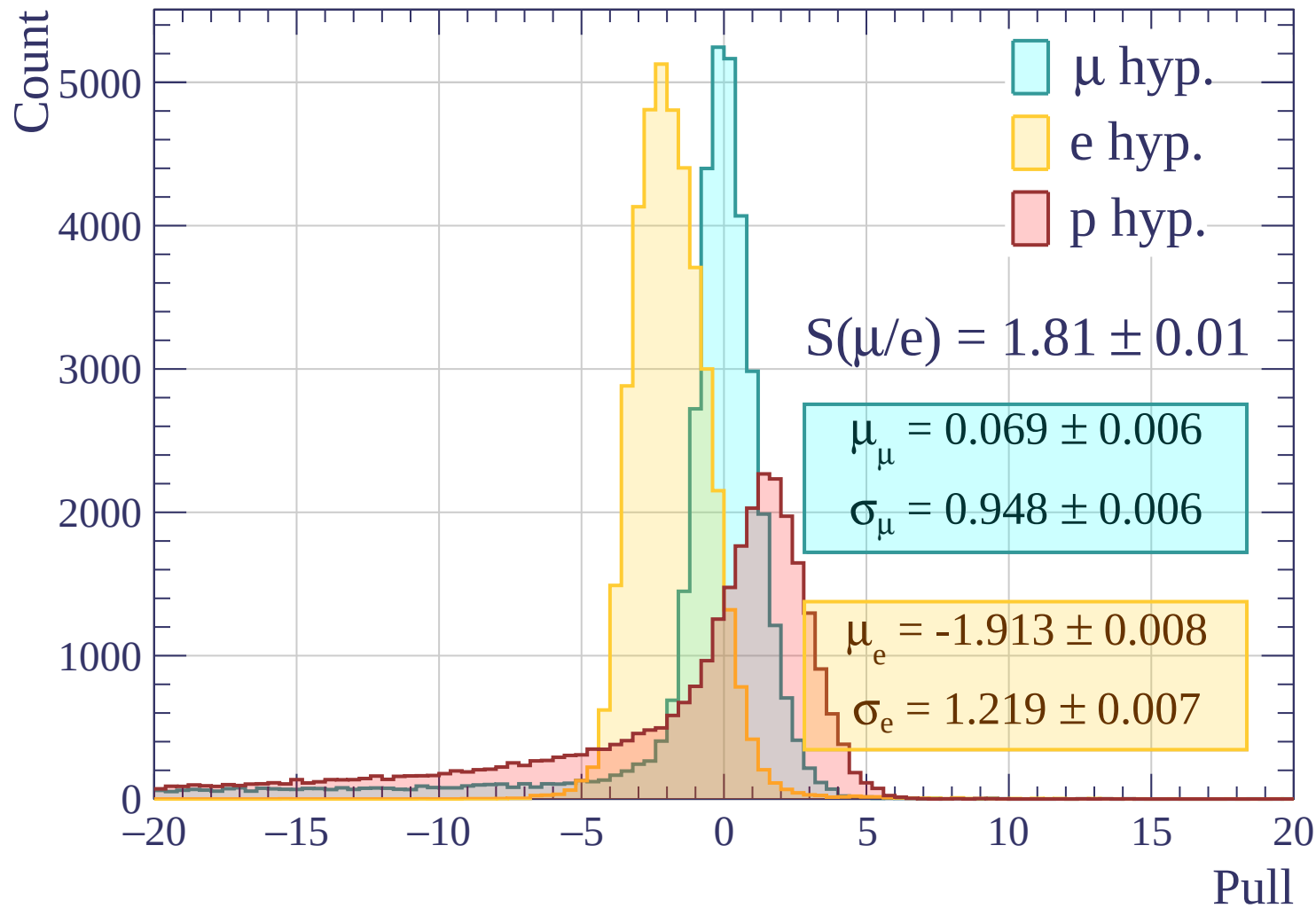


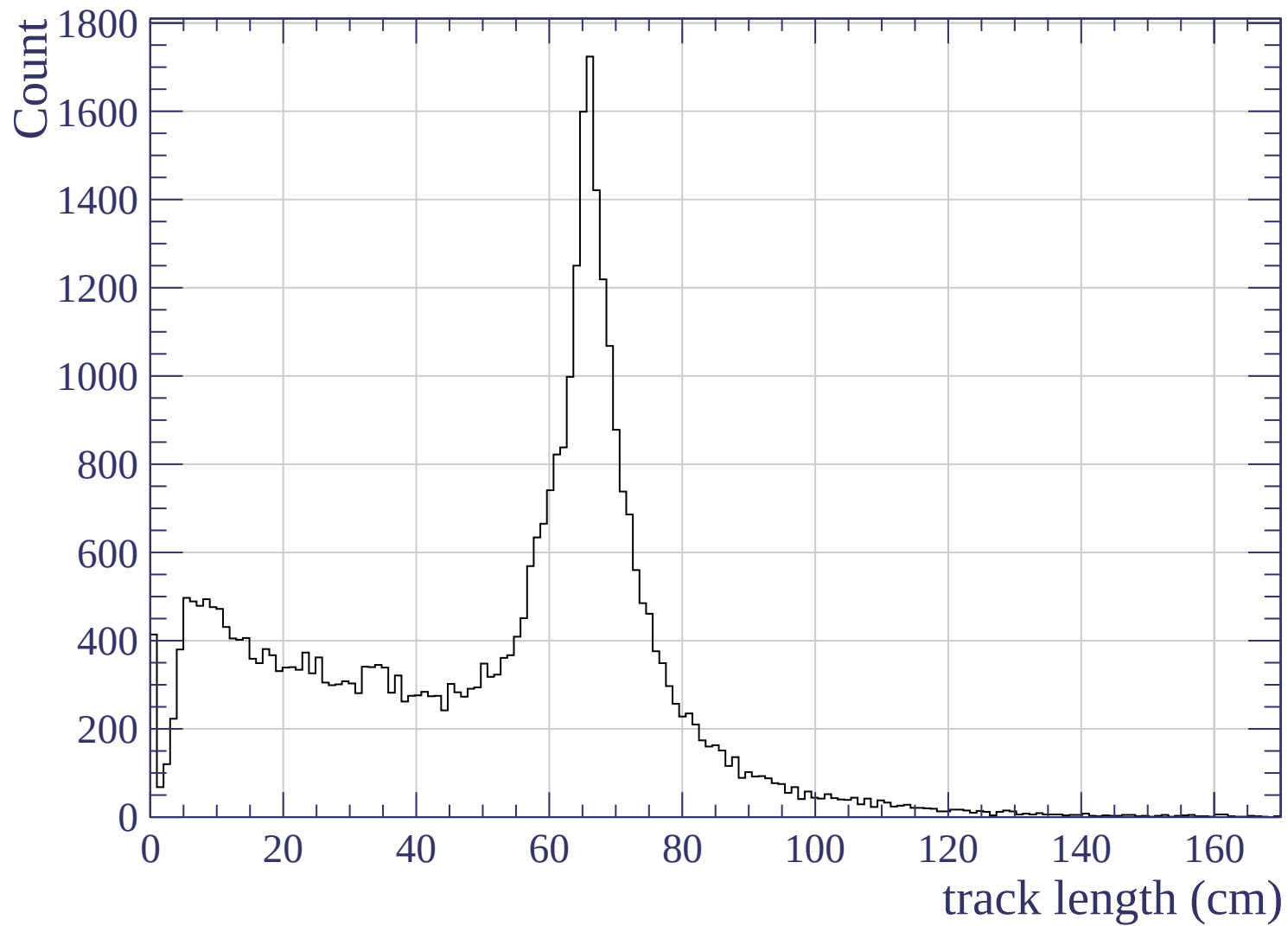


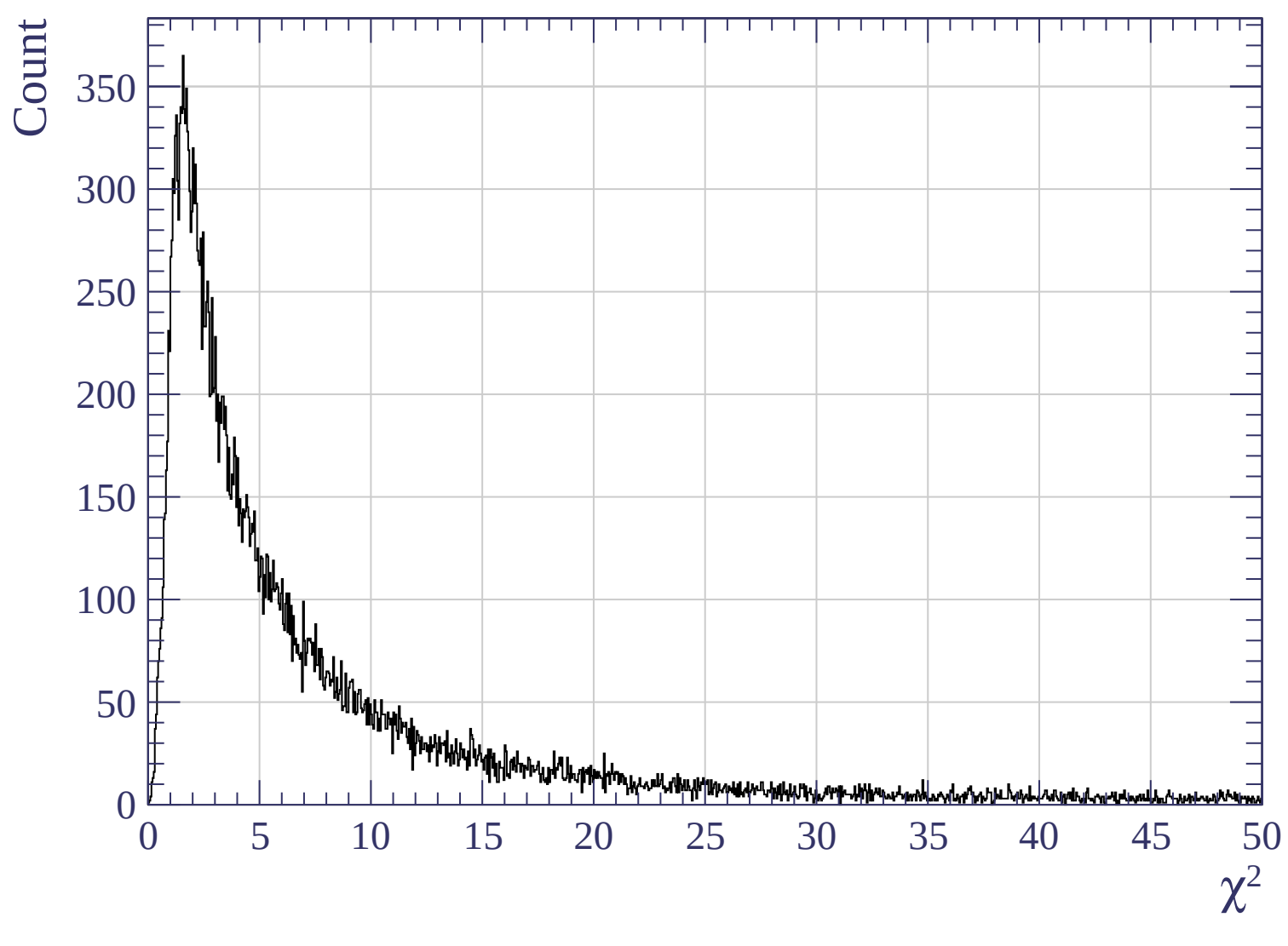


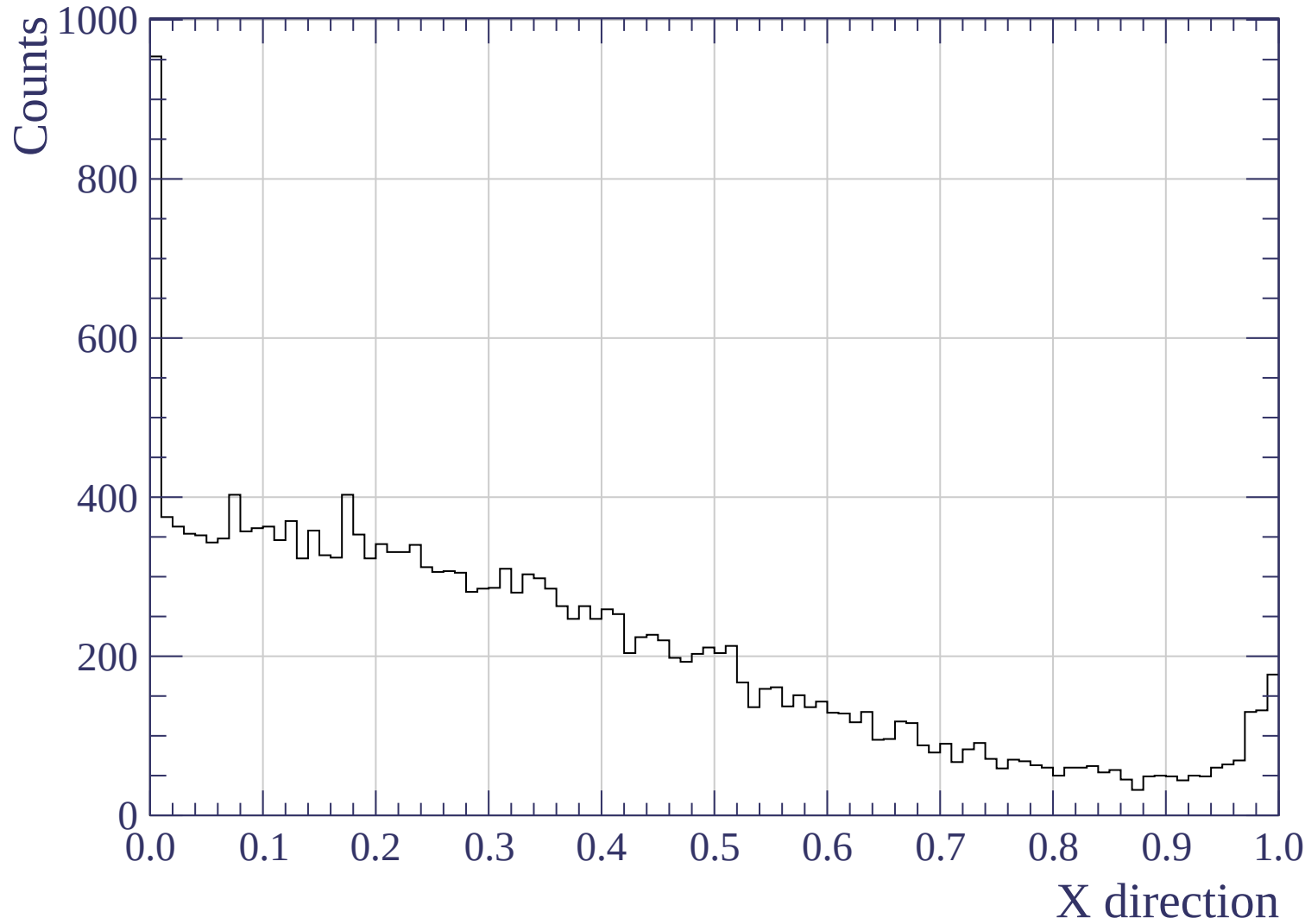


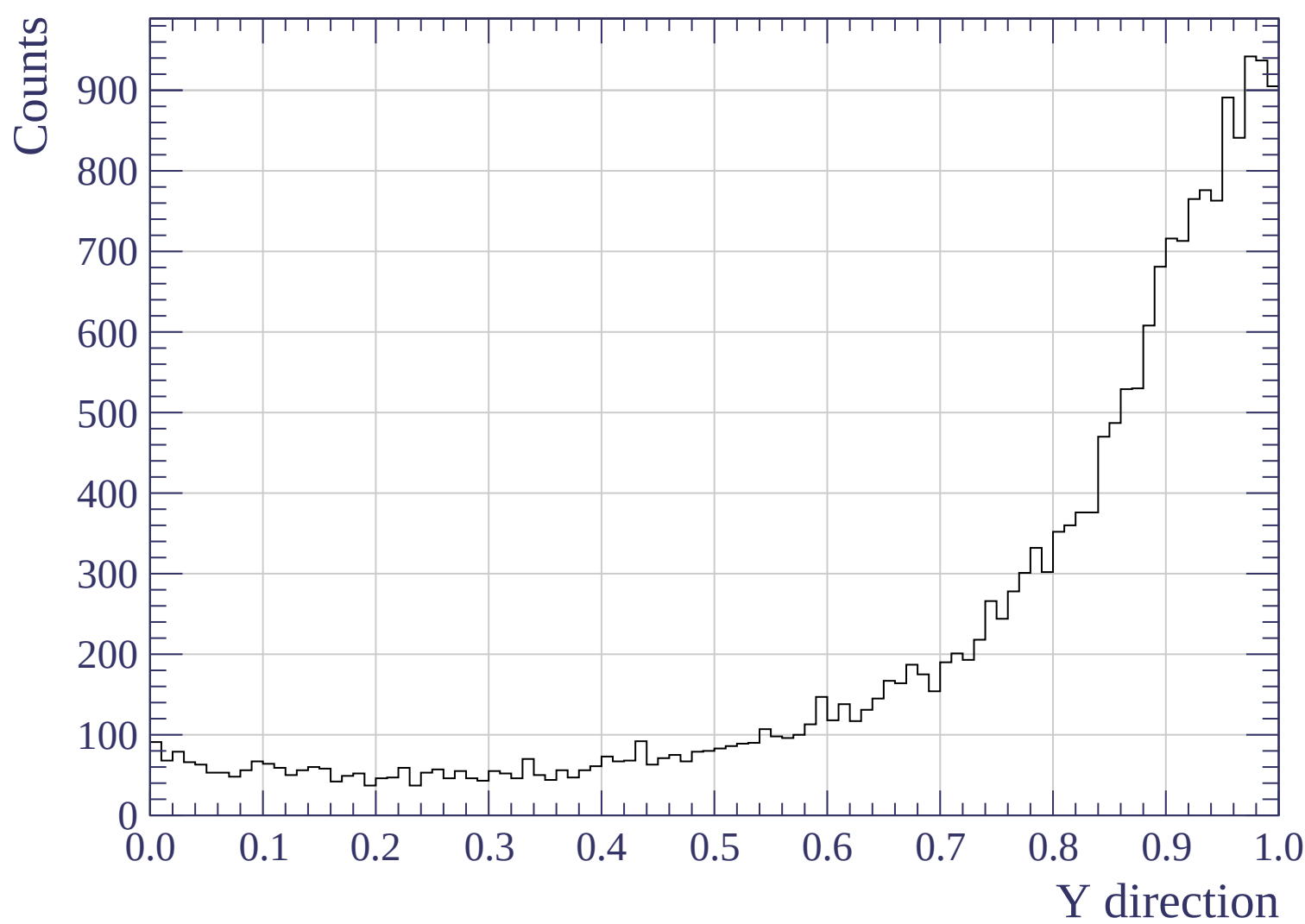


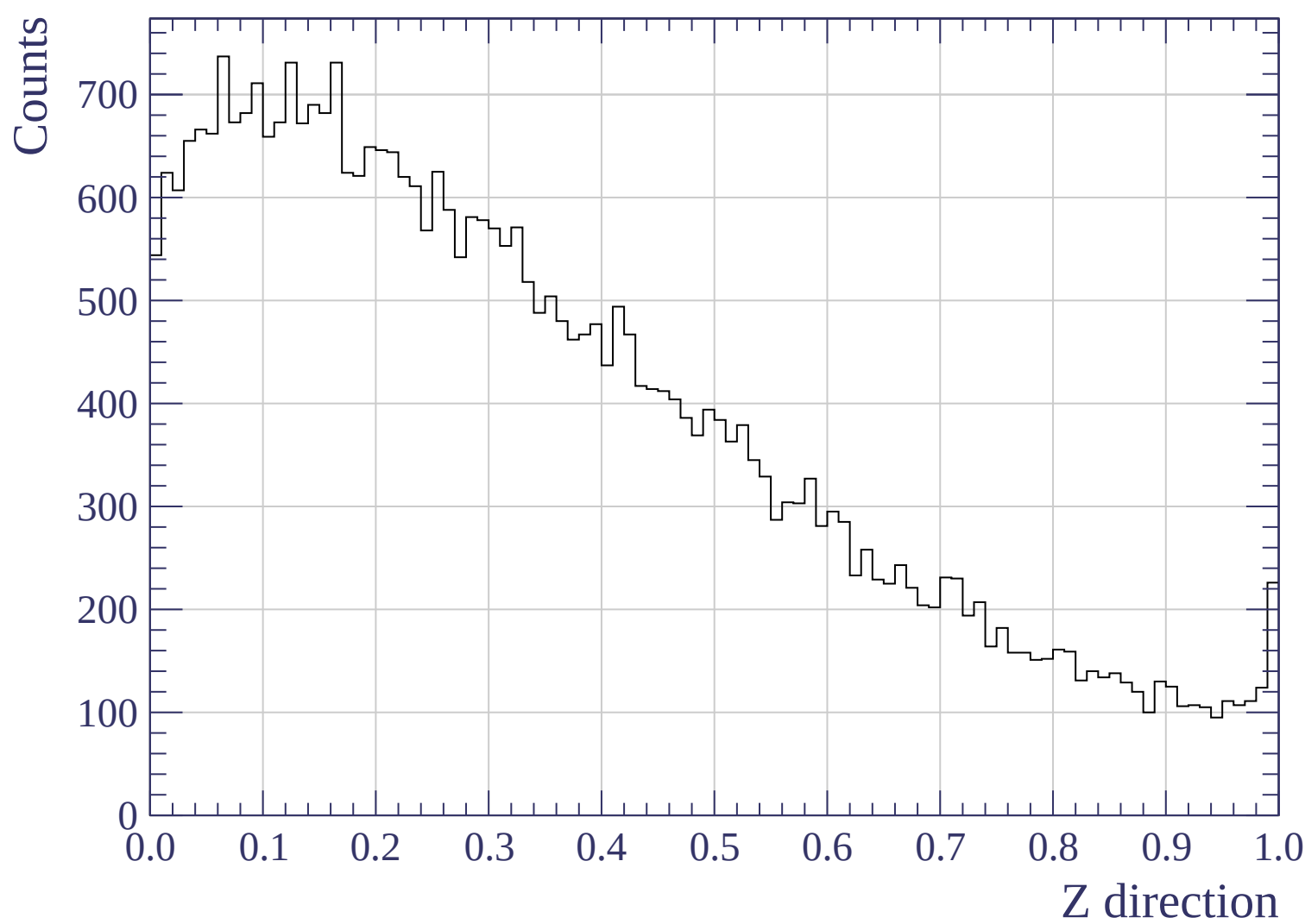


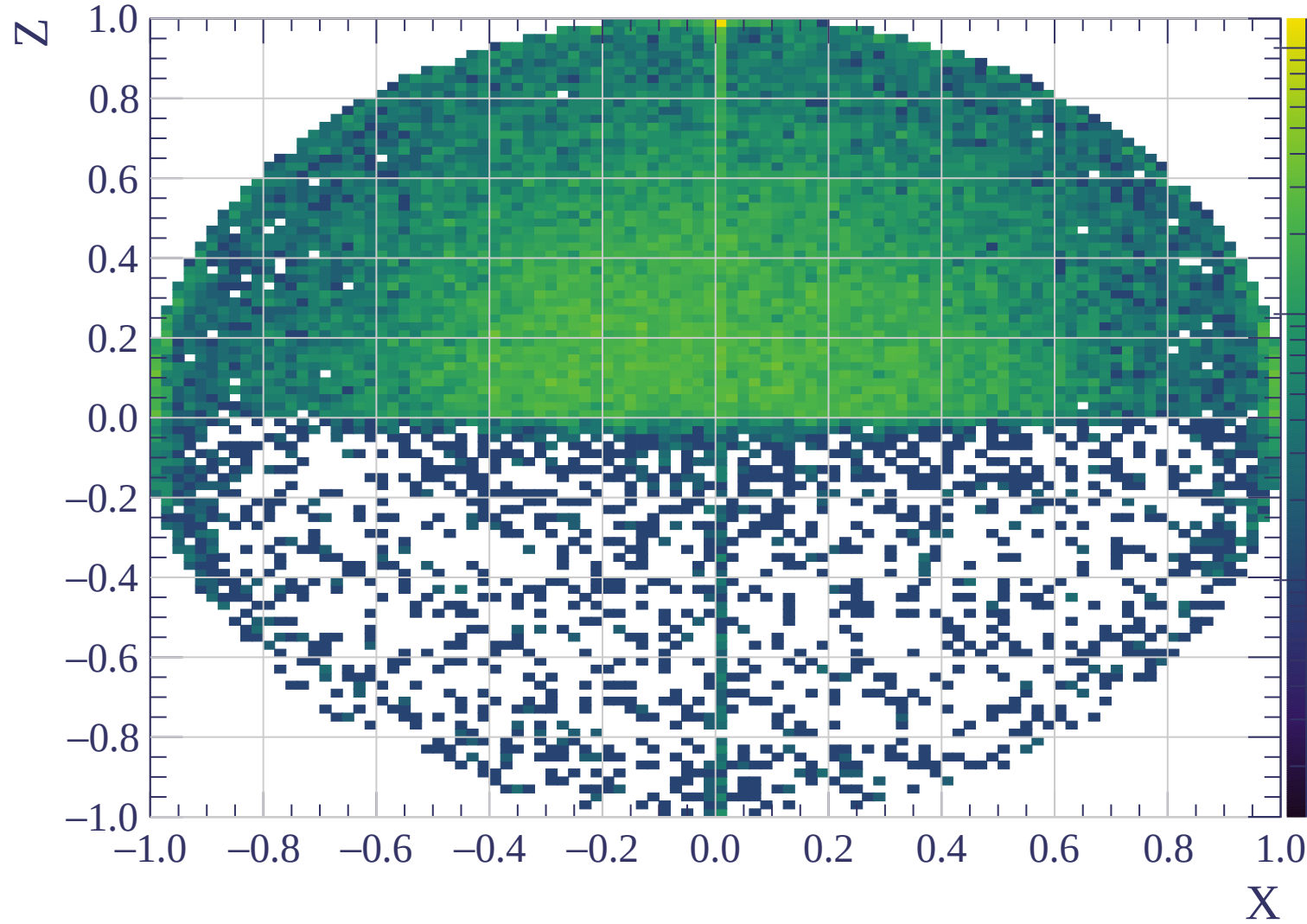


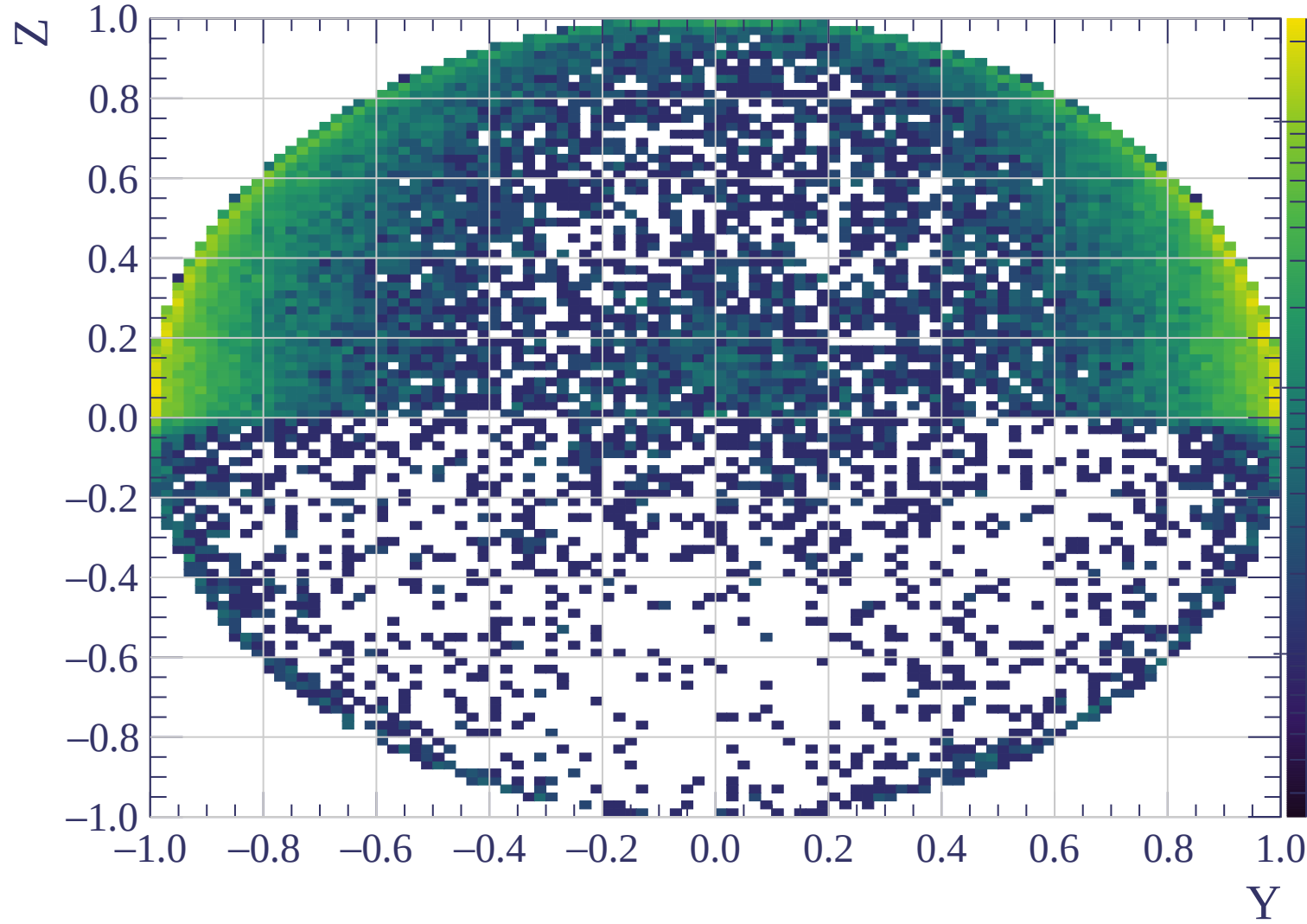


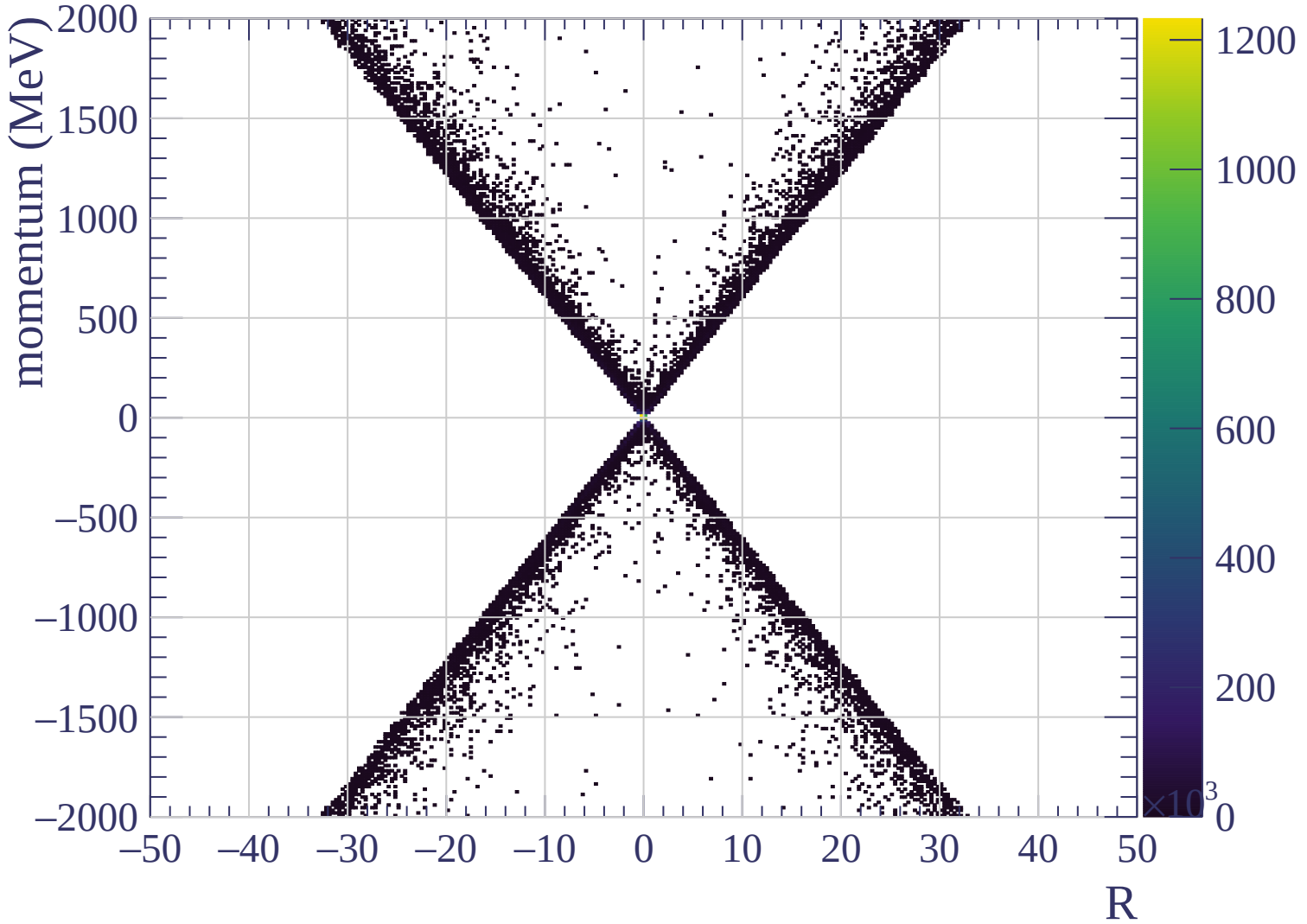


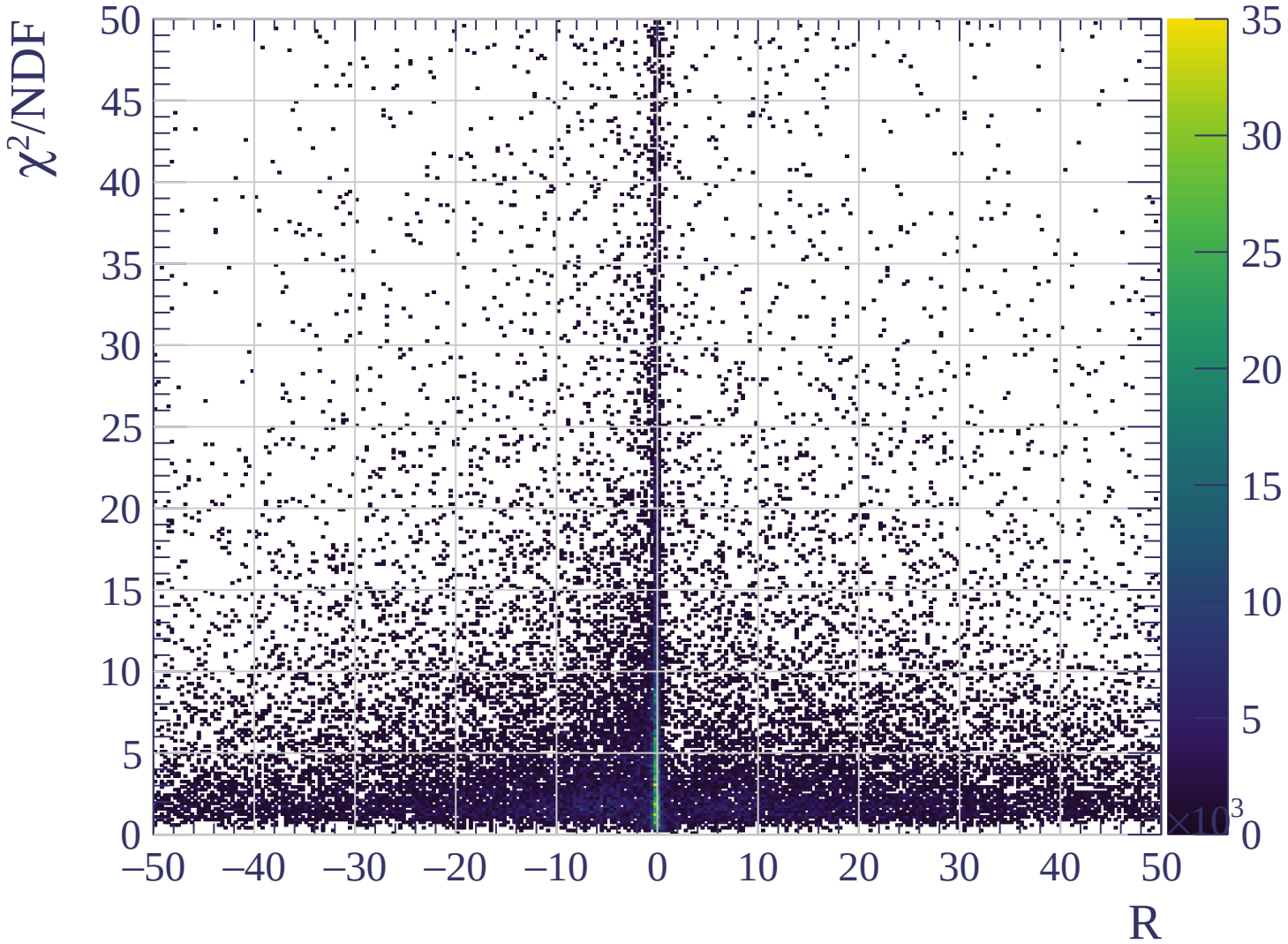






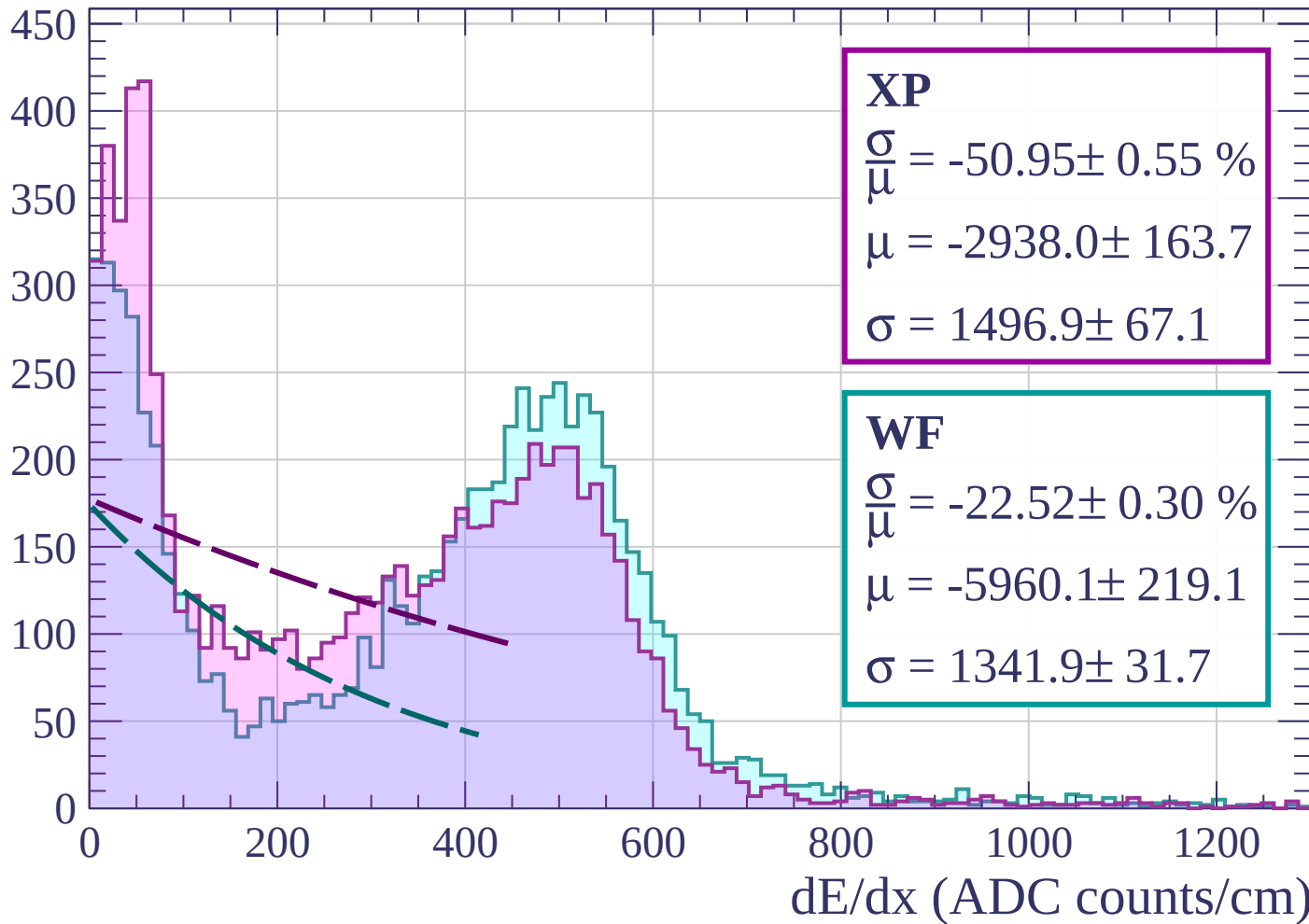






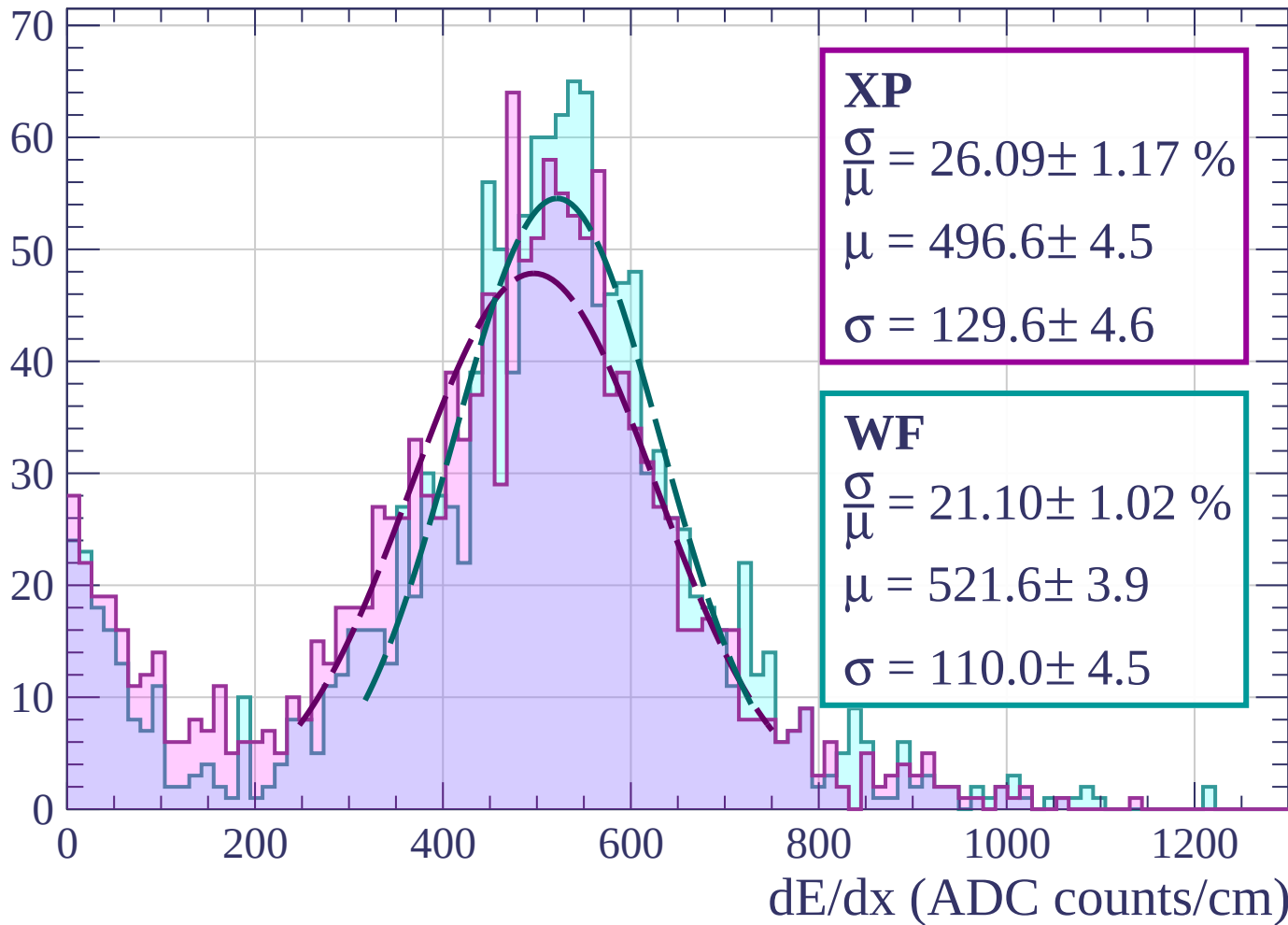
Energy loss | $0 < p < 80$

Count



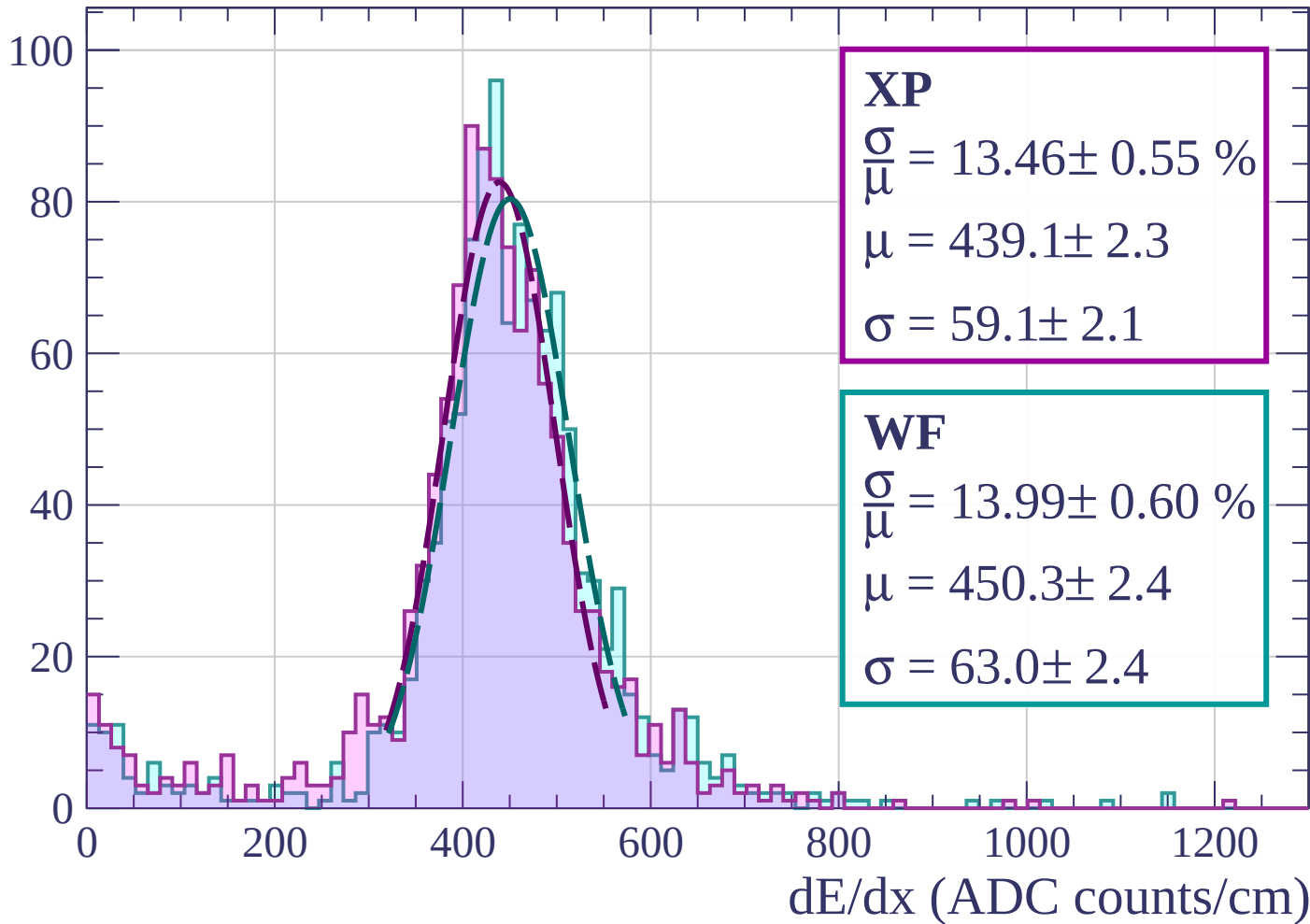
Energy loss | $80 < p < 160$

Count



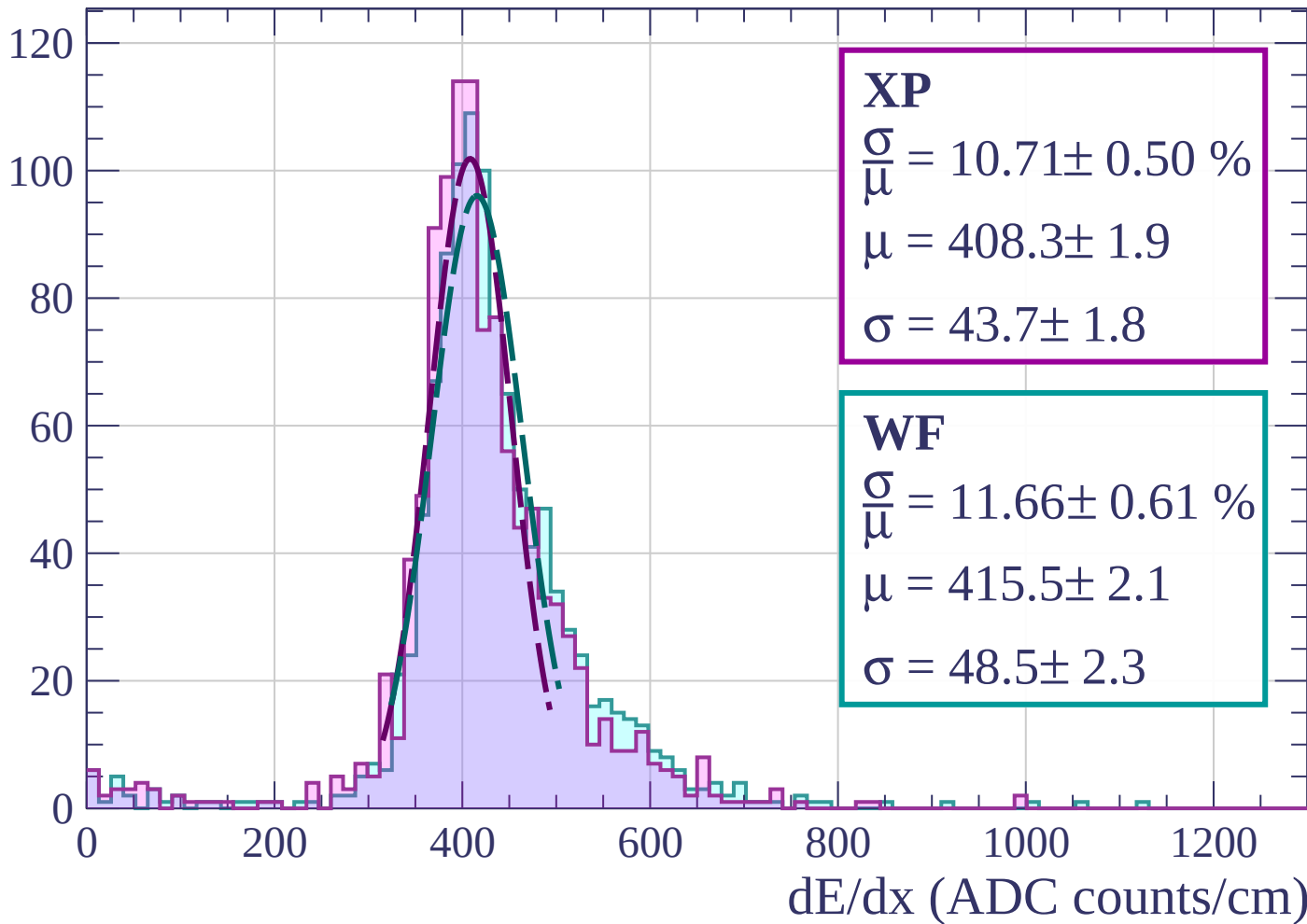
Energy loss | $160 < p < 240$

Count



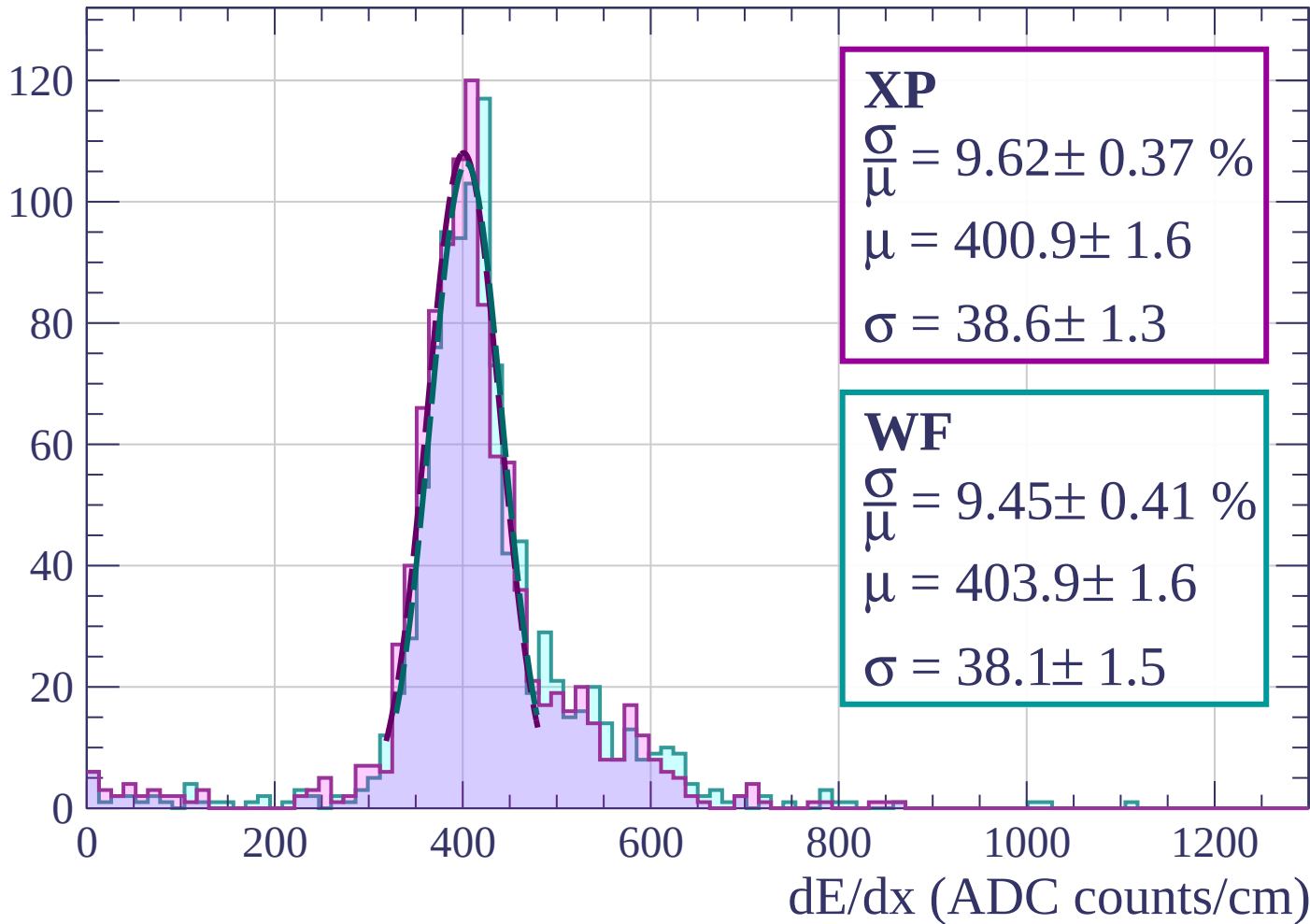
Energy loss | $240 < p < 320$

Count



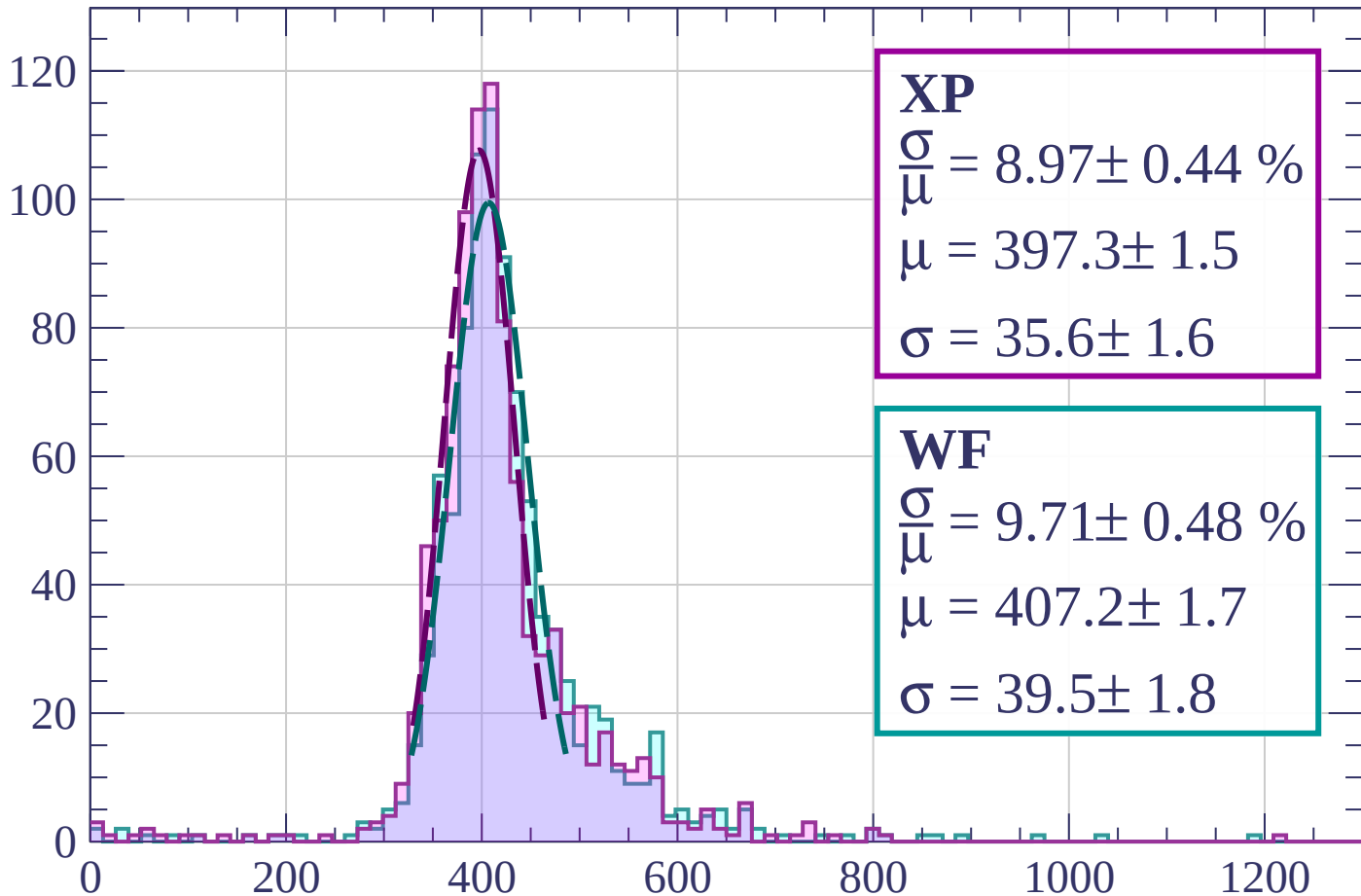
Energy loss | $320 < p < 400$

Count



Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.97 \pm 0.44 \%$$

$$\mu = 397.3 \pm 1.5$$

$$\sigma = 35.6 \pm 1.6$$

WF

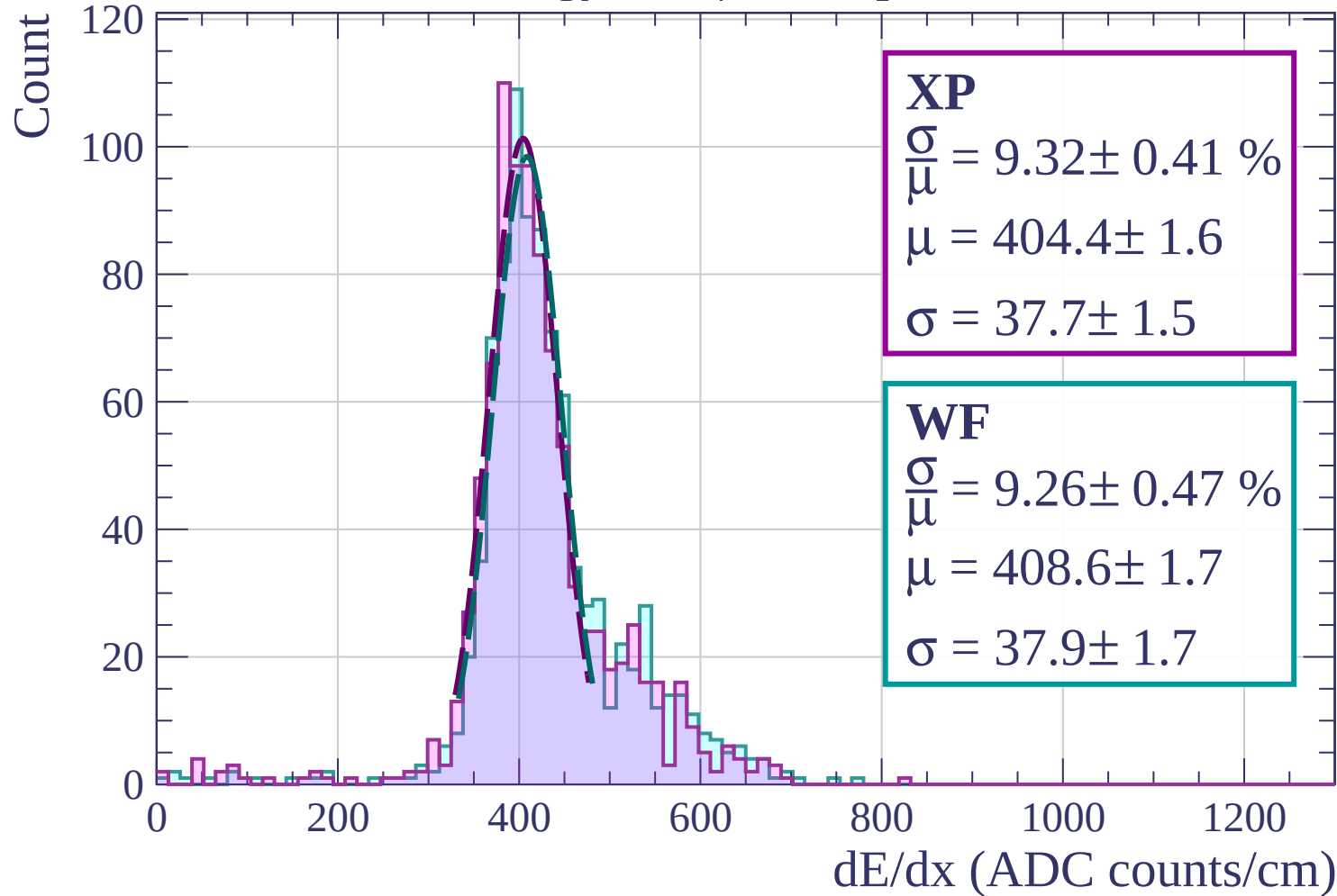
$$\frac{\sigma}{\mu} = 9.71 \pm 0.48 \%$$

$$\mu = 407.2 \pm 1.7$$

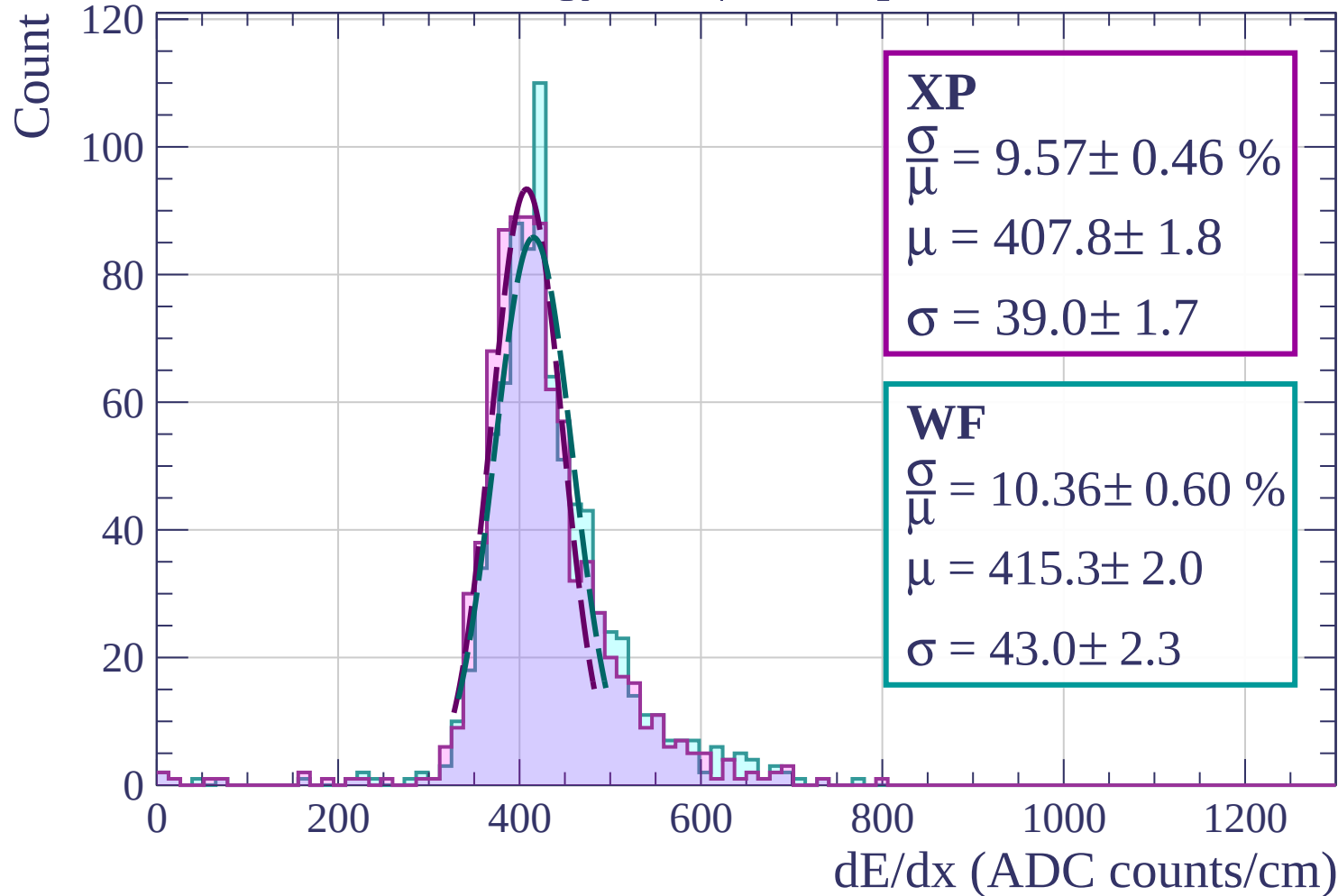
$$\sigma = 39.5 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

Count

XP

$$\frac{\sigma}{\mu} = 9.63 \pm 0.53 \%$$

$$\mu = 414.4 \pm 1.9$$

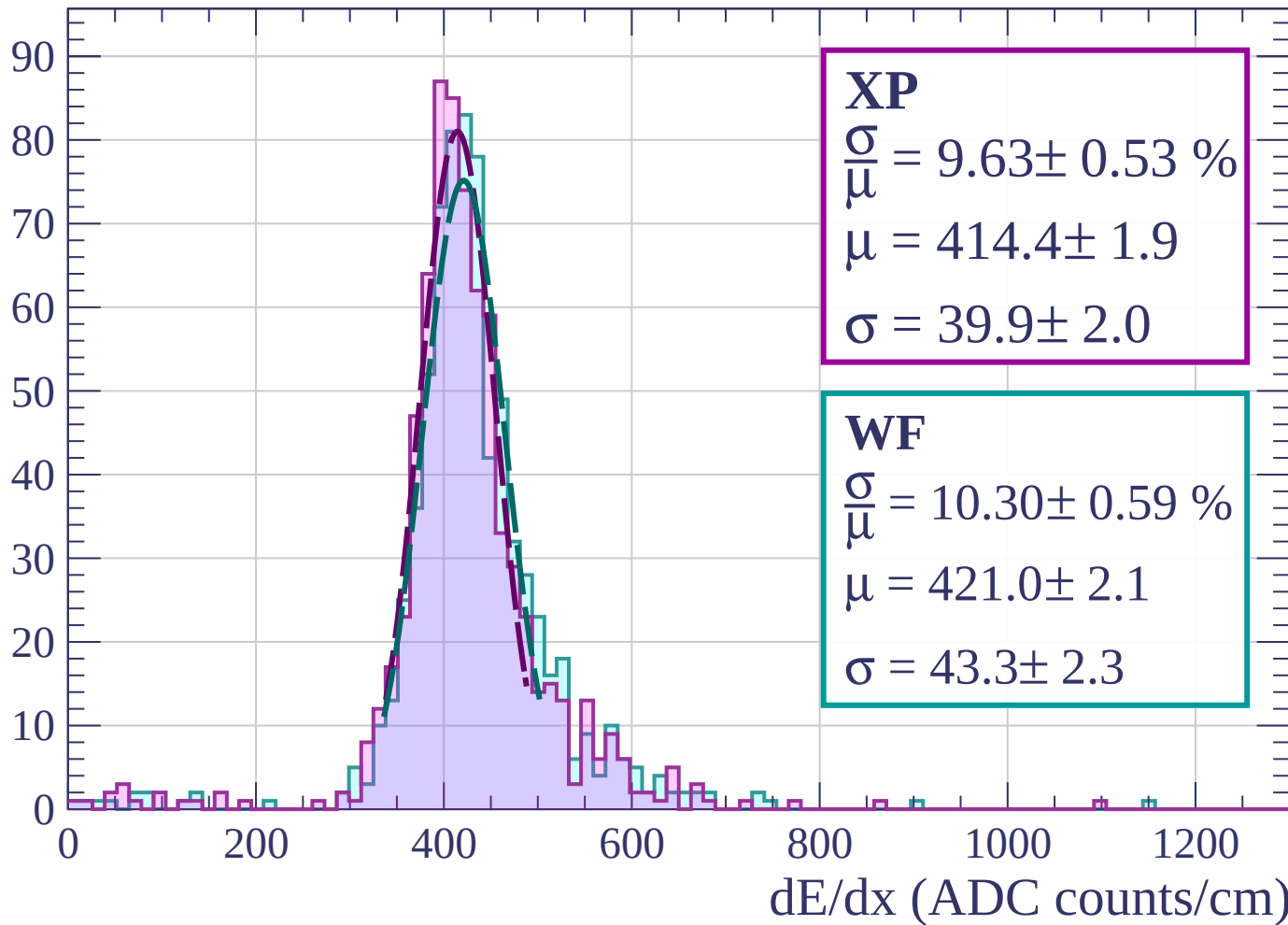
$$\sigma = 39.9 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.30 \pm 0.59 \%$$

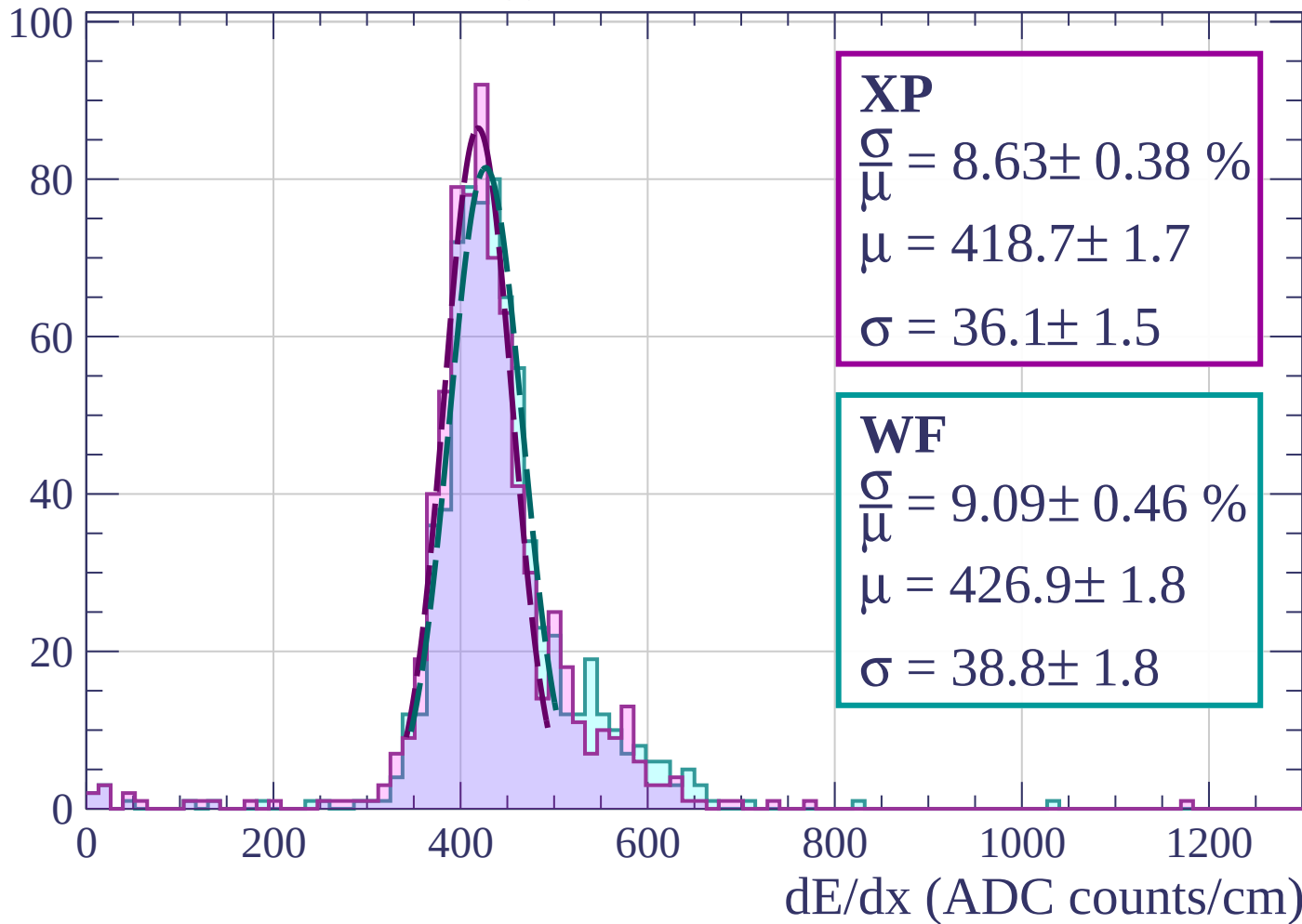
$$\mu = 421.0 \pm 2.1$$

$$\sigma = 43.3 \pm 2.3$$



Energy loss | $720 < p < 800$

Count



Energy loss | $800 < p < 880$

Count

XP

$$\frac{\sigma}{\mu} = 9.83 \pm 0.48 \%$$

$$\mu = 426.6 \pm 2.0$$

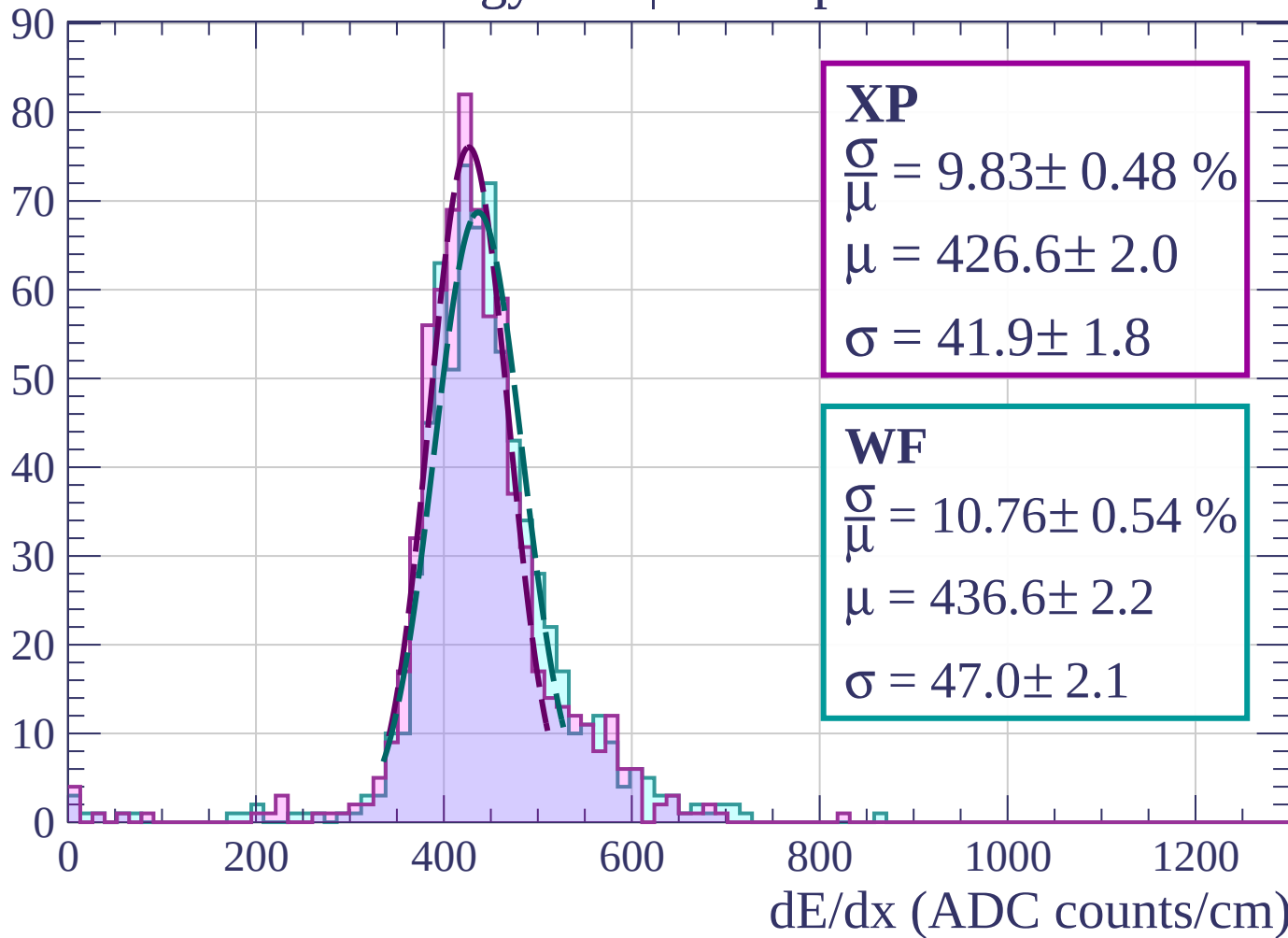
$$\sigma = 41.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.76 \pm 0.54 \%$$

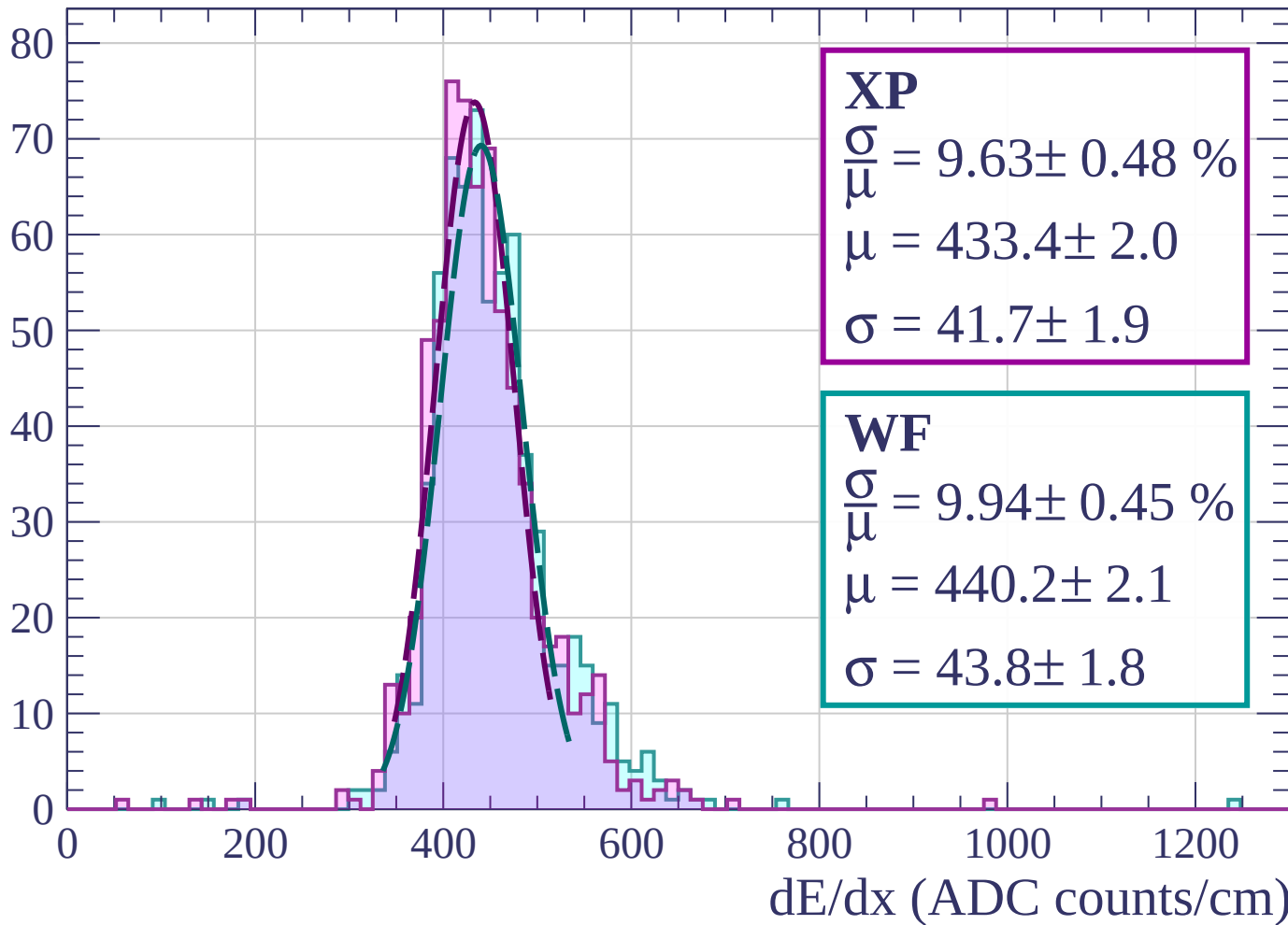
$$\mu = 436.6 \pm 2.2$$

$$\sigma = 47.0 \pm 2.1$$



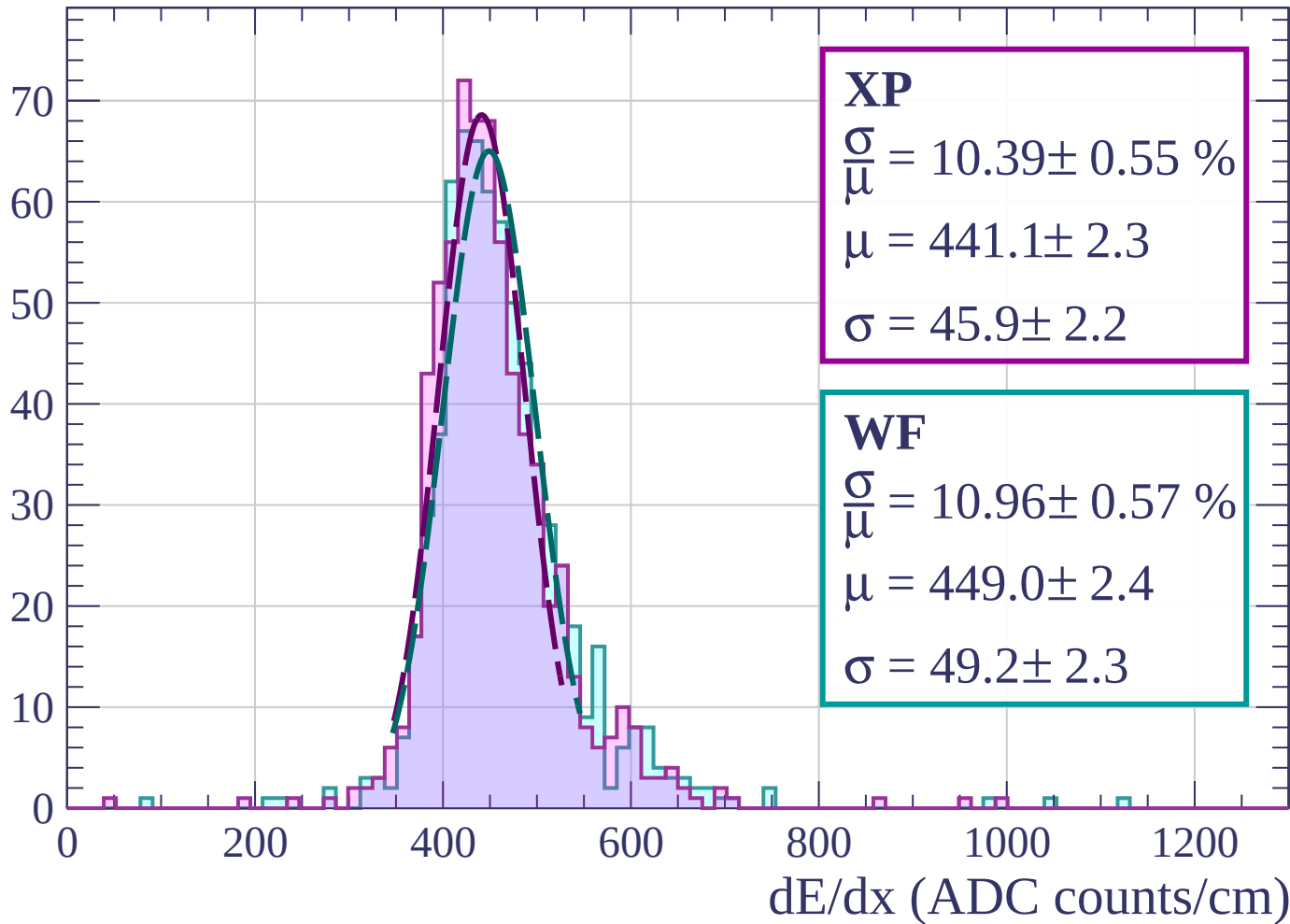
Energy loss | $880 < p < 960$

Count



Energy loss | $960 < p < 1040$

Count



Energy loss | $1040 < p < 1120$

Count

XP

$$\frac{\sigma}{\mu} = 10.14 \pm 0.50 \%$$

$$\mu = 441.4 \pm 2.3$$

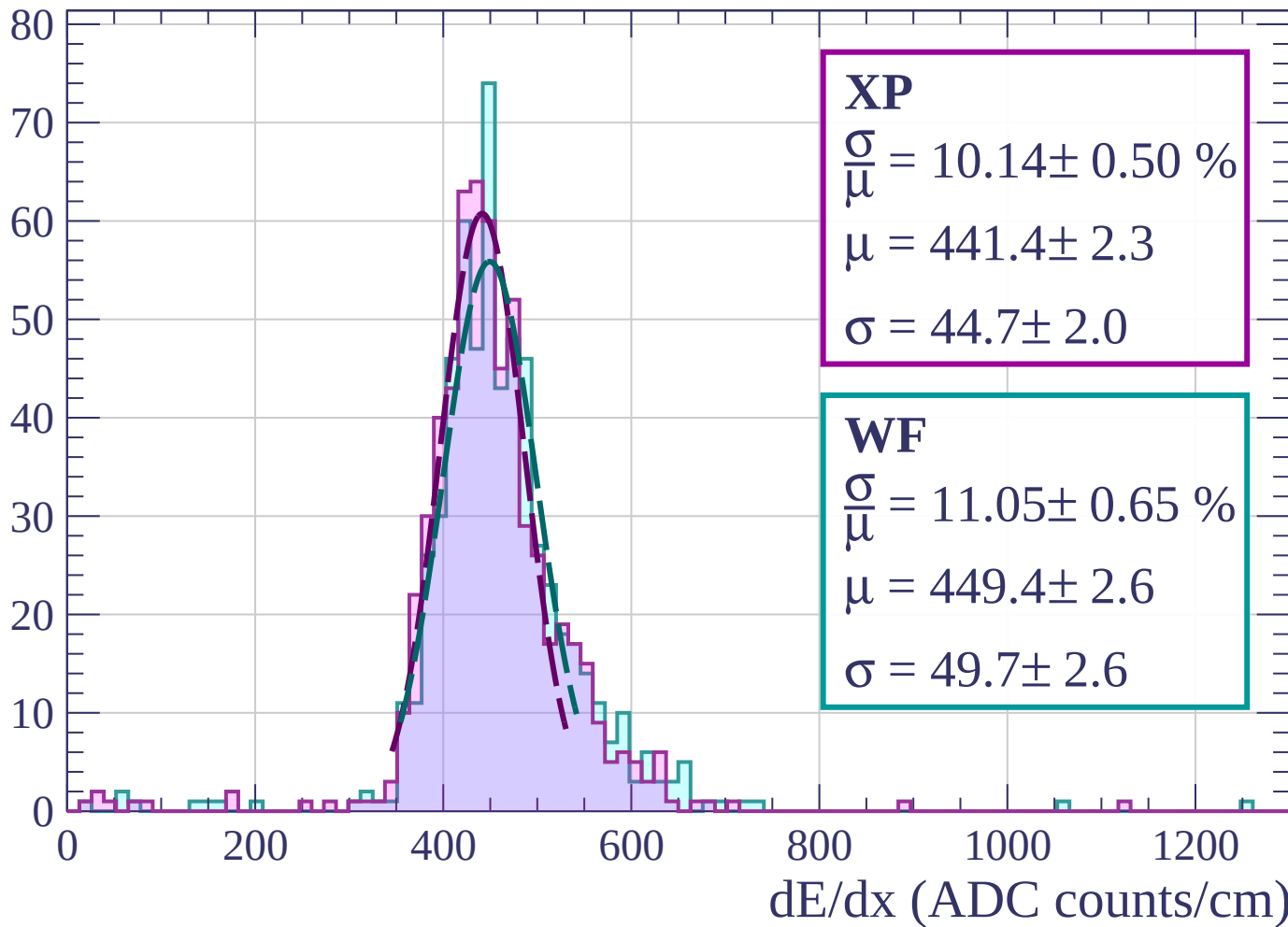
$$\sigma = 44.7 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 11.05 \pm 0.65 \%$$

$$\mu = 449.4 \pm 2.6$$

$$\sigma = 49.7 \pm 2.6$$



Energy loss | $1120 < p < 1200$

Count

XP

$$\frac{\sigma}{\mu} = 11.16 \pm 0.71 \%$$

$$\mu = 442.4 \pm 2.6$$

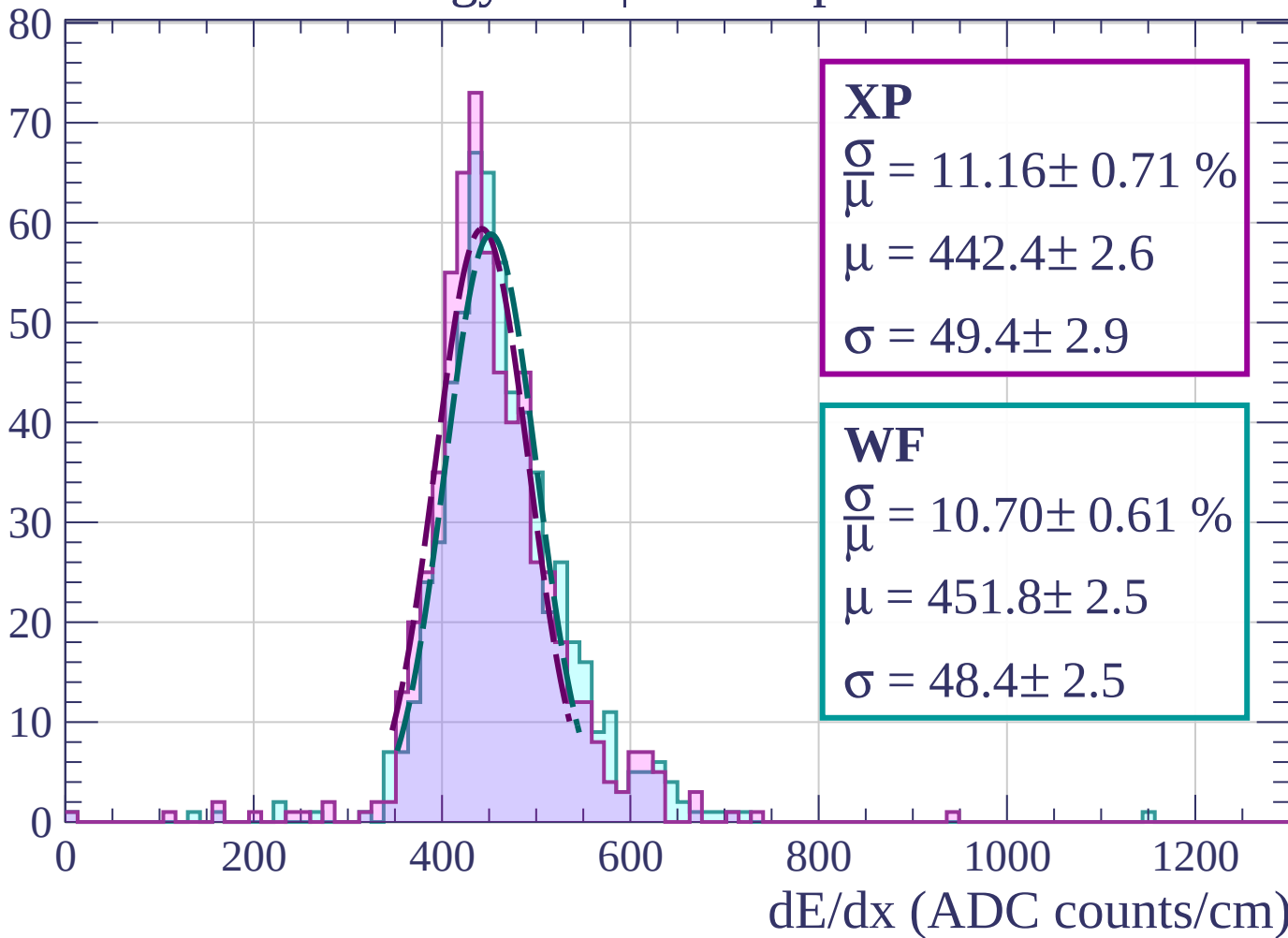
$$\sigma = 49.4 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 10.70 \pm 0.61 \%$$

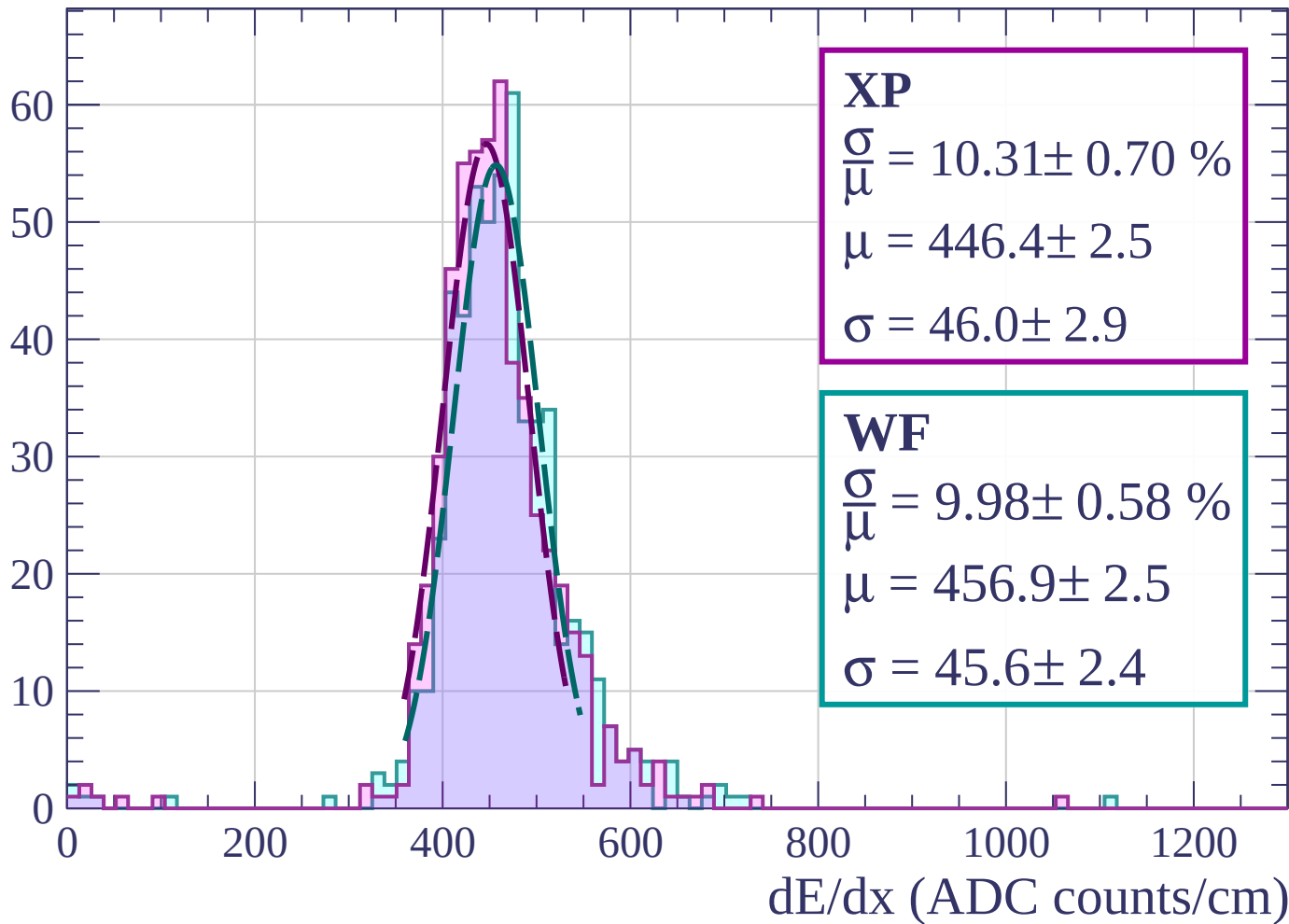
$$\mu = 451.8 \pm 2.5$$

$$\sigma = 48.4 \pm 2.5$$



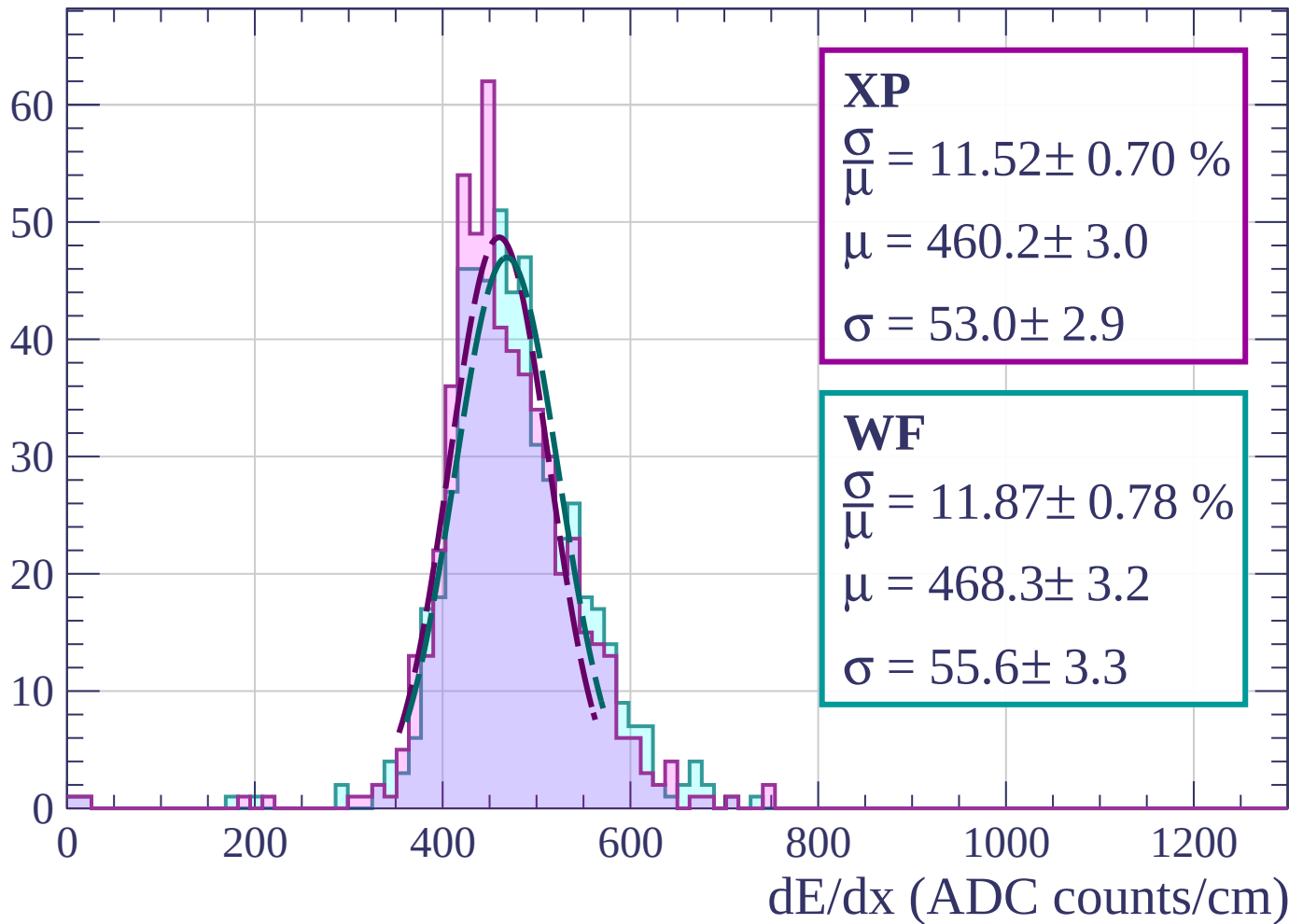
Energy loss | $1200 < p < 1280$

Count



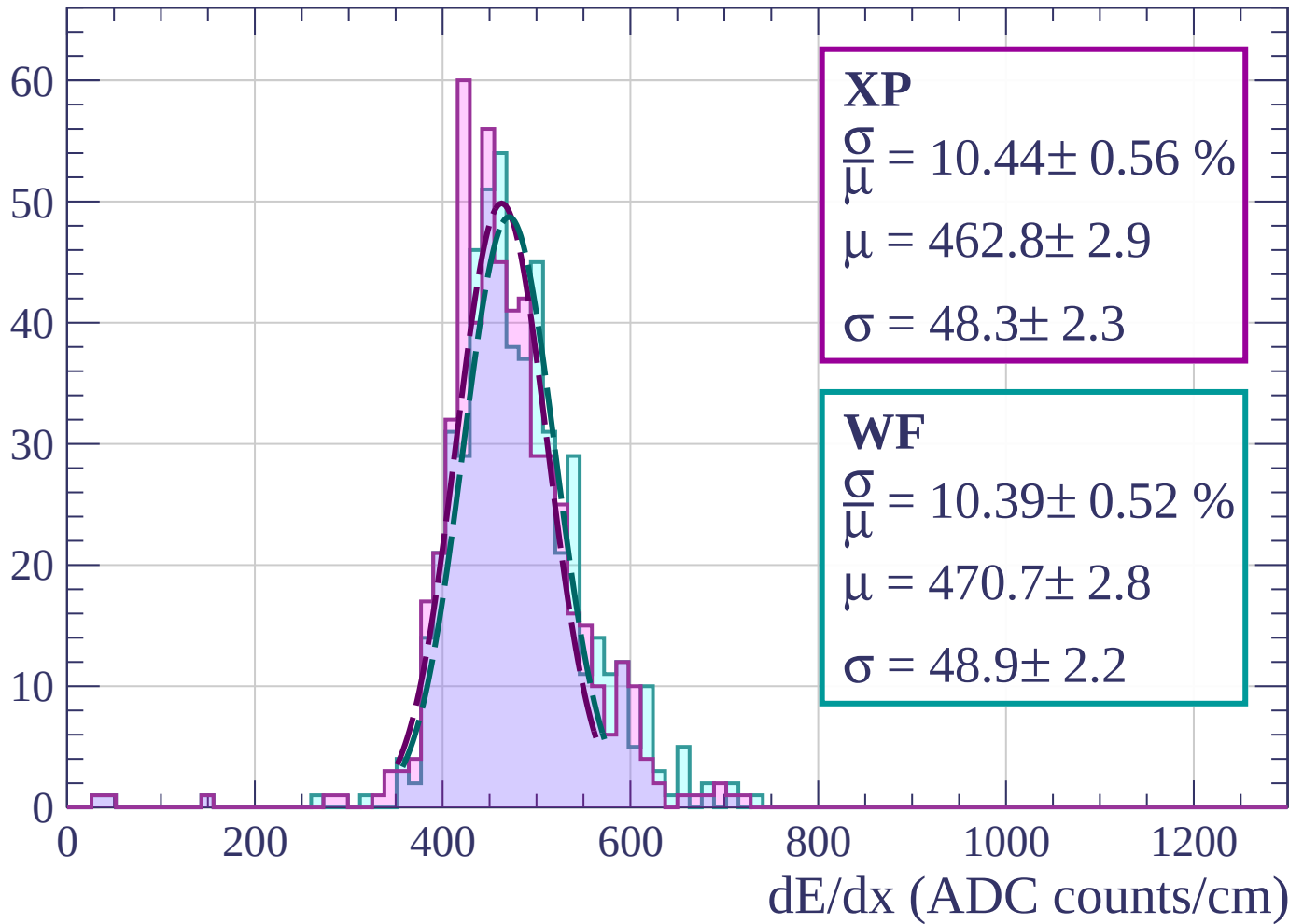
Energy loss | $1280 < p < 1360$

Count



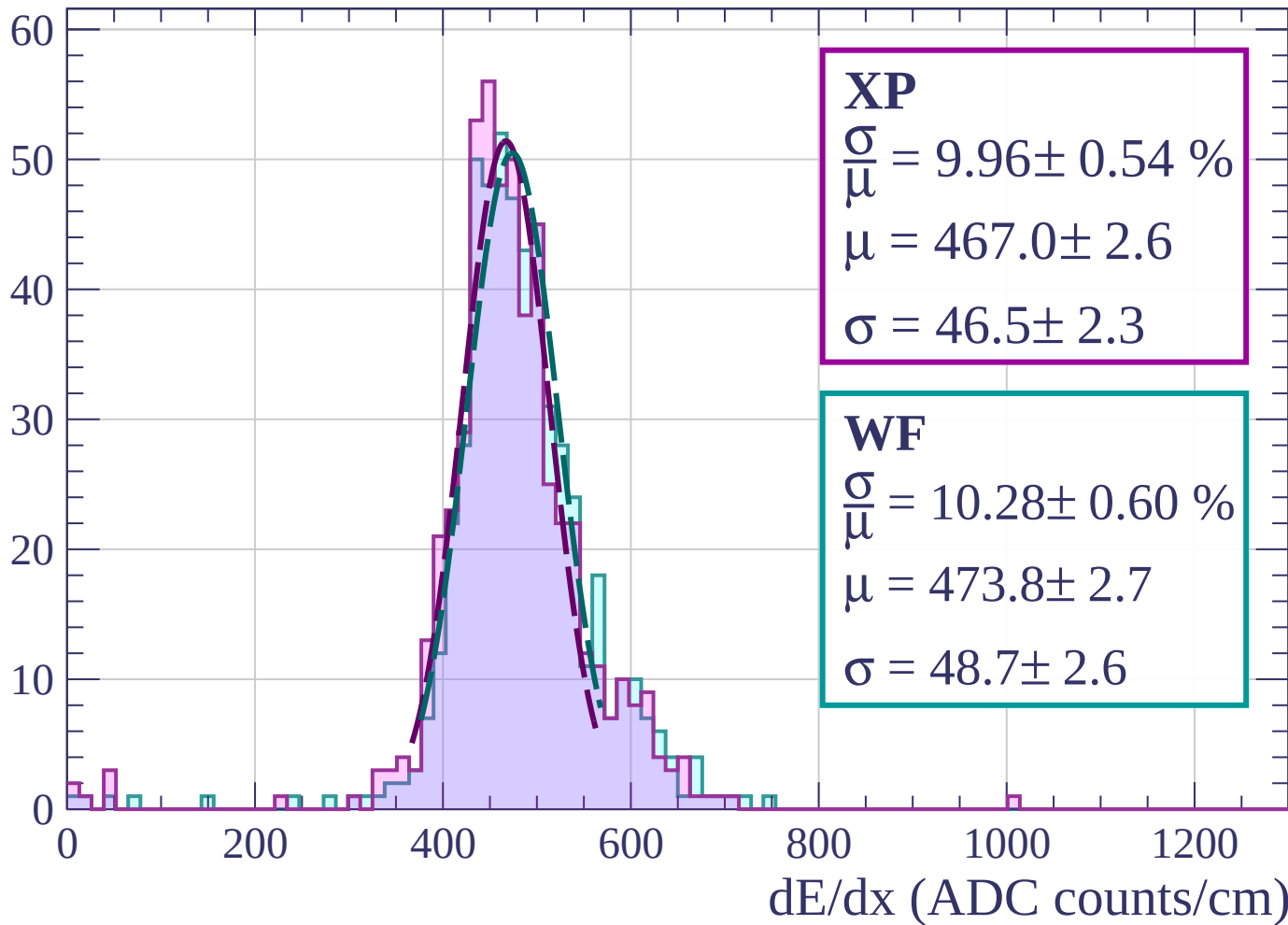
Energy loss | $1360 < p < 1440$

Count



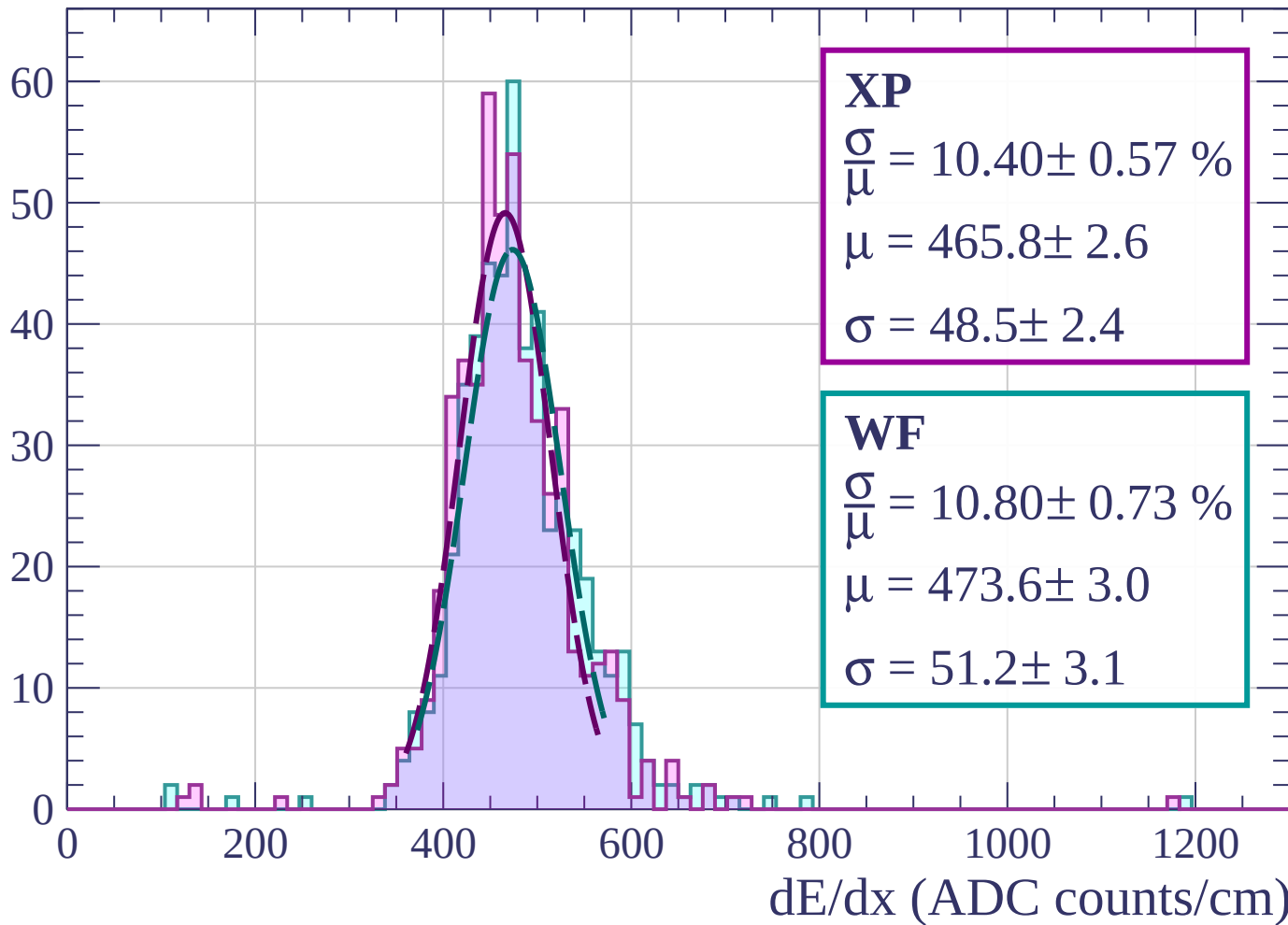
Energy loss | $1440 < p < 1520$

Count



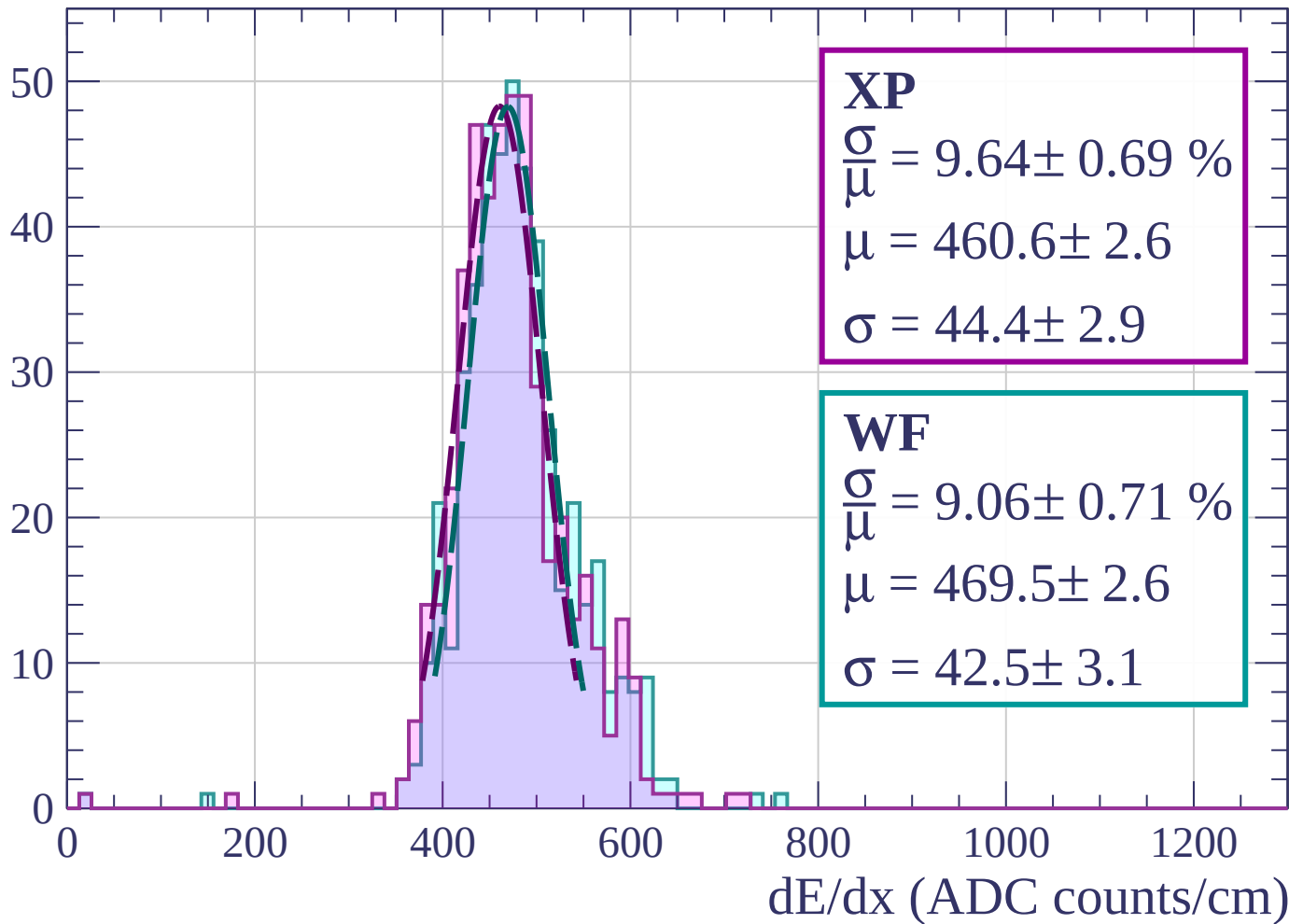
Energy loss | $1520 < p < 1600$

Count



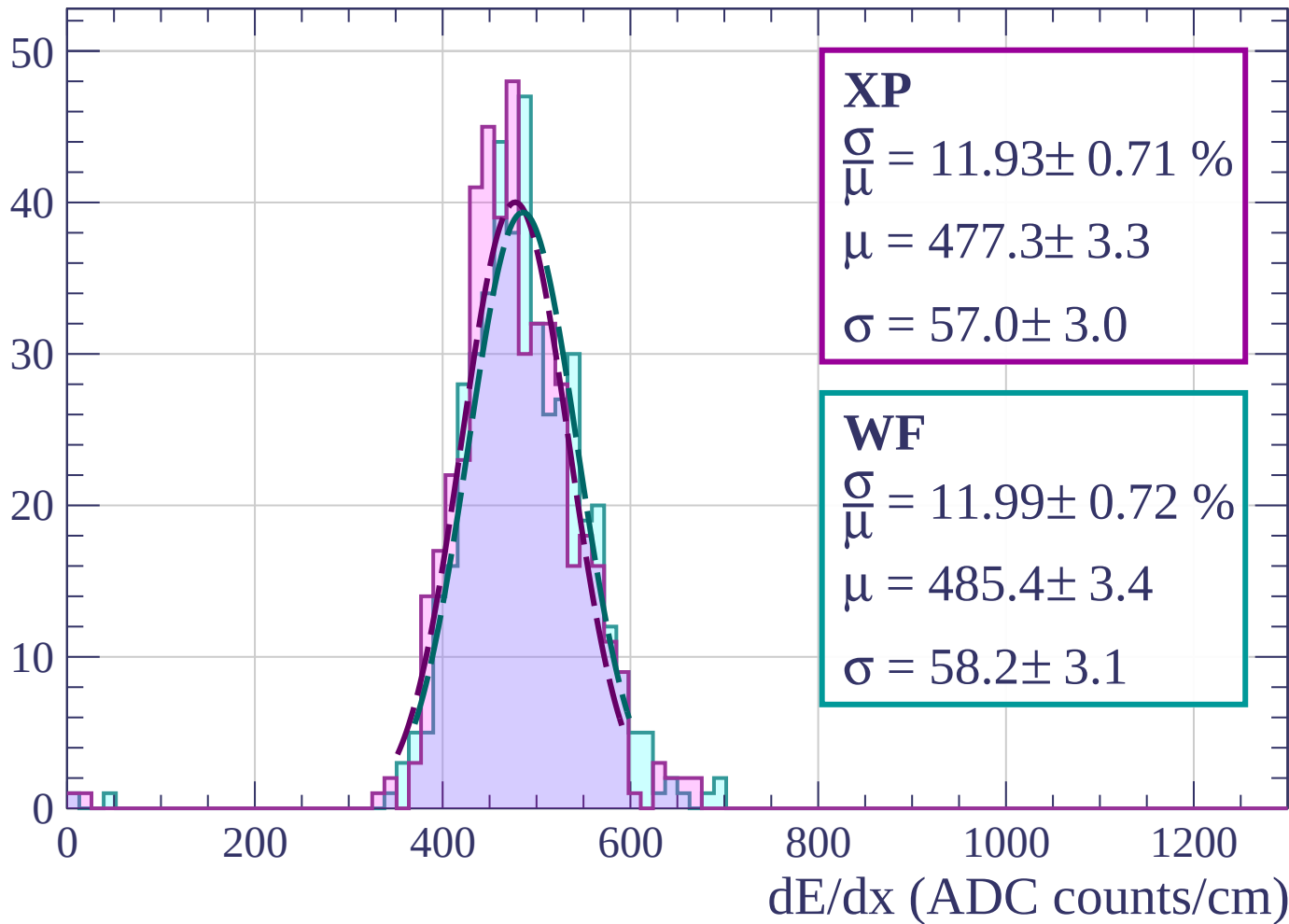
Energy loss | $1600 < p < 1680$

Count



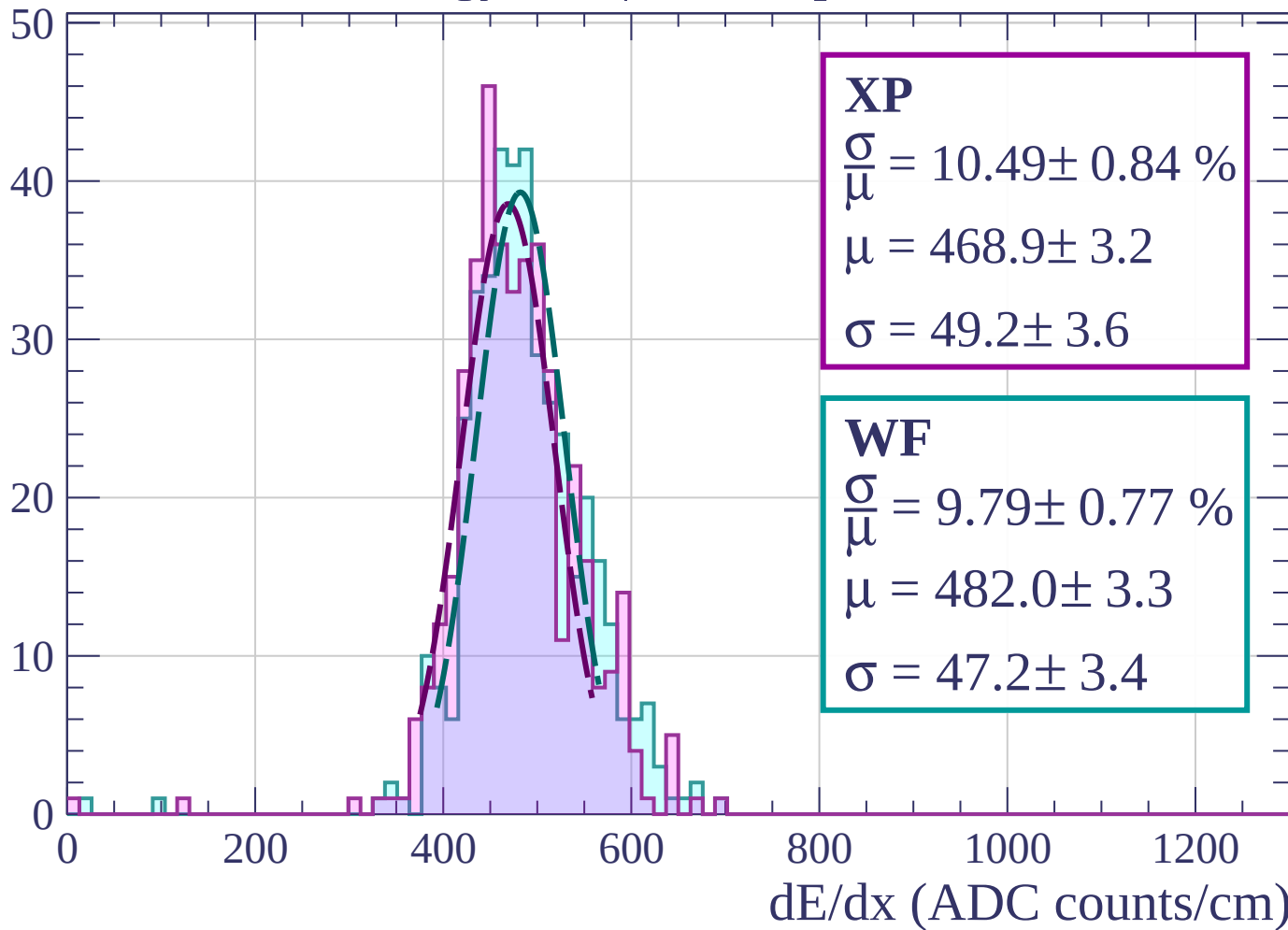
Energy loss | $1680 < p < 1760$

Count



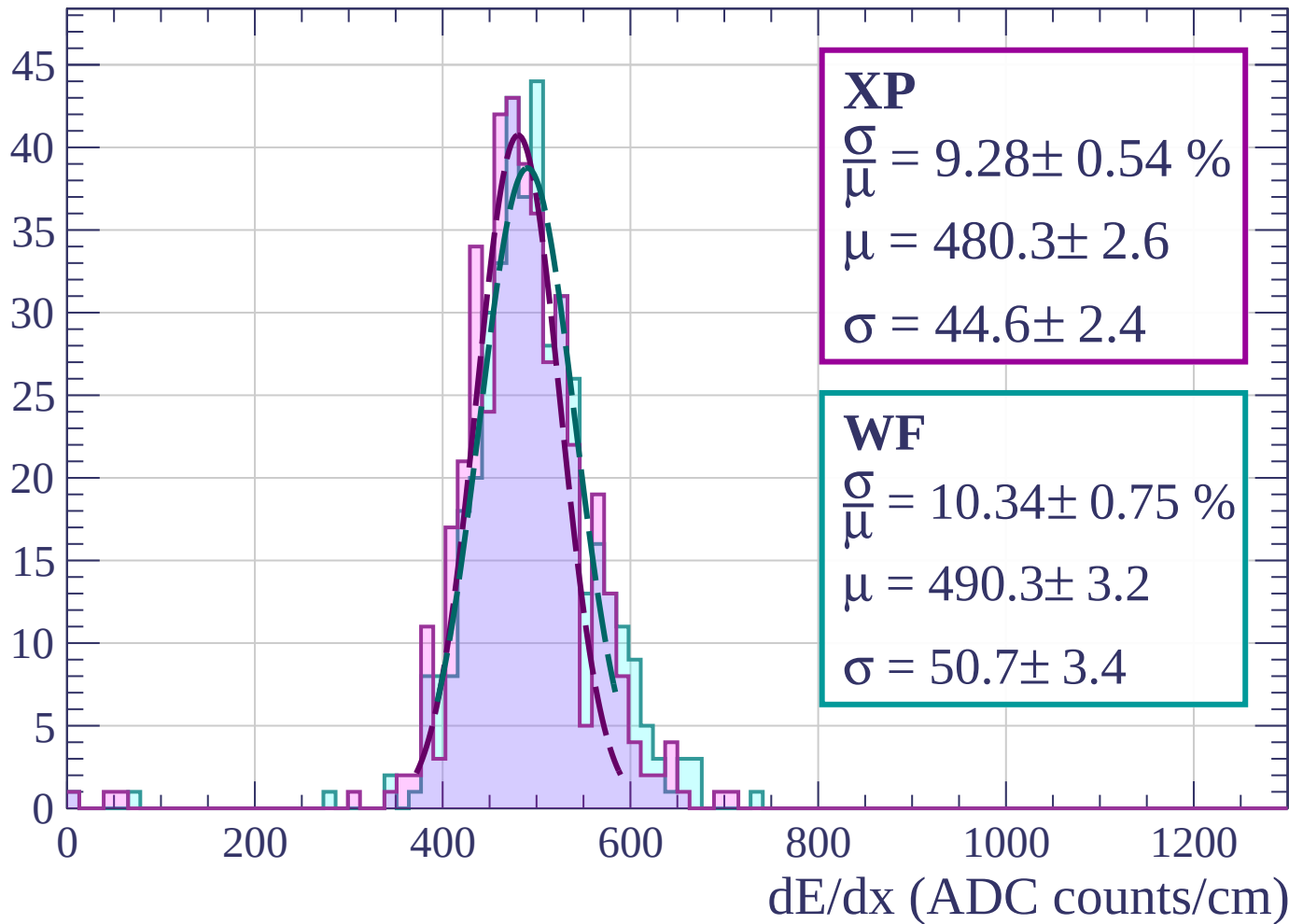
Energy loss | $1760 < p < 1840$

Count



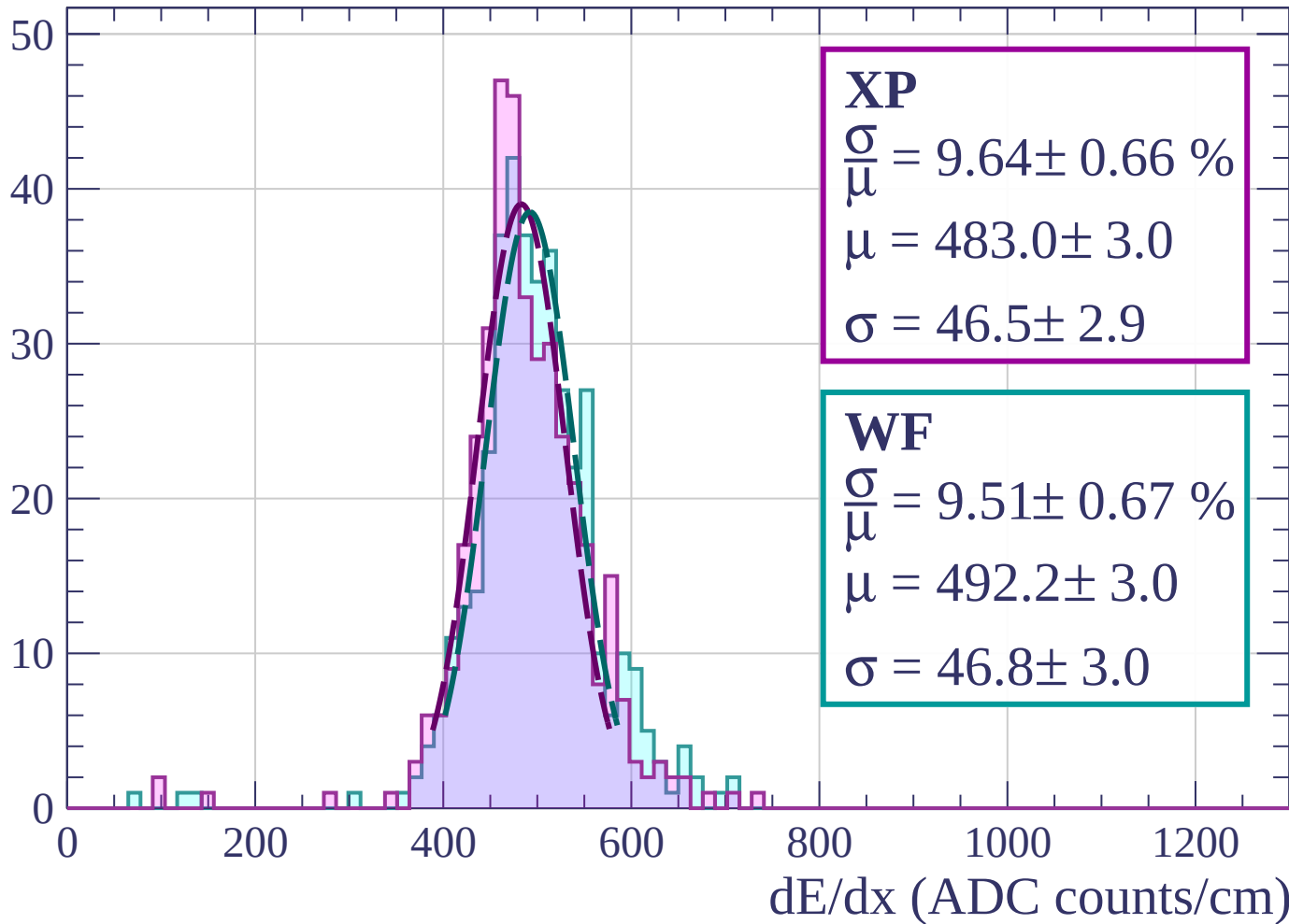
Energy loss | $1840 < p < 1920$

Count



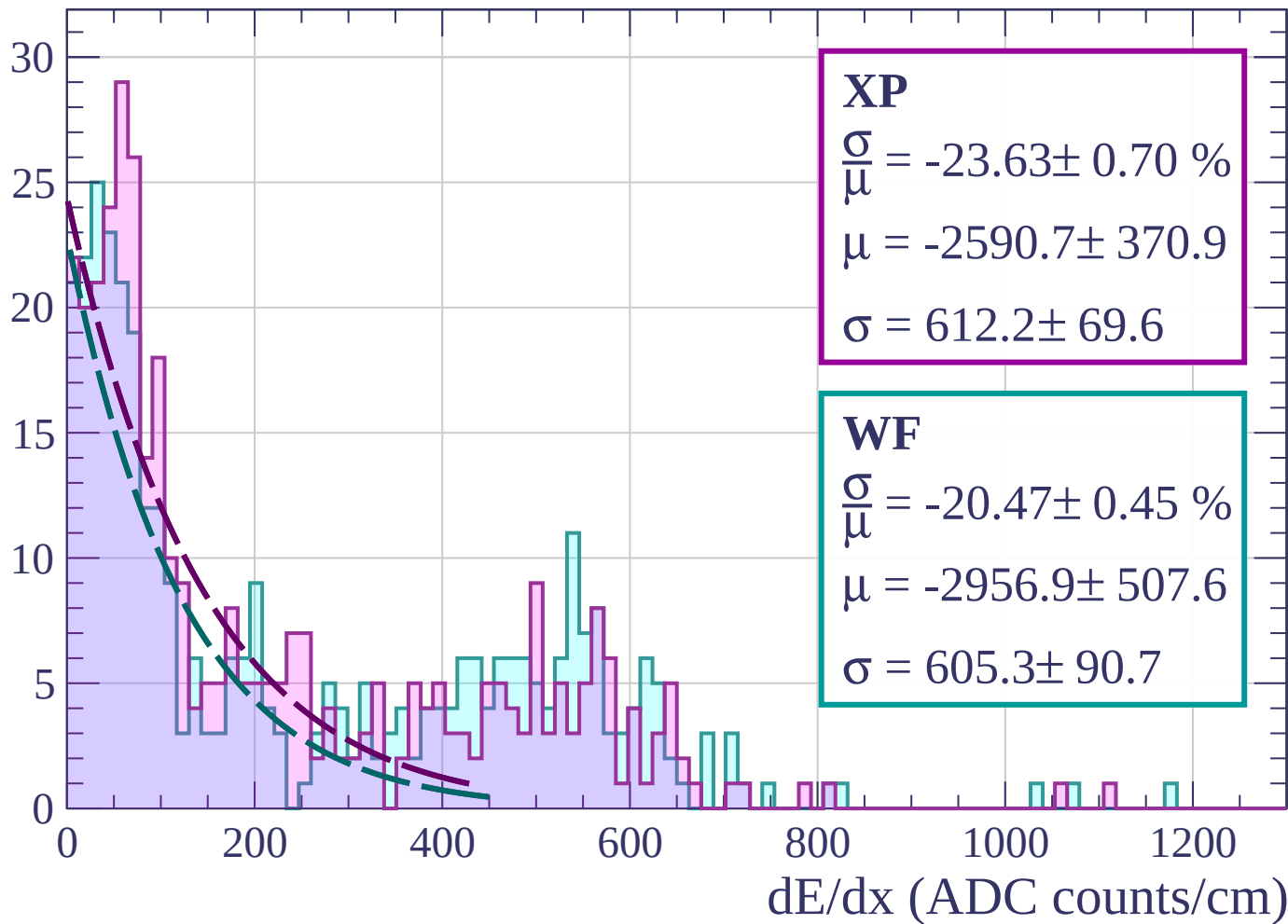
Energy loss | $1920 < p < 2000$

Count



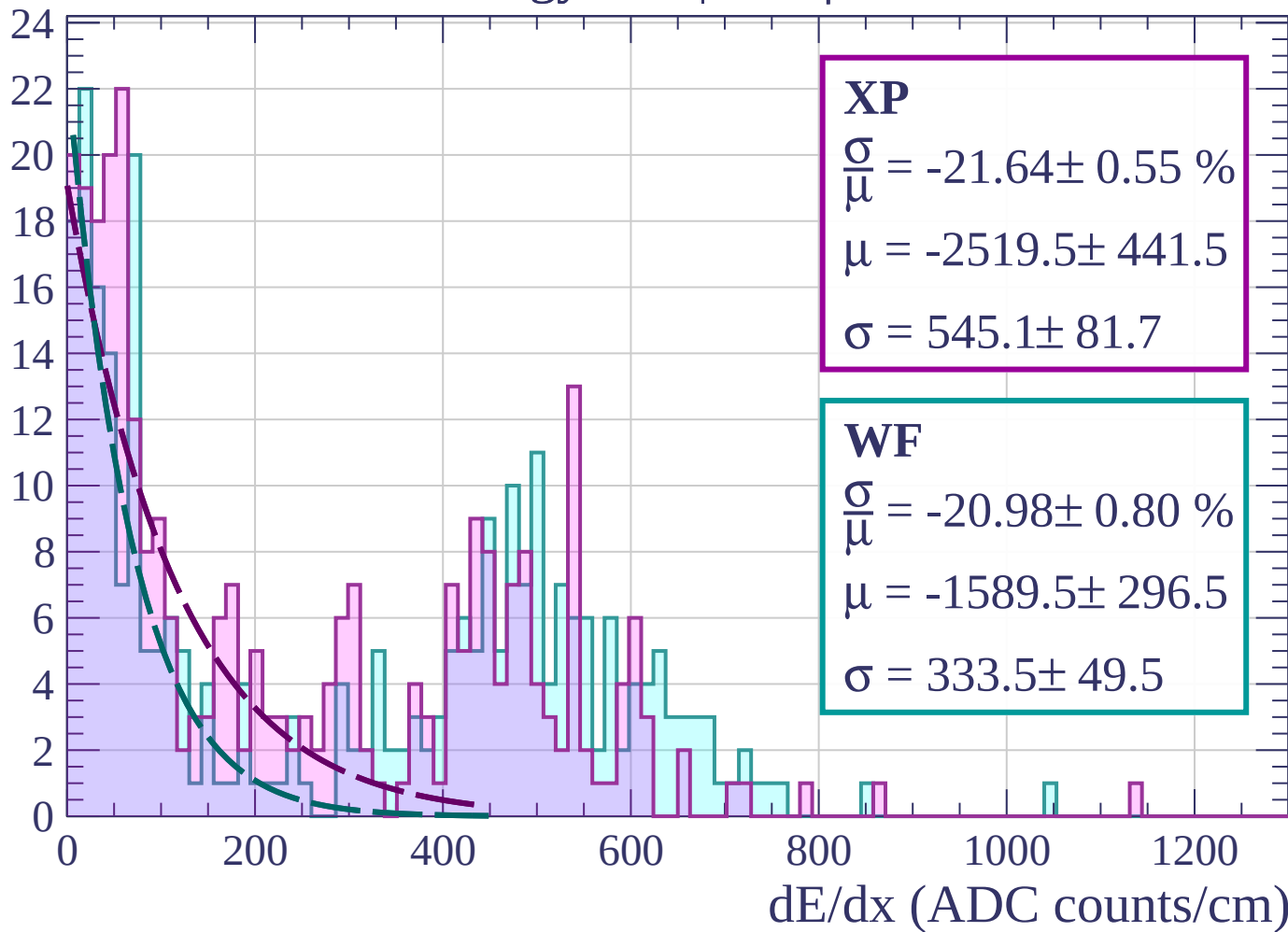
Energy loss | $0 < \phi < 3$

Count



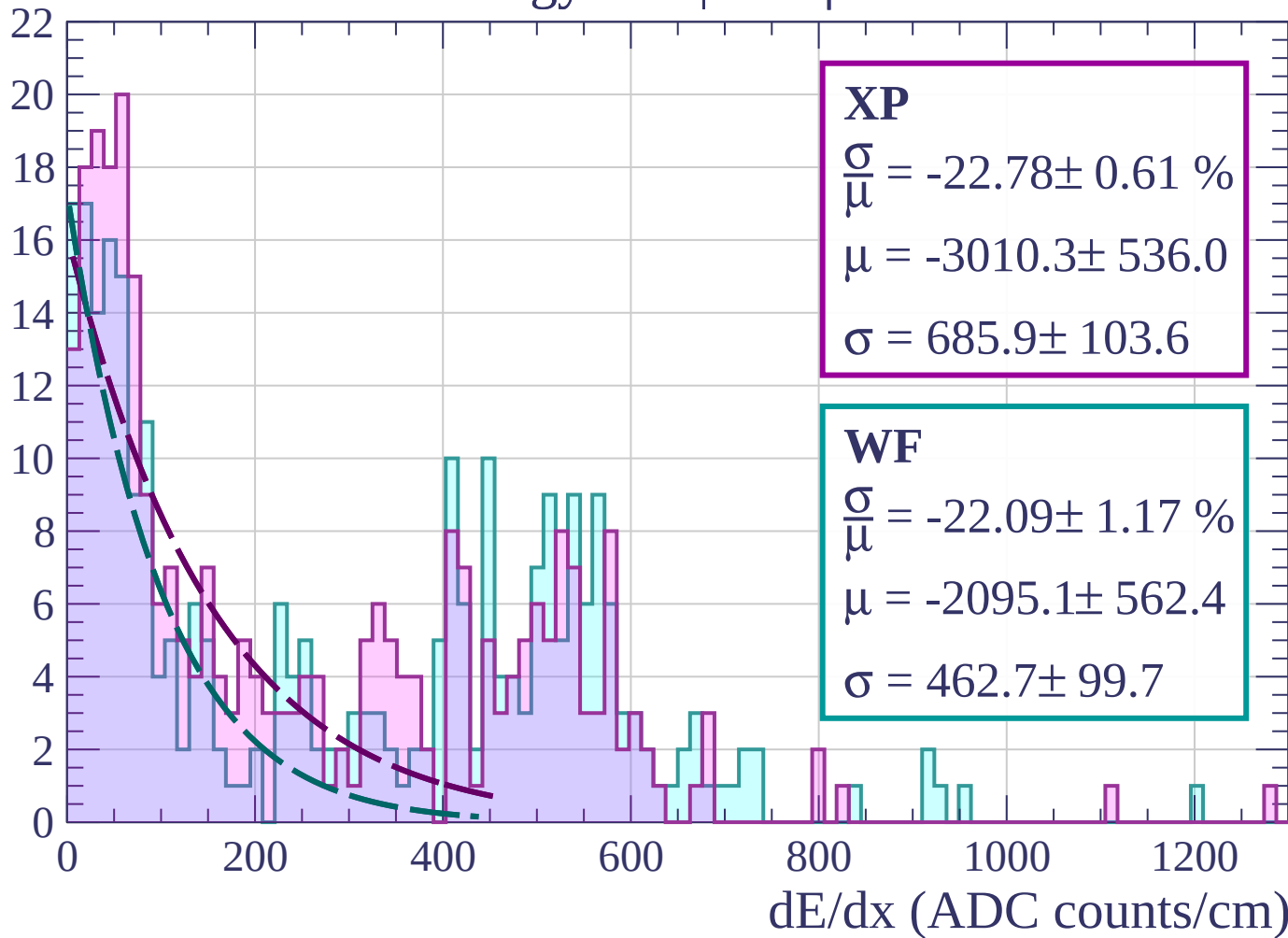
Energy loss | $3 < \phi < 6$

Count



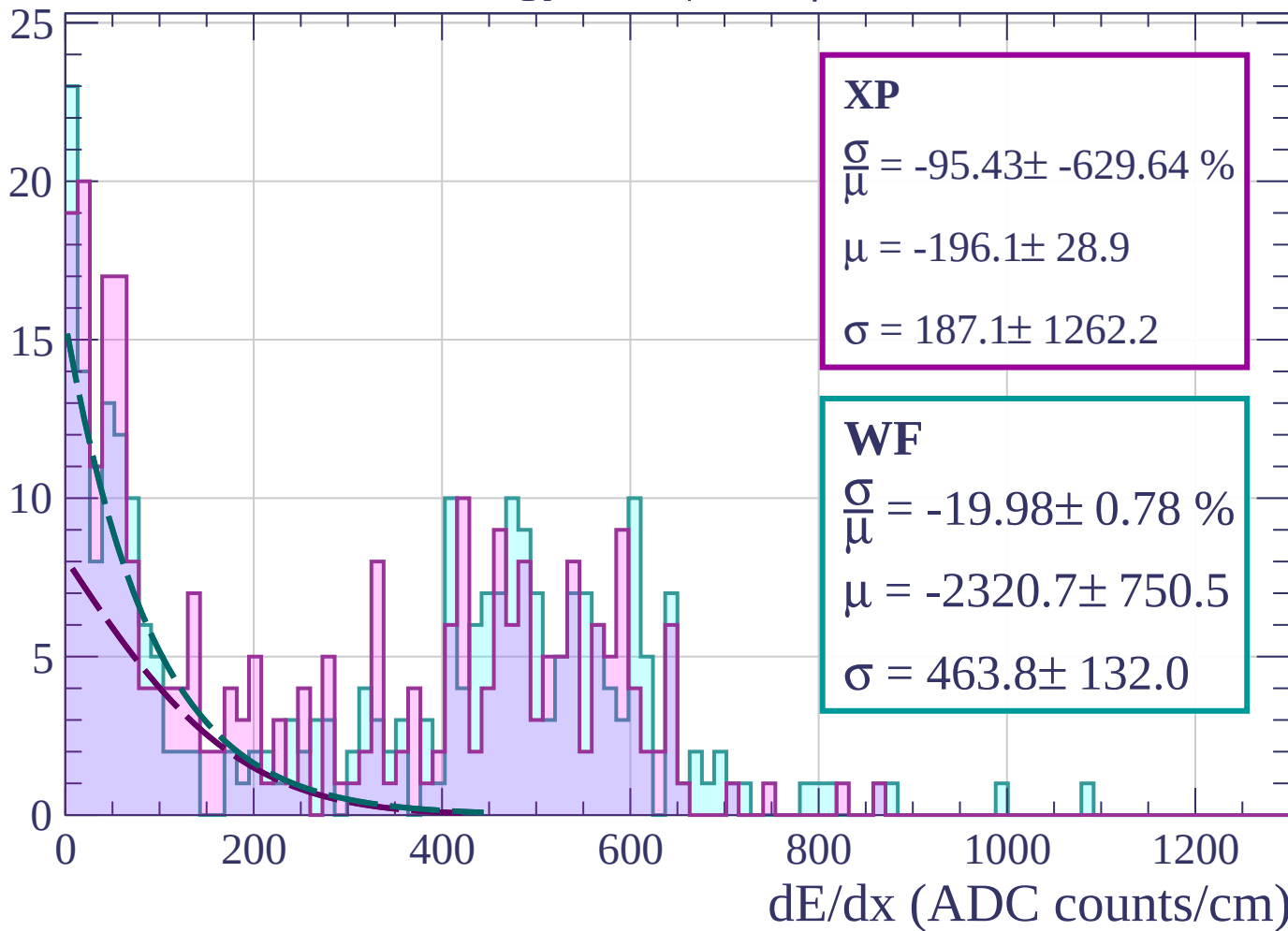
Energy loss | $6 < \phi < 9$

Count



Energy loss | $9 < \phi < 12$

Count



Energy loss | $12 < \phi < 15$

Count

XP

$$\frac{\sigma}{\mu} = -23.93 \pm 1.37 \%$$

$$\mu = -4741.9 \pm 1319.2$$

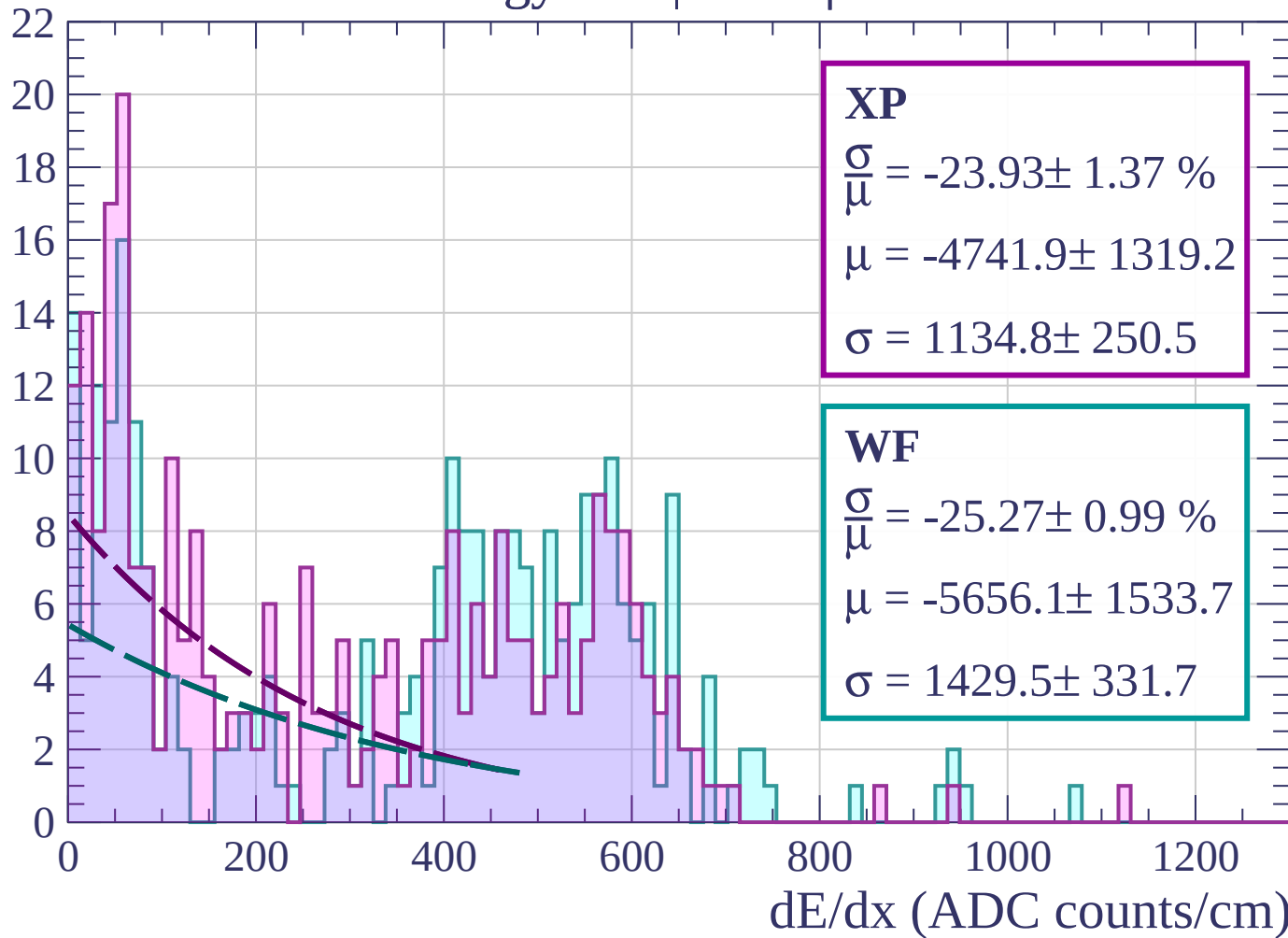
$$\sigma = 1134.8 \pm 250.5$$

WF

$$\frac{\sigma}{\mu} = -25.27 \pm 0.99 \%$$

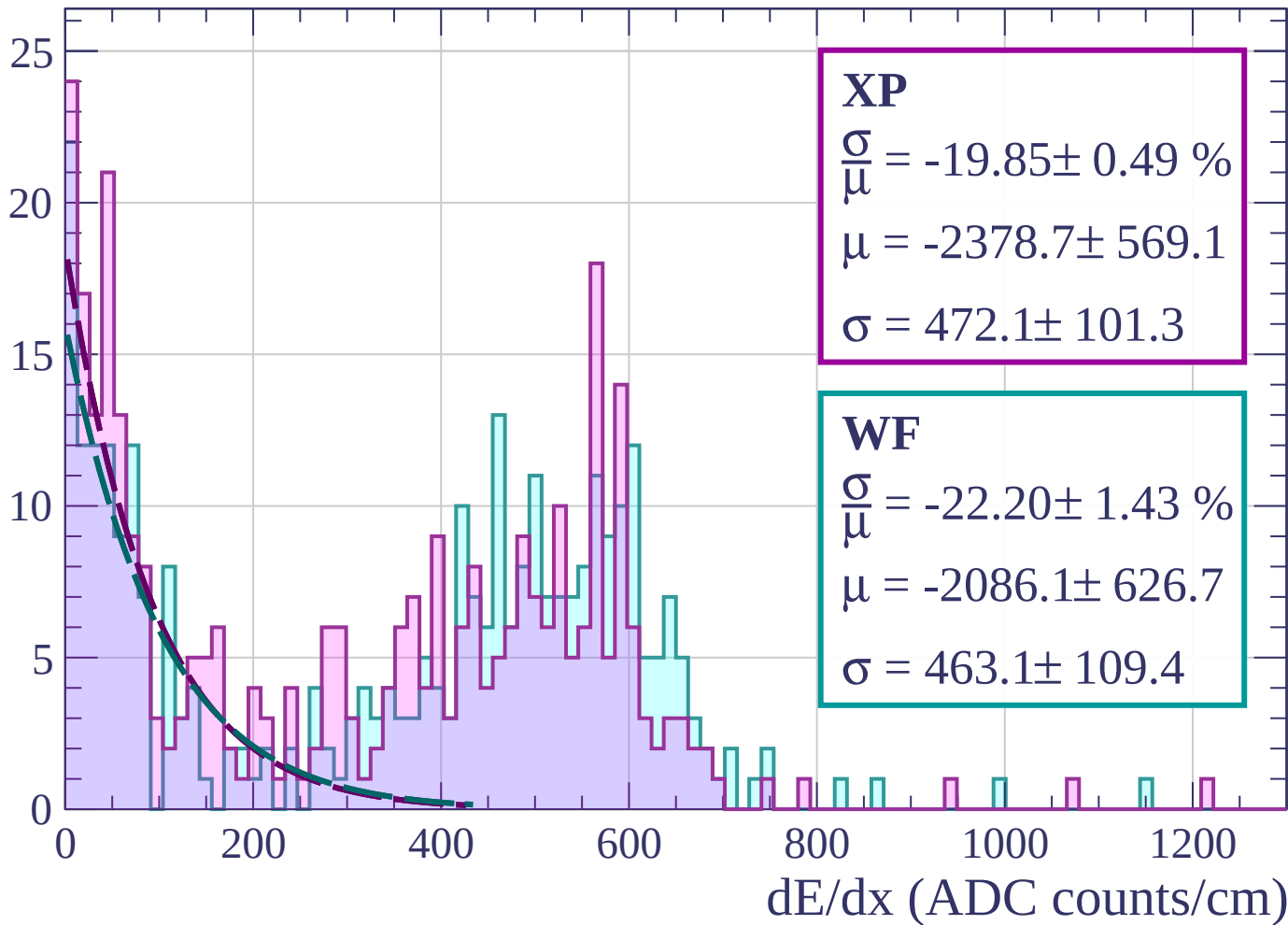
$$\mu = -5656.1 \pm 1533.7$$

$$\sigma = 1429.5 \pm 331.7$$



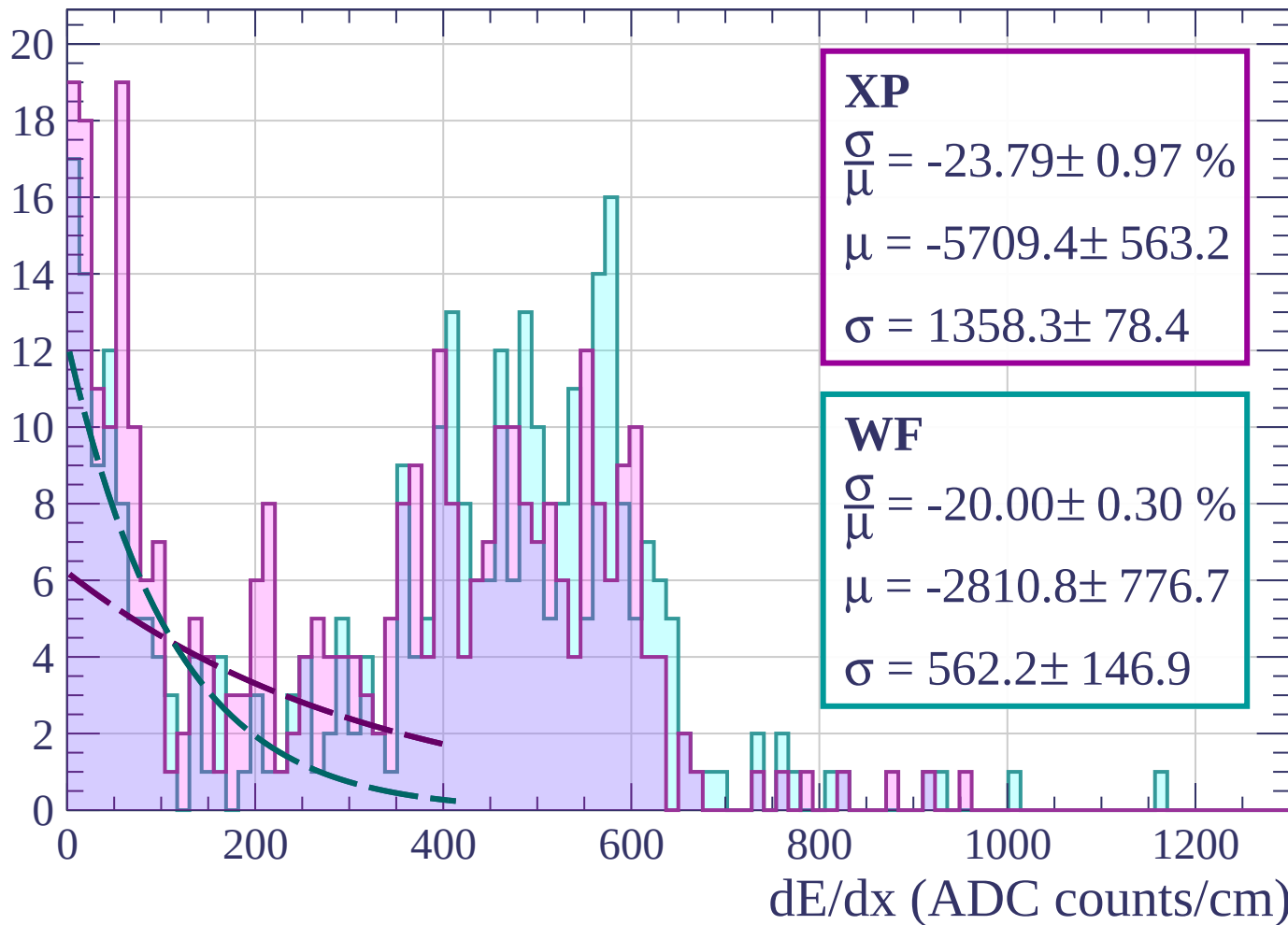
Energy loss | $15 < \phi < 18$

Count



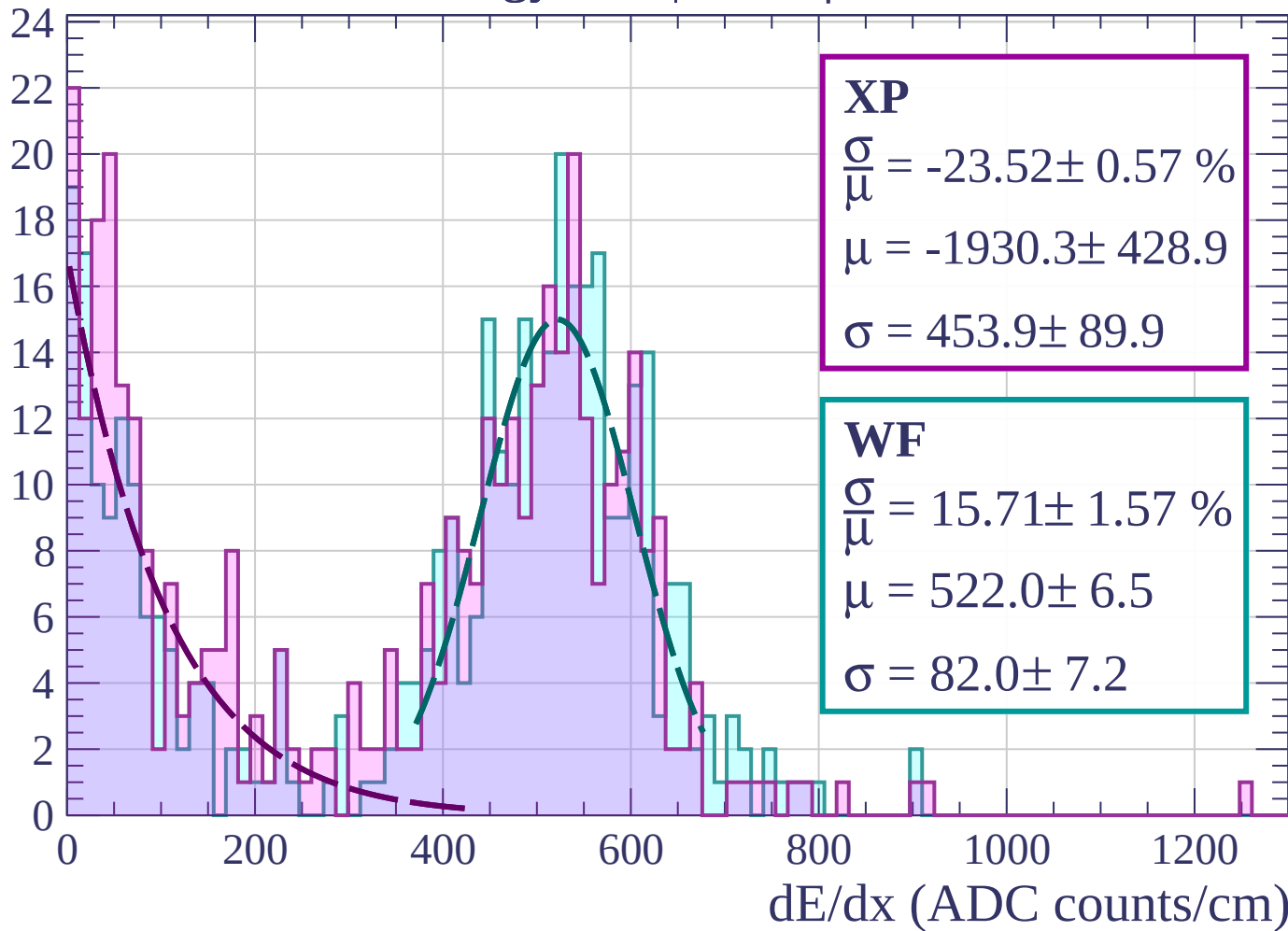
Energy loss | $18 < \phi < 21$

Count



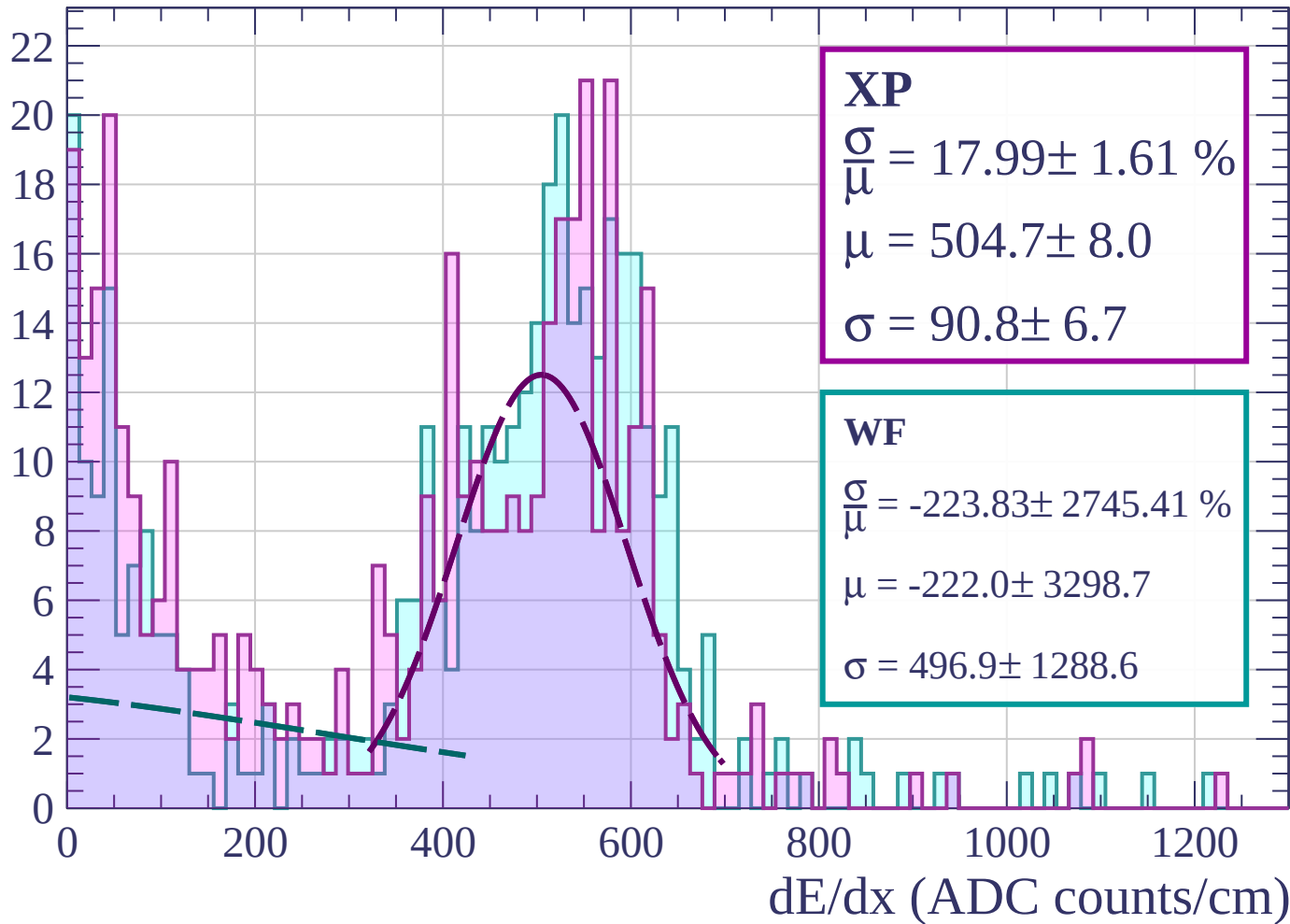
Energy loss | $21 < \phi < 24$

Count



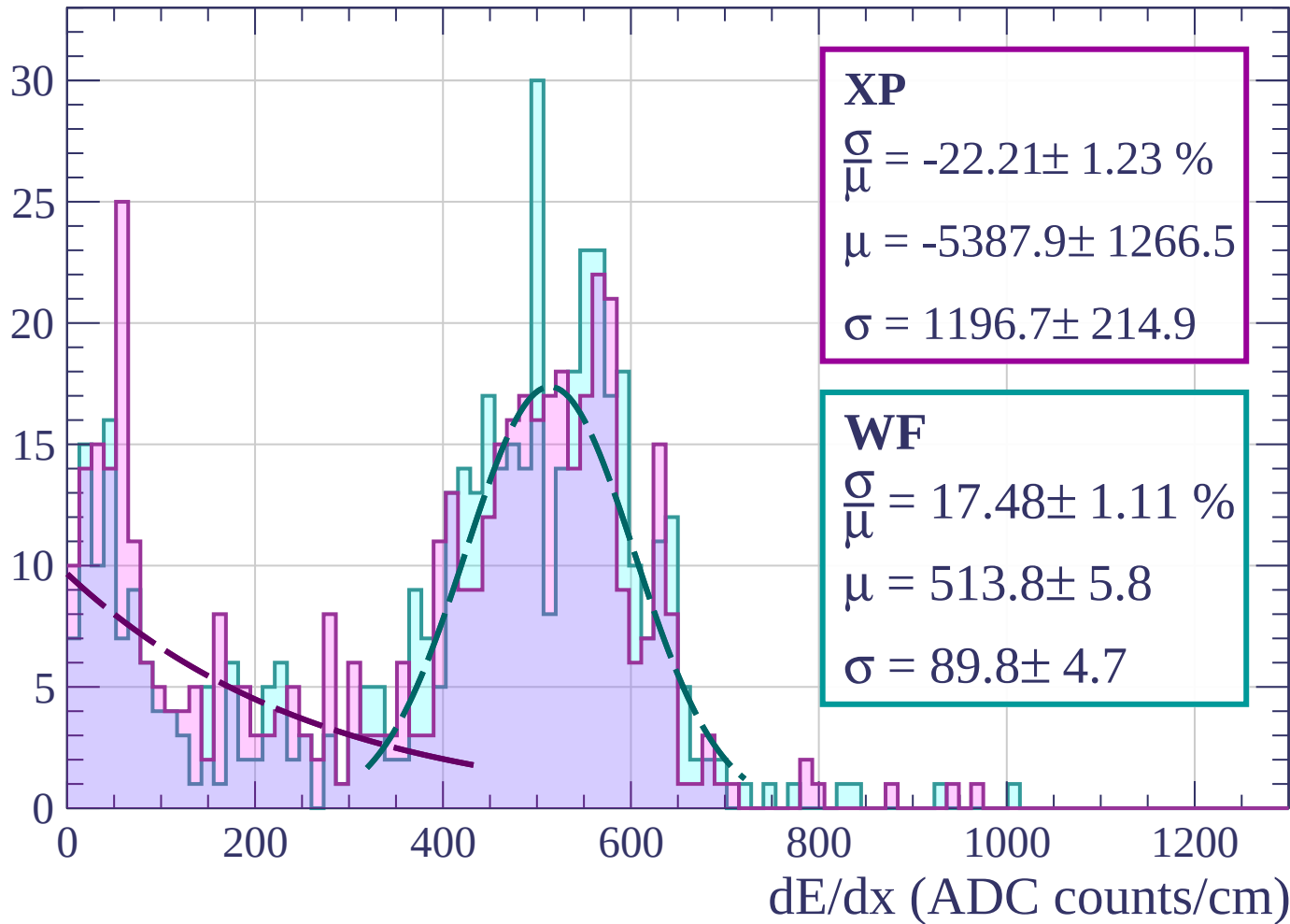
Energy loss | $24 < \phi < 27$

Count



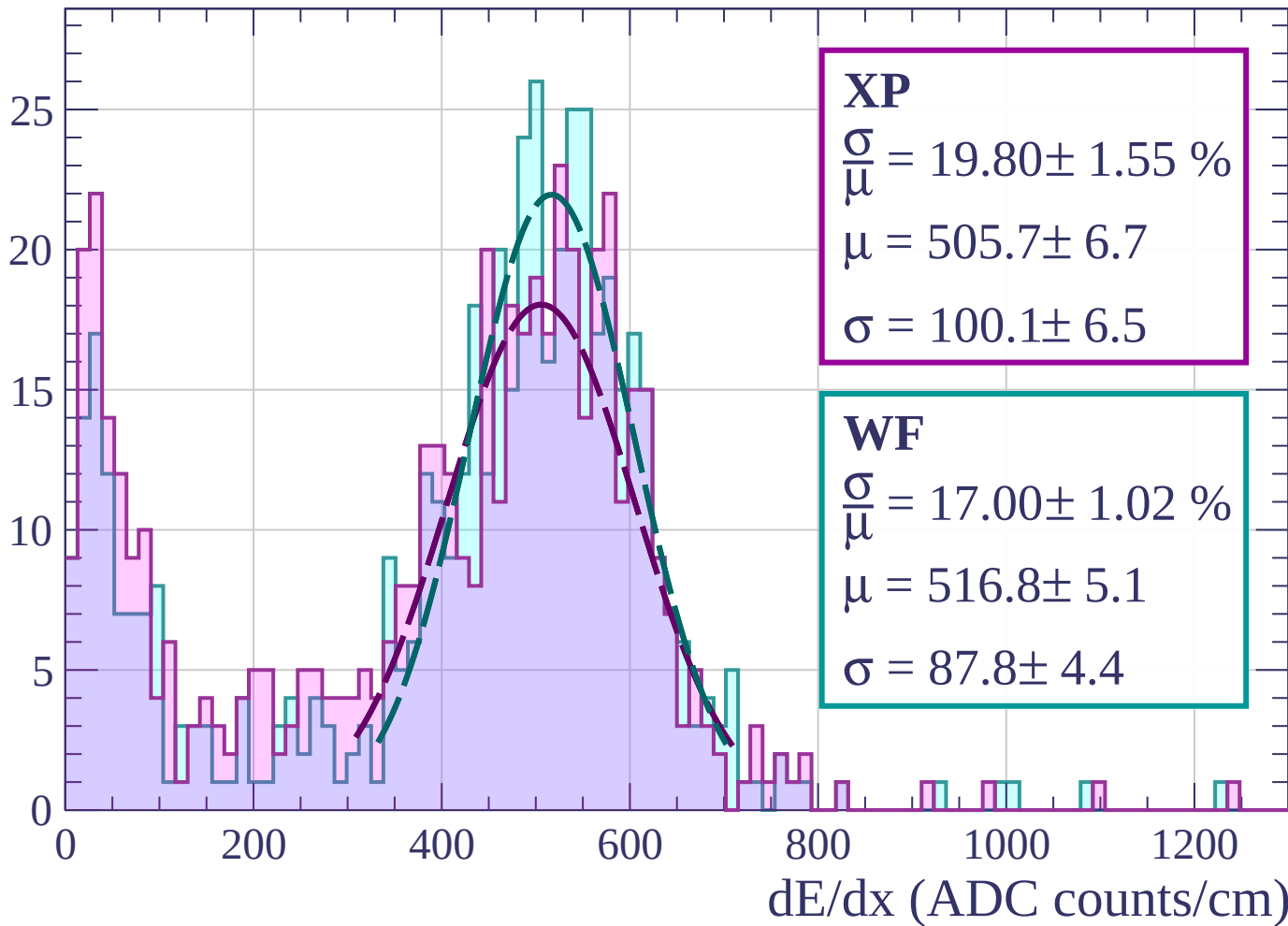
Energy loss | $27 < \phi < 30$

Count



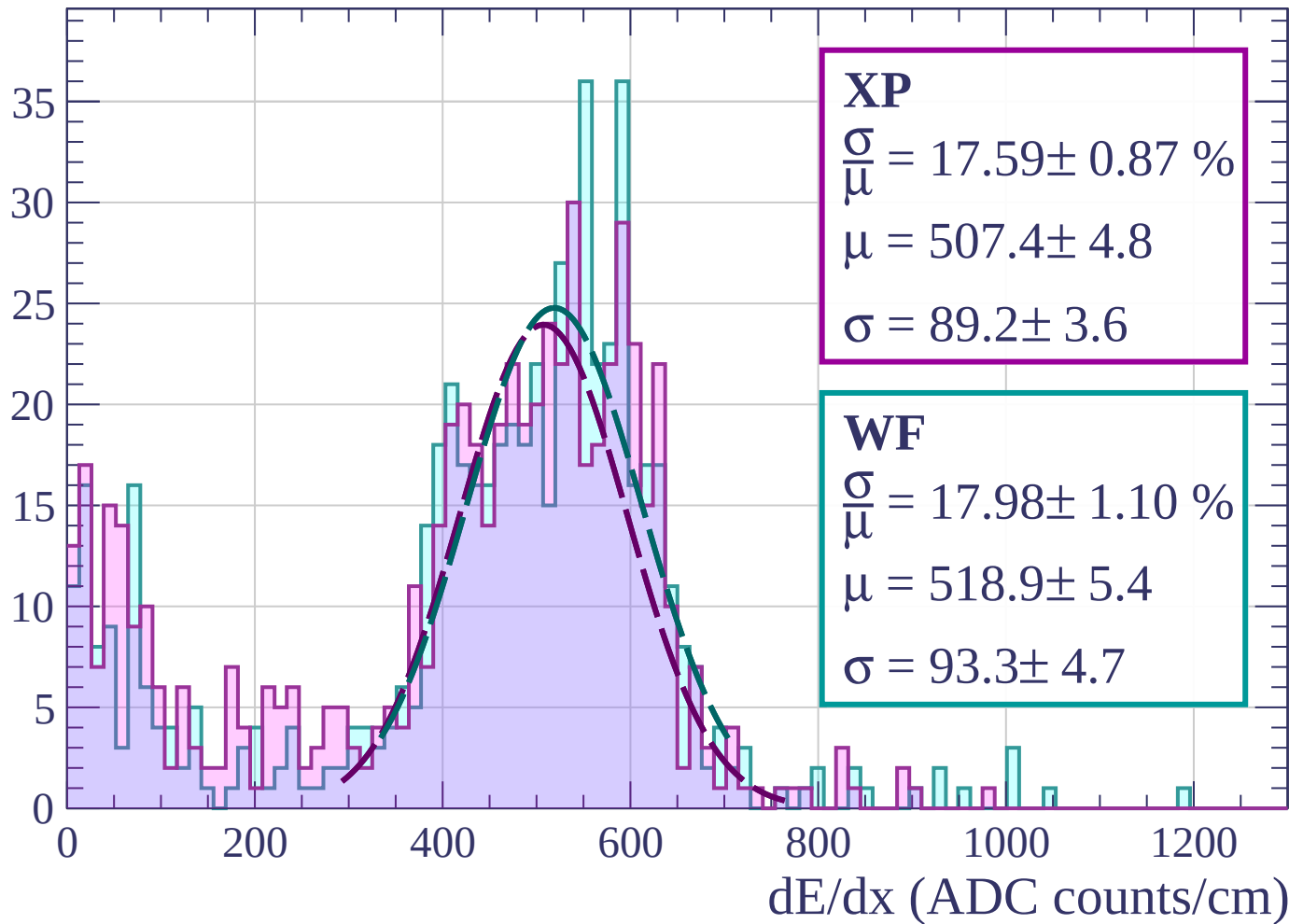
Energy loss | $30 < \phi < 33$

Count



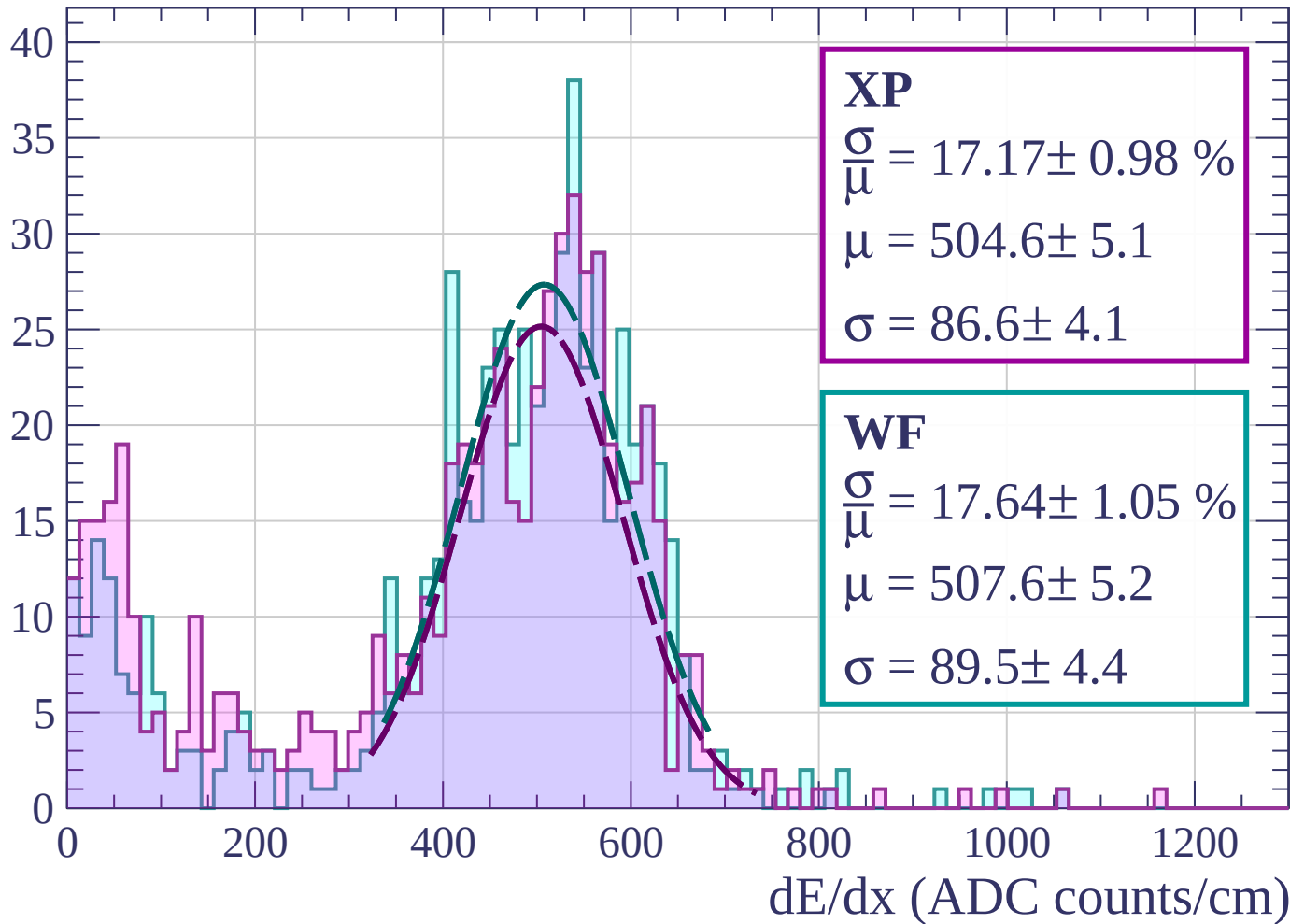
Energy loss | $33 < \phi < 36$

Count



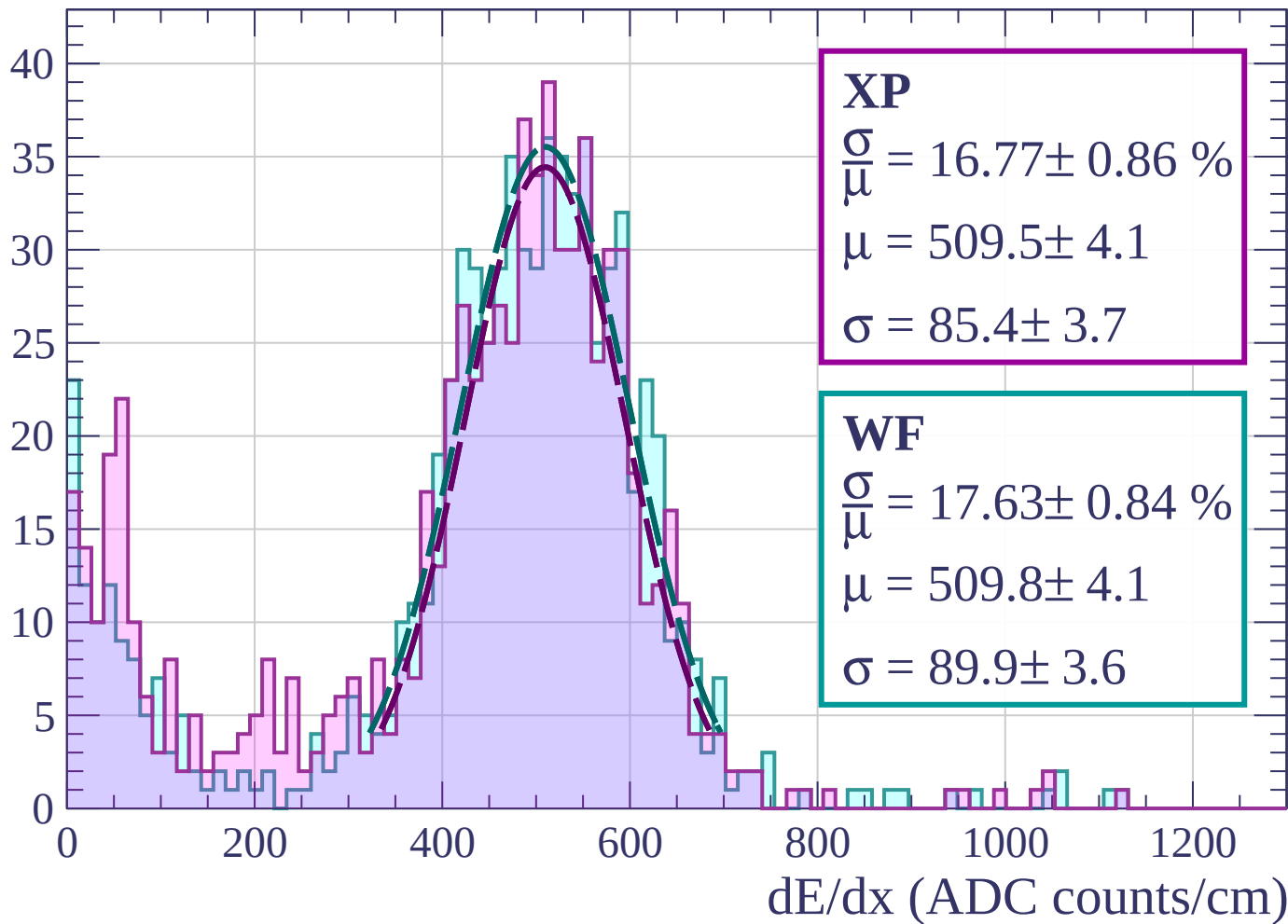
Energy loss | $36 < \phi < 39$

Count



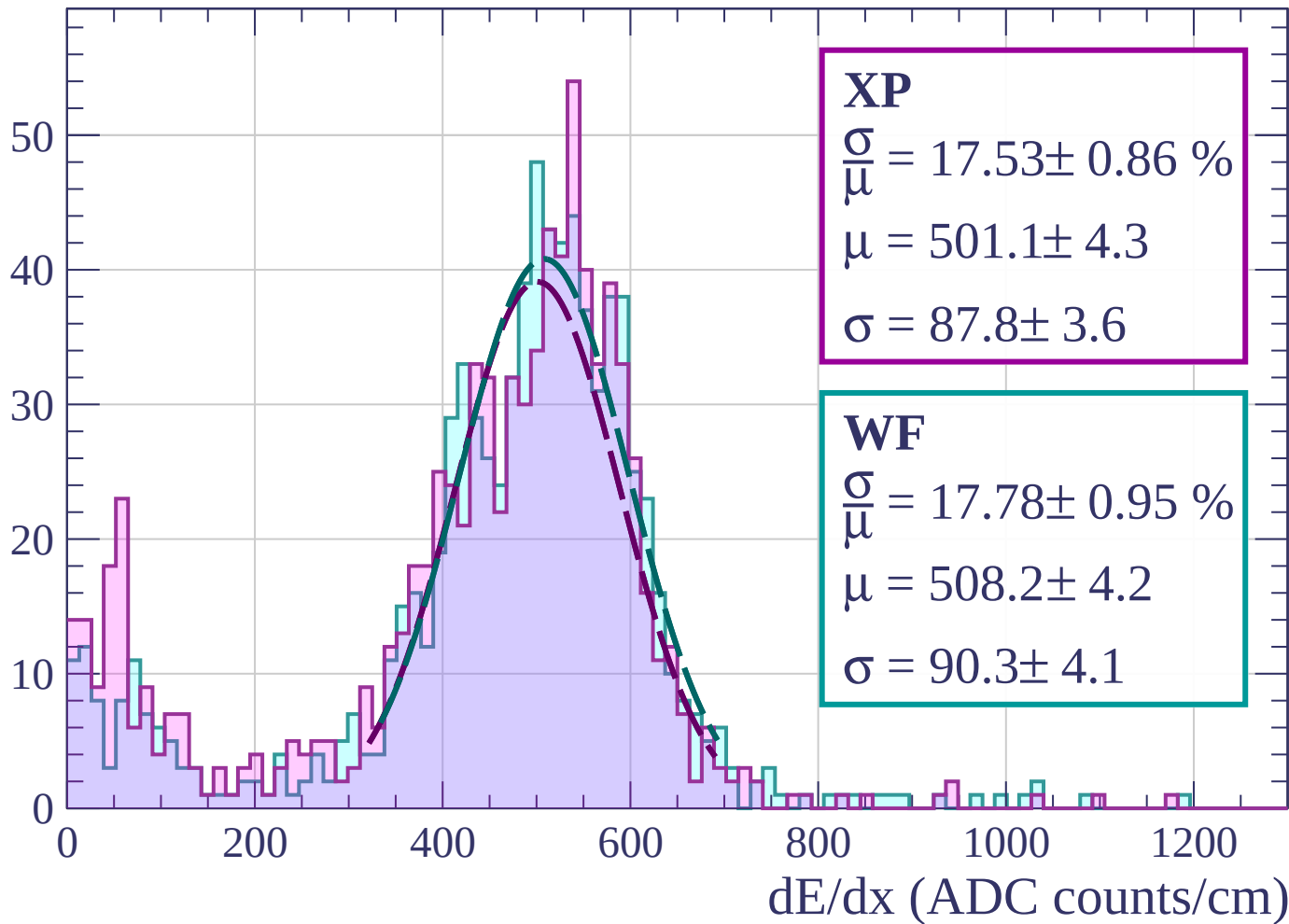
Energy loss | $39 < \phi < 42$

Count



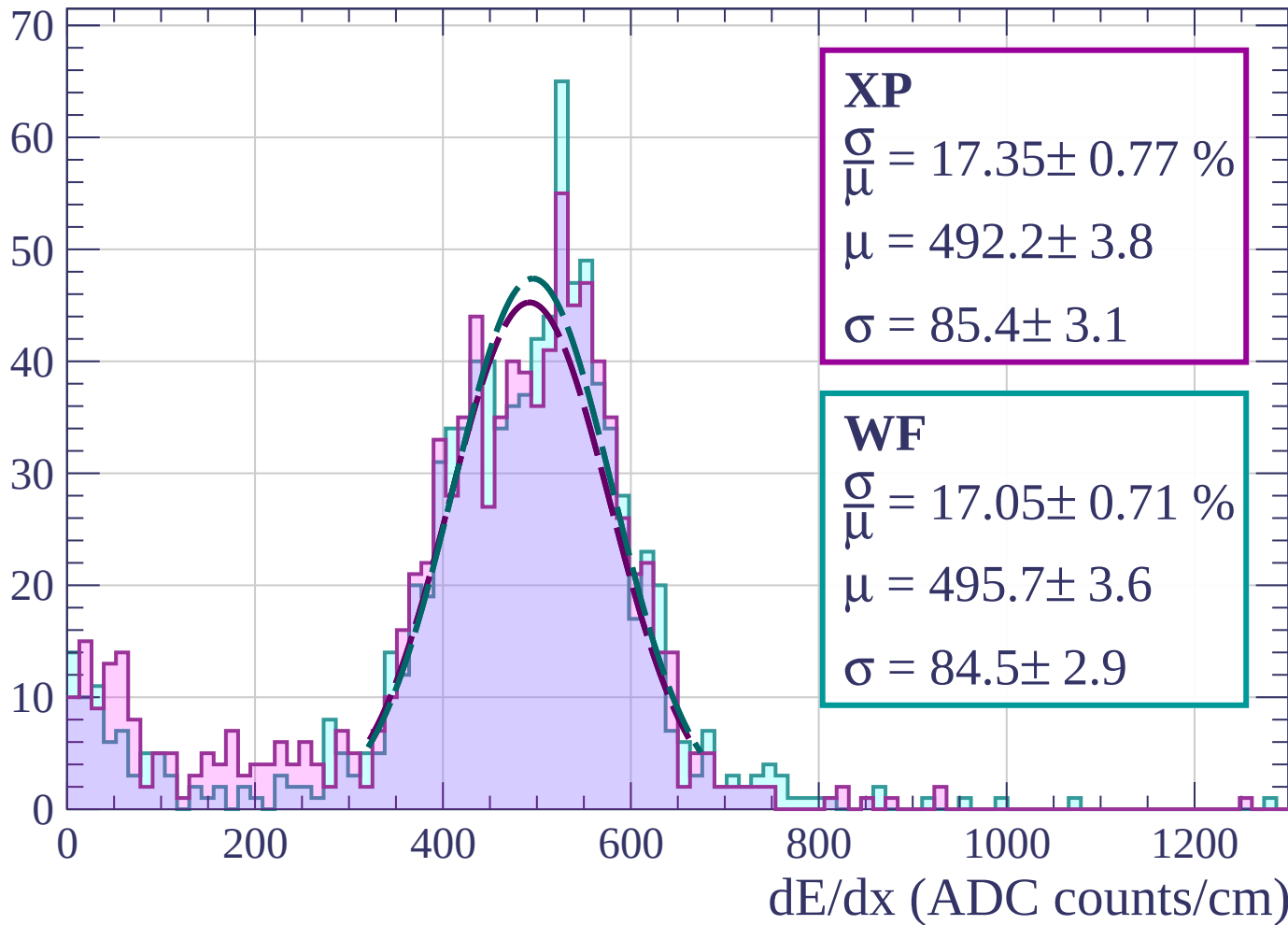
Energy loss | $42 < \phi < 45$

Count



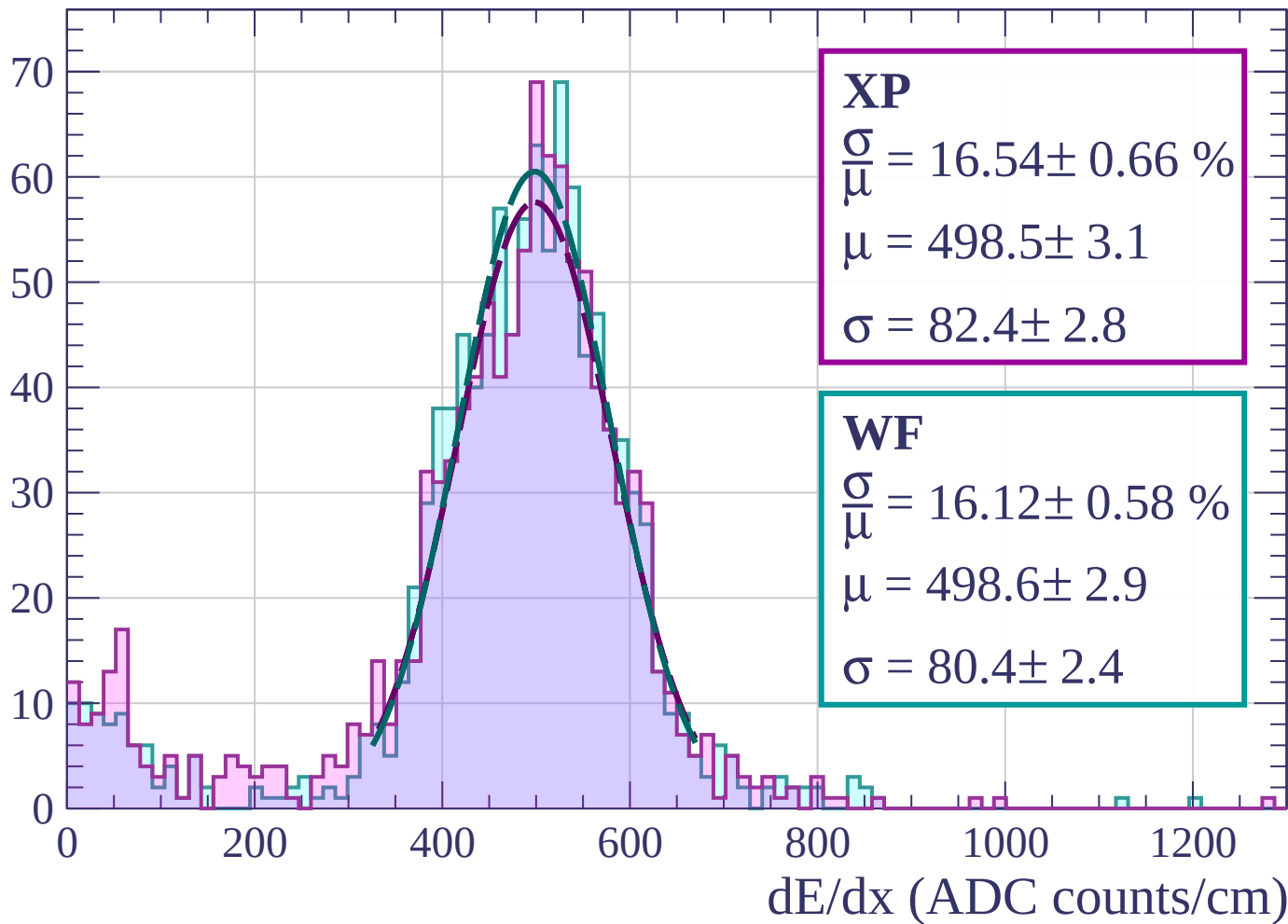
Energy loss | $45 < \phi < 48$

Count



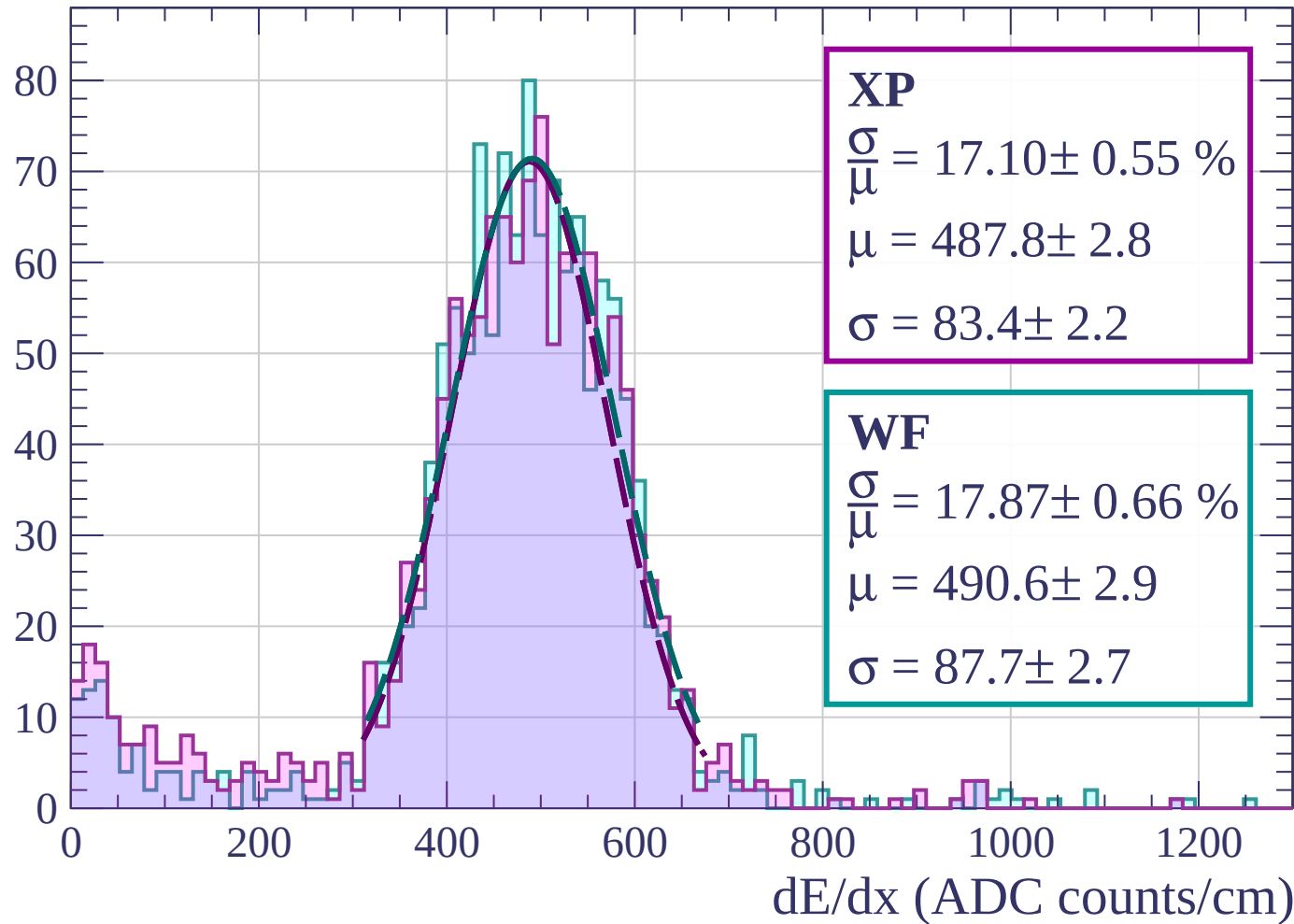
Energy loss | $48 < \phi < 51$

Count



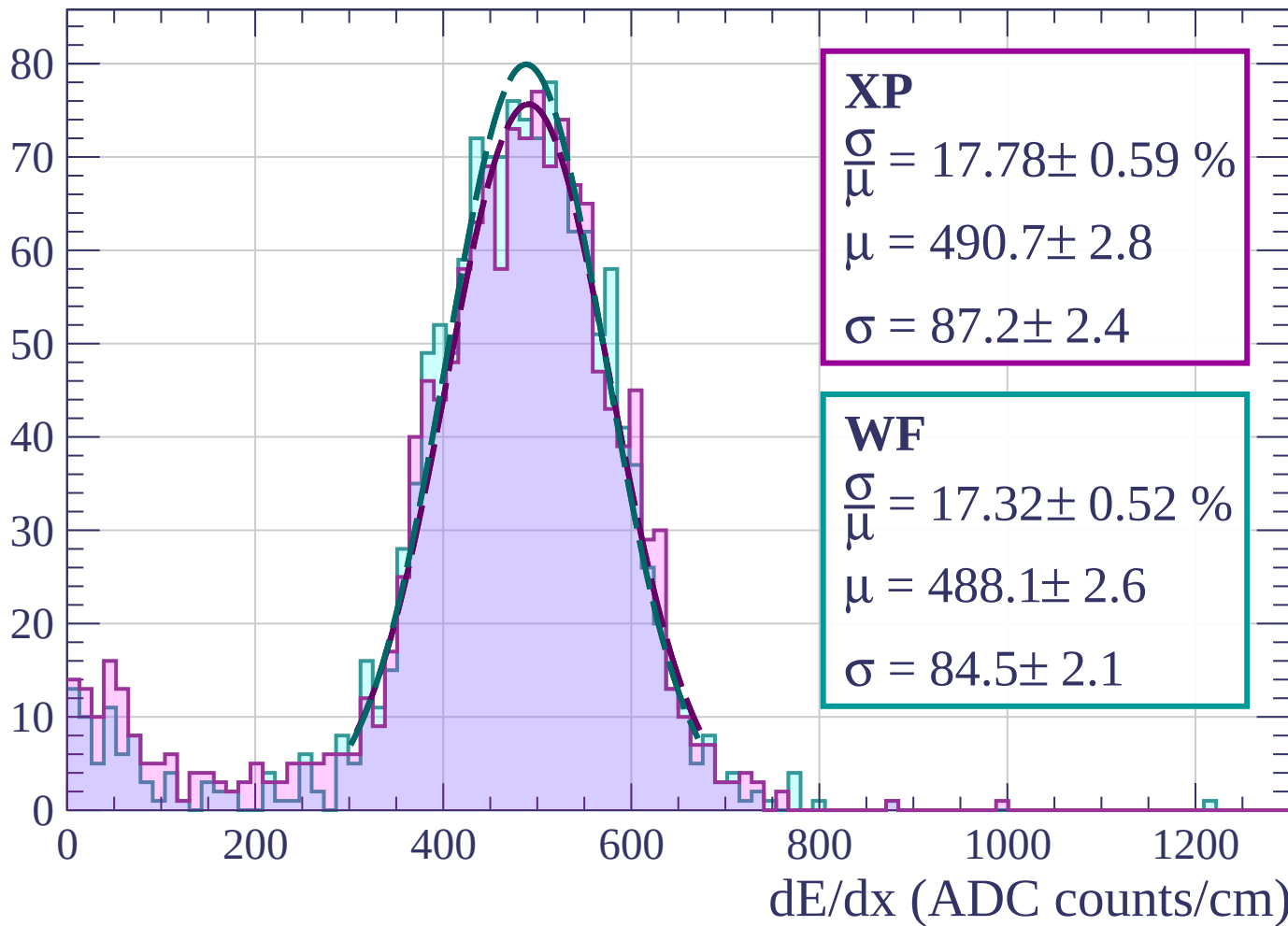
Energy loss | $51 < \phi < 54$

Count



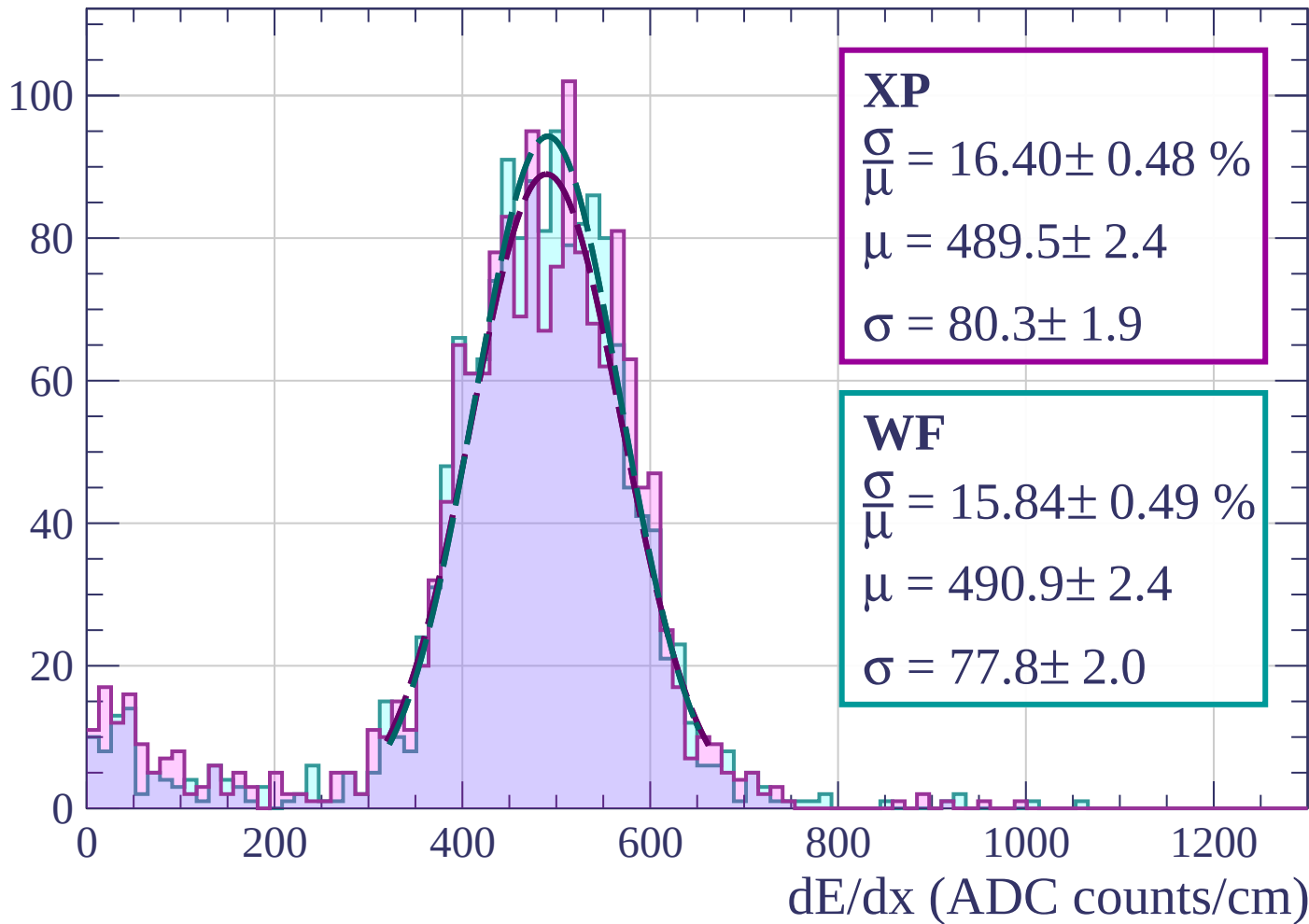
Energy loss | $54 < \phi < 57$

Count



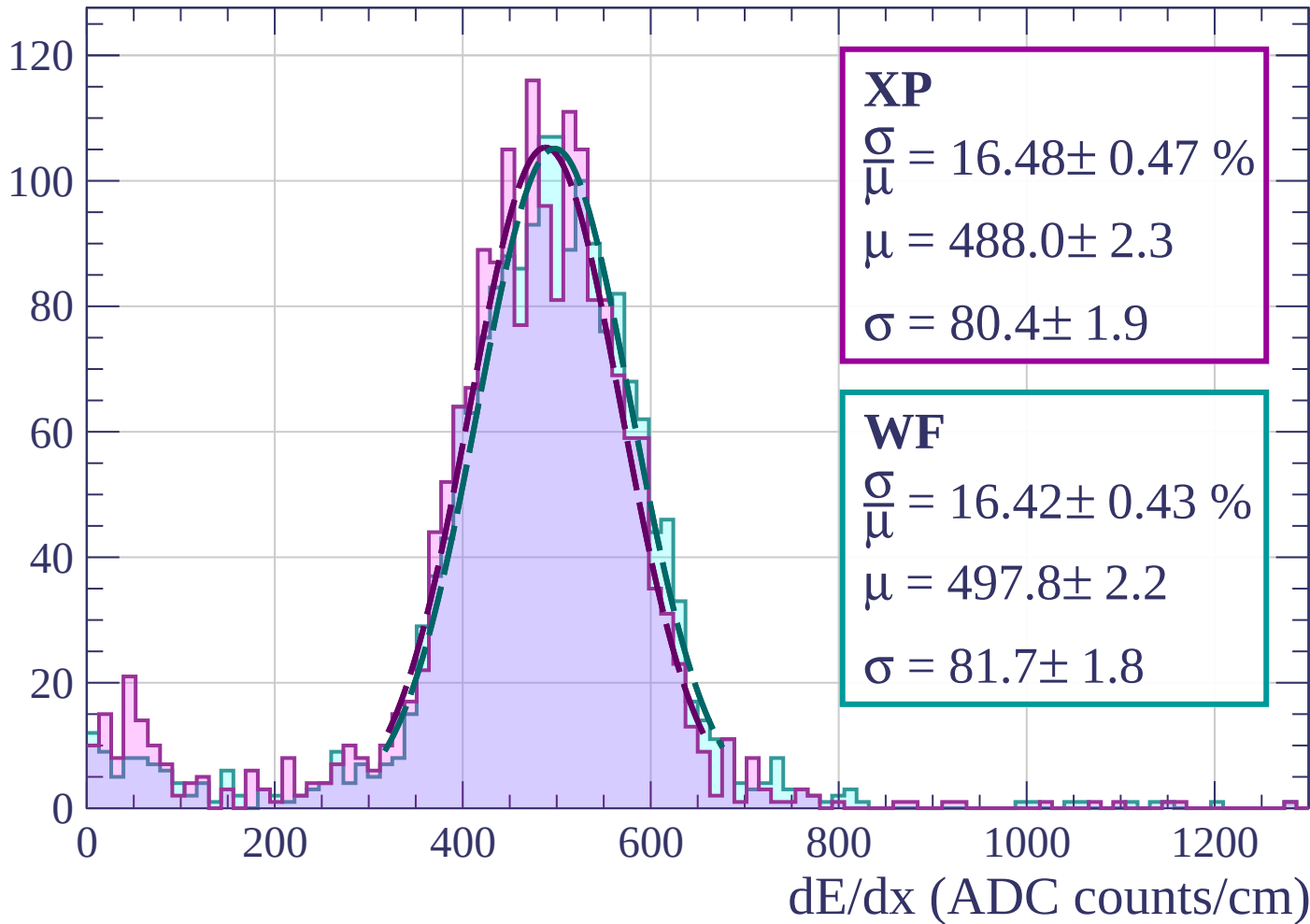
Energy loss | $57 < \phi < 60$

Count



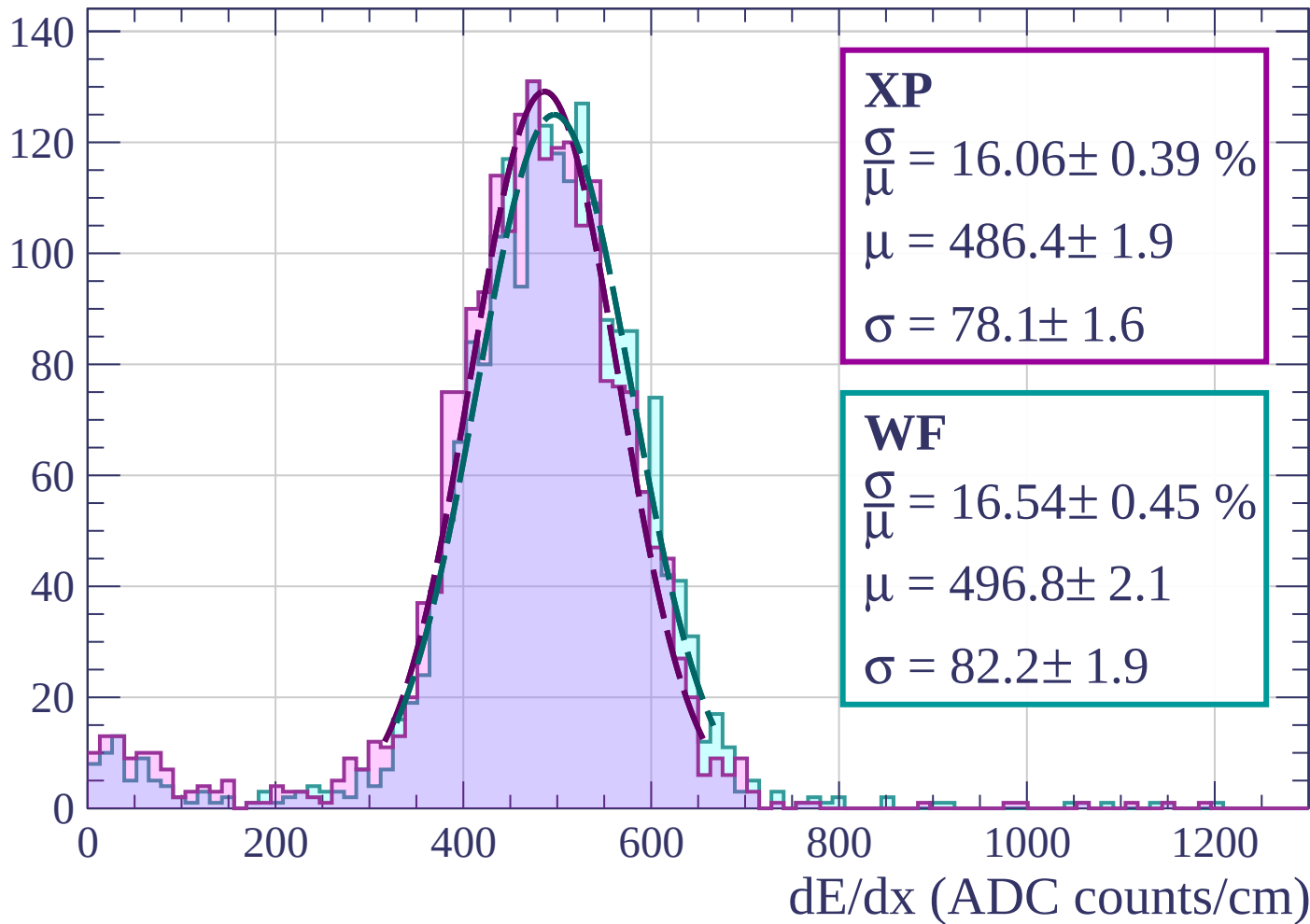
Energy loss | $60 < \phi < 63$

Count



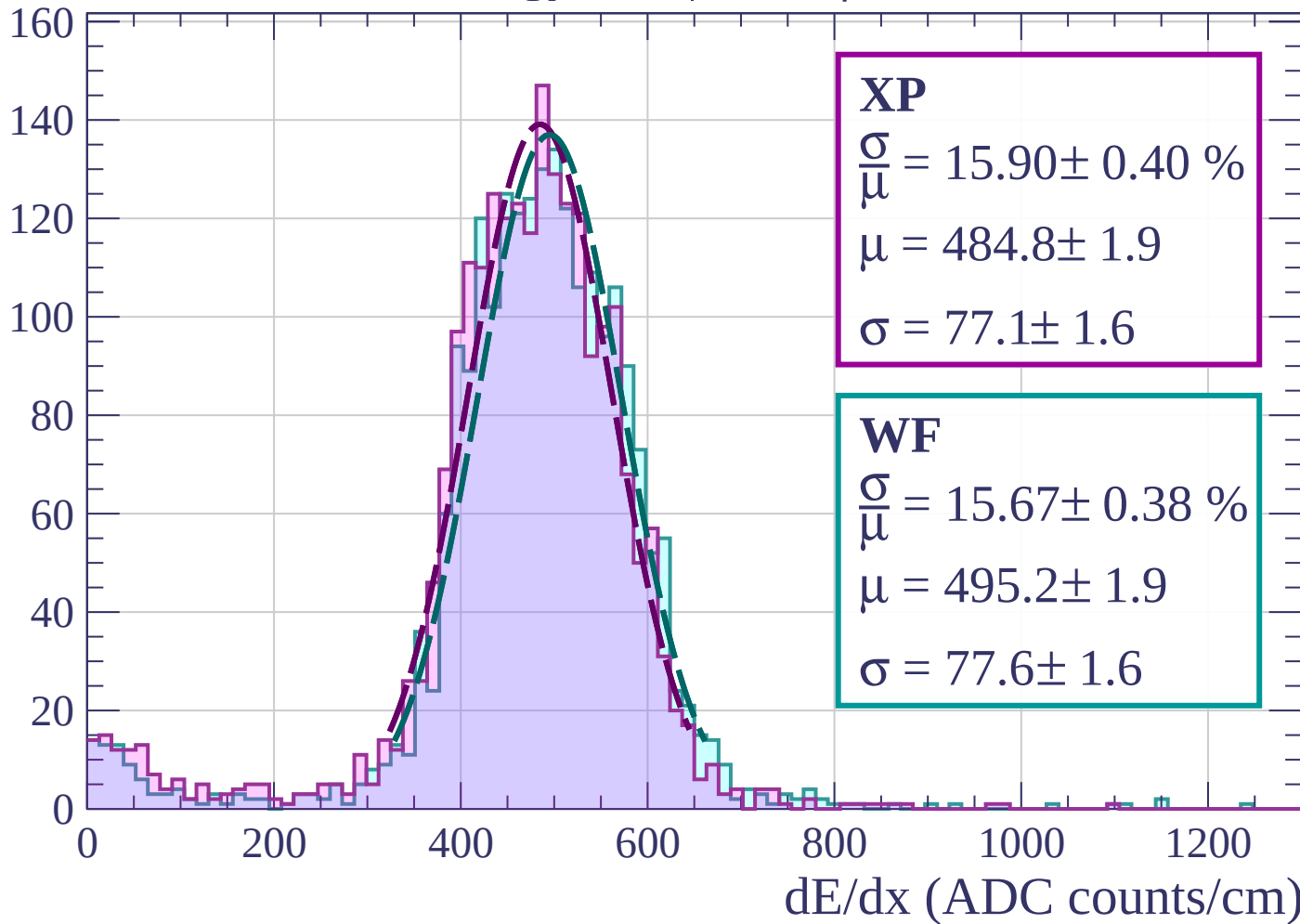
Energy loss | $63 < \phi < 66$

Count

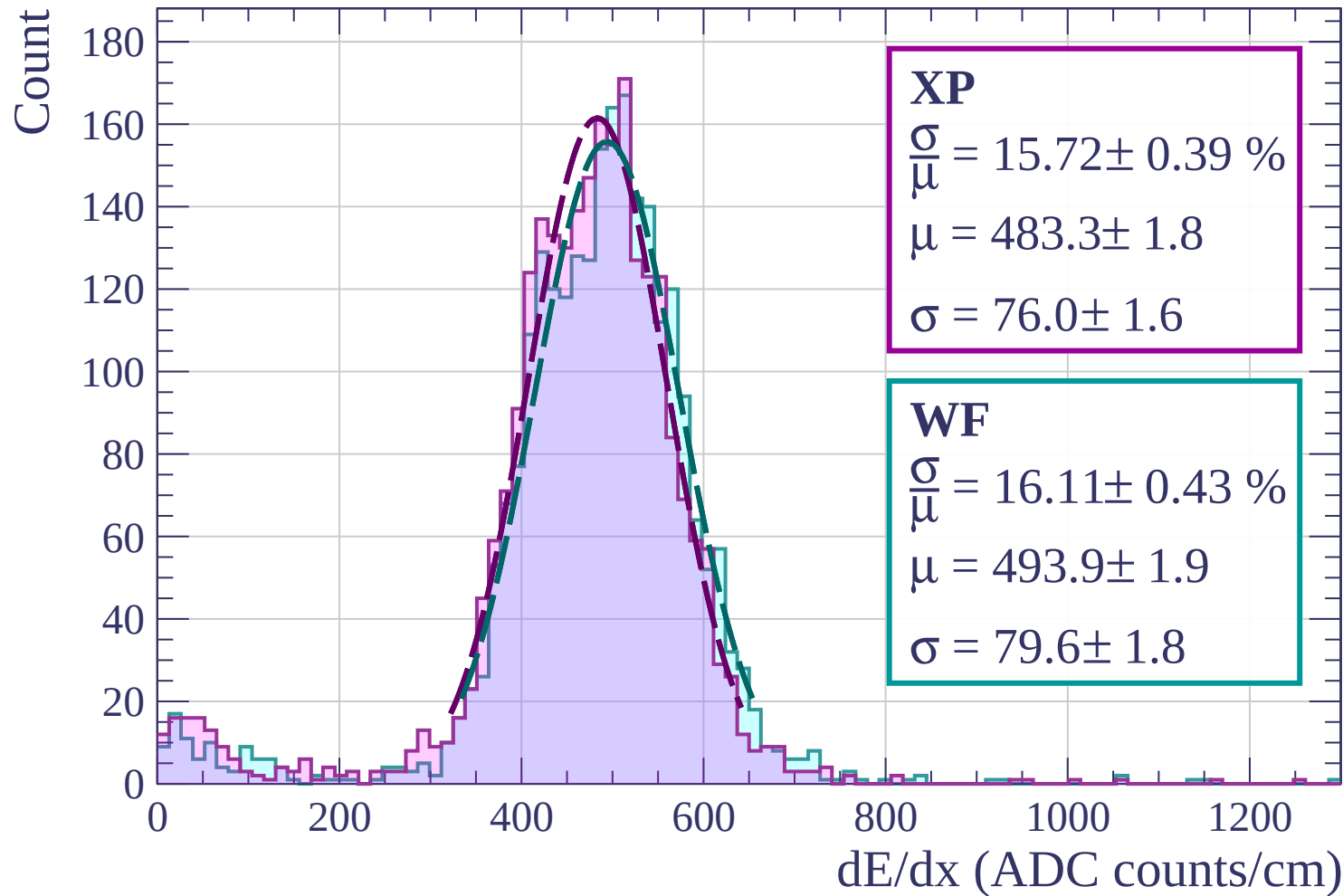


Energy loss | $66 < \phi < 69$

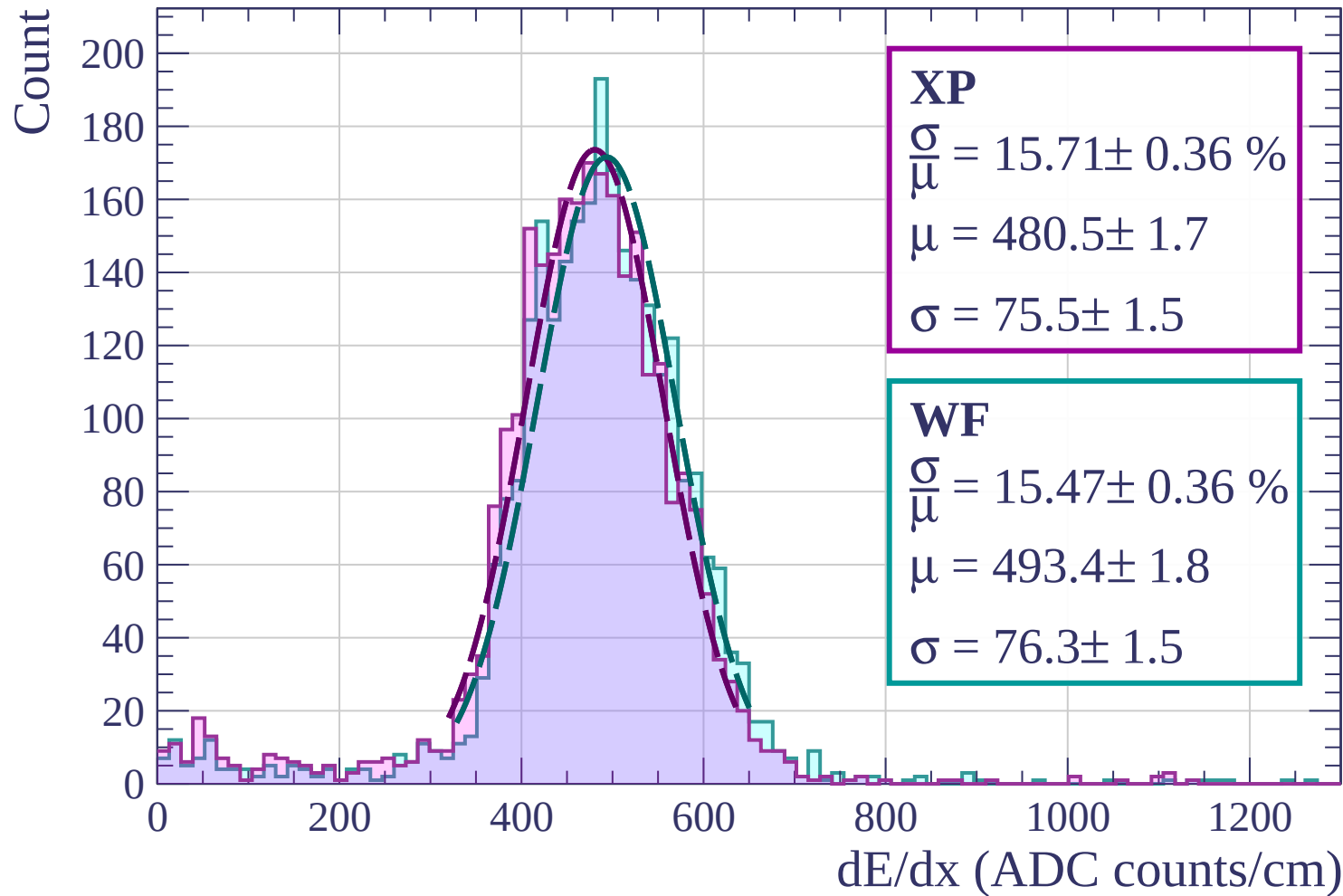
Count



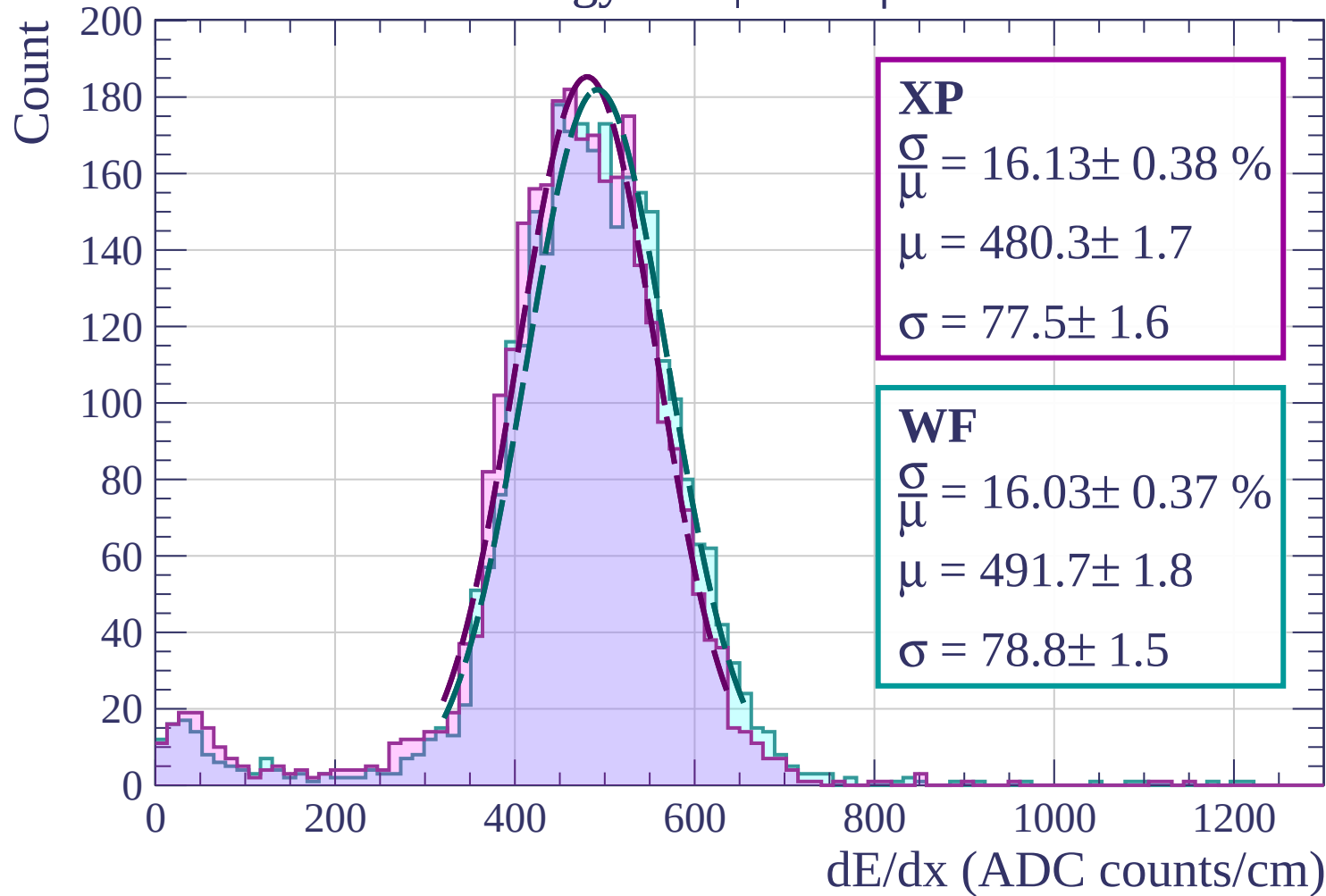
Energy loss | $69 < \phi < 72$



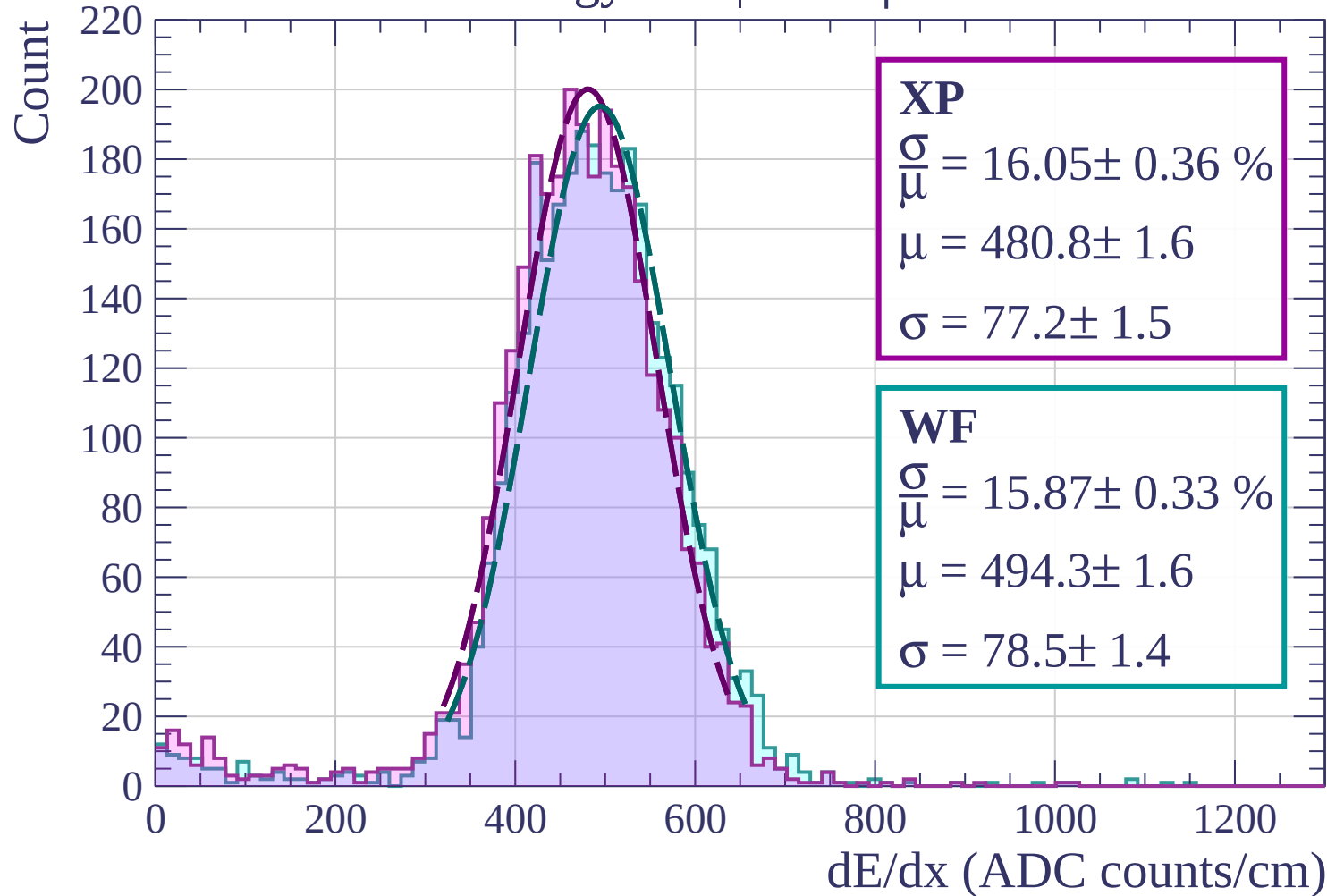
Energy loss | $72 < \phi < 75$



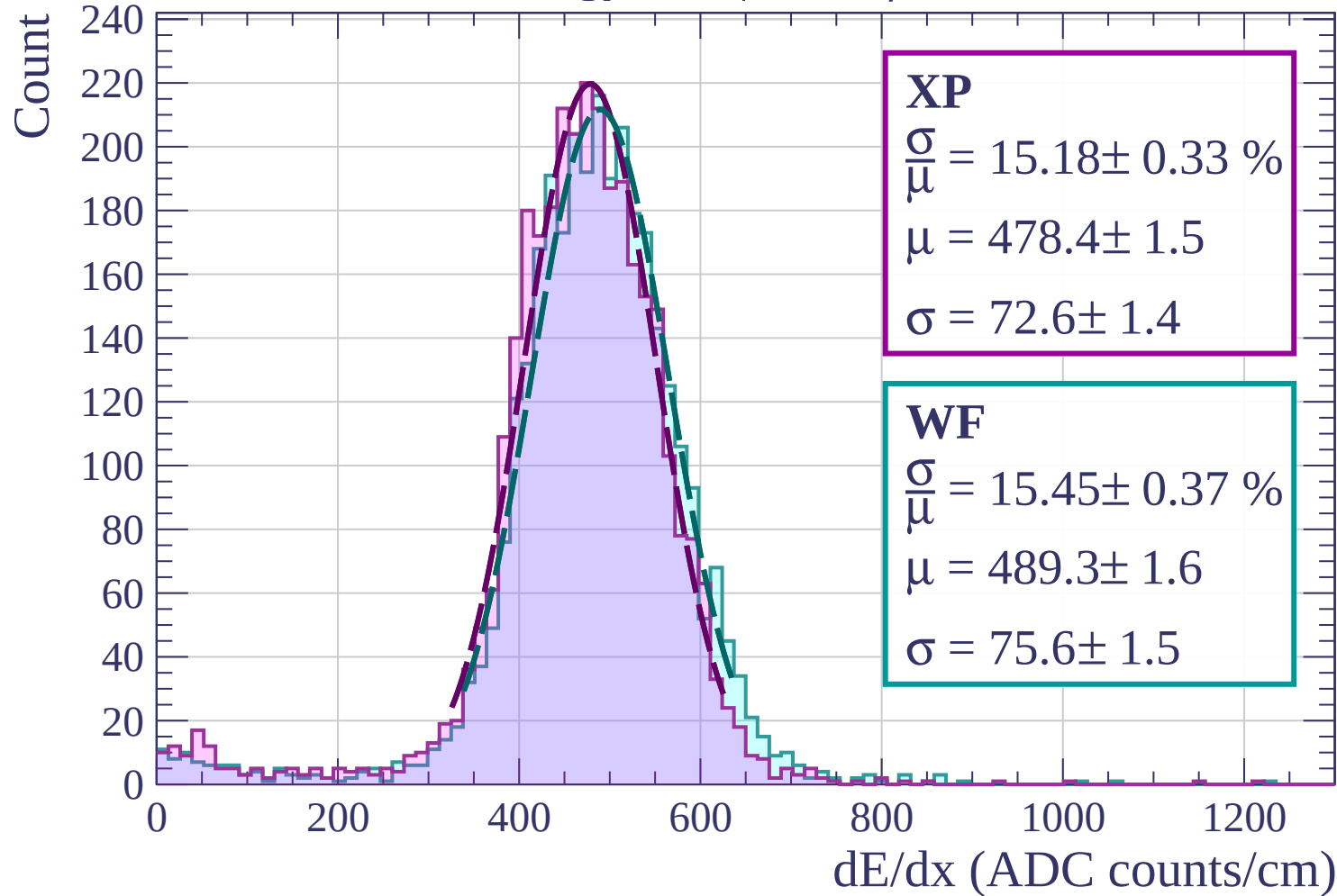
Energy loss | $75 < \phi < 78$



Energy loss | $78 < \phi < 81$

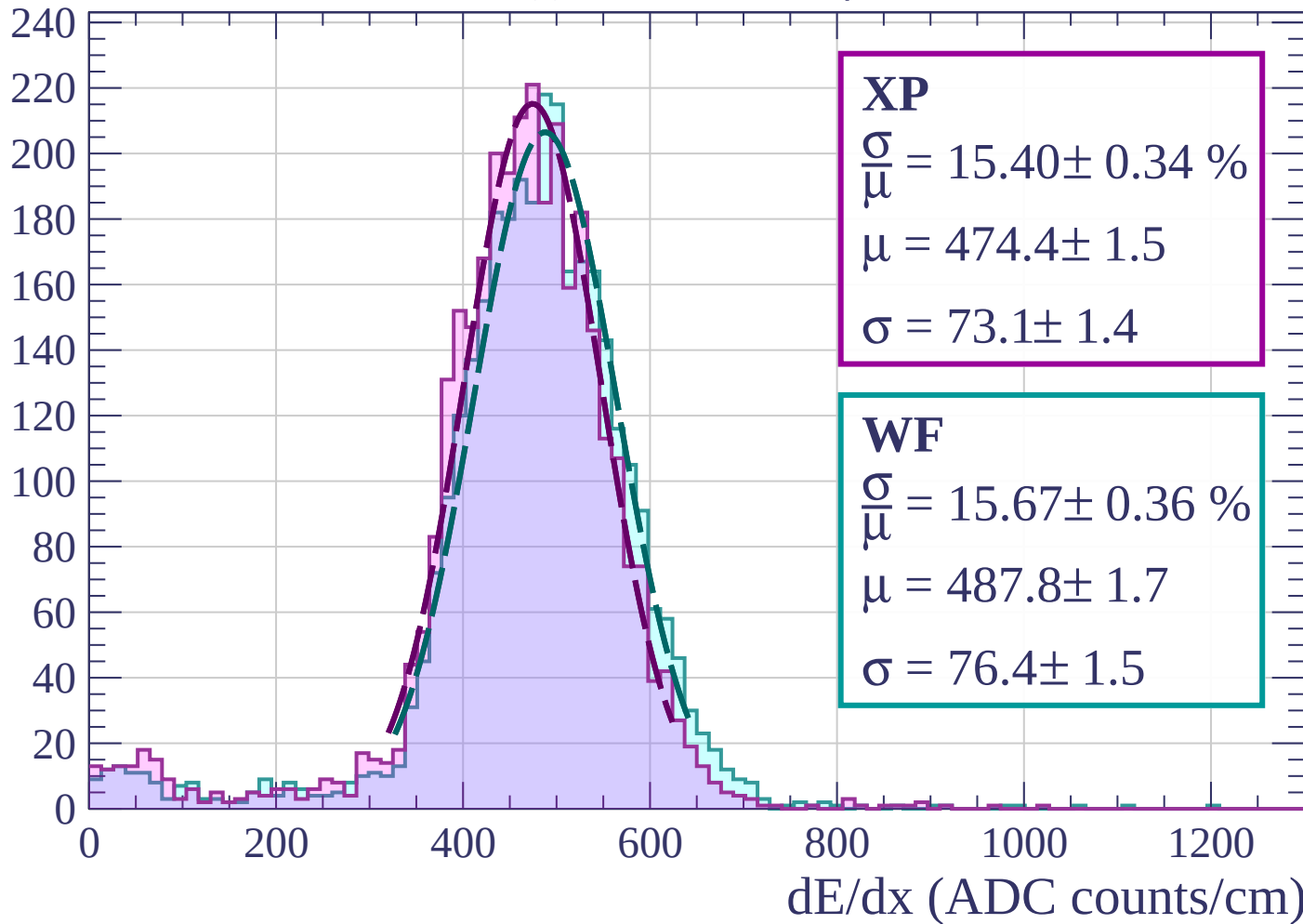


Energy loss | $81 < \phi < 84$



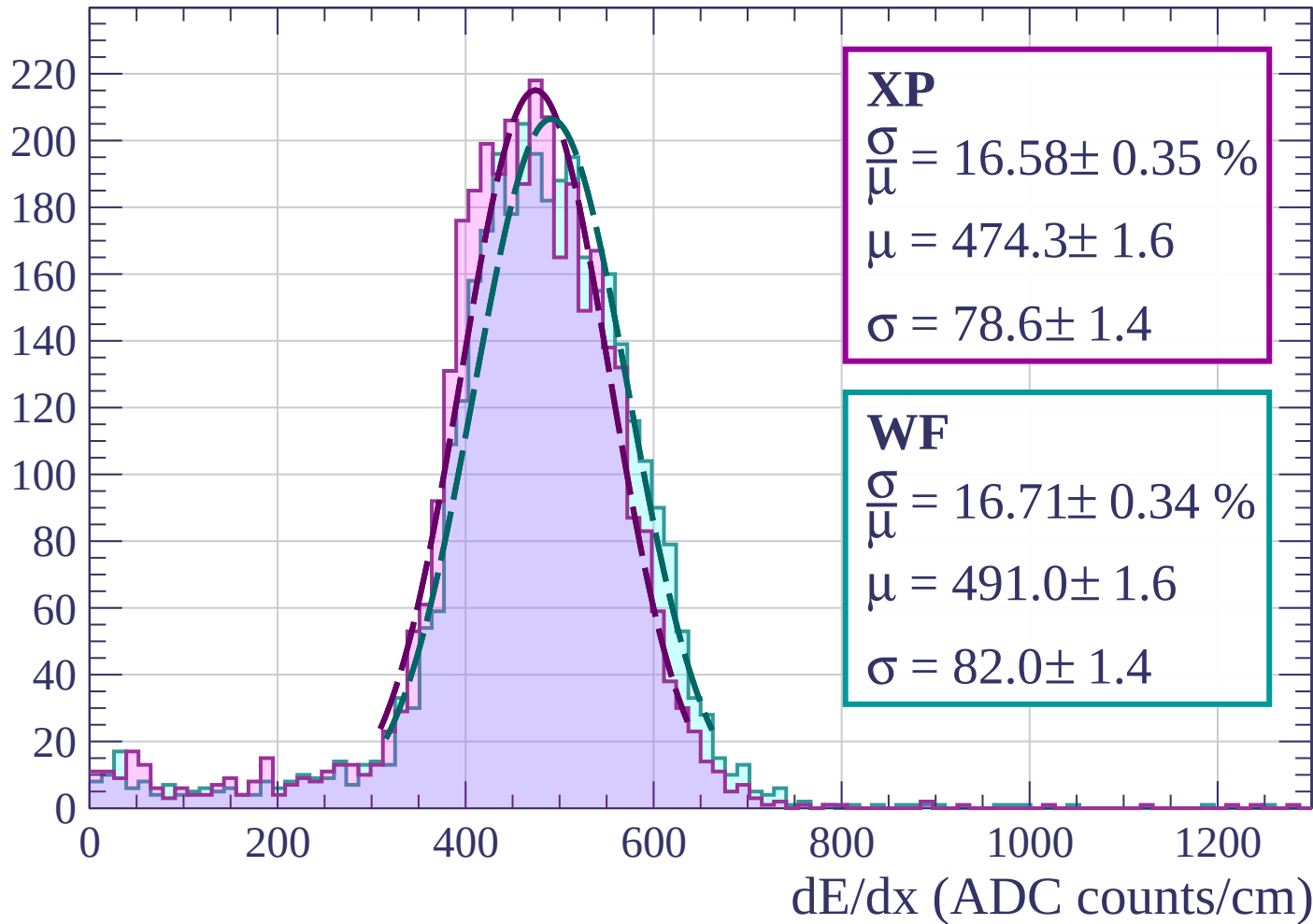
Energy loss | $84 < \phi < 87$

Count



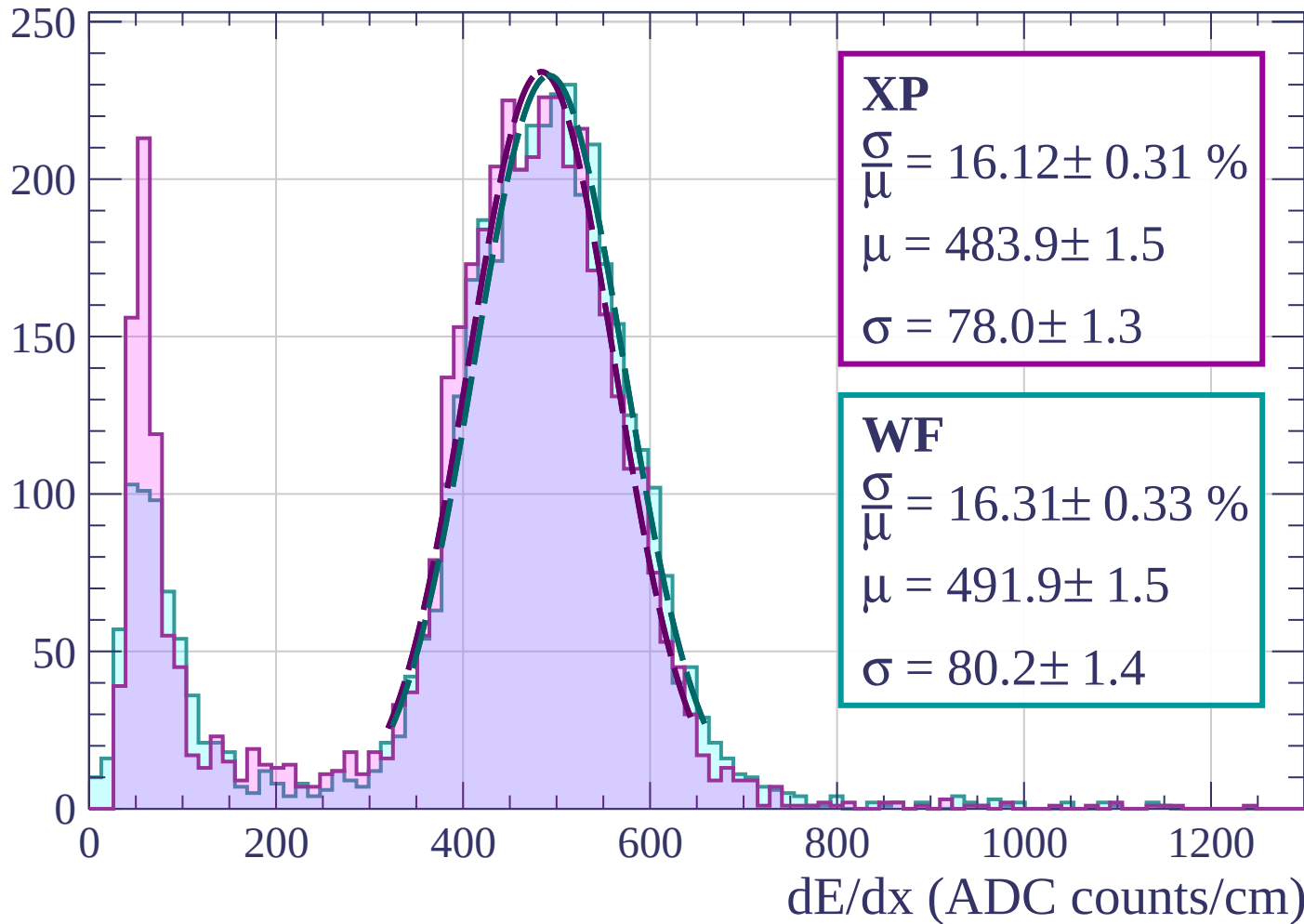
Energy loss | $87 < \phi < 90$

Count

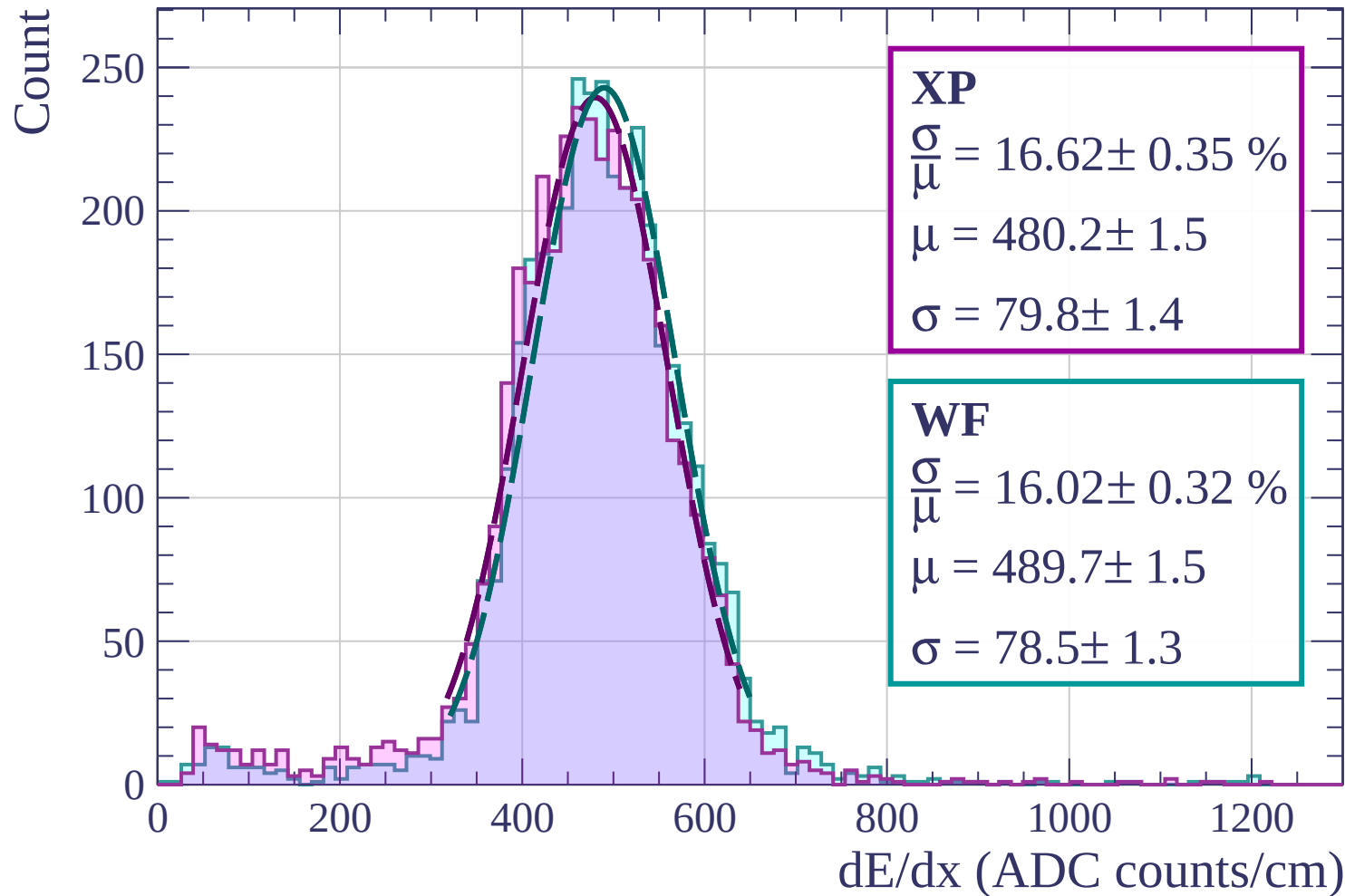


Energy loss | $0 < \theta < 3$

Count

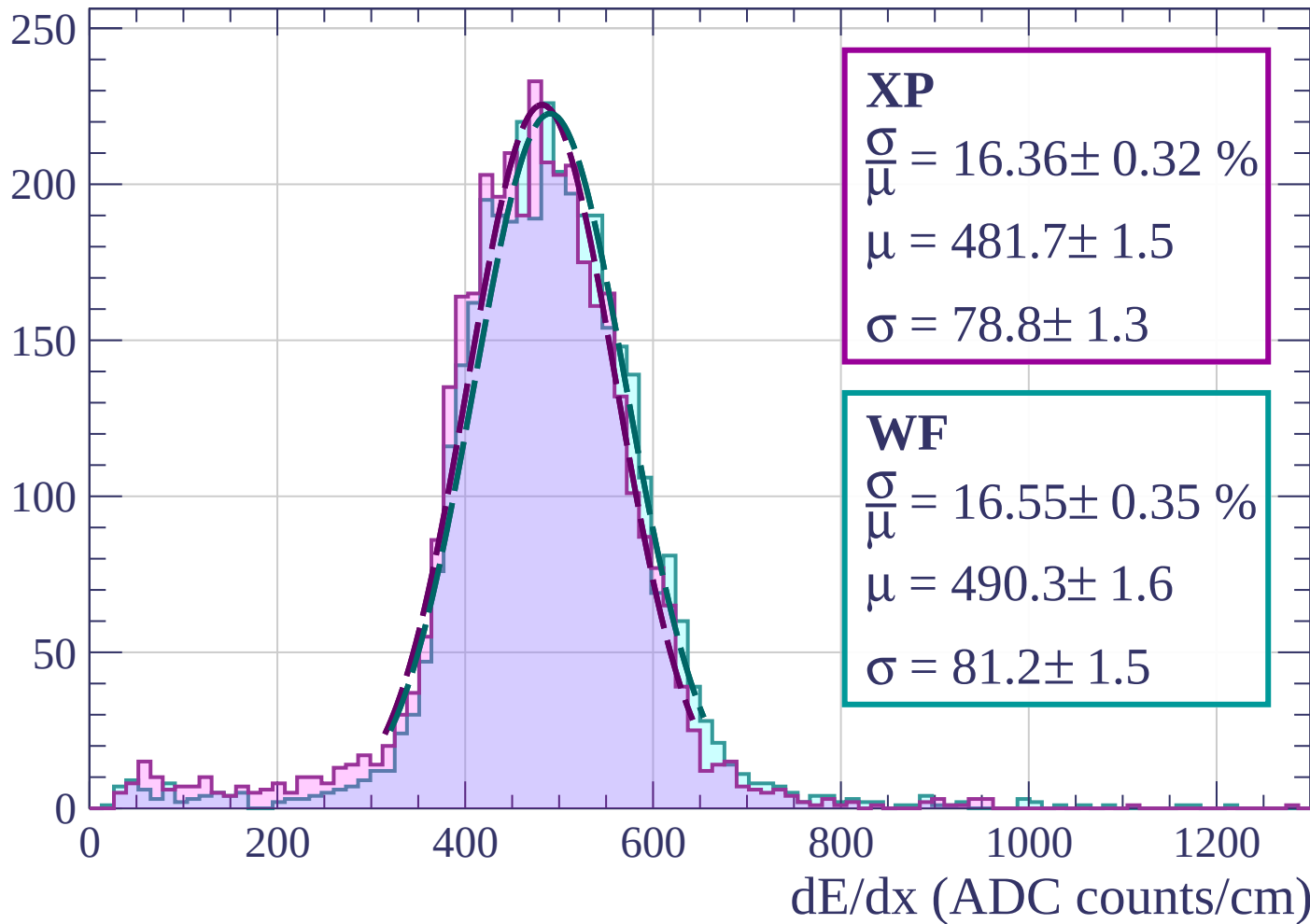


Energy loss | $3 < \theta < 6$

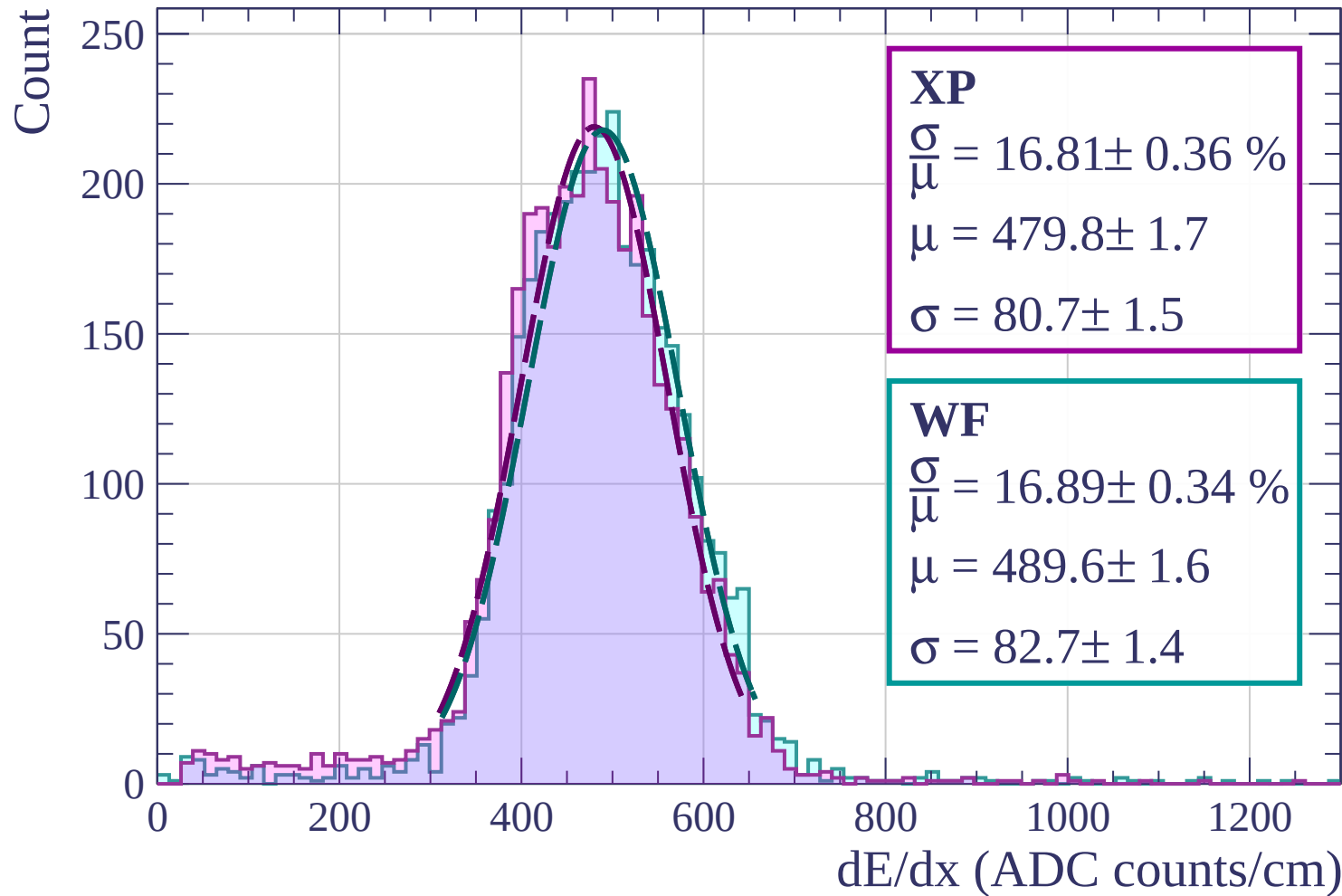


Energy loss | $6 < \theta < 9$

Count

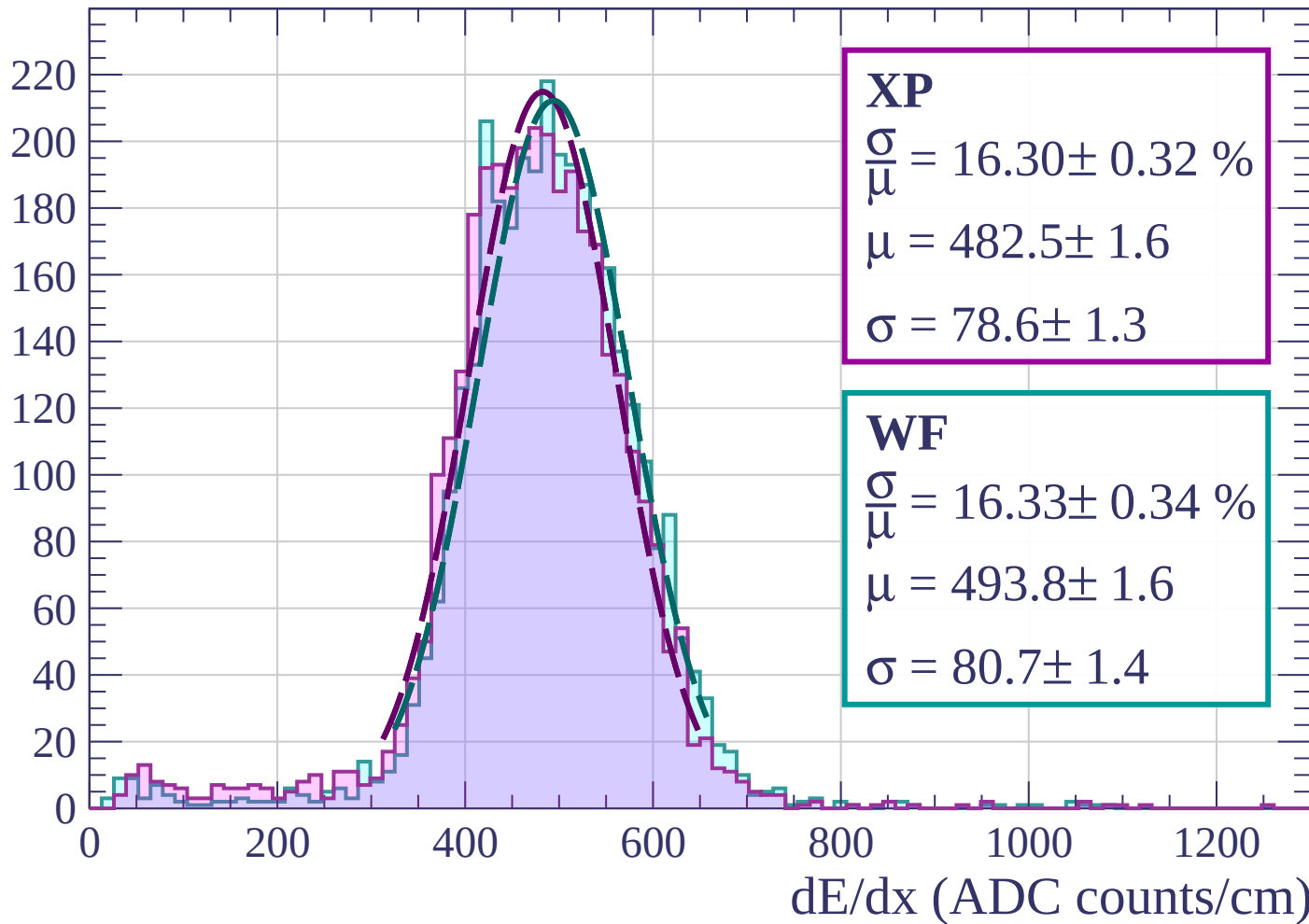


Energy loss | $9 < \theta < 12$

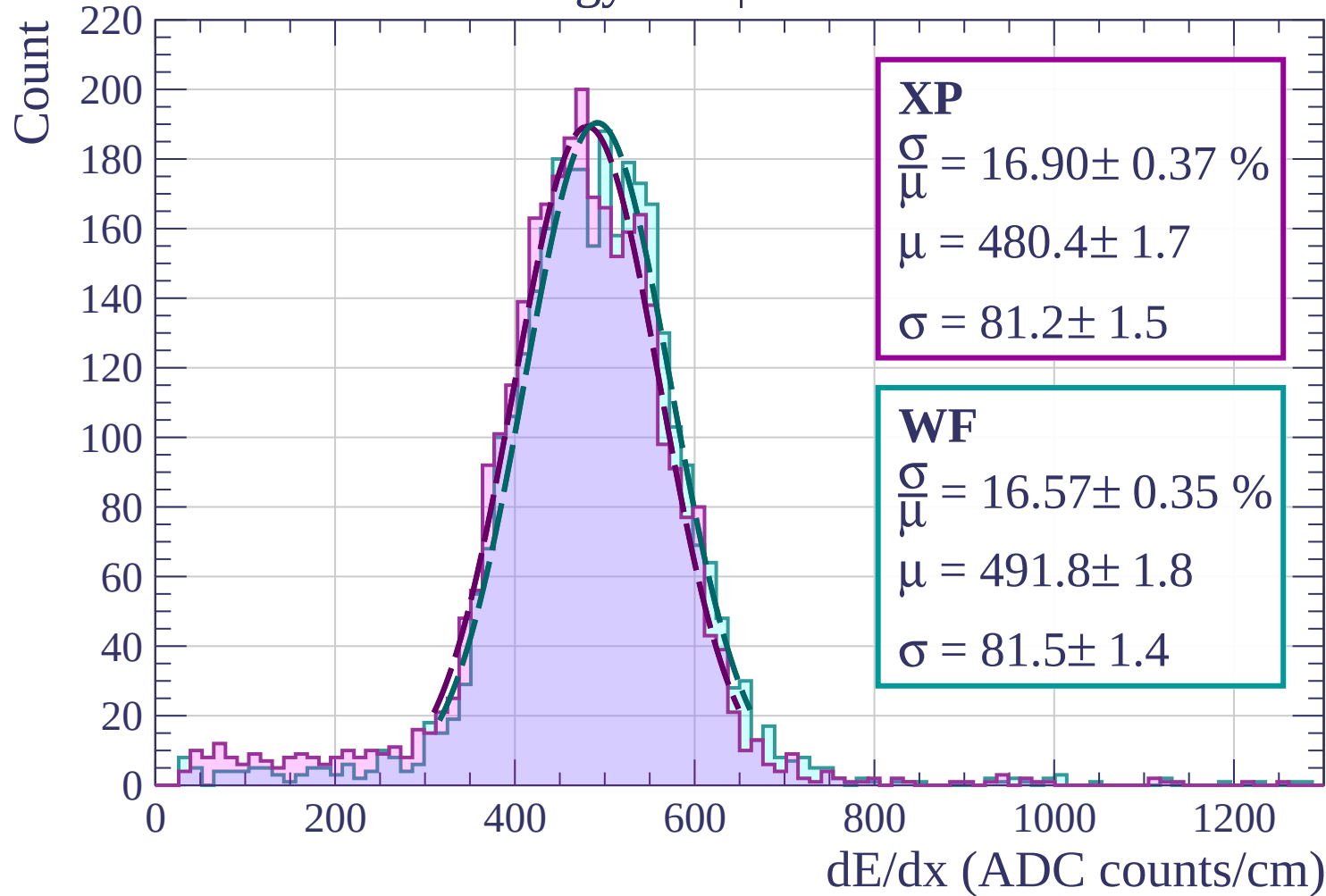


Energy loss | $12 < \theta < 15$

Count

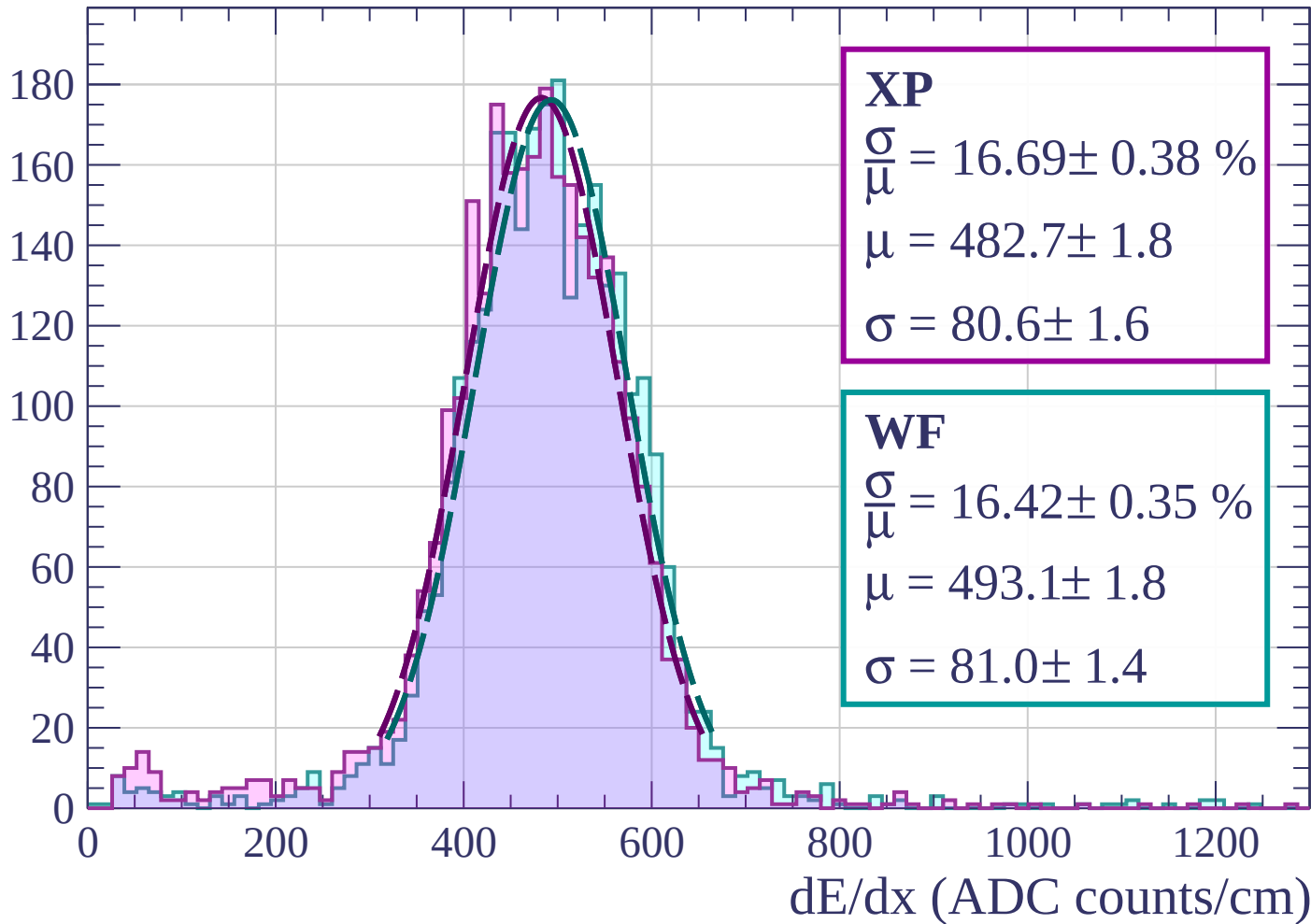


Energy loss | $15 < \theta < 18$



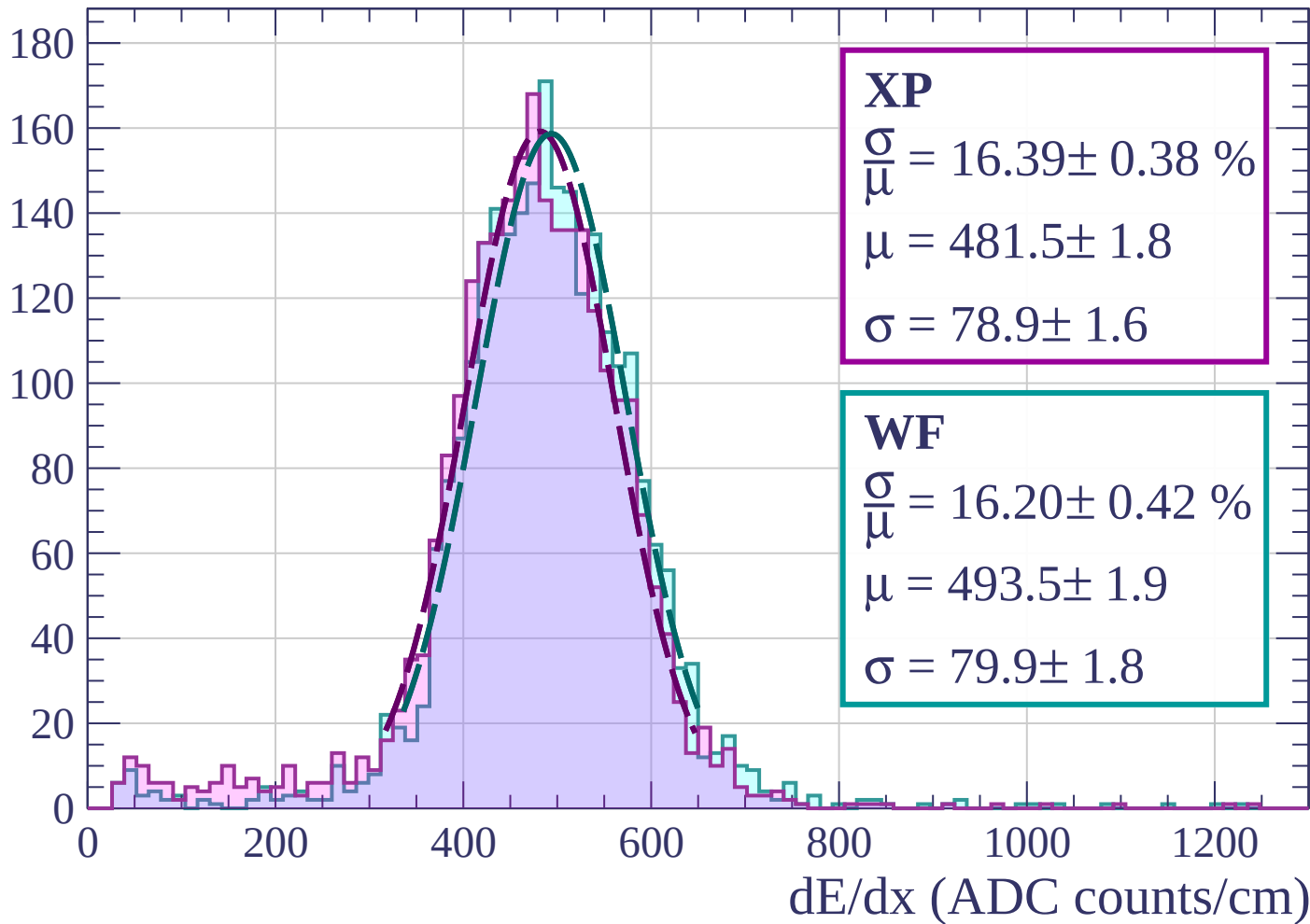
Energy loss | $18 < \theta < 21$

Count



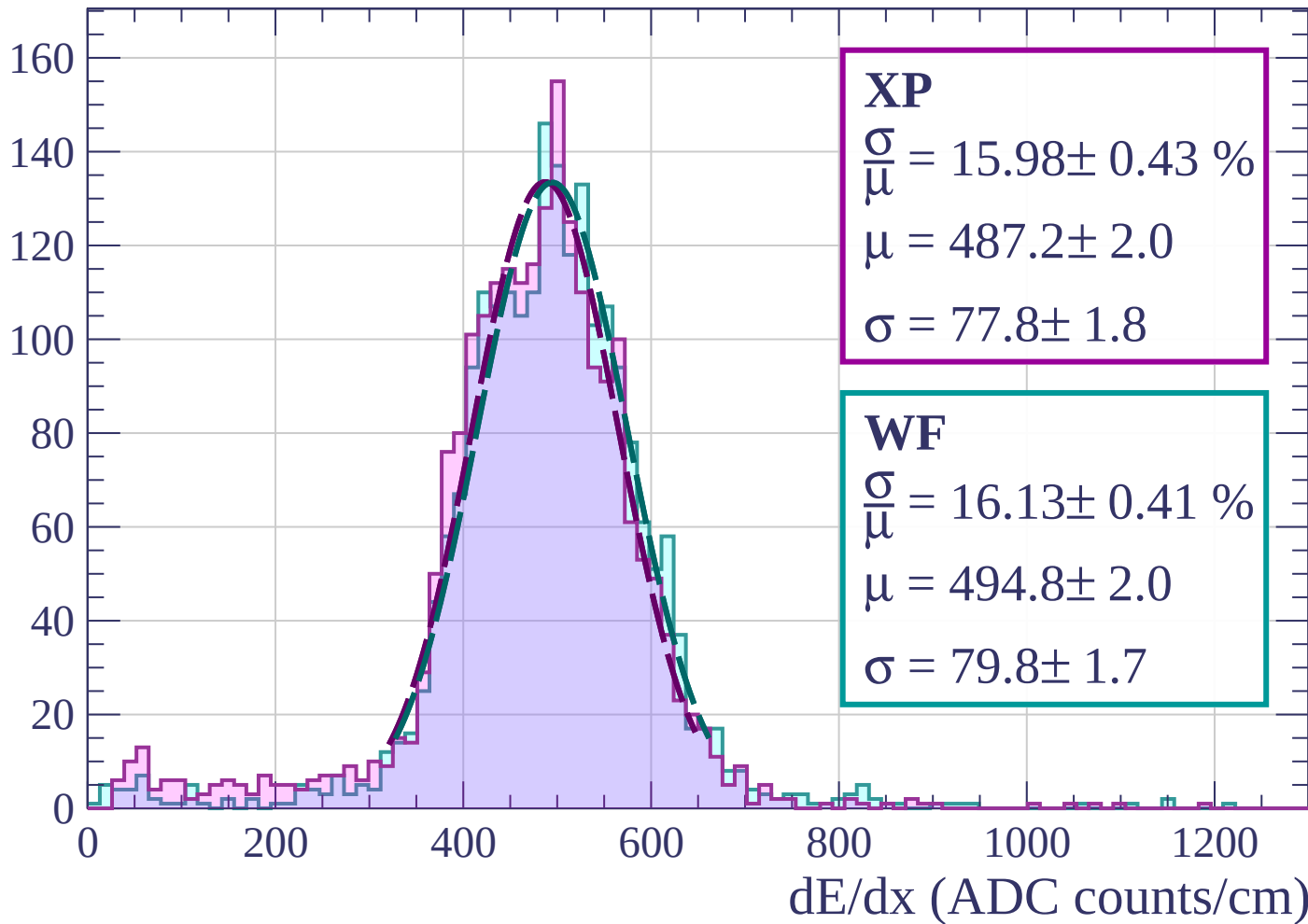
Energy loss | $21 < \theta < 24$

Count



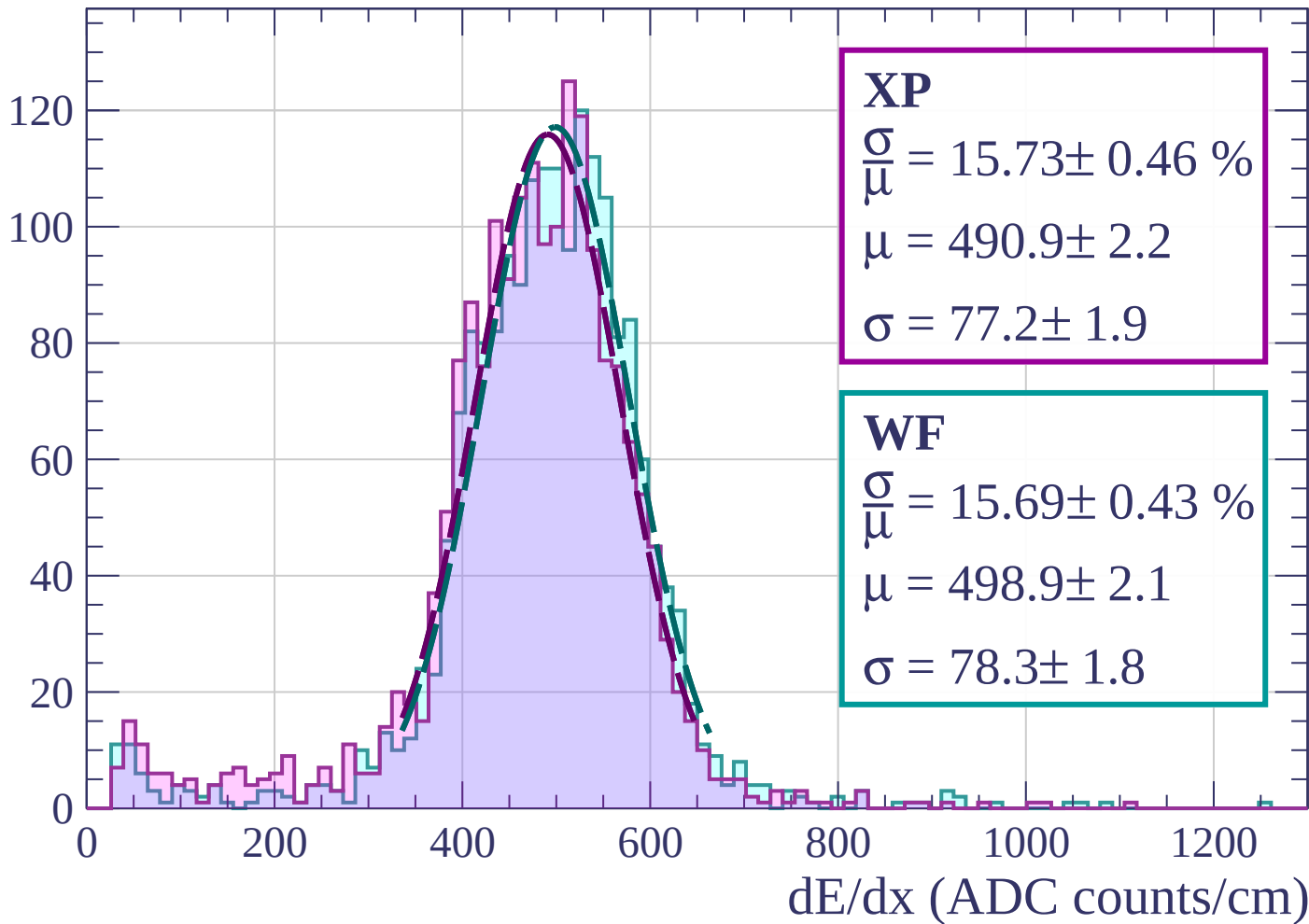
Energy loss | $24 < \theta < 27$

Count



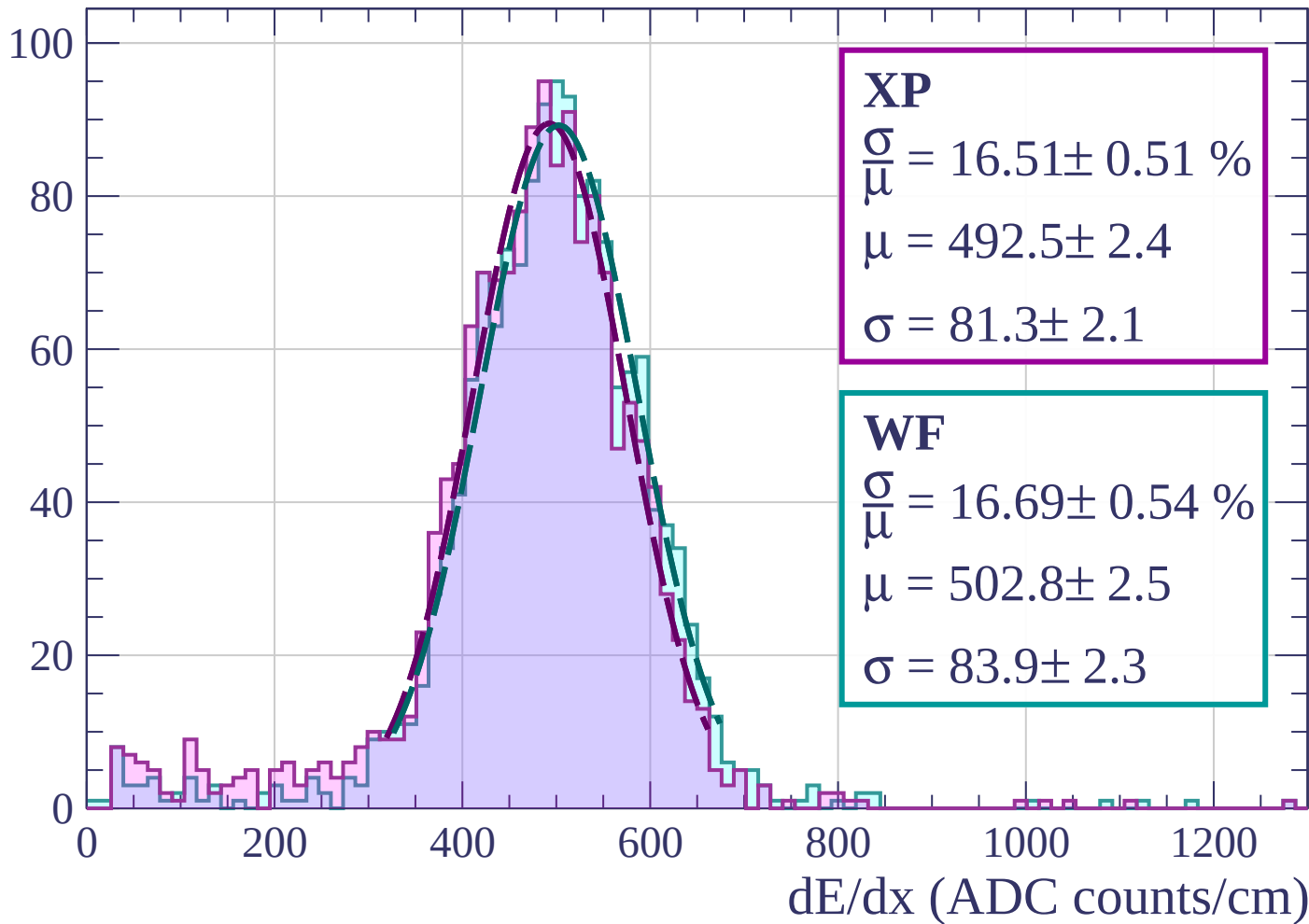
Energy loss | $27 < \theta < 30$

Count



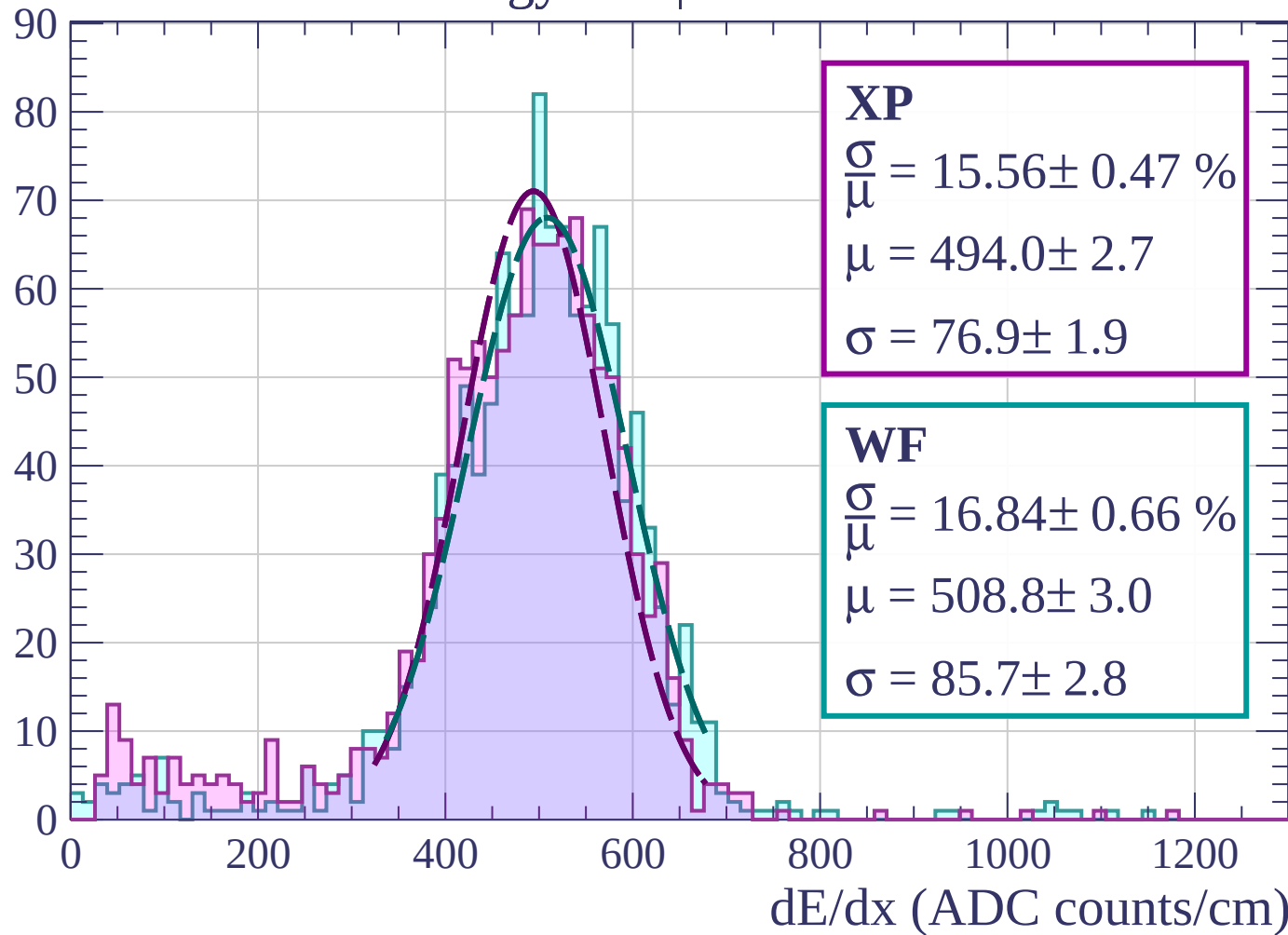
Energy loss | $30 < \theta < 33$

Count



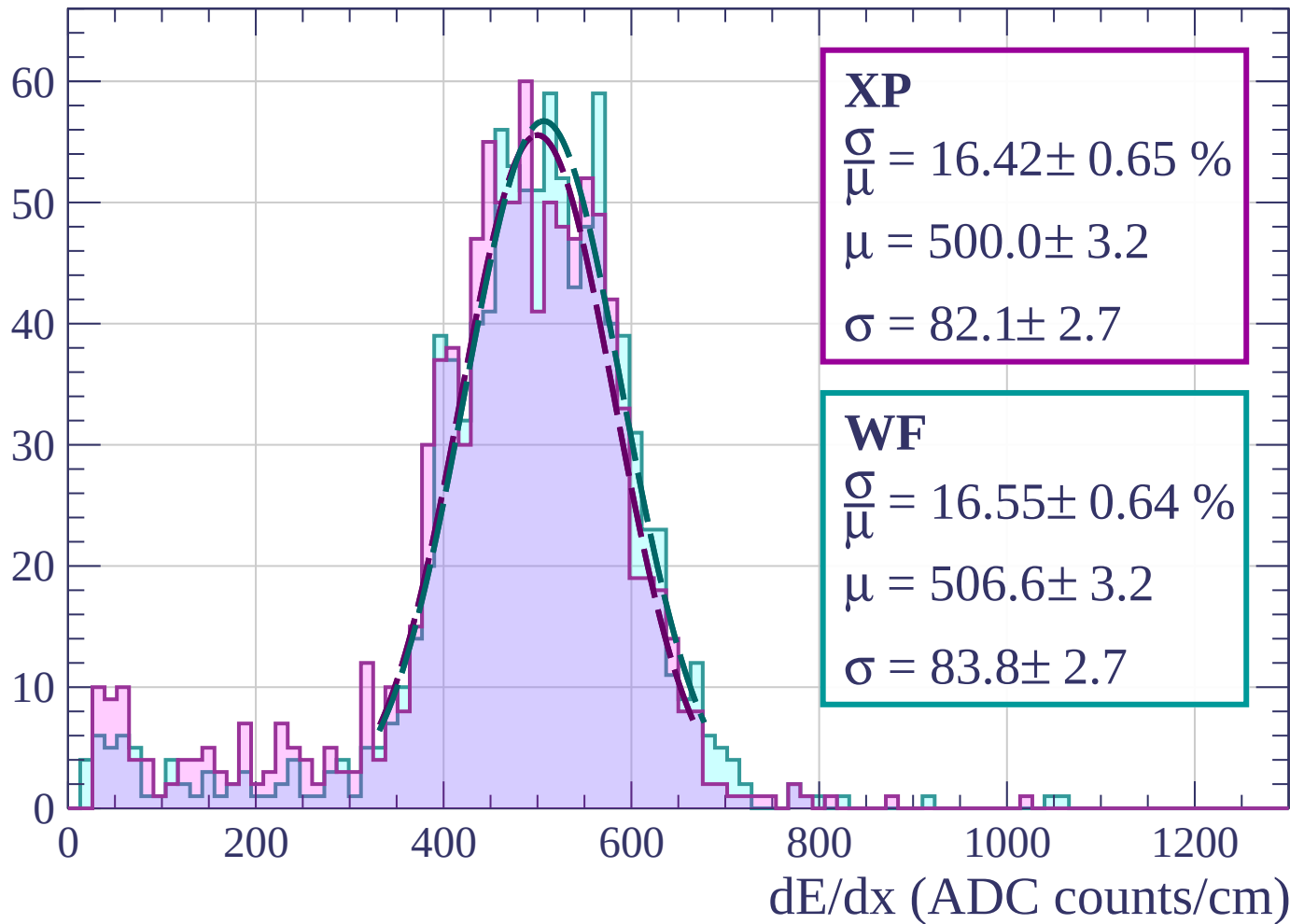
Energy loss | $33 < \theta < 36$

Count



Energy loss | $36 < \theta < 39$

Count



Energy loss | $39 < \theta < 42$

Count

XP

$$\frac{\sigma}{\mu} = 16.42 \pm 0.71 \%$$

$$\mu = 499.4 \pm 3.5$$

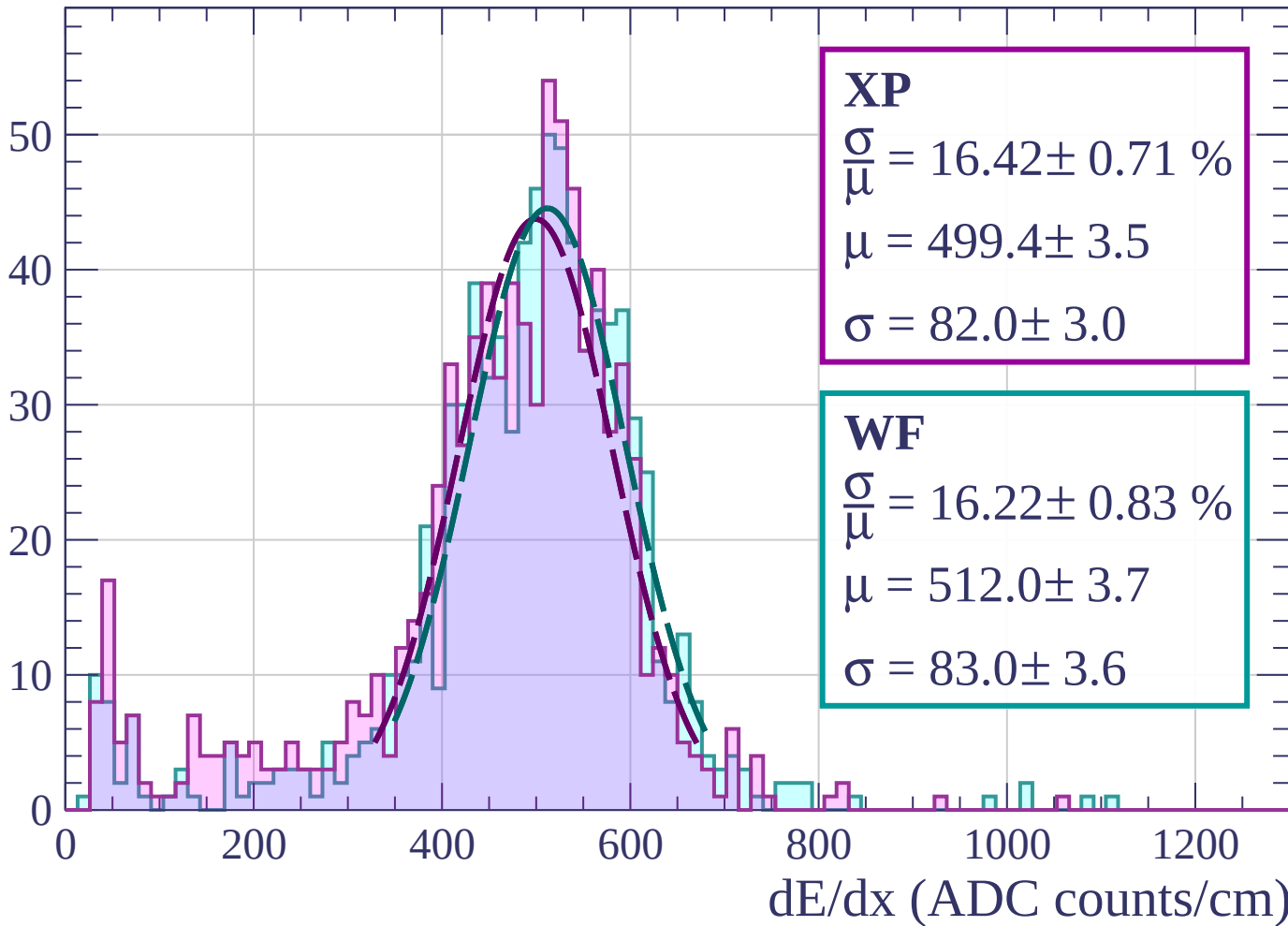
$$\sigma = 82.0 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 16.22 \pm 0.83 \%$$

$$\mu = 512.0 \pm 3.7$$

$$\sigma = 83.0 \pm 3.6$$



Energy loss | $42 < \theta < 45$

Count

XP

$$\frac{\sigma}{\mu} = 17.40 \pm 1.22 \%$$

$$\mu = 504.7 \pm 4.8$$

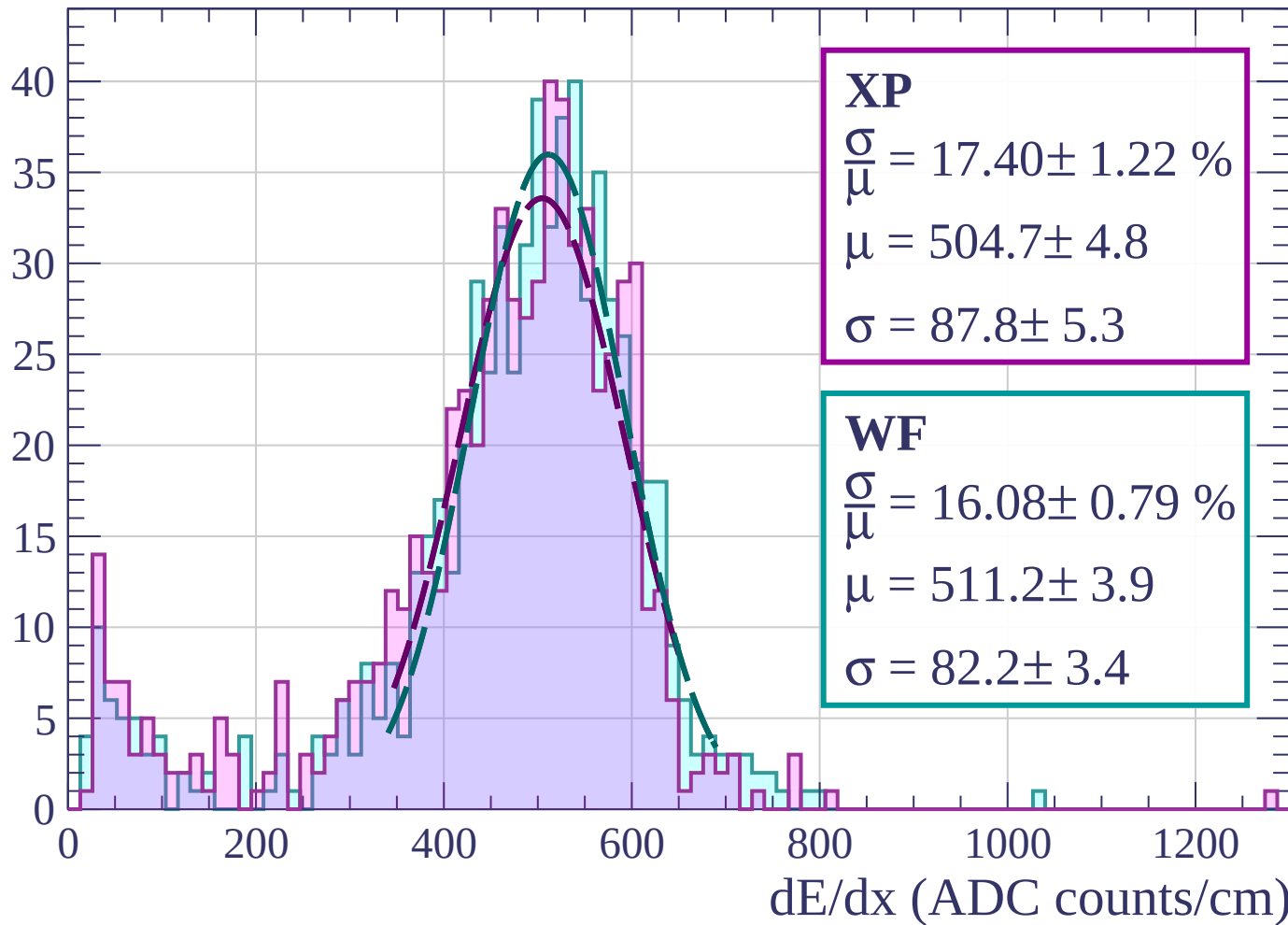
$$\sigma = 87.8 \pm 5.3$$

WF

$$\frac{\sigma}{\mu} = 16.08 \pm 0.79 \%$$

$$\mu = 511.2 \pm 3.9$$

$$\sigma = 82.2 \pm 3.4$$



Energy loss | $45 < \theta < 48$

Count

XP

$$\frac{\sigma}{\mu} = 18.94 \pm 1.09 \%$$

$$\mu = 504.3 \pm 5.2$$

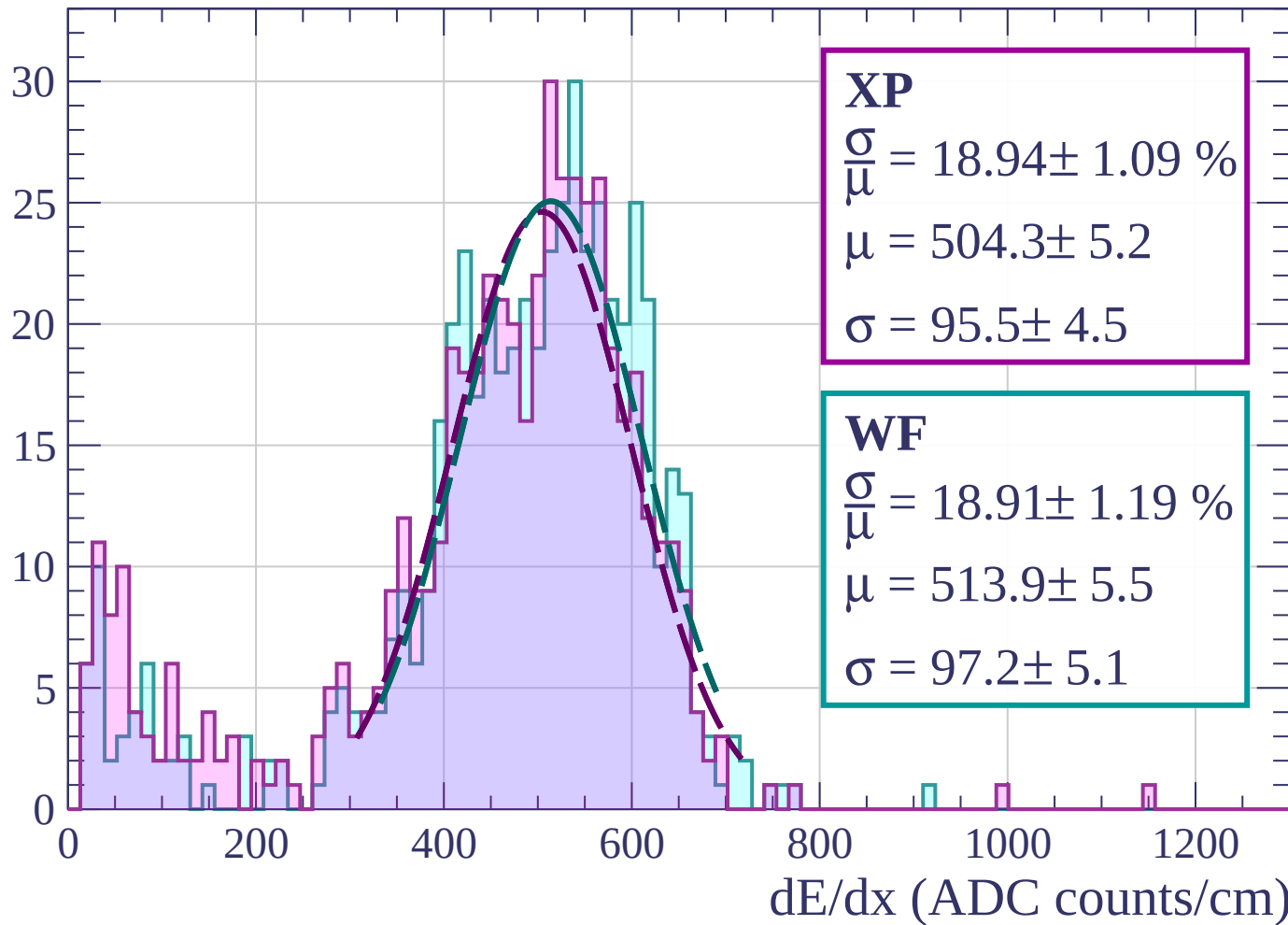
$$\sigma = 95.5 \pm 4.5$$

WF

$$\frac{\sigma}{\mu} = 18.91 \pm 1.19 \%$$

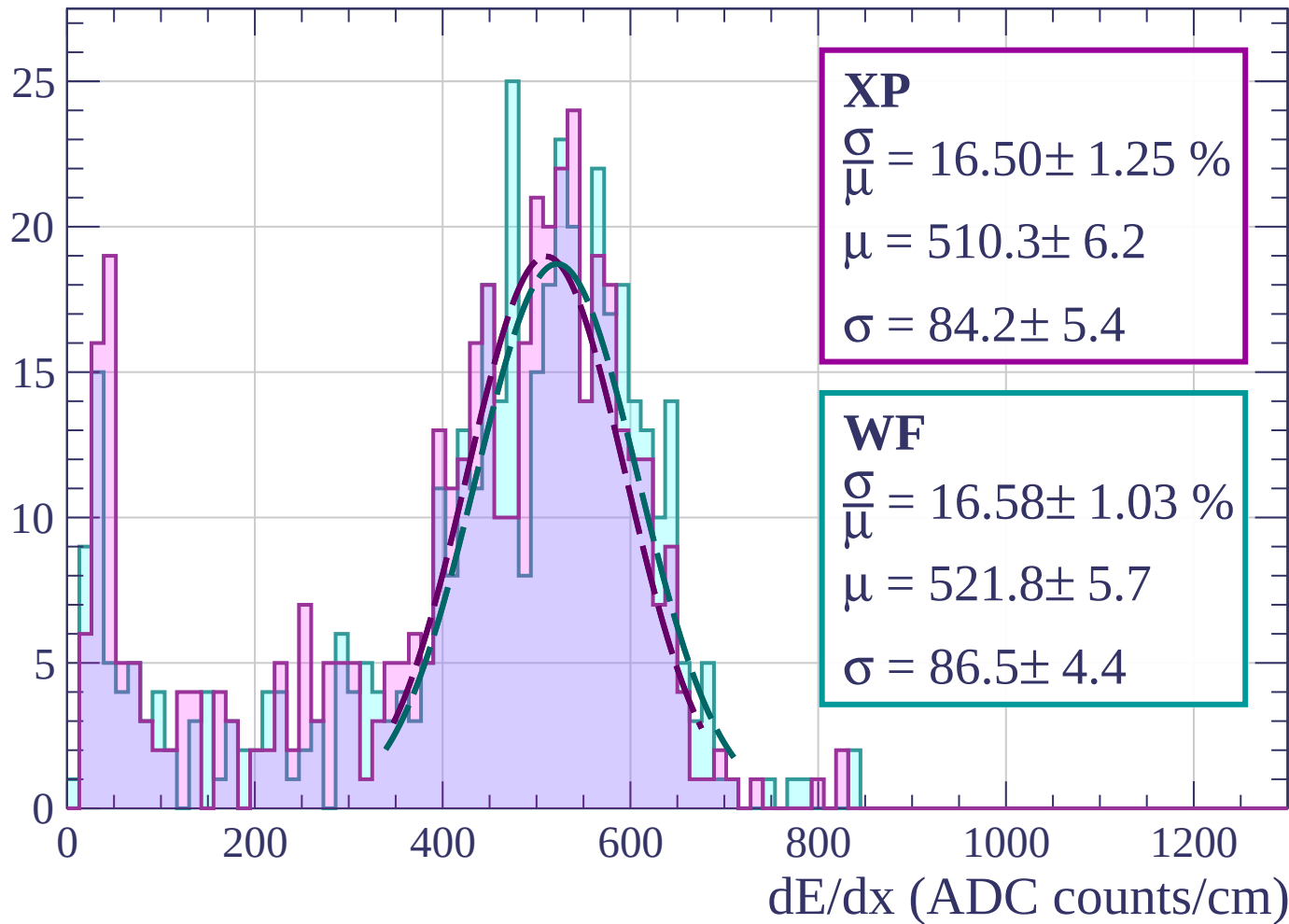
$$\mu = 513.9 \pm 5.5$$

$$\sigma = 97.2 \pm 5.1$$



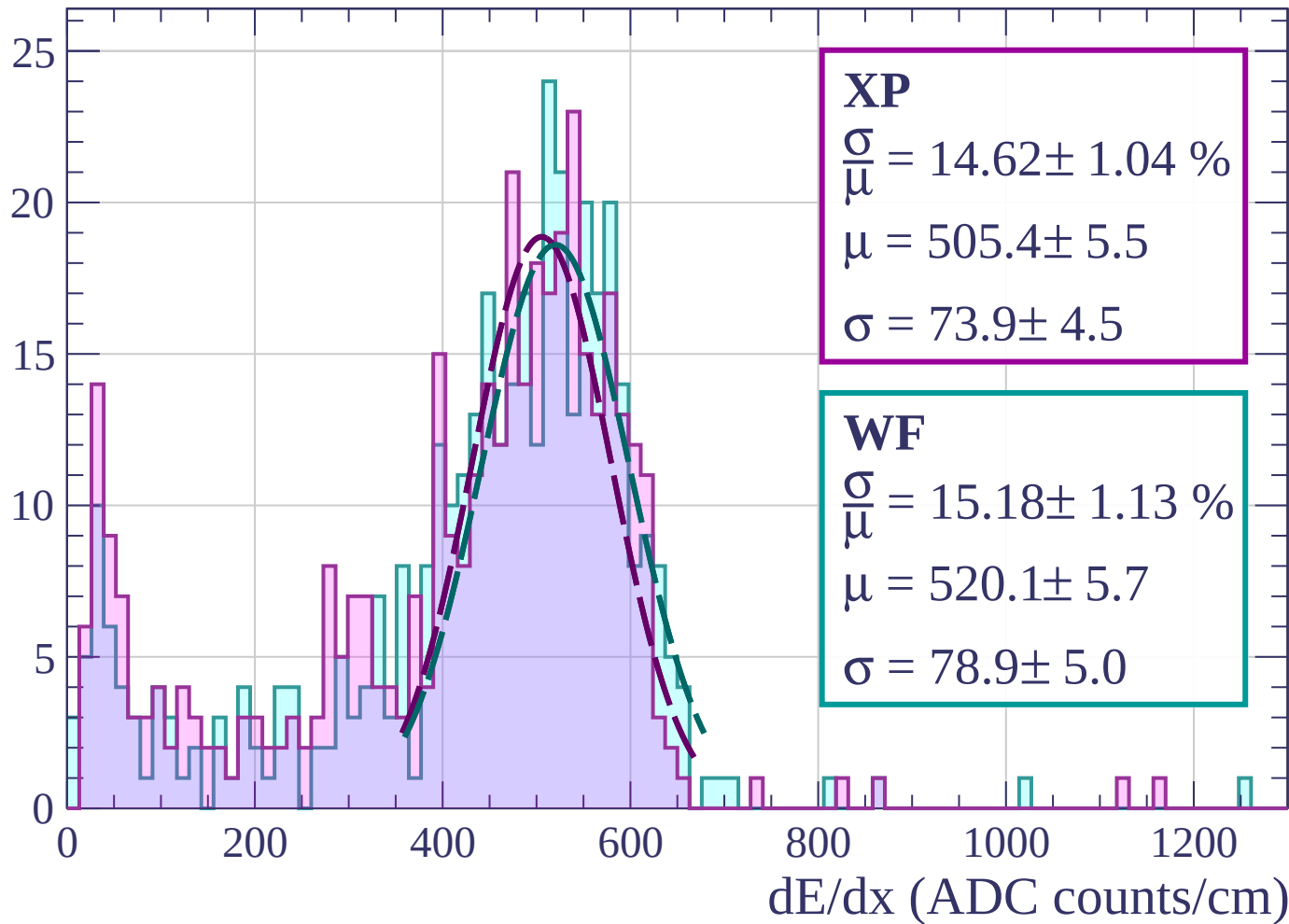
Energy loss | $48 < \theta < 51$

Count



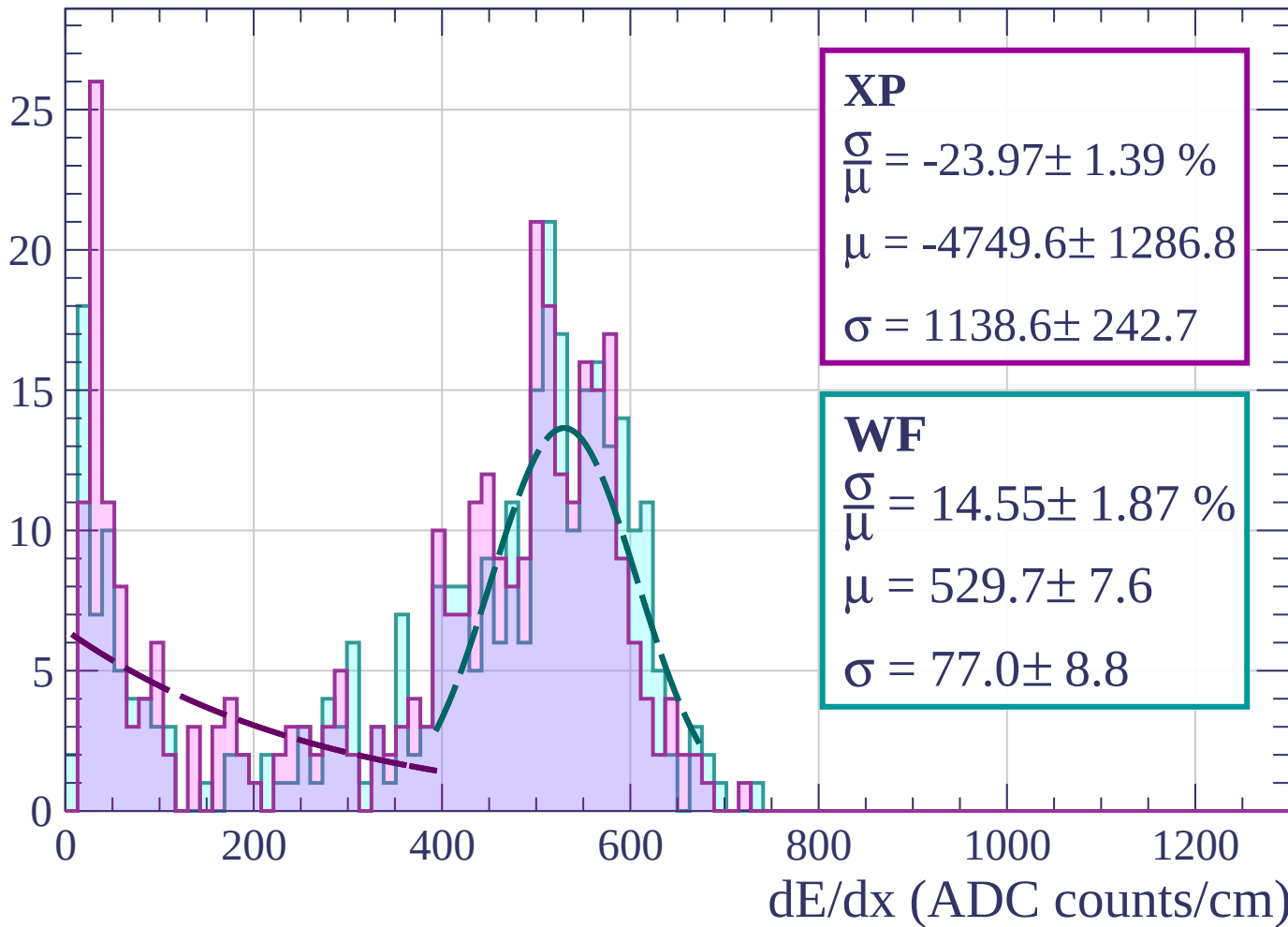
Energy loss | $51 < \theta < 54$

Count



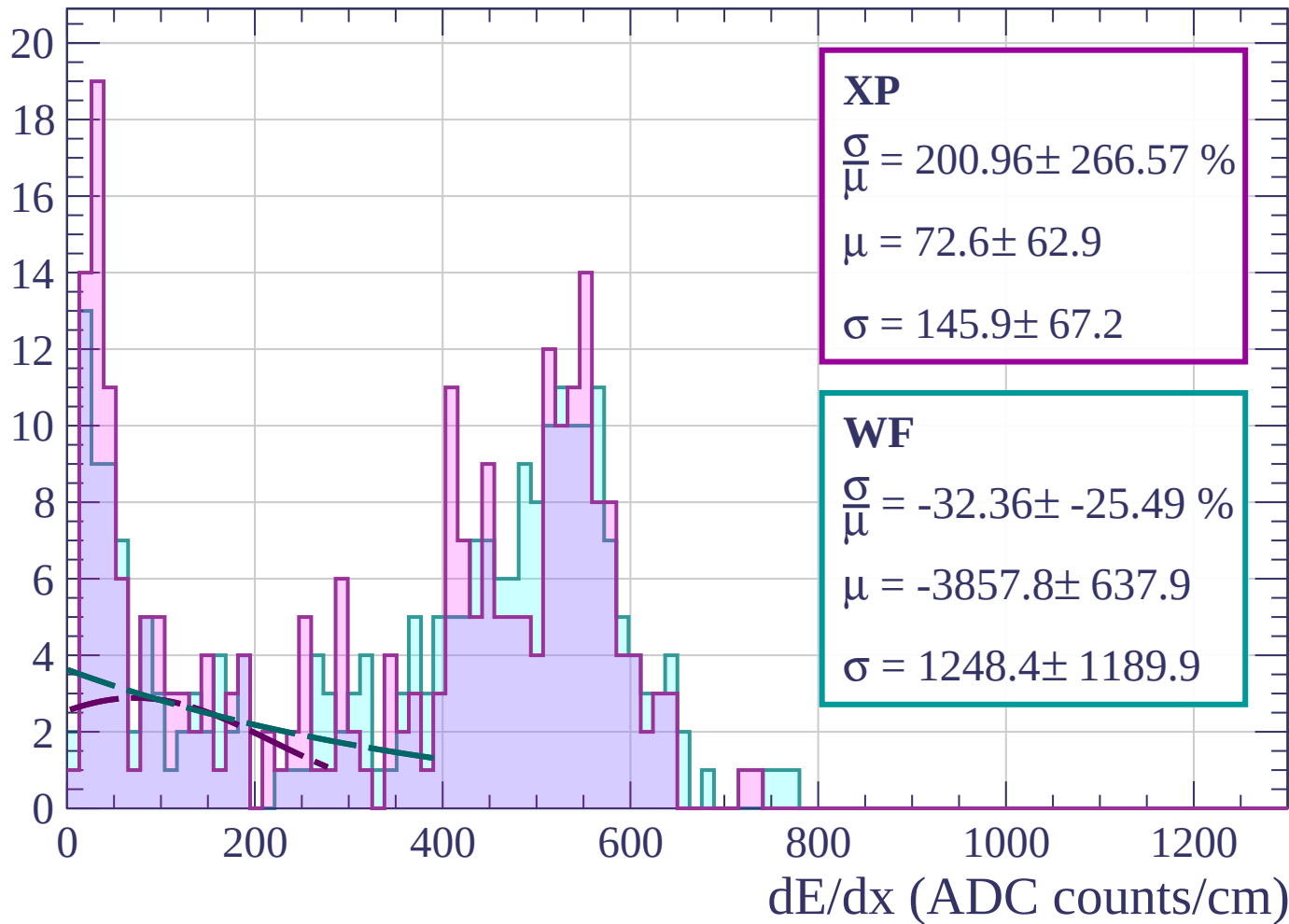
Energy loss | $54 < \theta < 57$

Count



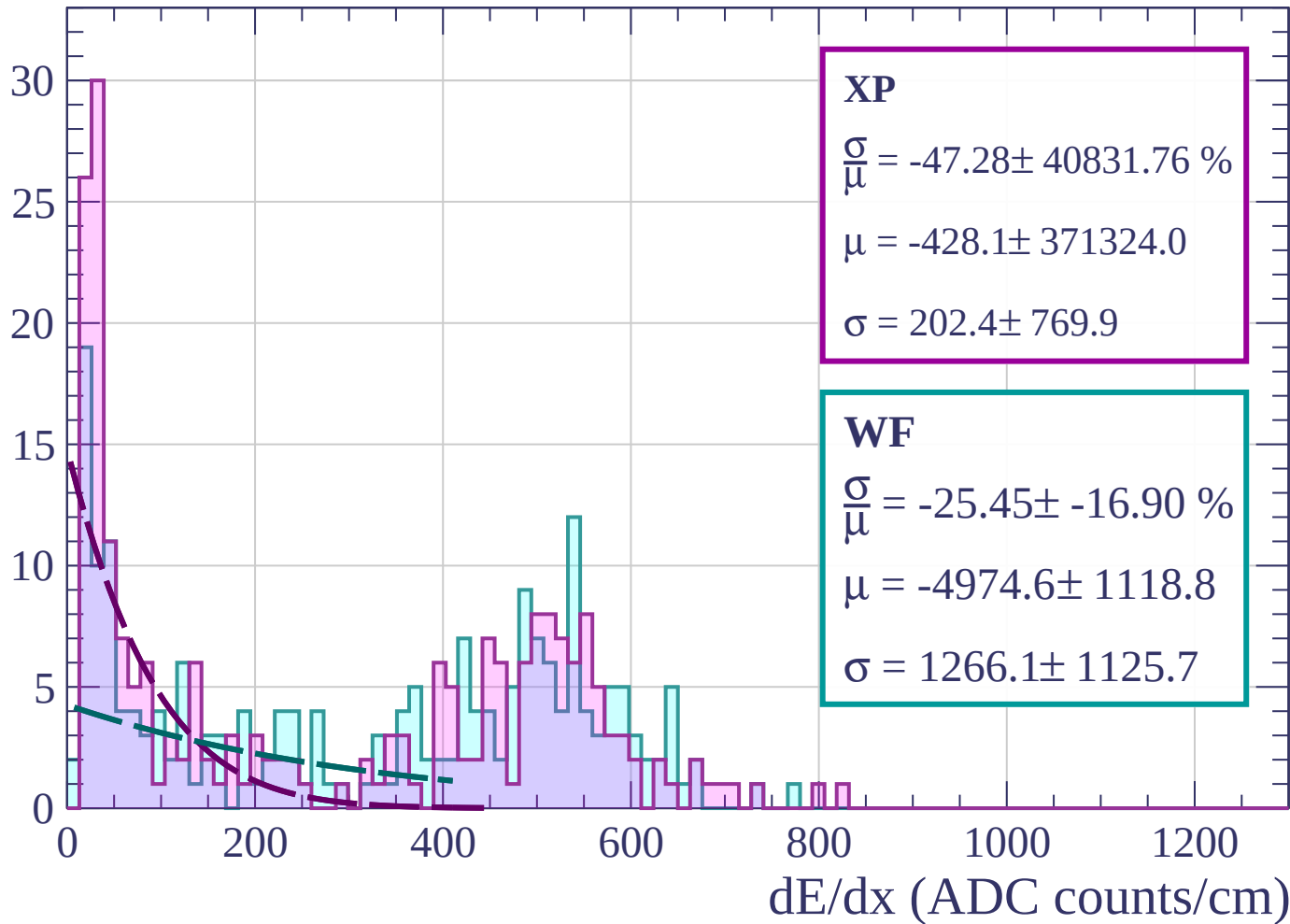
Energy loss | $57 < \theta < 60$

Count



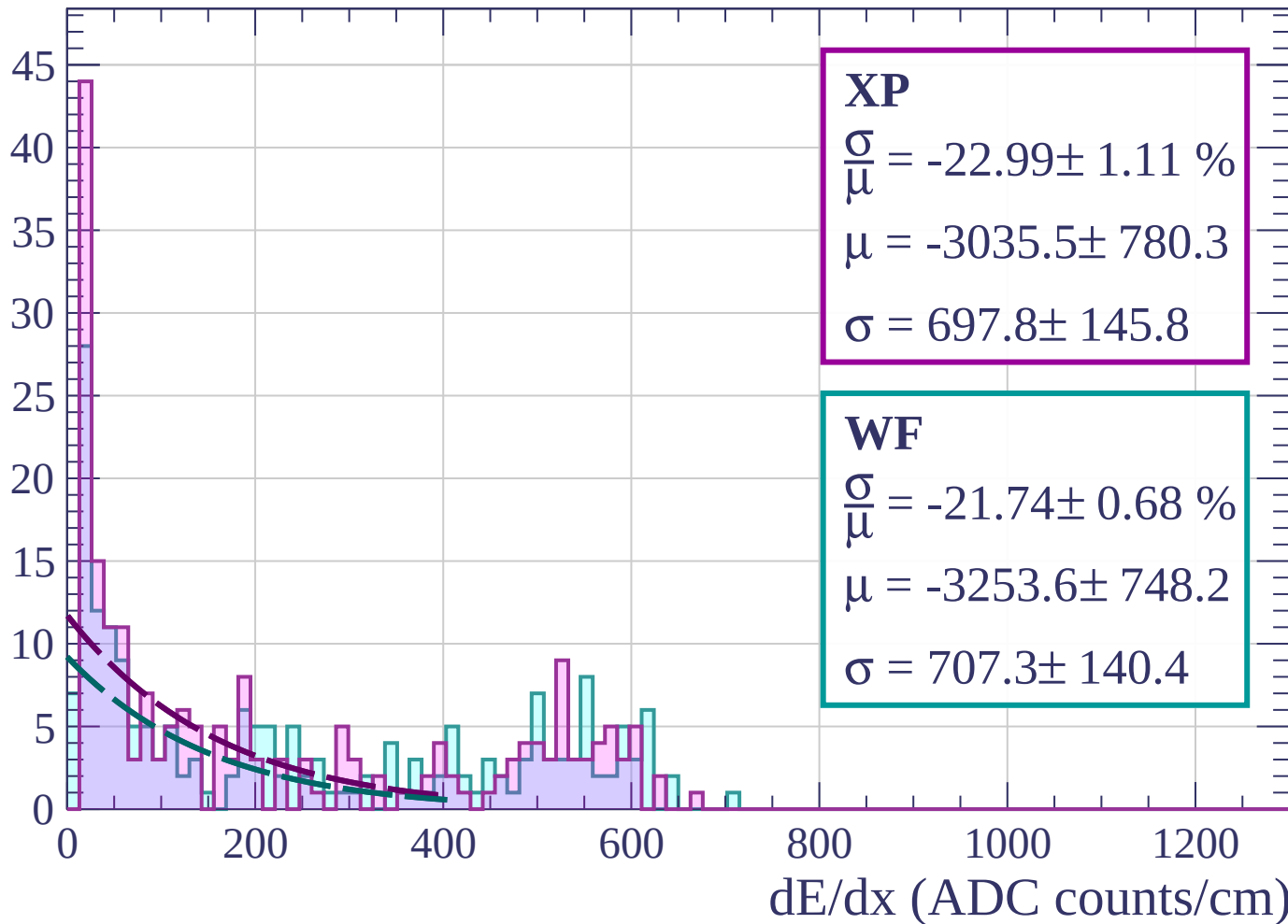
Energy loss | $60 < \theta < 63$

Count



Energy loss | $63 < \theta < 66$

Count



Energy loss | $66 < \theta < 69$

Count

XP

$$\frac{\sigma}{\mu} = 24.57 \pm 230.44 \%$$

$$\mu = 24.3 \pm 26.5$$

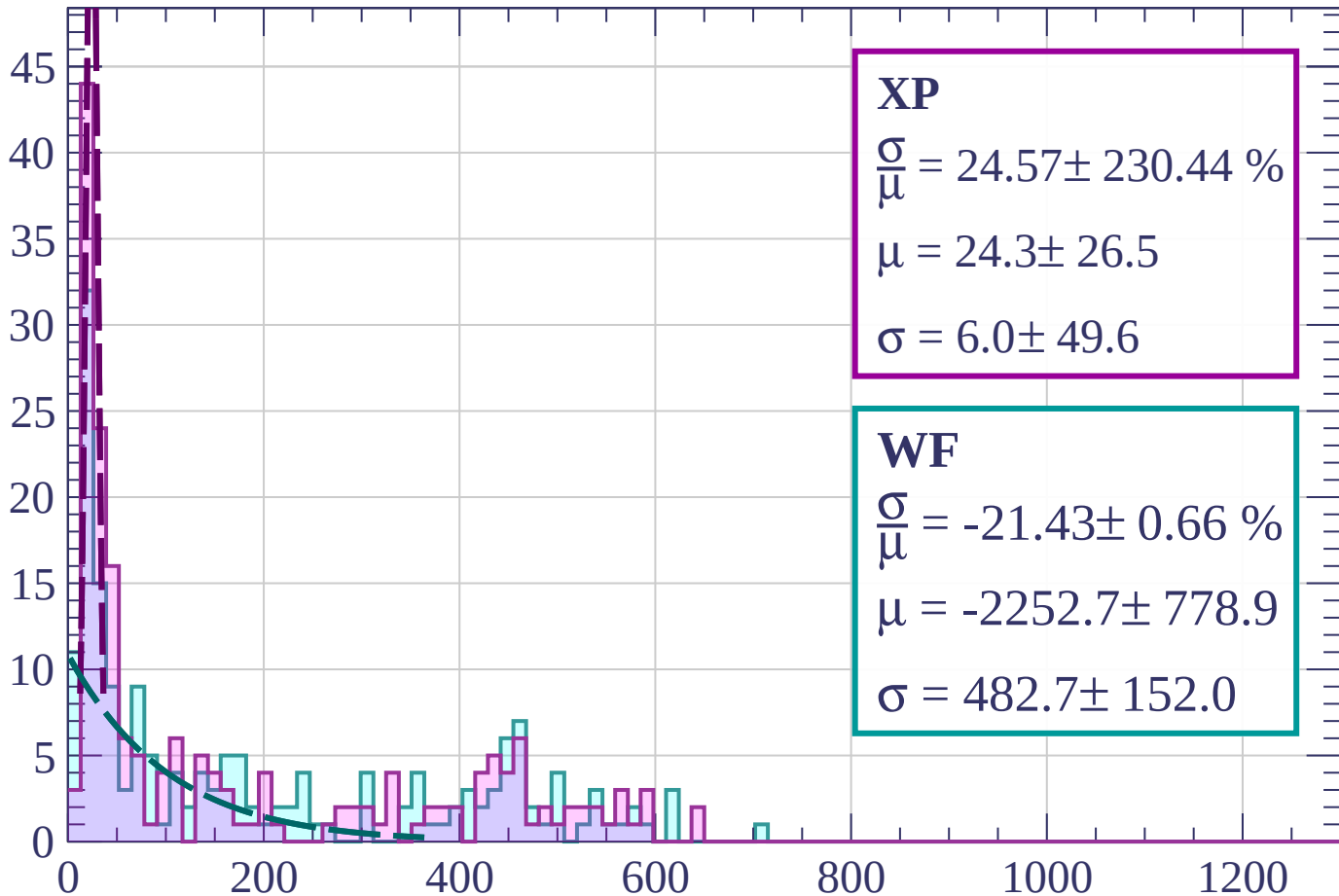
$$\sigma = 6.0 \pm 49.6$$

WF

$$\frac{\sigma}{\mu} = -21.43 \pm 0.66 \%$$

$$\mu = -2252.7 \pm 778.9$$

$$\sigma = 482.7 \pm 152.0$$



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

Count

XP

$$\frac{\sigma}{\mu} = 0.71 \pm 1.76 \%$$

$$\mu = 57.9 \pm 0.9$$

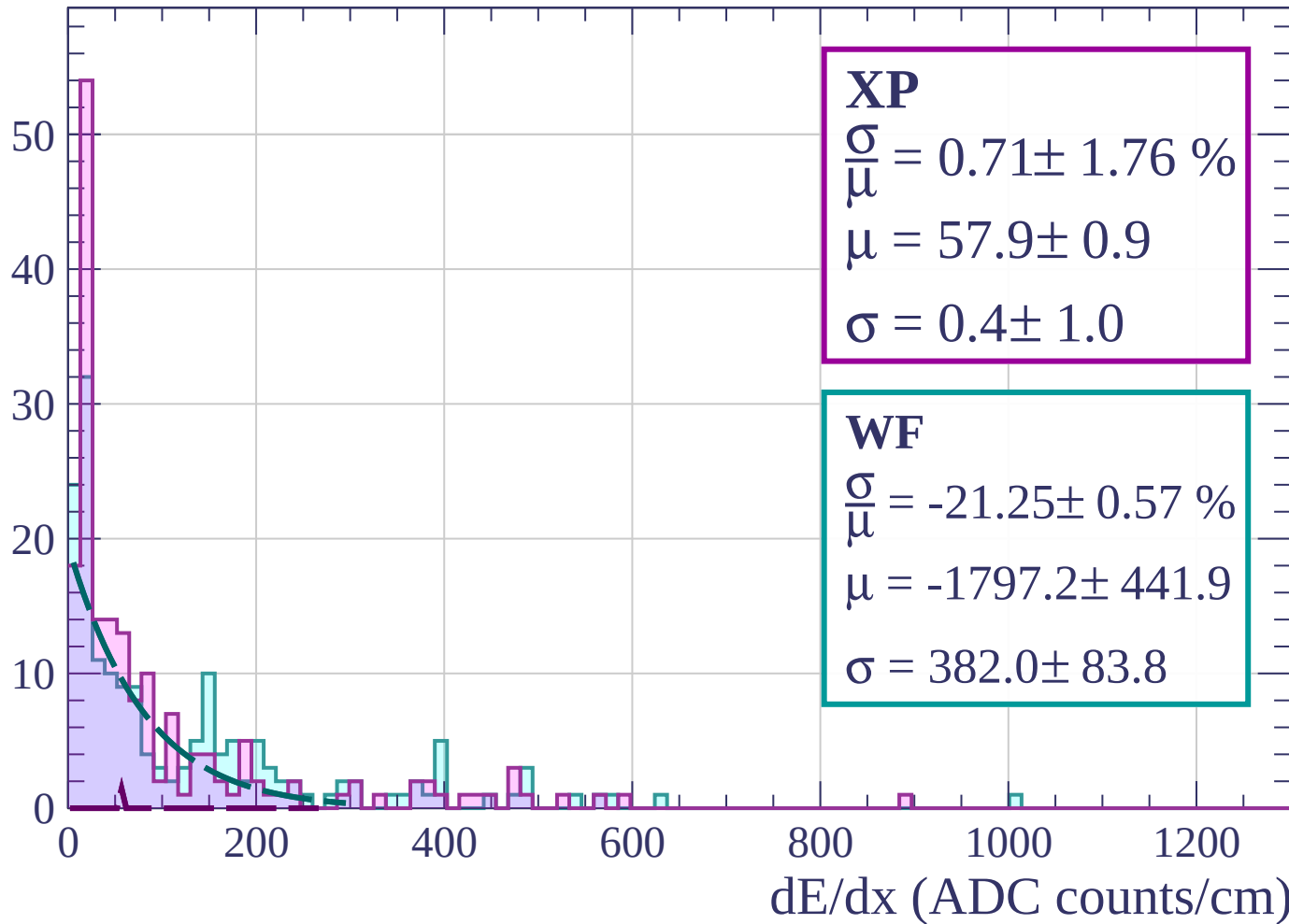
$$\sigma = 0.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = -21.25 \pm 0.57 \%$$

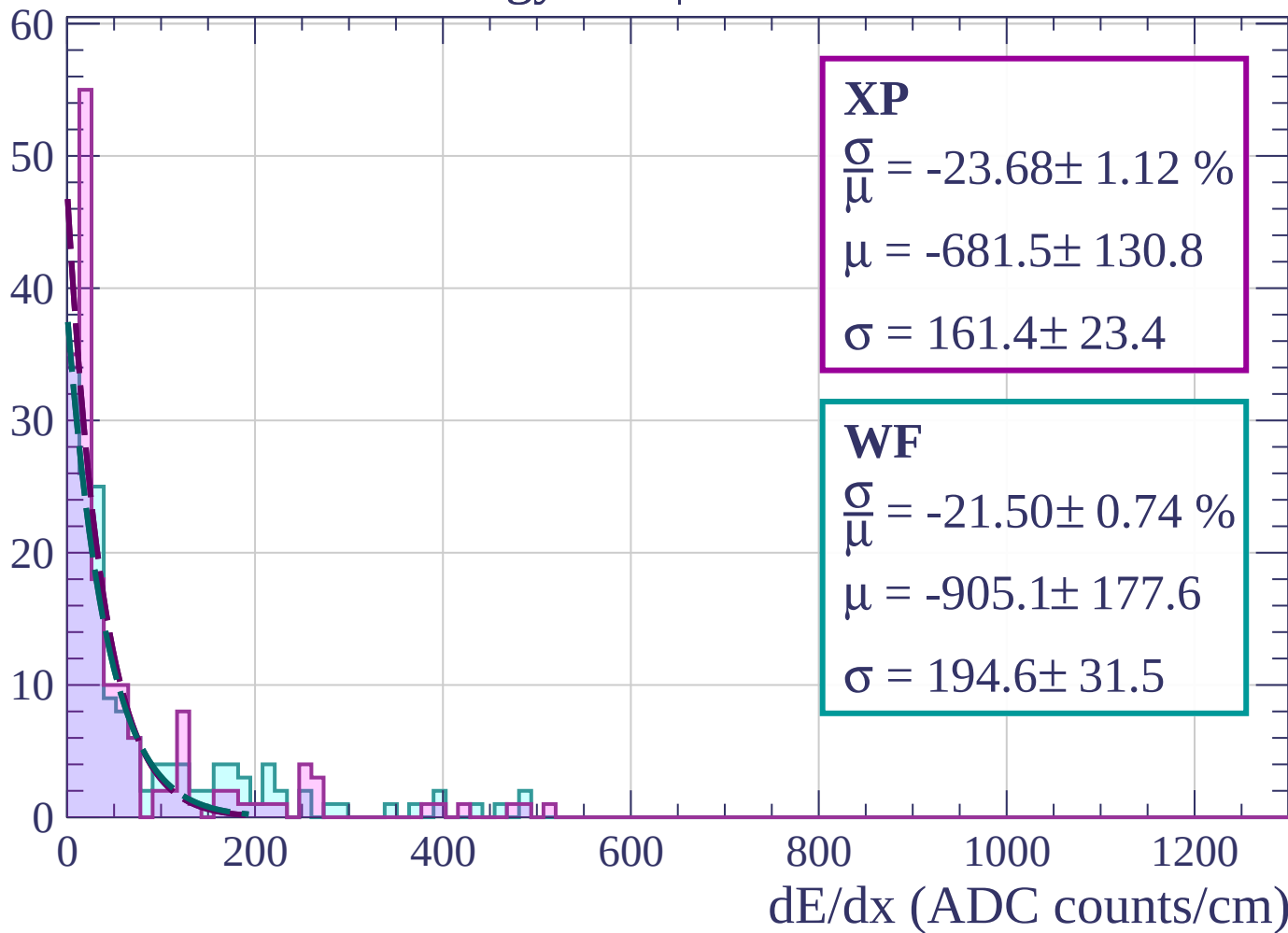
$$\mu = -1797.2 \pm 441.9$$

$$\sigma = 382.0 \pm 83.8$$



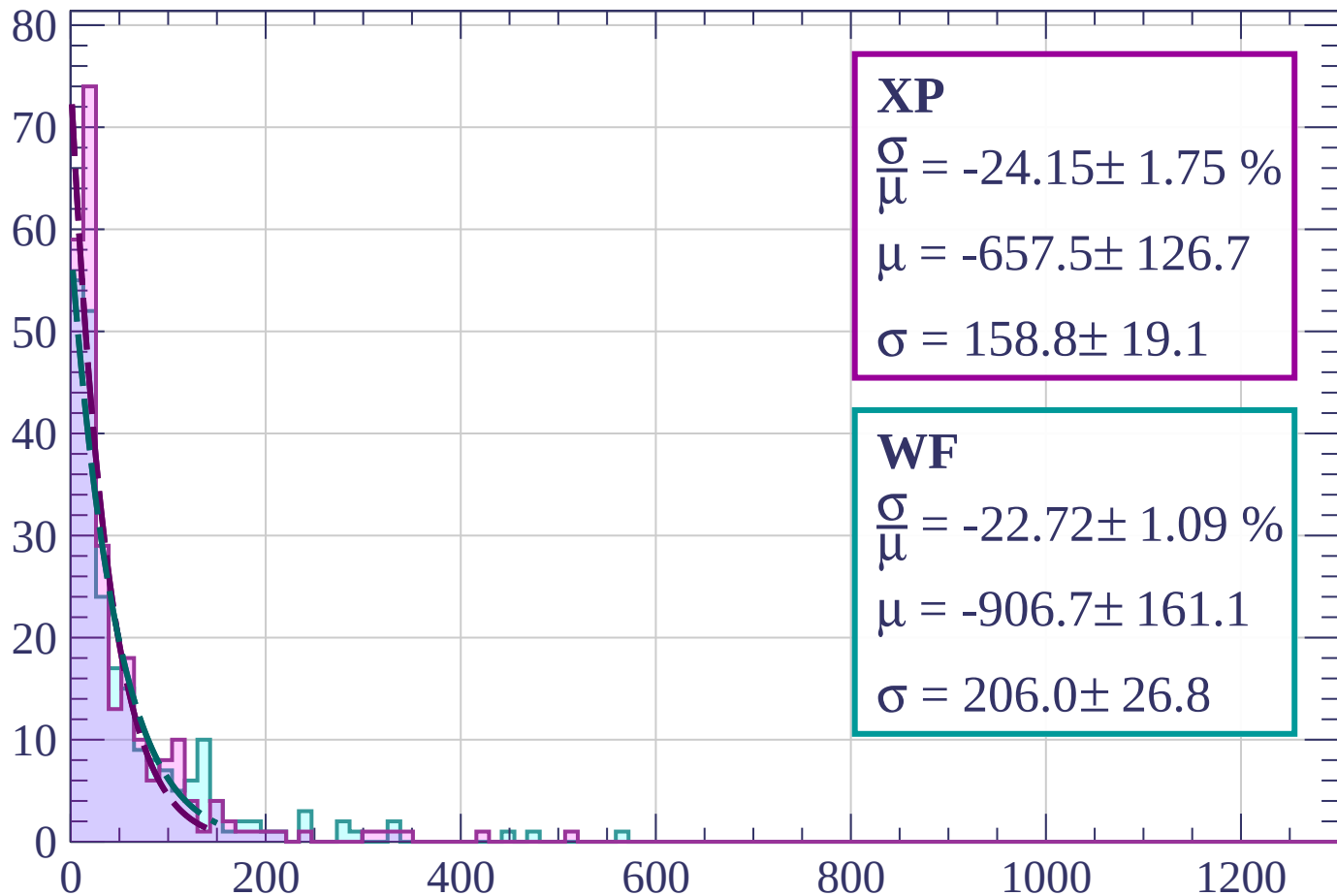
Energy loss | $72 < \theta < 75$

Count



Energy loss | $75 < \theta < 78$

Count



XP

$$\frac{\sigma}{\mu} = -24.15 \pm 1.75 \%$$

$$\mu = -657.5 \pm 126.7$$

$$\sigma = 158.8 \pm 19.1$$

WF

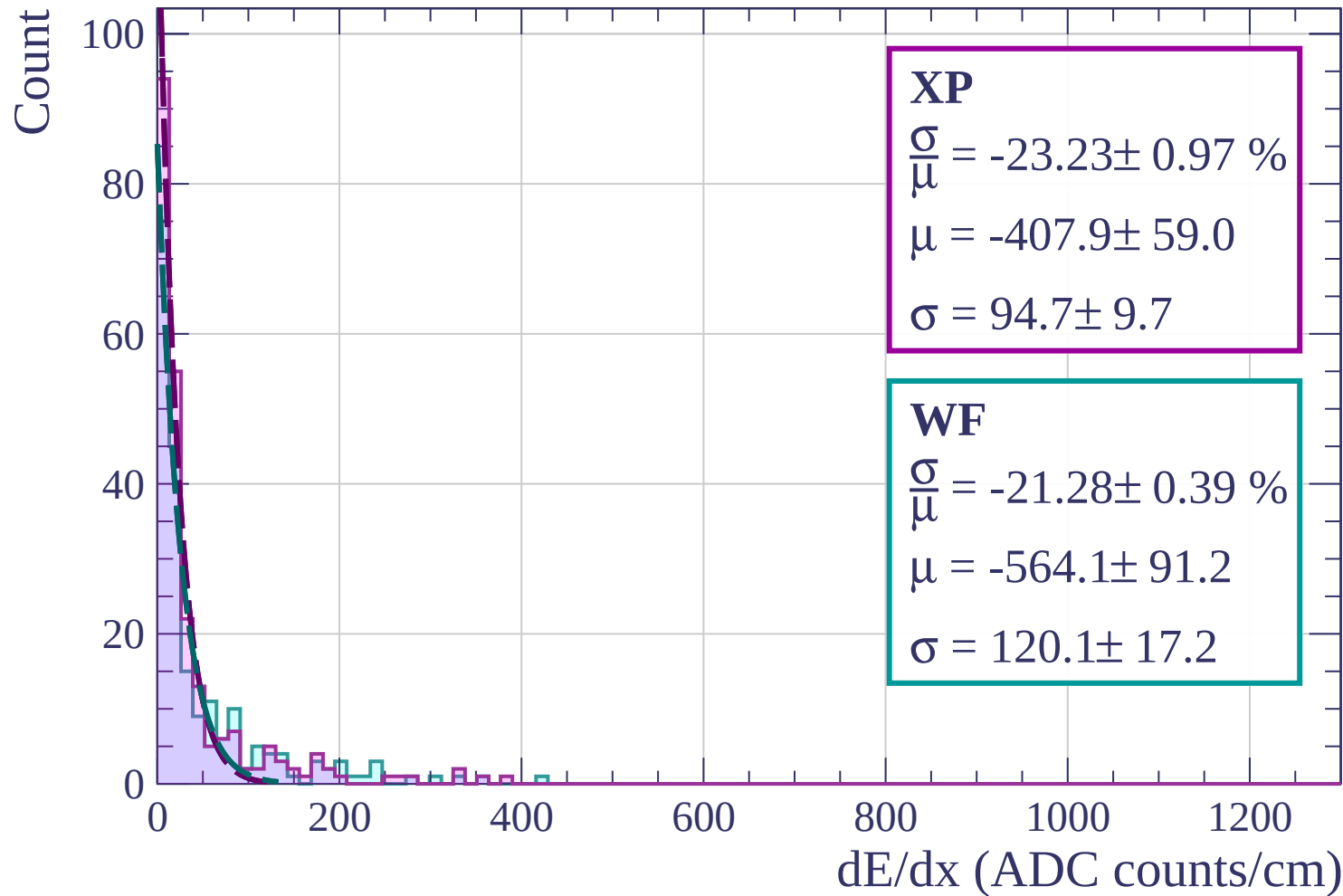
$$\frac{\sigma}{\mu} = -22.72 \pm 1.09 \%$$

$$\mu = -906.7 \pm 161.1$$

$$\sigma = 206.0 \pm 26.8$$

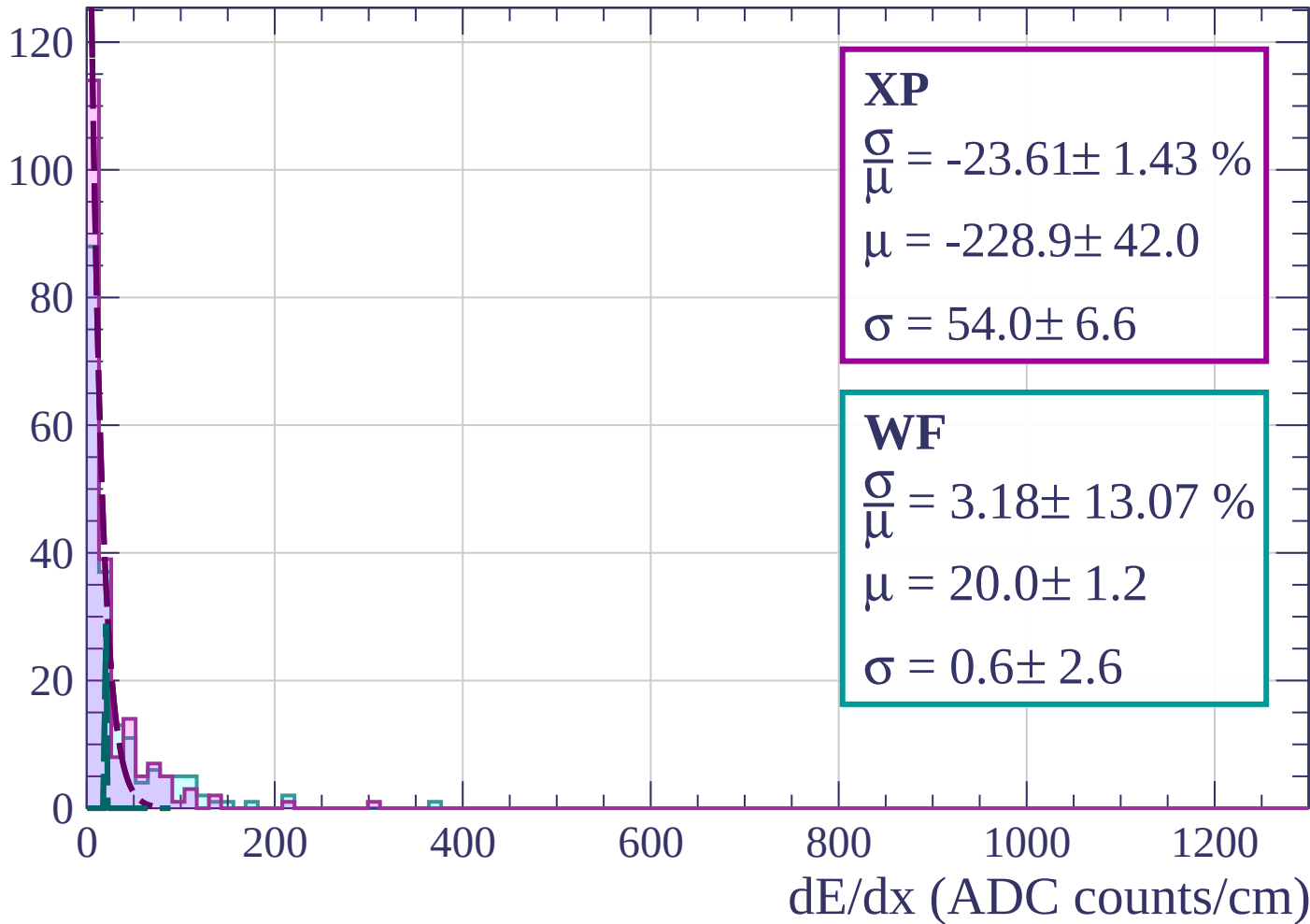
dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$



Energy loss | $81 < \theta < 84$

Count



Energy loss | $84 < \theta < 87$

Count

XP

$$\frac{\sigma}{\mu} = -24.33 \pm 3.66 \%$$

$$\mu = -148.9 \pm 51.1$$

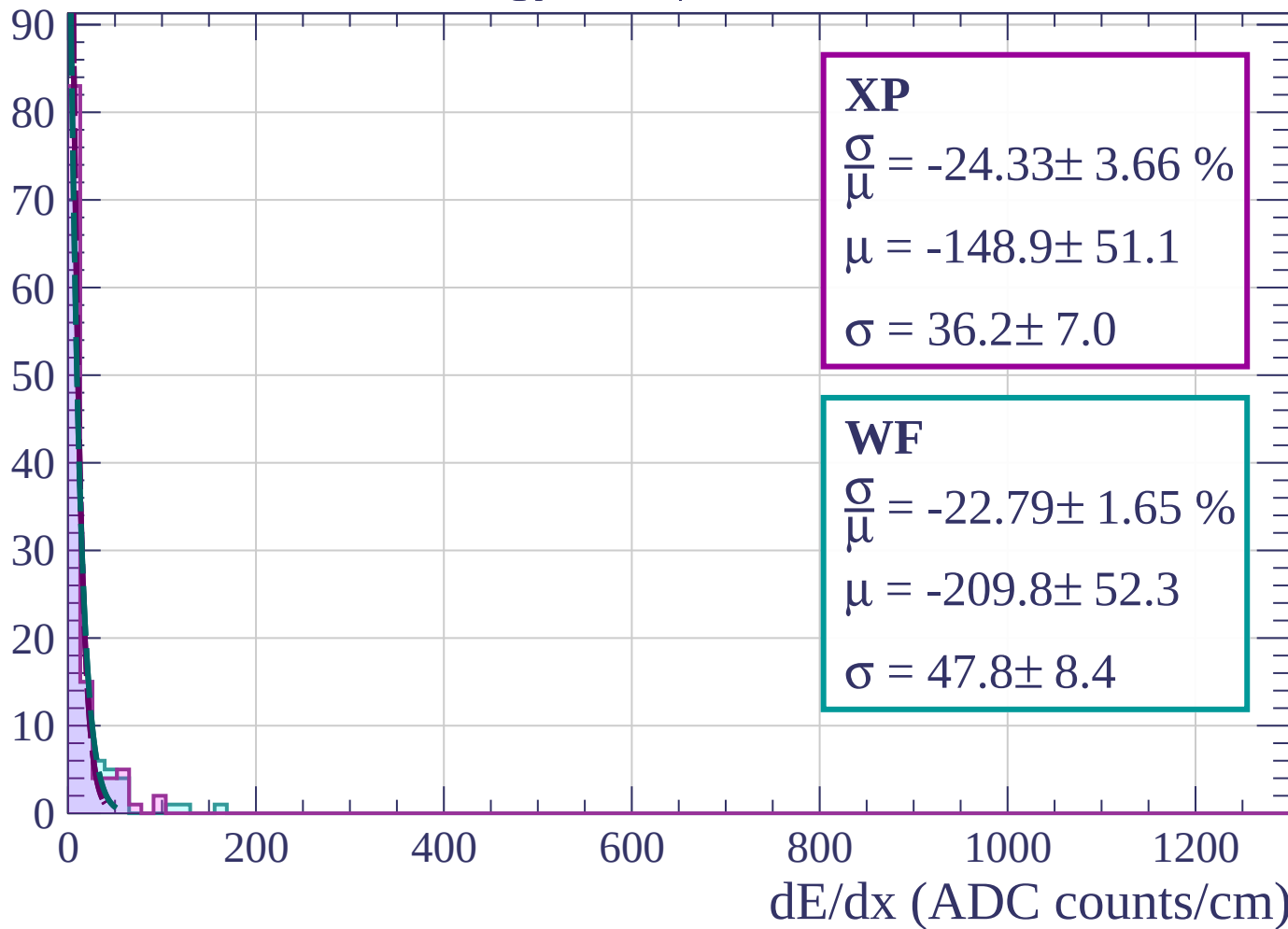
$$\sigma = 36.2 \pm 7.0$$

WF

$$\frac{\sigma}{\mu} = -22.79 \pm 1.65 \%$$

$$\mu = -209.8 \pm 52.3$$

$$\sigma = 47.8 \pm 8.4$$



Energy loss | $87 < \theta < 90$

Count

